

Contractibility of the complex of incompressible Seifert surfaces: the knot case of Kakimizu’s problem

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September 2, 2026

Abstract

Let $K \subset S^3$ be a knot and let $\text{IS}(K)$ be the simplicial complex whose vertices are the ambient isotopy classes of incompressible Seifert surfaces in the knot exterior $E(K)$, a finite set of distinct vertices spanning a simplex exactly when its classes admit simultaneously pairwise disjoint representatives; for an integer $\ell \geq g(K)$ let $\text{IS}_\ell(K)$ be the full subcomplex on the vertices of genus at most ℓ . Kakimizu asked whether $\text{IS}(K)$ is contractible, a question repeated by Przytycki and Schultens, who identified the obstruction: the projection used to prove connectedness is not known to be well defined on isotopy classes. We prove that $\text{IS}(K)$ and every $\text{IS}_\ell(K)$ are contractible, conditionally on the named theorems and the residual hypotheses listed in full in §12. The argument has two independent halves. The combinatorial half is unconditional apart from one homotopy-theoretic input isolated as (C1): every non-empty connected flag *exchange complex* is contractible, where an exchange complex is a complex carrying a complexity function subject to two axioms which require only that an exchanging vertex *exist*, never that one be selected — this is what circumvents the obstruction, and it uses no three-dimensional topology, no finiteness, no local finiteness, no bound on dimension or diameter. The geometric half verifies that $\text{IS}(K)$ is such a complex, through an exchange lemma for incompressible surfaces obtained by minimal-surface methods. Six appendices make the analytic and normal-surface inputs of that half self-contained: a version of the Hass–Scott area-minimisation theorem 6.12 for manifolds with boundary, a compactness and blow-up analysis for families of minimisers with fixed boundary, boundary regularity of the constructed minimiser, the Haken normalisation and the weight–area inequality, a self-contained treatment of divergence-form elliptic equations with Hölder coefficients, and the combinatorial lemmas. No claim of novelty or priority is made, and no formal verification is claimed; on how this manuscript was produced, see Appendix G.

2020 Mathematics Subject Classification. Primary 57K10; Secondary 57M50, 53A10, 35J15, 57Q35.

Key words and phrases. Kakimizu complex; incompressible Seifert surface; contractibility; flag complex; exchange complex; least-area surface; Haken normal surface.

1 Notation and statement

1.1 Knots, exteriors, Seifert surfaces

A *knot* is a smooth oriented embedding $K \hookrightarrow S^3$. Choose an open tubular neighbourhood $N(K)$ and write

$$E(K) = S^3 \setminus \text{Int } N(K)$$

for the compact *knot exterior*; its boundary is a torus. The *preferred longitude* is the oriented slope on $\partial E(K)$ whose linking number with K is zero.

A *Seifert surface in the exterior* is a compact, connected, orientable, neatly embedded surface $S \subset E(K)$ whose oriented boundary is a preferred longitude. Since S is connected with a single boundary component,

$$\chi(S) = 1 - 2g(S). \tag{1.1}$$

The *genus of the knot* is $g(K) = \min\{g(S)\}$.

A *compressing disc* for S is an embedded disc $D \subset E(K)$ with $D \cap S = \partial D$ such that ∂D is essential on S . If no compressing disc exists, S is *incompressible*. The definition of a vertex does *not* require boundary incompressibility, and no step of the chain below uses it.

An *ambient isotopy* is a smooth family of diffeomorphisms $H_t : E(K) \rightarrow E(K)$ with $H_0 = \text{id}$ and $H_1(S) = S'$. A *normal isotopy* is a smooth family $f_t : S \rightarrow E(K)$ of neat embeddings. That the restriction of an ambient isotopy is a normal isotopy is immediate; the converse is Lemma 4.1, so the two equivalence relations coincide here.

Two surfaces determine the same vertex if and only if they are ambient isotopic in $E(K)$. **The boundary longitude is allowed to move on the boundary torus**; the isotopy is not required to fix any chosen boundary curve pointwise.

The exterior is irreducible: an embedded sphere in $E(K)$, put back into S^3 , bounds two balls (Alexander–Schoenflies: a smoothly embedded 2-sphere in S^3 bounds two 3-balls; textbook-level, recorded in ledger B of §13), and the connected knot K misses the sphere and lies on one side, so the ball on the other side lies entirely inside $E(K)$.

1.2 The two complexes

The vertices of $\text{IS}(K)$ are the ambient isotopy classes of incompressible Seifert surfaces. A finite set of pairwise distinct vertices spans a simplex if and only if its classes have *simultaneously* pairwise disjoint representatives. For an integer $\ell \geq g(K)$,

$$\text{IS}_\ell(K) = \text{IS}(K) [\{ [S] : g(S) \leq \ell \}]$$

is the *full* subcomplex on the vertices of genus at most ℓ . In particular $\text{IS}_{g(K)}(K)$ is the minimal-genus Kakimizu complex $\text{MS}(K)$.

Write $d(u, w)$ for the graph distance in the 1-skeleton. A complex is *flag* if every finite clique in its 1-skeleton spans a simplex.

Remark (Two notions of disjointness that must be kept apart). An *edge* speaks only of a single pair of vertices, and different edges may use mutually incompatible representatives; a *simplex of dimension* ≥ 2 requires representatives chosen *once and for all* that are pairwise disjoint as a family. The passage from the former to the latter is supplied by Proposition 2.2, whose proof uses (E) and (U); it is never implicit.

1.3 The main theorem

Theorem 1.1 (Main theorem, conditional form). *Let $K \subset S^3$ be a knot and let $\ell \geq g(K)$ be an integer. Conditionally on the inputs listed in the table below — whose exact forms and provenance are given in the two tables of §12.2 — both $\text{IS}(K)$ and $\text{IS}_\ell(K)$ are contractible.*

<i>Input</i>	<i>Level</i>	<i>Supplied by</i>
(U) universal disjointness of relative area minimisers	research-level, cited (Schultens, Thm. 4 / Thm. 12; statement verified verbatim)	§12.2 (U); ledger A of §13
(E) the infimum of area in an isotopy class is attained by a smooth surface	proved here: Theorem B.3 (Appendix B), conditional on (H-sm), T3/T4/T7/T9 and Lemma 4.1	§12.2, entries H3, H5
(R1) the minimiser is $C^{1,1}$ up to $\partial E(K)$ and neat	proved here: Theorem C.3 (Appendix C), for the surface constructed in Theorem A.1, conditional on (H-cov), GP, Δ_2 , S402-G3	§12.2, entries H5–H7
the named theorems of Appendices A–E	research-level / textbook-level, listed one by one	first table of §12.2; §13
(C1) HEP + homotopy equivalence $\Rightarrow$ strong deformation retract	not proved here (combinatorial part, Lemma 9.5(1))	§12.2, entry H9

Theorem 1.2 (Combinatorial theorem). *Every non-empty connected flag exchange complex is contractible (Theorem 10.1). This statement depends on no geometric input and contains no finiteness, local finiteness, finite dimensionality or finite diameter hypothesis; it is conditional on (C1) (entry H9 of §12.2) and on the axiom of choice. The appeal to Whitehead’s theorem made in the earlier version is covered by Lemma F.7 (Appendix F).*

The conclusion is *contractibility*, not merely simple connectedness. Concerning the three items (U), (E), (R1):

- (U) is the universal reading of entry 2 of ledger A in §13. Its statement has been checked verbatim against the source (§12.2).
- (E) and (R1) were explicit hypotheses in the earlier version of this work. Here they are theorems: Theorem B.3 (compactness (G-b) $\Rightarrow$ (X₁) $\Rightarrow$ (E), Appendix B) and Theorem C.3 (boundary regularity of the surface constructed in Theorem A.1, Appendix C). Both are themselves conditional on the named theorems and residual conditions of the two tables of §12.2. The reading of (R1) adopted throughout — “the minimiser of (E)” means the surface F' constructed by Theorem A.1 over an optimal leaf ℓ_* — is a convention, stated and discussed in §C.5; under the alternative reading, Appendices B and C remain valid as conditional proofs and (E), (R1) revert to hypotheses.
- Given (R1) and the strictly convex boundary, transversality (R2) is a *corollary* rather than a hypothesis (Lemma 5.3; for F' it is immediate from Theorem C.3(a)).

Przytycki and Schultens posed exactly this question and located the obstruction:

“Kakimizu asks if $IS(L)$ is contractible as well. He proves that $IS(L)$ is connected, using a composition of the projection π with an additional operation, **which we do not know how to make well-defined on the set of isotopy classes of surfaces.**”

— *Trans. Amer. Math. Soc.* **364** (2012), no. 3, pp. 1492–1493.

The present paper answers the knot specialisation. It does *not* construct the well-defined choice of projection that they lack; the way around it is to write the exchange axiom so that it **requires only existence, never selectability** (§8).

1.4 The scope of the proof

The chain has two parts, of different degrees of self-containment. The self-containment record is kept as *two separate ledgers* in §13; the two ledgers must not be added together.

- **The combinatorial part (§§9–10) is self-contained (Appendix F).** Theorem 10.1, “every non-empty connected flag exchange complex is contractible”, uses only the two combinatorial axioms and the axiom of choice. It contains no three-dimensional topology, depends on none of the conclusions of §§1–8, and can be quoted and formalised on its own. The appeal to Whitehead’s theorem in the earlier version is covered by Lemma F.7, which proves the special case needed. The gluing step of Lemma 9.5(1) is conditional on **(C1)** — HEP together with a homotopy equivalence yields a strong deformation retract — which is *not proved here* and is the sole residual condition of the combinatorial part.
- **The geometric part (§§1–8) is conditional, with its external inputs re-audited.** It verifies that the complex of Seifert surfaces is an exchange complex. The eight external references of the earlier version have been made self-contained in Appendices A–E where possible; what remains is split in §13 into ledger A (research-level citations, listed by point of use) and ledger B (textbook-level facts, flagged explicitly). (E) and (R1) are supplied by Theorems B.3 and C.3.

§12.2 writes out the exact form of every residual condition. Its first table lists the named theorems (the “still conditional on” sections of the six appendices, merged and deduplicated); its second table lists the residual hypotheses H1–H14. Where a table entry records that the numbering in the original source has not been checked, that entry may not be used to claim closure.

No claim of novelty or priority is made, and no formal verification is claimed. On how this manuscript was produced, and on what has and has not been independently audited, see Appendix G.

2 Connectedness and flagness

Proposition 2.1 (Unknot and global connectedness). *$\text{IS}(\text{unknot})$ and $\text{IS}_\ell(\text{unknot})$ are single points. For non-trivial K , $\text{IS}(K)$ is connected.*

Remark. This proposition asserts only the connectedness of $\text{IS}(K)$ as a whole; it asserts nothing about the connectedness of any truncation $\text{IS}_\ell(K)$. Connectedness of the truncations is Corollary 9.3.1, derived internally from this proposition together with Theorem 9.3, and therefore *conditional* on the geometric part ((E), (R1), (U) and Theorem 7.1). An earlier draft ran the implication the other way, citing “every truncation is connected” from outside and deducing global connectedness from the nesting of truncations. That citation is no longer in the chain (entry 1 of §13 has been re-sourced), and keeping the old wording would close the chain into a cycle: Proposition 2.1 $\leftarrow$ §13 entry 1 $\leftarrow$ Corollary 9.3.1 $\leftarrow$ Theorem 9.3 $\leftarrow$ (N1),(N2) $\leftarrow$ Proposition 2.1. The direction must be *global* $\Rightarrow$ *truncated*, never the reverse.

Proof. For the unknot, $E(K)$ is a solid torus V (Corollary D.19: K unknotted $\Rightarrow E(K) \cong S^1 \times D^2$, Appendix D; for the converse see Theorem D.18), and the preferred longitude is a meridian of V .

Let $S \subset V$ be connected, orientable and incompressible with that single boundary curve. If some essential circle on S were null-homotopic in V , take a null-homotopy transverse to S with the least number of intersection curves and perform innermost-circle and outermost-arc surgery; this produces an embedded disc bounded by that circle, that is, a compressing disc. Hence incompressibility makes $\pi_1(S) \rightarrow \pi_1(V) \cong \mathbb{Z}$ injective. If $g(S) > 0$ then $\pi_1(S)$ is free of rank $2g(S) \geq 2$ and cannot embed in a cyclic group. So $g(S) = 0$, S is a disc, and its boundary is a meridian. Two meridian discs can be isotoped to have the same boundary; after minimising the number of intersection curves, innermost-circle surgery (using irreducibility of the solid torus as a 3-manifold) makes them disjoint, and the sphere assembled from the two discs bounds a ball (Alexander–Schoenflies, textbook-level, ledger B of §13), which gives an isotopy between them.

For non-trivial K : connectedness of the 1-skeleton of $\text{IS}(K)$ is *quoted* (entry 1 of §13, Kakimizu’s connectedness theorem, with *no* atoroidality hypothesis). $\square$

Remark (This citation is the single point of failure of the connectedness chain). Every demand for connectedness in this paper reduces, in the end, to that one sentence: Theorem 10.1 requires the complex to be connected, and connectedness of the truncations returns to it through Corollary 9.3.1. Consequently, *if Kakimizu’s original theorem gives connectedness only for $\text{MS}(K)$, the minimal-genus complex, and not for $\text{IS}(K)$, then the connectedness input of this paper does not exist* — and the old route “truncations connected $\Rightarrow$ globally connected” fails at the same time, its external source having been removed under the revision of §13 entry 1. What supports the statement at present is the verbatim report quoted in §1.3 from Przytycki–Schultens (*Trans. AMS* **364**, 2012, pp. 1492–1493): “He proves that $\text{IS}(L)$ is connected.” This is a published paper’s direct statement of Kakimizu’s conclusion, and it says IS , not MS . What is still to be checked in Kakimizu’s paper (*Hiroshima Math. J.* **22** (1992), 225–236) is the theorem number of that conclusion, whether it is stated for links or for knots, and whether the definition of a vertex there requires boundary incompressibility. A report is not the original statement; until the original has been read, entry 1 of §13 is recorded as “source to be verified” and must not be written up as though it had been.

Proposition 2.2 (Flagness). $\text{IS}(K)$ is flag, and hence so is every full subcomplex $\text{IS}_\ell(K)$.

Proof. Let $\{v_1, \dots, v_n\}$ be a clique in the 1-skeleton. The clique is finite and its vertices are distinct, so the corresponding surfaces are pairwise non-isotopic and all incompressible. By Theorem B.3 (Appendix B; this is (E), conditional on the list in §B.4.2), fix *once and for all*, for each i , a relative area minimiser A_i realising $A(v_i)$. For any pair $i \neq j$: $\{v_i, v_j\}$ is an edge, that is, the two isotopy classes *contain* disjoint representatives, so (U) applies (in the universal form of §12.2, taking the pairwise case, Thm. 4 of entry 2 of §13) and yields that A_i and A_j are either disjoint or coincide; and $v_i \neq v_j$ makes them non-isotopic, which excludes coincidence, so $A_i \cap A_j = \emptyset$. The A_i were *fixed before any comparison* and do not depend on the pair (i, j) , so $\{A_1, \dots, A_n\}$ is a family of *simultaneously* pairwise disjoint representatives, which is exactly the simplex required by flagness. Fullness makes the truncations retain the same simplices. $\square$

Remark (This proposition is conditional, and in the earlier draft it was not). The argument above uses Theorem B.3 (existence of a minimiser; its conditions are the named theorems of Appendix B together with (H-sm), see §12.2) and (U) (any two minimisers are disjoint). Flagness is therefore now *conditional on the named theorems of Appendix B and on (U)*. An earlier draft cited entry 5 of §13 instead (Neumann–Swarup, Lemma 2.3: “pairwise separable $\Rightarrow$ simultaneously separable”), which is a *purely topological, unconditional* input. Deleting entry 5 buys one fewer citation at the cost of turning an unconditional conclusion into a conditional one. Under the bookkeeping used here that trade is favourable, since self-containment is scored by counting external inputs; but it

must not be taken for an unqualified improvement. Should unconditional flagness be wanted later, restoring entry 5 to the chain suffices, and the rest of the argument is unaffected.

Proposition 2.3 (Non-emptiness). *For every knot K , $\text{IS}(K) \neq \emptyset$; and for every integer $\ell \geq g(K)$, $\text{IS}_\ell(K) \neq \emptyset$.*

Proof. Take any regular projection diagram of K and run Seifert's algorithm on it: smooth the crossings compatibly with the orientation to obtain disjoint Seifert circles, fill the circles with disjoint horizontal discs using the nesting depth as height, and attach a half-twisted band at each original crossing. The resulting $S_0 \subset S^3$ is a compact oriented embedded surface with $\partial S_0 = K$; the discs and bands are joined into one piece by the connectedness of the diagram, so S_0 is connected. Put S_0 transverse to $N(K)$, meeting it in an annulus; then $S = S_0 \cap E(K)$ is a compact connected oriented neatly embedded surface in $E(K)$ whose oriented boundary is the curve on $\partial E(K)$ with linking number zero with K , that is, a preferred longitude. So S is a Seifert surface in the sense of §1.1.

If S is incompressible, $[S]$ is already a vertex. Otherwise compress along an essential circle and keep the component carrying the longitudinal boundary; Lemma 3.1 says that each such compression strictly lowers the genus, and the genus is a non-negative integer, so after finitely many steps the process terminates at an incompressible Seifert surface S' whose boundary is still a preferred longitude. Then $[S'] \in \text{IS}(K)^{(0)}$, so $\text{IS}(K) \neq \emptyset$.

For the truncations: among all Seifert surfaces take one of least genus, $S_{\min}$, so that $g(S_{\min}) = g(K)$ by the definition of $g(K)$ (the set is non-empty by the previous paragraph). Then $S_{\min}$ is incompressible, for a compression would give a Seifert surface of smaller genus, contradicting minimality. So $[S_{\min}]$ is a vertex of genus $g(K)$ and lies in every $\text{IS}_\ell(K)$ with $\ell \geq g(K)$. $\square$

3 Compression and robustness of neighbours

Throughout this section A and B are transverse, neatly embedded compact surfaces. Cutting both along $A \cap B$, the resulting components are called *patches*, and the intersection curves and arcs duplicated on their boundaries are called *exchange seams*. “A disc component of $A \triangle B$ ” means a *disc patch*, not a component of the set-theoretic symmetric difference of two point sets.

Lemma 3.1 (Euler characteristic and compression). *Let A, B be compact and transverse. Cut along their intersection 1-manifold and reglue by the orientation-compatible normal exchange; let P_{tot} denote the union of all output components. Then*

$$\chi(P_{\text{tot}}) = \chi(A) + \chi(B). \quad (3.1)$$

If $P_{\text{tot}} = P \sqcup C_1 \sqcup \cdots \sqcup C_m$, then

$$\chi(P) = \chi(A) + \chi(B) - \sum_j \chi(C_j). \quad (3.2)$$

Moreover, compressing a connected Seifert surface along an essential circle and keeping the component that carries the longitudinal boundary strictly lowers the genus; iterating must therefore terminate, after finitely many steps, at an incompressible Seifert surface.

Proof. Triangulate $A \cup B$ so that each intersection circle and arc is a subcomplex. Cutting duplicates exactly the cells of the intersection trace, and the normal regluing identifies the same cells again, so the Euler correction is the same before and after the exchange; this gives (3.1), and (3.2) follows by additivity over components.

When one compresses along a non-separating circle, the surgery removes a handle and keeps the unique boundary component, so the genus drops by one. When one compresses along a separating essential circle, one side is closed and the other carries the original longitude; the closed side has positive genus (otherwise the circle would bound a disc), so discarding it strictly lowers the genus of the component with boundary. The genus is a non-negative integer, so such a sequence of compressions is finite. When no compression is available, the surface is incompressible by definition. $\square$

Lemma 3.2 (Prescribed compressions performed away from a fixed neighbour). *Let M be an irreducible orientable 3-manifold, let $P \subset M$ be connected, neatly embedded, with non-empty boundary, and let $Q \subset M$ be incompressible, neatly embedded and disjoint from P . Given any finite compression sequence for P (each step keeping the component with boundary), there is a compression sequence whose intermediate and final ambient isotopy classes agree with the given ones one by one, and all of whose discs are disjoint from Q . In particular the final surface has a representative disjoint from Q .*

Proof. First clean a single prescribed compressing disc D . Put D transverse to Q . Since $\partial D \subset P$ and $P \cap Q = \emptyset$, the intersection $D \cap Q$ consists of circles only. Each such circle bounds a subdisc of D and is therefore null-homotopic in M ; incompressibility of Q makes it null-homotopic in Q , hence it bounds a disc in Q as well. Choose an innermost such disc $E \subset Q$, so that $\text{Int } E \cap D = \emptyset$. The same circle γ bounds in D a unique subdisc F not containing ∂D , and by irreducibility the sphere $E \cup F$ bounds a ball B .

That sphere is disjoint from P : the ball lies in the interior of M , while P is connected with its boundary on ∂M , so P lies outside the ball. Push F across B to a parallel copy of E . This is an isotopy of D rel ∂D , during which the interiors of the discs stay disjoint from P , and it removes γ together with every intersection circle inside F . Repeating yields a compressing disc D' disjoint from Q . Since D' and D are isotopic rel boundary through compressing discs, the surface obtained by compressing along D' is *ambient isotopic* to the one obtained by compressing along D .

Now induct along the prescribed sequence: at each step transport the next prescribed disc to the current representative, clean it as above, and compress. The component with boundary stays disjoint from Q , has the prescribed next ambient isotopy class, and furnishes the input for the next step. The discarded closed components play no role. $\square$

Corollary 3.3 (Robustness of neighbours). *If a connected coarse exchange surface P is disjoint from an incompressible surface Q , then every ambient isotopy class obtained by compressing P down to an incompressible descendant with boundary is, in $\text{IS}(K)$, equal or adjacent to $[Q]$.*

Proof. Apply Lemma 3.2 to the chosen compression sequence; its final representative is disjoint from Q . If the two classes are the same they are equal, and otherwise their disjoint representatives span an edge. $\square$

4 Normal isotopies extend to ambient isotopies

Lemma 4.1 (Isotopy extension). *Let M be a compact smooth manifold with boundary, let F be a compact smooth manifold with boundary, and let*

$$f : F \times [0, 1] \longrightarrow M, \quad f_t(x) = f(x, t)$$

be a smooth isotopy through neat embeddings, that is, $f_t^{-1}(\partial M) = \partial F$ and $f_t(F)$ meets ∂M transversally along its boundary. Then there is a smooth ambient isotopy $H_t : M \rightarrow M$ with $H_0 = \text{id}_M$ such that

$$H_t \circ f_0 = f_t \quad (0 \leq t \leq 1). \quad (4.1)$$

Neither orientability nor any dimension condition relating F and M is needed.

Proof. Form the track

$$\hat{F} : F \times [0, 1] \longrightarrow M \times [0, 1], \quad \hat{F}(x, t) = (f_t(x), t).$$

It is an embedding: the second coordinate records t , and its differential is injective because its restriction in the F direction is df_t . The image W is compact and neat along the lateral boundary $\partial M \times [0, 1]$.

On W define the vertical velocity $v(f_t(x), t) = \partial_t f_t(x)$. If $x \in \partial F$ then the curve $t \mapsto f_t(x)$ lies in ∂M , so

$$v(f_t(x), t) \in T_{f_t(x)} \partial M. \quad (4.2)$$

Extend v to a vertical vector field X on $M \times [0, 1]$ tangent to the lateral boundary. Near the boundary take coordinates $(y_1, \dots, y_{m-1}, r)$ with $r \geq 0$ and $\partial M = \{r = 0\}$, and write $v = \sum a_i \partial_{y_i} + a_r \partial_r$ on W . By (4.2), a_r vanishes on $W \cap \{r = 0\}$. Neatness makes $r|_W$ a boundary defining function, so Hadamard's lemma yields a smooth b on W with $a_r = rb$. Extend the a_i and b off W and use rb as the normal coefficient of the extension: it agrees with v on W and is tangent to $\{r = 0\}$.

Patch by a partition of unity. **The partition of unity must be chosen so that every local section contributing at a point of W equals v on W ;** tangency to the boundary is a linear condition, so patching preserves it. Then

$$X|_W = v, \quad X(p, t) \in T_p \partial M \quad (p \in \partial M). \quad (4.3)$$

This step is the one delicate point of the lemma: multiplying the local sections by bump functions supported in small neighbourhoods would destroy $X|_W = v$. One must use directly a partition of unity subordinate to a cover whose sections already satisfy the matching condition, with no truncation. Such a partition exists: W is compact, so take a finite open cover $\{U_\alpha\}$ of $M \times [0, 1]$ in which every U_α meeting W carries a local section built by the Hadamard construction above so as to equal v on $U_\alpha \cap W$, while every U_α disjoint from W carries an arbitrary section tangent to the lateral boundary. For any partition of unity $\{\varphi_\alpha\}$ subordinate to that cover, the field $X = \sum_\alpha \varphi_\alpha X_\alpha$ satisfies, at each point p of W ,

$$X(p) = \sum_{\alpha: p \in U_\alpha} \varphi_\alpha(p) v(p) = v(p),$$

because every φ_α non-zero at p corresponds to a U_α containing p , whose section already equals $v(p)$ there, and $\sum_\alpha \varphi_\alpha \equiv 1$. Tangency to the boundary is a pointwise linear condition and is preserved by convex combinations.

Write $X_t(p) = X(p, t)$. Since M is compact, the flow of the time-dependent field X_t exists on all of $[0, 1]$. Condition (4.3) makes the boundary invariant: in a boundary chart the normal component has the form rc , so a flow line satisfies

$$r(t) = r(0) \exp \left(\int_0^t c(y(s), r(s), s) ds \right),$$

whence boundary points stay in the boundary and interior points stay in the interior; solving backwards gives the inverse map, so each H_t is a diffeomorphism.

Fix $x \in F$. Both $t \mapsto H_t(f_0(x))$ and $t \mapsto f_t(x)$ solve $\dot{q} = X_t(q)$ with the same initial value $f_0(x)$ — the latter because $X|_W = v$. Uniqueness for the ODE gives $H_t(f_0(x)) = f_t(x)$. $\square$

Use. $M = E(K)$ is compact with boundary, every Seifert surface is compactly and neatly embedded, and every normal isotopy is a smooth family of such embeddings. The resulting H_t preserves $\partial E(K)$ *setwise*, allowing the longitudinal boundary curve to move. Consequently a *normal* isotopy class, as used in relative area minimisation, is exactly an *ambient* isotopy vertex.

Remark (No silent strengthening). Strengthening the last sentence to “fixing the boundary pointwise” gives a *false* statement. Counterexample: $M = S^1 \times [0, 1]$, where the rotational isotopy on the boundary cannot be realised by an ambient isotopy fixing the boundary pointwise. In citations and in formalisation — in Lean especially — this must be written *setwise*.

5 Complexity and its well-foundedness

Fix on $E(K)$ a *relative Riemannian metric* — one making $\partial E(K)$ strictly convex — and a *real-analytic* foliation J of the boundary torus by preferred longitudes. Such a choice exists: on the collar $[0, \varepsilon)_r \times \partial E(K)$, where r is the *inward* coordinate, $\partial E(K) = \{r = 0\}$ and $\varepsilon < 1$, take

$$dr^2 + (1 - r)^2 g_{\partial E(K)} \quad (5.0)$$

and extend smoothly; for the foliation take the parallel translate of the linear longitudinal foliation, which is real-analytic to begin with. The conclusions do not depend on the particular choice, but the structure may not be omitted from the statement.

Remark (Two details that must be got right, and that collapse the argument if got wrong). **(i) The sign.** The collar metric must be $dr^2 + \varphi(r)^2 g_{\partial E(K)}$ with $\varphi(0) = 1$ and $\varphi'(0) < 0$. Under such a metric $\nabla_X \partial_r = (\varphi'/\varphi)X$ for X tangent to a level set, so with respect to the *outward* normal $\nu = -\partial_r$ one has $\text{II}(X, X) = -(\varphi'/\varphi)|X|^2 > 0$: the boundary $\partial E(K)$ is *strictly convex*. This is exactly the situation of the ball model $B^3 \subset \mathbb{R}^3$ with $r = 1 - |x|$ and metric $dr^2 + (1 - r)^2 g_{S^2}$. Were one to take $\varphi = 1 + r$ instead, then $\varphi'(0) = +1$, II would be negative definite, the boundary would be strictly *concave*, and the barrier would point the other way: superharmonicity of $r|_\Sigma$ in Lemma 5.2, and the transversality deduced from it in Lemma 5.3, would both fail. This is not a matter of notation — strict convexity is the only source of a barrier in this paper.

(ii) The foliation may not be taken merely C^1 . The transverse coordinate y on the collar is the flow coordinate of a transverse field W carrying leaves to leaves. If J is only C^1 then W is only C^0 , the coordinates only C^1 , the metric components in (s, r, y) only C^0 , the leading coefficients of the elliptic difference equation (7.8) of §7.2 only C^0 , and the isothermal coordinates only bi-Hölder — at which point the similarity principle (7.10), (7.11) and the subtraction of exponents in (7.12) all fail together (see the remark opening §7.2). Taking the foliation real-analytic costs nothing, since the construction above is real-analytic anyway, and real-analytic $\Rightarrow$ smooth $\Rightarrow C^1$, so the smoothness hypothesis on the foliation in the result cited as entry 2 of §13 (whose Definition 3 requires a *smooth* foliation) is automatically met.

A neatly embedded surface with boundary in J is called *relatively least-area* if its area is least in the normal isotopy class subject to that boundary condition — least among *all* surfaces of the class; this is the reading used throughout the paper. For a vertex v put

$$c(v) = (g(v), A(v)), \quad (5.1)$$

where $A(v)$ is the least relative area in the normal isotopy class determined by v , ordered *lexicographically*: genus first, then area.

Remark ($A(v)$ is a minimum, not an infimum). By Theorem B.3 ((G-b) $\Rightarrow$ (E), Appendix B) and Theorem C.1 (existence of an optimal leaf ℓ_* , Appendix C), $A(v)$ is attained by the surface

constructed over ℓ_* by Theorem A.1 (Appendix A, the boundary version of 6.12; for the reading involved see §C.5). All of §7 works with a smooth surface S *realising* $A(v)$, exchanging it, rounding its corners and comparing areas; if $A(v)$ were only an infimum, then “we have produced a surface of the same class with smaller area” would not be a contradiction. The statement of (E) and the named theorems it is conditional on are in §12.2. This is not a notational nicety: if one redefined $A(v)$ as the least area *in the moduli space of stable minimal surfaces*, where existence is free, then well-foundedness and disjointness would still hold, but the strict descent of §7 would fail at once, because the corner-rounded surface is not a minimal surface.

That $A(v)$ is well defined follows from the fact that an ambient isotopy restricts to a normal isotopy; Lemma 4.1 gives the converse, so normal isotopy classes may be used to count ambient isotopy vertices.

Remark (Essential). The order used below is the order on the *set of values actually taken*, $c(\text{IS}(K)^{(0)})$, and *not* the order on all of $\mathbb{N} \times \mathbb{R}_{>0}$. The latter is not well-founded. This distinction is the only reason the whole chain admits transfinite induction.

Lemma 5.1 (Well-foundedness on the set of attained values). *For every integer $h \geq 0$ and every $a > 0$ there are only finitely many vertices v of $\text{IS}(K)$ with $g(v) = h$ and $A(v) \leq a$. Consequently the lexicographic order on the set of attained values $\{c(v)\}$ has no infinite strictly decreasing sequence.*

Proof. Fix the genus h , so the topological type of a connected orientable surface with one boundary component is fixed. The least-area representative S of any such vertex is a minimal surface of that fixed topological type, with boundary in J , of area at most a , and incompressible (by the definition of a vertex). K is non-trivial, so under the setting (S) of Appendix D, Theorem D.50 ((AW), Appendix D) produces a normal surface N , ambient isotopic to S through triangulation-preserving isotopies, whose weight satisfies $w(N) \leq Ca + D$ with constants C, D depending only on the triangulation, the metric and J — here only “ h fixed and a bounded $\Rightarrow w$ bounded” is used. Theorem D.54 (Haken finiteness, Appendix D) asserts that the normal surfaces of weight at most $Ca + D$ fall into *finitely many* ambient isotopy classes (Corollary D.56). This is a statement about the isotopy relation itself; it does not require joining two surfaces by a smooth family, and it needs no intermediate step of the form “a continuous family can be smoothed”. Hence the least-area representatives of genus h and area at most a fall into at most finitely many normal isotopy classes. By Lemma 4.1, distinct ambient isotopy vertices give distinct normal isotopy classes. So there are only finitely many such vertices. (With this argument the word “stable” is no longer needed here; for S402-G3 see the table of §12.2.)

Suppose $c(v_1) > c(v_2) > \dots$. The genus coordinate is a non-decreasing sequence of non-negative integers read downwards, hence eventually constant, equal to some h from v_N on. From there the areas strictly decrease, but they are drawn from the finitely many area values of genus h and area at most $A(v_N)$ — a contradiction. This proves well-foundedness on the set of *attained values* only, and asserts nothing about $\mathbb{N} \times \mathbb{R}_{>0}$. $\square$

Remark (For the W of this paper the well-ordering is direct: no choice is needed here). What was just proved is “no infinite strictly decreasing sequence”, whereas what is used from §8 on is “every non-empty subset has a least element”. On a general total order the passage from the former to the latter goes through dependent choice (DC) (see the remark in §8). For the $W = c(\text{IS}(K)^{(0)})$ actually occurring here, the latter is *provable directly in ZF*. Let $\emptyset \neq S \subseteq W$. Put $h := \min\{g : (g, a) \in S\}$ ($\mathbb{N}$ is well-ordered, no choice needed). Pick any a_1 with $(h, a_1) \in S$; by the finiteness just proved there are only *finitely many* vertices of genus h and area at most a_1 , so $\{a : (h, a) \in S, a \leq a_1\}$ is a finite non-empty set, and its least element is $\min\{a : (h, a) \in S\}$; hence $(h, \min\{a : (h, a) \in S\}) = \min S$. In this paper the axiom of choice is genuinely used only in Lemma 9.1, to break ties.

Lemma 5.2 (A strictly convex boundary is a barrier: $r|_\Sigma$ is superharmonic). *Let the collar metric be $dr^2 + \varphi(r)^2 g_{\partial E(K)}$ with $\varphi(0) = 1$, $\varphi'(0) < 0$, that is, (5.0). Let $\Sigma \subset E(K)$ be an immersed surface, C^2 in its interior, with identically vanishing mean curvature ($\vec{H}_\Sigma \equiv 0$). Then $\Delta_\Sigma r \leq 0$ inside the collar, that is, $r|_\Sigma$ is superharmonic; consequently the interior of Σ is disjoint from $\partial E(K)$.*

Remark (The regularity hypothesis is not decorative). The proof computes the trace of $\text{Hess } r$ on $T_q \Sigma$ and applies the strong minimum principle, and both require Σ to be at least C^2 near interior points. In this paper the lemma is applied only to relatively least-area surfaces, whose interior smoothness is the standard interior regularity of minimal surfaces and is unrelated to boundary regularity (R1): *this lemma does not depend on (R1)*. Lemma 5.3 does.

Proof. From $g = dr^2 + \varphi(r)^2 g_{\partial E(K)}$ one gets $\text{Hess } r = \frac{1}{2} \partial_r h_r = \frac{\varphi'}{\varphi} h_r$ with $h_r = \varphi(r)^2 g_{\partial E(K)}$; in particular $\text{Hess } r(\partial_r, \cdot) = 0$, while in the horizontal directions $\text{Hess } r$ is $\frac{\varphi'}{\varphi} < 0$ times a positive definite form, hence *negative semidefinite*. Take an orthonormal basis e_1, e_2 of $T_q \Sigma$ and split each e_i into its horizontal part e_i^{lev} and its ∂_r part:

$$\Delta_\Sigma r = \sum_{i=1}^2 \text{Hess } r(e_i, e_i) + \langle \nabla r, \vec{H}_\Sigma \rangle = \frac{\varphi'}{\varphi} \sum_{i=1}^2 |e_i^{\text{lev}}|^2 \leq 0,$$

the last step using $\vec{H}_\Sigma = 0$. If some interior point q of Σ lay on $\partial E(K)$, then $r|_\Sigma \geq 0$ would attain its minimum 0 at an interior point; the strong minimum principle for superharmonic functions would give $r \equiv 0$ on the connected component containing q , so that component would be contained in the two-dimensional $\partial E(K)$, contradicting that Σ is a neatly embedded surface in a 3-manifold. $\square$

Remark. The sign carries weight: with $\varphi = 1 + r$ (so $\varphi'(0) > 0$) one gets $\Delta_\Sigma r \geq 0$, $r|_\Sigma$ subharmonic, and the argument above, together with Lemma 5.3 below, fails in the same direction. See remark (i) at the start of §5.

Lemma 5.3 (A relatively least-area surface is transverse to $\partial E(K)$ along the boundary). *Let Σ be as in Lemma 5.2, and let Σ be the surface constructed in Theorem C.3 (Appendix C): a neatly embedded surface with boundary which is C^∞ up to $\partial E(K)$, in particular $C^{1,1}$ near $p \in \partial \Sigma$, that is, satisfying (R1) (statement in §12.2). Then*

$$T_p \Sigma \neq T_p \partial E(K).$$

Proof. Suppose $T_p \Sigma = T_p \partial E(K) =: \Pi$. Choose coordinates (x_1, x_2, y) at p with $\Pi = \{y = 0\}$, $E(K)$ locally $\{y \geq \sigma(x)\}$ and $\partial E(K) = \{y = \sigma(x)\}$, where σ is smooth, $\sigma(0) = 0$, $D\sigma(0) = 0$. By (R1), supplied by Theorem C.3, and by the assumed tangency, Σ is near p a graph $\{y = f(x) : x \in D'\}$ with $f \in C^{1,1}(\overline{D'})$, $0 \in \partial D'$, $f(0) = 0$, $Df(0) = 0$, and $\partial D'$ a $C^{1,1}$ curve near 0.

Put $w := f - \sigma$. Then $\Sigma \subset E(K)$ gives $w \geq 0$ on D' ; $\partial \Sigma \subset \partial E(K)$ gives $w = 0$ on the corresponding arc of $\partial D'$; and $w(0) = 0$. Let $\mathcal{M}[\phi]$ denote the mean curvature operator (in divergence form) of the graph $\{y = \phi\}$ with respect to the *inward* normal. Minimality of Σ gives $\mathcal{M}[f] = 0$; strict convexity of $\partial E(K)$ gives $\mathcal{M}[\sigma] \geq c_0 > 0$ on a neighbourhood of p (continuity and compactness). Subtracting and applying the mean value theorem to $\mathcal{M}$ — legitimate, since $f, \sigma \in C^{1,1}$ and hence $Dw \in L^\infty$:

$$Lw := \partial_i (\tilde{a}^{ij} \partial_j w) + \tilde{b}^i \partial_i w + \tilde{c} w = \mathcal{M}[f] - \mathcal{M}[\sigma] \leq -c_0 < 0,$$

with $\tilde{a}$ uniformly elliptic and $\tilde{b}, \tilde{c} \in L^\infty$. Moreover $\tilde{a} \in C^{0,1}$: from the mean value representation on the previous line, $\tilde{a}^{ij}(x) = \int_0^1 A^{ij}(D\sigma(x) + s Dw(x)) ds$, where $A^{ij}(\xi) = \partial(\xi \mapsto \xi / \sqrt{1 + |\xi|^2})$ is a

smooth function and $D\sigma, Df \in C^{0,1}$. A smooth function composed with a Lipschitz map and then integrated in s is again Lipschitz, so $\tilde{a} \in C^{0,1}(\overline{D'})$. This line carries weight: $\tilde{a} \in C^{0,1}$ is what turns the pair “Hopf boundary point lemma (Theorem E.31) + strong minimum principle (Theorem E.29)” from a citation of Gilbarg–Trudinger into a self-contained proof (Appendix E), which directly affects the self-containment ledger.

Interior strict positivity $w > 0$. The Hopf boundary point lemma requires w to be *strictly* above its boundary minimum in the interior; $w \geq 0$ is not enough, since if w also vanished at an interior point the directional derivative in the conclusion could be 0. This is supplied as follows. Move the zeroth-order term to the right-hand side, writing $\tilde{L}w := \partial_i(\tilde{a}^{ij}\partial_j w) + \tilde{b}^i\partial_i w$; then

$$\tilde{L}w = Lw - \tilde{c}w \leq -c_0 + \|\tilde{c}\|_\infty w.$$

Take $D'' \subset D'$ a neighbourhood of 0 small enough that $w < c_0/(2\|\tilde{c}\|_\infty + 1)$ there (w is continuous and $w(0) = 0$, so such a neighbourhood exists). Then on D''

$$\tilde{L}w \leq -c_0/2 < 0,$$

that is, w is a strict supersolution of the uniformly elliptic divergence-form operator $\tilde{L}$, which has *no* zeroth-order term. The strong minimum principle for such operators (divergence form: Theorem E.29, Appendix E) says that a non-constant supersolution does not attain its minimum in the interior. If w did attain the minimum value 0 at an interior point of D'' , then w would be identically 0 on that component of D'' , whence $\tilde{L}w = 0$, contradicting $\tilde{L}w \leq -c_0/2 < 0$. So $w > 0$ in the interior of D'' .

Now D' is a $C^{1,1}$ domain, so it satisfies the interior ball condition at $0 \in \partial D'$, and the interior ball may be taken inside D'' . Thus $w > 0$ in the ball, $w(0) = 0$ and $\tilde{L}w < 0$, so the Hopf boundary point lemma — in its *divergence-form* version, Theorem E.31 of Appendix E, and *not* Gilbarg–Trudinger Lemma 3.4, which is stated in non-divergence form and does not match the shape of L here — gives $\partial_\nu w(0) > 0$ for the inward derivative. But $Dw(0) = Df(0) - D\sigma(0) = 0$: a contradiction. $\square$

Remark (Strict convexity is exactly the source of transversality, and without it the conclusion is false). In the half-space $\{y \geq 0\} \subset \mathbb{R}^3$, whose boundary is flat and not strictly convex, the least-area surface spanning a planar circle in $\{y = 0\}$ is that flat disc, *everywhere tangent* to the boundary, and not one word of the graph representation (7.7) of §7.2 could be written down. Lemma 5.3 demotes the transversality needed in step 1 of §7.2 from an extra hypothesis to a *corollary* of (R1).

6 The cyclic-cover frontier construction

Lemma 6.1 (Frontier construction). *Let S, S_+, S_- be connected Seifert surfaces in $E(K)$ with S disjoint from $S_+ \cup S_-$ while S_+ meets S_- . Assume the latter pair transverse, with disjoint boundaries, and with no disc component in their symmetric difference. Then there are connected Seifert surfaces $P_\uparrow, P_\downarrow$ such that:*

1. *both are disjoint from each of S, S_+, S_- , and from each other;*
2. *every compact surface Z disjoint from $S_+ \cup S_-$ is disjoint from some choice of representatives of $P_\uparrow, P_\downarrow$;*
3. *writing $T_\uparrow, T_\downarrow$ for the full (possibly disconnected) frontier outputs containing $P_\uparrow, P_\downarrow$,*

$$\chi(T_\uparrow) + \chi(T_\downarrow) = \chi(S_+) + \chi(S_-), \tag{6.1}$$

and every component discarded from either T is a closed surface of Euler characteristic ≤ 0 .

Proof. Let $p : \tilde{E} \rightarrow E(K)$ be the infinite cyclic cover determined by the linking-number homomorphism $\pi_1(E(K)) \twoheadrightarrow \mathbb{Z}$, and let τ generate its positive deck group. Take a lift S_0 of S and put $S_1 = \tau S_0$. A Seifert surface represents a primitive generator of $H_2(E(K), \partial E(K))$, hence is non-separating; equivalently, cutting the exterior along S yields the fundamental domain C between S_0 and S_1 , and

$$p|_C : C \longrightarrow E(K) \setminus S$$

is a homeomorphism. Use the unique inverse of this homeomorphism to lift S_+ and S_- into C , obtaining S_0^+, S_0^- .

The infinite cyclic cover can equally be obtained by cutting along S_+ , or along S_- , and stacking. Since each represents the primitive class dual to the covering map, the chosen lift is a wall in that stacking, and its complement has exactly two components. Write $M_\uparrow^+, M_\downarrow^+$ for the two sides of S_0^+ and $M_\uparrow^-, M_\downarrow^-$ for the two sides of S_0^- , in each case taking the side containing S_1 to be the upper one. Define the cornered frontiers

$$\tilde{B}_\downarrow = \text{fr}(M_\downarrow^- \cap M_\downarrow^+), \quad \tilde{B}_\uparrow = \text{fr}(M_\uparrow^- \cap M_\uparrow^+). \quad (6.2)$$

They are assembled from complementary subsurfaces of $S_0^+ \cup S_0^-$ and are *piecewise smooth surfaces with corners*, their non-smooth locus being exactly $\Gamma := S_0^+ \cap S_0^-$ — in the transverse case a finite family of disjoint smooth closed curves, see (7.3.0). Round their corners by Lemma 7.0.A to obtain smooth neatly embedded surfaces $\hat{B}_\downarrow, \hat{B}_\uparrow$, then push each off into the indicated open region at scale σ by Lemma 7.0.B, obtaining embedded surfaces $\tilde{T}_\downarrow(\sigma), \tilde{T}_\uparrow(\sigma)$. The two regions are opposite, so after the push-off the surfaces are disjoint from the input lifts and from each other. They lie in C , on which p is injective, so they project homeomorphically to embedded surfaces $T_\downarrow(\sigma), T_\uparrow(\sigma)$ disjoint from S, S_+, S_- . By clause 3 of Lemma 7.0.B, all $\sigma \in (0, \sigma_0]$ give mutually isotopic outputs, so below σ labels only the representative and not the isotopy class, and is suppressed when not needed.

Each frontier in (6.2), being the frontier of a region in an oriented cover, is two-sided and carries a boundary orientation, and the local orientations on its input patches agree across each resolved seam. The two original boundary curves are disjoint oriented preferred longitudes; on the lifted boundary annulus one lies on the upper frontier and the other on the lower. Equivalently, orient each frontier by its region side: if $\tilde{\gamma}$ is a lift of an oriented meridian, its algebraic intersection number with each frontier boundary is $+1$, as is its intersection number with each input longitude.

The two frontiers together consume the two input boundary curves, so each frontier has exactly one boundary component. Hence each T has exactly one component with boundary; that component is a connected oriented Seifert surface, denoted $P_\uparrow$ resp. $P_\downarrow$, and all other components are closed.

If Z is compact and disjoint from $S_+ \cup S_-$, its distance to that compact union is positive, so the push-off in (6.2) can be taken small enough to avoid Z . Two sufficiently small push-offs on the same side are isotopic through the collar between them, so this adjustment changes only representatives, not the isotopy class of either P .

Finally, the upper and lower frontiers between them exhaust the cut-open $S_0^+ \cup S_0^-$. Additivity of the Euler characteristic — or Lemma 3.1 applied to this exchange — gives (6.1). If some output disc or sphere carried no exchange seam, it would already be a component of one of the inputs with the intersection curves removed: in the disc case this contradicts the hypothesis of no disc component, and the sphere case is impossible since the inputs are connected with boundary. Otherwise an innermost seam on such a component bounds a disc patch containing no further seam, and that is a disc component of $S_0^- \triangle S_0^+$, again contradicting the hypothesis. So no output component is a disc or a sphere, and every discarded component is a closed orientable surface of Euler characteristic ≤ 0 . $\square$

Remark (Relation to the construction in the literature). The corresponding construction in entry 2 of §13 requires *minimal genus* both in its hypotheses and in its conclusion. What is taken above is the part that does *not* depend on that hypothesis: this lemma asserts no minimal genus for either output, and everything needed comes from the Euler identity (6.1).

7 The exchange lemma

7.1 Two construction lemmas: rounding and pushing off

The frontier construction (Lemma 6.1) produces a *piecewise smooth surface with corners*, whereas the complexity $A(\cdot)$ of §5 is an infimum over smooth neatly embedded surfaces. Two steps therefore separate a frontier from a legitimate competitor: first *round the corners* to get a smooth surface, then *push it off* to make it disjoint from the inputs. Each step changes the area, and this subsection keeps the two accounts apart — their order is the pivot of the whole quantifier structure of §7.

Lemma 7.0.A (Corner rounding). *Let Γ be a compact C^2 curve without boundary in a compact 3-manifold (or a finite family of disjoint such curves), let $\tilde{B}$ be a piecewise smooth embedded surface with a corner along Γ , and let U be the open region it bounds; assume that in the normal slice at each $x \in \Gamma$ the trace of $\tilde{B}$ consists of the two edges of a wedge with vertex x and opening angle $\alpha(x)$, and that*

$$\theta_0 \leq \alpha(x) \leq \pi - \theta_0 \quad \text{for every } x \in \Gamma \text{ and some } \theta_0 > 0. \quad (7.0.1)$$

Then there are $\eta_0 > 0$, depending only on the normal injectivity radius of Γ , on θ_0 and on the ambient metric, and $g_0 = g_0(\theta_0) > 0$ depending only on θ_0 , such that for every $\eta \in (0, \eta_0]$ there is a compact C^∞ neatly embedded surface $\hat{B} = \hat{B}(\eta)$ with:

1. $\hat{B}$ coincides pointwise with $\tilde{B}$ outside the η -tubular neighbourhood of Γ ; in particular $\partial\hat{B} = \partial\tilde{B}$;
2. $\hat{B} \subset \bar{U}$, and there is a one-sided collar $c : \hat{B} \times [0, \epsilon_1) \rightarrow \text{the ambient manifold}$ with $c(x, 0) = x$ and $c(x, s) \in U$ for $s > 0$;
3. $\hat{B}$ is isotopic to $\tilde{B}$ by an isotopy supported in that η -tubular neighbourhood;
4. $\text{Area}(\hat{B}) \leq \text{Area}(\tilde{B}) - g_0 \eta \text{length}(\Gamma)$.

More generally, replacing the constant width η by any smooth function $\eta(\cdot) \leq \eta_0$, conclusion 4 becomes $\text{Area}(\hat{B}) \leq \text{Area}(\tilde{B}) - g_0 \int_\Gamma \eta(s) ds$; in particular the gain is non-negative slice by slice, so one may estimate the gain on one arc of Γ and discard the rest.

Proof. (i) The profile function. Take an odd C^∞ function $\chi : [-1, 1] \rightarrow [-1, 1]$ with $0 \leq \chi \leq 1$ on $[0, 1]$, $\chi \equiv 1$ on $[1 - \delta_1, 1]$ and $\chi(0) = 0$. Put

$$\Theta(t) = c_0 + \int_0^t \chi, \quad c_0 := 1 - \int_0^1 \chi. \quad (7.0.2)$$

Since χ is odd, $\int_0^t \chi$ is even, so Θ is even; $\Theta(\pm 1) = 1$, $\Theta'(\pm 1) = \pm 1$, and $\Theta^{(k)}(\pm 1) = 0$ for $k \geq 2$ because χ is constant ± 1 near the endpoints; hence Θ joins $|t|$ in a C^∞ fashion at $|t| \geq 1$. Also $|\Theta'| = |\chi| \leq 1$. For $t \in [0, 1]$ put $F(t) := \Theta(t) - t = c_0 + \int_0^t (\chi - 1)$: since $\chi \leq 1$, F is non-increasing, while $\int_0^1 (\chi - 1) = -c_0$ gives $F(1) = 0$; hence $F \geq 0$, that is, $\Theta(t) \geq t$ on $[0, 1]$, and by evenness $\Theta \geq |\cdot|$ on $[-1, 1]$. From $\chi(0) = 0$ we get $\Theta'(0) = 0$. (And $c_0 \geq 0$, since $\int_0^1 \chi \leq 1$.)

For $m > 0$ put

$$g(m) := \int_{-1}^1 \left(\sqrt{1 + m^2} - \sqrt{1 + m^2 \Theta'(\tau)^2} \right) d\tau. \quad (7.0.3)$$

The integrand is ≥ 0 because $|\Theta'| \leq 1$, and it is > 0 near $\tau = 0$ by $\Theta'(0) = 0$ and continuity; so $g(m) > 0$, and g is continuous. By (7.0.1), $m(x) = \cot(\alpha(x)/2)$ takes values in the compact interval $[\tan(\theta_0/2), \cot(\theta_0/2)] \subset (0, \infty)$, so

$$2g_0 := \min \{g(m) : m \in [\tan(\theta_0/2), \cot(\theta_0/2)]\} > 0. \quad (7.0.4)$$

(ii) Local construction. Take Fermi coordinates (s, ξ_1, ξ_2) along Γ , in which the metric is $ds^2 + |d\xi|^2 + O(|\xi|)$. Place the wedge in the normal slice with vertex at the origin and bisector along $+\xi_1$, its two edges being $\xi_1 = m(s)|\xi_2| + O(|\xi|^2)$ with $m(s) = \cot(\alpha(s)/2)$ — the wedge $\{\xi_1 \geq m|\xi_2|\}$ of opening α has half-angle $\arctan(1/m) = \alpha/2$. On $|\xi_2| \leq \eta$ replace the two sheets by

$$\xi_1 = m(s)\eta\Theta(\xi_2/\eta) \quad (\text{plus the original } O(|\xi|^2) \text{ correction, matched smoothly in } |\xi_2|). \quad (7.0.5)$$

Since $\Theta \geq |\cdot|$, the new sheet lies in the closed wedge $\bar{U}$, and lies in the open wedge where $|\xi_2| < \eta$; since Θ meets $|\cdot|$ to infinite order at $|\xi_2| = \eta$, the new sheet agrees pointwise with the original frontier outside the tube. Take $\eta_0 < \frac{1}{2} \min$ (normal injectivity radii of the components) so that the tubes are disjoint; this gives $\hat{B}$ and clauses 1 and 3. For clause 2: where $|\xi_2| < \eta$, $\hat{B}$ lies in the interior of U ; at all other points $\hat{B}$ coincides with $\text{fr}(U)$ and U lies locally on one side; compactness of $\hat{B}$ gives a uniform ϵ_1 .

(iii) The area account. In one normal slice the two replaced segments have total length $\int_{-\eta}^{\eta} \sqrt{1+m^2} d\xi_2 = 2\eta\sqrt{1+m^2}$, while the new profile has length $\eta \int_{-1}^1 \sqrt{1+m^2\Theta'(\tau)^2} d\tau$; so the decrease in that slice is exactly $\eta g(m(s)) \geq 2g_0\eta$, and it is non-negative *for every* η . Integrating in s by Fubini — the area element is $\sqrt{1+m^2\Theta'^2} d\xi_2 ds \cdot (1 + O(\eta))$ — and taking η_0 small enough that the multiplicative error lies in $[\frac{1}{2}, 2]$ gives conclusion 4. $\square$

Remark (Circular arcs must not be used). A natural but *false* way to write this step is “replace the broken line by the circular arc with the same endpoints; the arc is shorter”. In the two-dimensional model, the broken line with vertex at the origin, opening α and edges of length η has total length 2η , while the arc of radius η centred at the vertex has length $\eta\alpha$; the decrease is $\eta(2 - \alpha)$, which is *negative once* $\alpha > 2$ (radians, about 114.6°) — and all that is known here is $\alpha \leq \pi - \theta_0$, which may well exceed 2. The correct comparison is with the *chord*: $2\eta \sin(\alpha/2) < 2\eta$ for every $\alpha < \pi$, and $\alpha \leq \pi - \theta_0$ yields the uniform gain $\geq 2\eta(1 - \cos(\theta_0/2)) > 0$. The smooth profile Θ above is precisely a smoothed version of the chord, and g_0 in (7.0.4) is its quantitative form.

Lemma 7.0.B (Pushing off). *Let $\hat{B} \subset \tilde{E}$ be a compact C^∞ neatly embedded surface satisfying clause 2 of Lemma 7.0.A, whose boundary $\partial\hat{B}$ is a leaf of the (lifted) longitudinal foliation $\tilde{J}$. Then there are $\sigma_0 > 0$ and $C = C(\hat{B}) < \infty$ such that for every $\sigma \in (0, \sigma_0]$ there is a compact C^∞ neatly embedded surface $T(\sigma)$ with:*

1. $T(\sigma) \subset U$;
2. $\partial T(\sigma)$ is a leaf of $\tilde{J}$;
3. $T(\sigma)$ is isotopic to $\hat{B}$, and any $\sigma, \sigma' \in (0, \sigma_0]$ give mutually isotopic $T(\sigma), T(\sigma')$;
4. $|\text{Area}(T(\sigma)) - \text{Area}(\hat{B})| \leq C\sigma$.

Proof. **(i) The push-off field.** Let ν be the unit normal of $\hat{B}$, with the side indicated by the collar taken positive, and let n be the outward unit normal of $\partial\tilde{E}$, extended smoothly through the boundary collar. Put $X := \nu - \psi(r)\langle \nu, n \rangle n$, where r is the distance to $\partial\tilde{E}$ and ψ equals 1 near $r = 0$ and is supported in the collar. Along $\partial\tilde{E}$ one has $\langle X, n \rangle = 0$, so X is tangent to $\partial\tilde{E}$; also

$\langle X, \nu \rangle = 1 - \psi \langle \nu, n \rangle^2$, and neatness $(\widehat{B} \cap \partial \widetilde{E})$ together with compactness gives $\langle \nu, n \rangle^2 \leq 1 - c$, whence $\langle X, \nu \rangle \geq c > 0$. On $\partial \widetilde{E}$ modify X into a field Y transverse to the foliation whose local flow carries leaves to leaves, oriented towards the U side, and interpolate in r by a cut-off; shrink the collar so that $\langle X, \nu \rangle \geq c/2 > 0$ still holds.

(ii) The flow. X has compact support, so its flow Ψ_s exists for $s \in [0, \sigma_0]$, preserves $\partial \widetilde{E}$ and carries leaves to leaves. Put $T(\sigma) := \Psi_\sigma(\widehat{B})$. Transversality of X to $\widehat{B}$ makes $(x, s) \mapsto \Psi_s(x)$ an embedding on $\widehat{B} \times [0, \epsilon)$ with image inside the one-sided collar of clause 2 of Lemma 7.0.A; shrinking $\sigma_0 < \epsilon$ gives clause 1. Clause 2 follows from the leaf-preserving property of Y . For clause 3: Ψ is itself an ambient isotopy, and $\Psi_{\sigma'} \circ \Psi_\sigma^{-1}$ gives an isotopy between two push-offs.

(iii) Area. Let $K_1 := \sup |\nabla X|$ on a compact neighbourhood of $\widehat{B}$. The Jacobian J of $\Psi_s|_{\widehat{B}}$ satisfies $\partial_s \log J = \operatorname{div}_{T\Psi_s(\widehat{B})} X$, of absolute value $\leq 2K_1$, so $e^{-2K_1 s} \leq J \leq e^{2K_1 s}$ and therefore

$$|\operatorname{Area}(T(\sigma)) - \operatorname{Area}(\widehat{B})| \leq (e^{2K_1 \sigma} - 1) \operatorname{Area}(\widehat{B}) \leq C\sigma, \quad (7.0.6)$$

with $C := 2K_1 e^{2K_1 \sigma_0} \operatorname{Area}(\widehat{B})$, independent of σ . $\square$

Remark (What C depends on, and why the order of the two steps is forced). C depends only on $\widehat{B}$ — its area, its second fundamental form, its angle of transversality with $\partial \widetilde{E}$ — and on a C^1 bound for the ambient metric on a fixed neighbourhood of $\widehat{B}$. And $\widehat{B}$ is determined by the cornered frontier together with the once-chosen rounding width η . Hence C and the rounding gain are both fixed before σ is.

Why the rounding must come first: a normal push-off applied directly to a cornered surface makes the two sheets' normals cross at the corner and does not produce an embedded surface, while using instead “the frontier of the inner parallel set” produces another cornered surface, not a smooth competitor. Rounding first removes the corner, after which the push-off is a smooth push-off of a smooth surface — which is exactly what yields $|J - 1| \leq C\sigma$.

7.2 The exchange lemma for incompressible surfaces

Theorem 7.1 (Incompressible exchange lemma). *Let v, v_+, v_- be vertices of $\operatorname{IS}(K)$ with*

$$d(v, v_+) = d(v, v_-) = 1, \quad d(v_-, v_+) = 2. \quad (7.1)$$

Then there are vertices $w_\uparrow, w_\downarrow$ such that:

- (a)** *each is equal or adjacent to each of v, v_+, v_- ;*
- (b)** *every vertex z equal or adjacent to both v_+ and v_- is equal or adjacent to both outputs;*
- (c)** $g(w_\uparrow) + g(w_\downarrow) \leq g(v_+) + g(v_-)$; (7.2)
- (d)** *if (7.2) is an equality, then $A(w_\uparrow) + A(w_\downarrow) < A(v_+) + A(v_-)$.* (7.3)

Proof, transverse case. Take relatively least-area representatives S, S_+, S_- of the three vertices (existence by Theorem B.3 (Appendix B) and Theorem C.1 (Appendix C); see the remark after (5.1) in §5). Apply (U) (Schultens, Theorem 12; research-level citation) to the separable pairs (v, v_+) and (v, v_-) to obtain *one and the same* choice with $S \cap S_+ = S \cap S_- = \emptyset$. Since v_+, v_- are at distance 2, their representatives neither coincide nor are disjoint.

Remark (The form in which the cited result is used). The previous sentence uses the *universal* form of (U) (Schultens, Theorem 12; statement in §12.2): for pairwise disjoint, pairwise non-isotopic

incompressible surfaces, *any* relatively least-area representatives of their isotopy classes are again pairwise disjoint. Only this form licenses “one and the same choice” — first take a least-area representative of S , then require it to be compatible with the representatives produced by the two separate applications to (v, v_+) and to (v, v_-) . The whole geometric part rests on this form; §12.2 writes out its exact content.

Assume first that S_+, S_- are transverse with disjoint boundaries. Lemma 1.2 of entry 3 of §13 rules out a product region between them, and a disc component of $S_- \triangle S_+$ would yield exactly such a product region. Hence the symmetric difference has no disc component, and Lemma 6.1 produces coarse surfaces $P_\uparrow, P_\downarrow$.

By (6.1) and the fact that the discarded closed components have $\chi \leq 0$,

$$\chi(P_\uparrow) + \chi(P_\downarrow) \geq \chi(S_+) + \chi(S_-). \quad (7.4)$$

Substituting $\chi = 1 - 2g$ for the four connected surfaces with a single boundary component,

$$2 - 2(g(P_\uparrow) + g(P_\downarrow)) \geq 2 - 2(g(v_+) + g(v_-)),$$

that is,

$$g(P_\uparrow) + g(P_\downarrow) \leq g(v_+) + g(v_-). \quad (7.5)$$

Compress each P along essential circles, always keeping the component carrying the longitudinal boundary, until it is incompressible, and call the resulting vertices $w_\uparrow, w_\downarrow$. Lemma 3.1 gives $g(w_\uparrow) \leq g(P_\uparrow)$ and $g(w_\downarrow) \leq g(P_\downarrow)$, which together with (7.5) gives (c).

Lemma 6.1 makes each coarse P disjoint from each of S, S_+, S_- . Applying Corollary 3.3 *separately* to these three incompressible surfaces gives (a). Adjacency is a condition on a *pair*, so the three applications are independent of one another and may use different representatives of the descendant; (a) does *not* assert the existence of a family of representatives spanning a simplex. Indeed there cannot be one: $d(v_+, v_-) = 2$ forces $S_+ \cap S_- \neq \emptyset$, so $S \sqcup S_+ \sqcup S_-$ is not even a neatly embedded surface to which Corollary 3.3 could be applied.

Next let z be distinct from v_+, v_- and equal or adjacent to both, with relatively least-area representative Z . Entry 2 of §13 makes Z disjoint from both S_+ and S_- . By the small push-off clause of Lemma 6.1 the coarse outputs have representatives disjoint from Z , and Corollary 3.3 gives (b). If z equals an endpoint, (a) already gives the conclusion. That the push-off scale may depend on z causes no quantifier problem: all sufficiently small push-offs represent the same coarse classes, and the chosen compressed descendant is transported along that isotopy.

Now suppose (7.2) is an equality. Then (7.5) and the compression inequalities are all equalities; in particular no genus-lowering compression occurred, so each coarse P was already incompressible and represents the corresponding w .

(0) The corner structure of the frontier. Since $S_\pm$ are neat, $S_\pm \cap \partial E(K) = \partial S_\pm$; and if $x \in \partial S_+ \cap S_-$ then $x \in \partial S_+ \cap \partial S_- = \emptyset$. Hence

$$\begin{aligned} \Gamma &:= S_+ \cap S_- \text{ is a compact 1-submanifold without boundary contained in } \text{Int } S_+ \cap \text{Int } S_-, \\ &\text{that is, finitely many disjoint smooth closed curves; } \ell_0 := \text{length}(\Gamma) \in (0, \infty). \end{aligned} \quad (7.3.0)$$

In the normal slice at $x \in \Gamma$ the two tangent planes cut the normal plane into four sectors of angles $\alpha, \pi - \alpha, \alpha, \pi - \alpha$ in turn, and $M_\downarrow^- \cap M_\downarrow^+$ and $M_\uparrow^- \cap M_\uparrow^+$ pick out a pair of *opposite* sectors; so at each corner point both frontiers have the same wedge angle $\alpha(x) \in (0, \pi)$, and both are strictly convex. By transversality and compactness of Γ , $\theta_0 := \min_{x \in \Gamma} \min\{\alpha(x), \pi - \alpha(x)\} > 0$, that is $\theta_0 \leq \alpha \leq \pi - \theta_0$, which is exactly hypothesis (7.0.1) of Lemma 7.0.A.

(i) The area identity for the cornered frontiers. The cornered frontiers use up the pieces of $S_+ \cup S_-$ exactly: after cutting along Γ , each patch is assigned to the upper or to the lower frontier *exactly once*; the two-dimensional Hausdorff measure on $S_+ \cup S_-$ is additive over this finite partition; and Γ itself is one-dimensional and of measure zero. Hence

$$\text{Area}(\tilde{B}_\uparrow) + \text{Area}(\tilde{B}_\downarrow) = \text{Area}(S_+) + \text{Area}(S_-). \quad (7.3a)$$

Remark. (7.3a) is a statement about the cornered frontiers themselves, not about the surfaces obtained after pushing off. The push-off changes area in general, and its cost is accounted for separately in (iii). An earlier draft wrote (7.3a) directly for the pushed-off $T_\uparrow, T_\downarrow$, which is unfounded.

(ii) The rounding gain, fixed before the push-off. By (0) and Lemma 7.0.A, choose a width $\eta \in (0, \eta_0]$ *once and for all* and round both frontiers along Γ , obtaining smooth neatly embedded surfaces $\hat{B}_\uparrow, \hat{B}_\downarrow$ with

$$\text{Area}(\hat{B}_\uparrow) + \text{Area}(\hat{B}_\downarrow) \leq \text{Area}(\tilde{B}_\uparrow) + \text{Area}(\tilde{B}_\downarrow) - 2\varepsilon_0, \quad \varepsilon_0 := g_0(\theta_0) \cdot \eta \cdot \ell_0 > 0, \quad (7.3a')$$

with $g_0(\theta_0) > 0$ from (7.0.4), depending only on θ_0 . Note that ε_0 depends only on the fixed S_+, S_- and on this one choice of η ; it depends on no later choice, and in particular not on the push-off scale σ .

(iii) The cost of pushing off. Apply Lemma 7.0.B to the smooth surfaces $\hat{B}_\uparrow, \hat{B}_\downarrow$: there are $\sigma_0 > 0$ and $C < \infty$, depending only on $\hat{B}_\uparrow, \hat{B}_\downarrow$ and the ambient metric and hence *fixed before* σ , such that for $\sigma \in (0, \sigma_0]$

$$\text{Area}(T_\uparrow(\sigma)) + \text{Area}(T_\downarrow(\sigma)) \leq \text{Area}(\hat{B}_\uparrow) + \text{Area}(\hat{B}_\downarrow) + C\sigma. \quad (7.3b)$$

Choose $\sigma \in (0, \sigma_0]$ with $C\sigma < \varepsilon_0$. Combining (7.3a), (7.3a') and (7.3b),

$$\text{Area}(T_\uparrow(\sigma)) + \text{Area}(T_\downarrow(\sigma)) \leq \text{Area}(S_+) + \text{Area}(S_-) - \varepsilon_0. \quad (7.3c)$$

(iv) Discarding the closed components. $P_\uparrow, P_\downarrow$ are the components with boundary of $T_\uparrow(\sigma), T_\downarrow(\sigma)$; the remaining components are closed and of positive area (non-empty embedded closed surfaces), so

$$\text{Area}(P_\uparrow) + \text{Area}(P_\downarrow) \leq \text{Area}(T_\uparrow(\sigma)) + \text{Area}(T_\downarrow(\sigma)). \quad (7.3d)$$

(v) Conclusion. By (7.3c) and (7.3d), this family of smooth neatly embedded representatives — with boundary a leaf of J , by clause 2 of Lemma 7.0.B — consists of legitimate competitors in the two P -classes, whose areas sum to at most $\text{Area}(S_+) + \text{Area}(S_-) - \varepsilon_0$. The relative least area is the infimum over the isotopy class, so

$$A(w_\uparrow) + A(w_\downarrow) \leq \text{Area}(P_\uparrow) + \text{Area}(P_\downarrow) \leq \text{Area}(S_+) + \text{Area}(S_-) - \varepsilon_0 < A(v_+) + A(v_-),$$

which is (7.3). □

Remark (Order of choices; no circularity).

$$S_\pm \Rightarrow (\Gamma, \theta_0, \ell_0) \Rightarrow \eta \Rightarrow \varepsilon_0 \Rightarrow (\hat{B}_\uparrow, \hat{B}_\downarrow) \Rightarrow C \Rightarrow \sigma.$$

Each step uses only the quantities before it. Shrinking σ further to avoid Z , as in clause 2 of Lemma 6.1 and in part (b) above, does not disturb this order: by clause 3 of Lemma 7.0.B all $\sigma \in (0, \sigma_0]$ give the *same* isotopy classes, while (7.3) is a statement about an infimum over an isotopy class and needs only one σ as a witness. The former only changes representatives, the latter only needs one witness, and the two do not interfere.

7.3 The non-transverse case: recovering the quantifiers

The order of quantifiers to be proved is

$$\begin{aligned} & \exists (w_{\uparrow}, w_{\downarrow}) \forall z : \quad z \text{ equal or adjacent to both endpoints} \\ \implies & \quad z \text{ equal or adjacent to both of these two fixed outputs.} \end{aligned} \tag{7.6}$$

Choosing the perturbation *after* the representative Z of z is known is not enough, unless the output vertices are independent of that after-the-fact perturbation. The four steps below prove exactly that independence. Put $A = S_-$, $B = S_+$, and let S still be the third least-area surface, disjoint from both.

Step 1: one family of perturbations that works for *all* small $t > 0$

We construct a *family* A_t , $0 \leq t \leq \delta$, with $A_0 = A$, such that:

- $A_t \cap S = \emptyset$ for all t ;
- for each $t > 0$, A_t is transverse to B in the interior and $\partial A_t \cap \partial B = \emptyset$;
- ∂A_t is a leaf of the fixed longitudinal foliation;
- the interior perturbation is of the *subtract-a-constant* type (7.13') — this is a hypothesis of Lemma 7.2.B and cannot be replaced;
- $A_t \rightarrow A$ in C^1 , so $\text{Area}(A_t) \rightarrow \text{Area}(A)$.

Remark (A requirement that has been deleted). An earlier draft also listed here: “a fixed transverse disc $D_- \subset A$, used for the rounding gain, is contained in every A_t ”. That clause was *never proved anywhere*, and it was at the time the sole load-bearing ingredient of the strict area decrease in step 4. Corollary 7.2.C now supplies a lower bound for the gain that is independent of t , so D_- is no longer needed, and the clause has been deleted rather than proved.

If the two boundary leaves are already disjoint, only the interior construction at the end of this step is needed. So suppose they coincide along a longitude L . Take a vector field on $\partial E(K)$ transverse to the longitudinal foliation whose local flow carries leaves to leaves, and extend it into the collar. After shrinking the collar use its flow direction as a coordinate y , use $r \geq 0$ as the inward boundary coordinate, and use s along L . The extension may be taken transverse along L to the tangent planes of both surfaces, so the two neat surfaces are graphs over one and the same half-cylinder:

$$\partial E(K) = \{r = 0\}, \quad B = \{y = 0\}, \quad A = \{y = u(s, r)\}, \quad u(s, 0) = 0. \tag{7.7}$$

The longitudinal leaves on $\{r = 0\}$ are $y = \text{const}$. Since both graphs are minimal in the interior and smooth up to the boundary (Theorem C.3, Appendix C), subtracting the two quasilinear minimal graph equations gives a smooth uniformly elliptic homogeneous linear equation for u , valid up to $r = 0$:

$$Lu = \partial_i(a^{ij}\partial_j u) + b^i\partial_i u + cu = 0. \tag{7.8}$$

To remove the possible zeroth-order term: on a sufficiently small half-disc solve $Lh = 0$ with boundary value 1; equivalently solve the zero boundary value problem for $h = 1 + w$, and after rescaling to a small disc the elliptic estimates of Appendix E (Theorems E.2 and E.5: existence and uniqueness for the Dirichlet problem, with an estimate involving only the data; Theorem E.13: $C^{1,\alpha}$

up to the boundary) give $\|w\|_\infty = O(R^2)$, so $h \geq 1/2$. Put $v = u/h$. Expanding $L(hv) = 0$ using $Lh = 0$ gives

$$\partial_i(h a^{ij} \partial_j v) + (a^{ij} \partial_j h + h b^i) \partial_i v = 0. \quad (7.9)$$

This equation has Lipschitz leading coefficients, bounded first-order coefficients and no zeroth-order term, and v vanishes on the flat boundary. Multiplying both sides by the reciprocal square root of the determinant of the positive definite leading matrix normalises that determinant; this changes neither the solution set nor anything but the bounded first-order coefficients.

Apply to (7.9) the reflected similarity principle (Theorem E.51, Appendix E; for the interface form see Corollary E.40). The isothermal, boundary-flattening coordinates it requires are supplied by Theorem E.34 (isothermal coordinates up to the boundary), in which the boundary term a^{12} is handled by Lemma E.35 (mollified shear) and *not* by Fermi coordinates. If $p \in L$ is a critical point of v , the factorisation takes the form

$$2 \partial_z v = e^{q(z)} G(z), \quad (7.10)$$

with q bounded and G holomorphic with a zero of finite order m at p . In the isothermal boundary-flattening coordinates z of Theorem E.34, and hence also in the original locally bi-Lipschitz coordinates (s, r) , at distance ρ from p ,

$$|dv| \asymp \rho^m, \quad |v| = O(\rho^{m+1}). \quad (7.11)$$

Since $u = hv$,

$$|du| = |h dv + v dh| \geq c_1 \rho^m - c_2 \rho^{m+1} > 0 \quad (7.12)$$

on a sufficiently small punctured half-neighbourhood of p . So every boundary critical point of u is isolated even among the interior critical points; at a non-critical boundary point continuity gives the same local absence of critical points. The alternative branch, “ u vanishes on an open set”, is excluded: strong unique continuation (Theorem E.48, Appendix E, obtained from the interior similarity principle Theorem E.43) would make the two connected minimal sheets coincide (Corollary E.55), contradicting $d(v_-, v_+) = 2$. As L is compact, we obtain a uniform collar containing only finitely many boundary critical points, all on the level $u = 0$.

Take a smooth cut-off $\chi(r)$, equal to 1 near $r = 0$ and supported in that collar, with its transition band placed where $u = 0$ is regular. Modify the graph (7.7) to

$$y = u(s, r) + t \chi(r). \quad (7.13)$$

Near the boundary its intersection with B is the regular level set $u = -t$, transverse for every sufficiently small non-zero t . On the compact transition band, if transversality failed for some $t_n \rightarrow 0$, the bad points would converge to a point with $u = 0$ and $du = 0$, contradicting the choice of the band. At $r = 0$ the boundary of (7.13) is $y = t$, a longitudinal leaf disjoint from L .

On the compact complement of this boundary collar the remaining non-transverse points are interior and finite in number: in an interior graph chart the same elliptic difference equation admits the similarity factorisation (7.10) without reflection, so critical points have finite order — unless the difference of the two graphs vanishes on an open set, which is excluded as above by strong unique continuation (Theorem E.48) and $d(v_-, v_+) = 2$ — and are therefore isolated; compactness makes the set finite. Call this set of interior tangency points $\mathcal{T}$, and put $N := \#\mathcal{T} < \infty$.

In pairwise disjoint small discs D_p around the points of $\mathcal{T}$, perturb only the A sheet, using a *subtract-a-constant* perturbation: take $D'_p \Subset D_p$ and a bump function ψ_p with $\psi_p \equiv 1$ on D'_p , supported in D_p , $0 \leq \psi_p \leq 1$, and change the graph of A over D_p to

$$y = u - t \psi_p. \quad (7.13')$$

This already yields the required transversality, with nothing cited from outside: on D'_p one has $A_t \cap B = \{u = t\}$, and the only critical point of u in D_p is p , with critical value 0 only (as $u(p) = 0$), so every $t \neq 0$ is a *regular value*, $\{u = t\}$ is regular, and $A_t \pitchfork B$. On the transition annulus $D_p \setminus D'_p$ one has $|du| \geq c > 0$ (compact, no critical points), hence $|d(u - t\psi_p)| \geq c/2 > 0$ for $t \leq c/(2\|d\psi_p\|_\infty)$, again transverse; and outside D_p , $A_t = A$ is unchanged. The D_p are pairwise disjoint, so take δ to be the least of the thresholds. The untouched compact part is already transverse, and openness of transversality handles it after one further shrinking of δ .

Remark (Sard is not needed here, and no perturbation family need be cited). The last paragraph used only the regular value theorem, and “the only critical value is 0” was read off directly from the fact that the only critical point of u in D_p is p . An earlier draft cited Lemma 1.5 of entry 3 of §13 (families of the form $\phi + t\psi$). That lemma guarantees transversality only, and gives no control whatever over the *topology* of the intersection curves — which is exactly what Lemma 7.2.B below needs. With (7.13') in place, *Lemma 1.5 of entry 3 of §13 is no longer used*.

The collar and the balls may be taken within positive distance of the compact surface S . Small graph modifications preserve embeddedness and neatness. All the listed properties now hold.

Fix a reference value $t_* > 0$. **Claim:** $A_{t_*} \triangle B$ has no disc patch.

Proof of the claim. Let Δ be a disc patch, that is, a disc component, with boundary made of exchange seams, arising when A_{t_*} and B are cut along $A_{t_*} \cap B$. Its boundary $\partial\Delta$ is a closed intersection curve of $A_{t_*} \cap B$: if it contained a boundary arc, the endpoints of that arc would lie in $\partial A_{t_*} \cap \partial B$, which step 1 has arranged to be empty; so $\partial\Delta$ is a closed circle. Now $\partial\Delta$ bounds a disc on both A_{t_*} and B : on one of them it is Δ itself, and on the other — call it Δ' — because $\partial\Delta$ is null-homotopic there, B being incompressible while $\partial\Delta = \partial\Delta'$ bounds the disc Δ in $E(K)$; the same argument applies to A_{t_*} , which differs from the incompressible A by a small graph modification and so is still incompressible. Since $E(K)$ is irreducible, the sphere $\Delta \cup \Delta'$ bounds a ball $\mathcal{B}$, which exhibits a *product region* $\Delta \times [0, 1]$ between A_{t_*} and B .

But A and B are relatively least-area in their respective normal isotopy classes.

Remark (One citation that may not be made here). That there is no product region between the unperturbed pair (A, B) is known in the literature (Lemma 1.2 of entry 3 of §13); but that result assumes both surfaces *minimal*, and A_{t_*} is *not* minimal: by (7.13), (7.13') it is a minimal graph plus a cut-off function unrelated to the solution, and the minimal graph equation is quasilinear, so a solution plus a non-constant function is in general no longer a solution; likewise $\text{Area}(A_{t_*}) > \text{Area}(A)$ makes it not relatively least-area in its isotopy class either. So Lemma 1.2 may not be applied to (A_{t_*}, B) . The argument below does not use it, and uses only the relative least-area property of A and of B separately. This must not, however, be reported as “Lemma 1.2 leaves the chain”: it is still used in §7.1 (the first case), where S_+ and S_- are *both* relatively least-area and thus fall squarely within its hypotheses. What leaves the chain is that one application to (A_{t_*}, B) , not the lemma.

The product region $\Delta \times [0, 1]$ is compact, with one boundary disc on B and one on A_{t_*} . Perform a *two-way* area exchange: replace Δ in A_{t_*} by a parallel copy of Δ' across $\mathcal{B}$, and simultaneously replace Δ' in B by a parallel copy of Δ , obtaining A', B' . The exchange takes place inside the ball $\mathcal{B}$, so A' is normally isotopic to A_{t_*} and B' to B , and the total area tends to $\text{Area}(A_{t_*}) + \text{Area}(B)$ as the push-off tends to zero. Along $\partial\Delta$ the surfaces A', B' have genuine corners (as $A_{t_*} \pitchfork B$), and rounding them by Lemma 7.0.A (variable-width version) gives smooth embedded surfaces A'', B'' , still with boundary a leaf of J , with

$$\text{Area}(A'') + \text{Area}(B'') \leq \text{Area}(A_{t_*}) + \text{Area}(B) - \mathcal{G}(t_*),$$

where $\mathcal{G}(t_*) > 0$ is the rounding gain along $\partial\Delta$. By the relative least-area property of A and of B separately, $\text{Area}(A'') \geq \text{Area}(A)$ and $\text{Area}(B'') \geq \text{Area}(B)$, hence

$$\mathcal{G}(t_*) \leq \text{Area}(A_{t_*}) - \text{Area}(A) =: e(t_*). \quad (7.15a)$$

And the C^1 convergence of step 1 gives $e(t) \rightarrow 0$. So it suffices to prove

$$\exists \delta' > 0, \mathcal{G}_0 > 0 : \quad \mathcal{G}(t) \geq \mathcal{G}_0 \quad \text{for all } t \in (0, \delta'], \quad (7.15b)$$

for then one may fix $\mathcal{G}_0$ *first* and *afterwards* choose $t_* \in (0, \delta']$ with $e(t_*) < \mathcal{G}_0$, contradicting (7.15a) and proving the claim. $\square$

Remark (Quantifiers). An earlier draft wrote here “the gain depends only on Δ and on the intersection angle, and is independent of t ”, which is *circular*: Δ exists only once t has been chosen, and both the length of $\partial\Delta$ and the angle along it vary with t , while $\varepsilon \geq e(t)$ must in turn be smaller than that gain. Statement (7.15b) puts what has to be proved into the correct quantifier order: $\mathcal{G}_0$ *must be pinned down before* t . The three results below prove exactly (7.15b).

Lemma 7.2.A (A tangency point is not a local extremum of u). *Let $p \in \mathcal{T}$ be an interior tangency point and let u be the difference function writing A as a normal graph over B near p . Then u takes both positive and negative values in every neighbourhood of p .*

Proof. Suppose instead $u \geq 0$ on some neighbourhood D_p (the case $u \leq 0$ is identical).

First remove the zeroth-order term — and this must be redone at p ; the h of step 1 cannot be reused. The function h constructed in step 1 was solved on a *boundary* half-disc with the flat boundary $L \subset \partial E(K)$, whereas p is an *interior* tangency point lying outside that half-disc, so the factorisation $u = hv$ does not exist near p . Redo the same construction at p : take the *disc* B_R of centre p and radius R inside the interior — L is a *two-dimensional* operator whose domain is a planar region, so “ball” would be a slip of notation — and solve $Lh_p = 0$ with $h_p|_{\partial B_R} = 1$. For small R the rescaled elliptic estimates (Theorems E.2 and E.5, and interior $C^{1,\alpha}$ regularity, Theorem E.9, Appendix E; divergence form) give $\|h_p - 1\|_\infty = O(R^2)$, so $h_p \geq 1/2$ on B_R . Put $v_p := u/h_p$; the same expansion as in (7.9), using $Lh_p = 0$, gives

$$\partial_i(h_p a^{ij} \partial_j v_p) + (a^{ij} \partial_j h_p + h_p b^i) \partial_i v_p = 0 \quad \text{on } B_R,$$

with Lipschitz leading term, bounded first-order term, *no zeroth-order term*, and uniform ellipticity.

Now apply the extremum principle. From $u \geq 0$ and $h_p \geq 1/2$ we get $v_p \geq 0$ on B_R , with $v_p(p) = u(p)/h_p(p) = 0$. The Hopf strong maximum principle for a uniformly elliptic operator without zeroth-order term says that a non-negative solution attaining the value zero at an interior point vanishes identically on that component; so $v_p \equiv 0$, that is $u \equiv 0$ on B_R . By strong unique continuation (Theorem E.48, Appendix E, obtained from the interior similarity principle (V1), Theorem E.43 — used only on interior open sets), u vanishes identically on that component of its domain, that is, A and B coincide on an open set; and two connected minimal surfaces coinciding on an open set coincide (Corollary E.55: unique continuation together with connectedness, Appendix E), contradicting $d(v_-, v_+) = 2$. $\square$

Remark (Why this repetition cannot be avoided). Applying the extremum principle to u directly does not work: the equation (7.8) satisfied by u carries a zeroth-order term c whose sign is not controlled, and the strong extremum principle with a zeroth-order term is available only when $c \leq 0$, and even then protects only one side of the zero level set. *The removal of the zeroth-order term has to be carried out afresh at each point where it is used*; having done it at the boundary does not mean it has been done in the interior.

Lemma 7.2.B (No small closed intersection curves near a tangency point). *There are $\rho_0 > 0$ and $\delta_1 > 0$ such that for every $t \in (0, \delta_1]$ every closed component γ of $A_t \cap B$ satisfies $\text{diam}(\gamma) \geq \rho_0$.*

Proof. Suppose instead that there are $t_n \rightarrow 0$ and closed components γ_n with $\text{diam}(\gamma_n) \rightarrow 0$. Since $A_t \rightarrow A$ in C^0 and the surfaces are compact, $A_{t_n} \cap B$ lies in an arbitrarily small neighbourhood of $A \cap B$ (otherwise a convergent subsequence would produce a limit point outside $A \cap B$, a contradiction). Take $x_n \in \gamma_n$ with $x_n \rightarrow x \in A \cap B$, and distinguish three cases.

(a) *x is a transverse point of $A \cap B$ outside the boundary collar.* In a fixed small ball at x one has $A \pitchfork B$, and the intersection is a single arc crossing that ball; in that ball A_t differs from A by t times a fixed C^1 function (or not at all), so by C^1 stability, for small t , $A_t \cap B$ is still a single arc crossing the ball, with *no closed component*. For large n the curve γ_n lies in that ball: a contradiction.

(b) *$x \in \mathcal{T}$ is an interior tangency point.* Take a small disc $D_x \subset B$ at x in which the only critical point of u is x — by (7.12), x has a punctured neighbourhood free of critical points, and critical points $q_k \rightarrow x$ would have to lie in it. For large n , γ_n lies in D_x and, by the Jordan curve theorem, bounds a disc D_n inside D_x . In D_x one has $A_t \cap B = \{u_t = 0\}$, where $u_t = u - t$ on the core and $u_t = u - t\psi_x$ with $|du_t| \geq c/2 > 0$ on the transition annulus. The function u_t is continuous on $\overline{D_n}$ and vanishes on $\partial D_n = \gamma_n$. If $u_t \equiv 0$ on D_n then $du_t \equiv 0$ on D_n , but the zeros of du_t in D_x number at most the single point x : a contradiction. Otherwise u_t attains a strictly positive maximum or a strictly negative minimum in the interior of D_n ; that point is critical, hence equals x ; but $u_t(x) = u(x) - t = -t$, and by Lemma 7.2.A, x is not a local extremum of u , hence not of $u_t = u - t$ either, the two differing by a constant on the core. A contradiction.

(c) *x lies in the boundary collar.* In the uniform collar produced by step 1 all critical points of u lie on $\{r = 0\}$, so u has no critical point in the open collar $\{0 < r < r_0\}$. There $u_t = u + t\chi(r)$ with $\chi \equiv 1$ on $\{r \leq r_1\}$: on $\{0 < r \leq r_1\}$ we have $du_t = du \neq 0$, and on the compact transition band $\{r_1 \leq r \leq r_0\}$ we have $|du| \geq c > 0$, hence $|du_t| \geq c/2$ for small t . The small disc D_n bounded by γ_n does not meet ∂B , so $D_n \subset \{r > 0\}$, where u_t has no critical point; hence both the maximum and the minimum of u_t on $\overline{D_n}$ are attained on ∂D_n and both equal 0, so $u_t \equiv 0$ on D_n and $du_t \equiv 0$, contradicting $|du_t| \geq c/2$. $\square$

Remark (This is stronger than it looks, and where its weight lies). The proof used only that γ_n is a small-diameter closed component of $A_t \cap B$; it made *no* use of which of the two surfaces carries the patch it bounds. So the conclusion covers both the case $\Delta \subset A_t$ and the case $\Delta \subset B$.

Case (b) depends throughout on step 1's having taken the interior perturbation in the *subtract-a-constant* form (7.13'). With a family that merely guarantees transversality, the fact that u_t and u differ by a constant on the core would be lost, Lemma 7.2.A could not be brought to bear, and this lemma would fail with it.

Corollary 7.2.C (Three positive lower bounds independent of t). *Choose $\rho_2 > 0$ such that the balls $B_{2\rho_2}(p)$, $p \in \mathcal{T}$, are pairwise disjoint and $4N\rho_2 < \rho_0/8$, and put $P := \bigcup_{p \in \mathcal{T}} B_{2\rho_2}(p)$ and $G := E(K) \setminus P$. Then there are $\theta_1 > 0$, $\eta_1 > 0$, $\delta_2 > 0$ such that for every $t \in (0, \delta_2]$ and every closed component γ of $A_t \cap B$:*

1. $\text{length}(\gamma \cap G) \geq \rho_0/8$;
2. the angle between A_t and B along $\gamma \cap G$ is at least θ_1 ;
3. the admissible rounding width may be taken to be the constant $\eta_1 > 0$ on $\gamma \cap G$.

Proof. 1. By Lemma 7.2.B, $\text{diam}(\gamma) \geq \rho_0$, so there are $a, b \in \gamma$ with $d(a, b) \geq \rho_0/2$. Let γ_{ab} be one of the two arcs of γ from a to b and put $L := \text{length}(\gamma_{ab} \cap G)$. Collapse each of the N connected

components of P to a point, giving a quotient pseudometric d' . Collapsing N sets of diameter $\leq 4\rho_2$ decreases distances by at most $4N\rho_2$, so $d(a, b) \leq d'(a, b) + 4N\rho_2$; and the image of γ_{ab} in the quotient is a path from $[a]$ to $[b]$ whose length is exactly $\text{length}(\gamma_{ab} \cap G) = L$, the portions inside P having been reduced to length zero. Hence

$$\rho_0/2 \leq d(a, b) \leq d'(a, b) + 4N\rho_2 \leq L + 4N\rho_2 < L + \rho_0/8,$$

so $L > \rho_0/2 - \rho_0/8 = 3\rho_0/8 > \rho_0/8$.

Remark (Why the quotient metric is necessary). A curve may enter and leave the same ball *repeatedly*, with no a priori bound on the number of visits, so the argument “at most N balls, each displacing by at most $4\rho_2$, hence total displacement $\leq 4N\rho_2$ ” is *wrong*. The quotient-metric formulation is immune to repeated visits: however many times the curve passes through, each component is collapsed only once.

2. Put $G' := E(K) \setminus \bigcup_p B_{\rho_2}(p)$. The set $A \cap B \cap G'$ is compact and everywhere transverse, so the angle has a positive infimum $2\theta_1 > 0$. For small t , $A_t \cap B$ lies in the $\rho_2/2$ -neighbourhood of $A \cap B$, and on G the C^1 distance from A_t to A is at most $C't$ (there the two differ by t times a fixed C^1 function). Hence the angle along $\gamma \cap G$ is at least θ_1 .

3. The rounding width is constrained only by the normal injectivity radius of γ inside the surface being rounded; on G the curve γ is a C^1 -small perturbation of $A \cap B \cap G'$, a fixed compact 1-manifold of positive normal injectivity radius, so a uniform $\eta_1 > 0$ exists. $\square$

Remark (Two uses). The corollary holds for every closed component of $A_t \cap B$, regardless of whether that component bounds a patch. In (7.15c) below it is applied to a patch boundary $\partial\Delta$; in step 4 it is applied to the *whole* curve $\Gamma_t = A_t \cap B$. The latter use is what makes the “fixed transverse disc D_- ” of the earlier draft superfluous — see the remark at the end of step 4.

Proof of (7.15b)

By the variable-width version of Lemma 7.0.A the rounding gain is non-negative slice by slice, so it suffices to account for it on $\partial\Delta \cap G$ (note $\partial\Delta$ is a closed component of $A_t \cap B$, so Corollary 7.2.C applies):

$$\mathcal{G}(t) \geq g_0(\theta_1) \cdot \eta_1 \cdot \text{length}(\partial\Delta \cap G) \geq g_0(\theta_1) \cdot \eta_1 \cdot \rho_0/8 =: \mathcal{G}_0 > 0 \quad (7.15c)$$

for every $t \in (0, \min\{\delta_1, \delta_2\}]$. Here $\mathcal{G}_0$ depends only on A, B and the once-chosen perturbation scheme, and *not on t* . So (7.15b) holds and the claim is proved.

Remark (Order). $A, B \Rightarrow (\rho_0, \rho_2, \theta_1, \eta_1) \Rightarrow \mathcal{G}_0 \Rightarrow t_*$. Each step uses only what precedes it. This is precisely the layer that was missing at the point identified by external audit as SP-1.

The next step transports this property from t_* to every other $t > 0$.

Step 2: a transverse paired isotopy over a positive parameter interval

Lemma (Paired isotopy lemma). *Let M be compact with boundary, let A_t ($t \in J = [a, b]$) be a smooth family of compact neat submanifolds, and let B, S be fixed compact neat submanifolds with $A_t \pitchfork B$, $\partial A_t \cap \partial B = \emptyset$ and $S \cap (A_t \cup B) = \emptyset$ for $t \in J$. Then for $t_0 \in J$ there is an ambient isotopy H_t with*

$$H_{t_0} = \text{id}, \quad H_t(A_{t_0}) = A_t, \quad H_t(B) = B, \quad H_t|_S = \text{id}, \quad (7.14)$$

and each H_t preserves ∂M .

Proof. Take embeddings $f_t : F \rightarrow M$ with image A_t and put $v_t(f_t(x)) = \partial_t f_t(x)$. At $p = f_t(x) \in A_t \cap B$, transversality makes $T_x F \rightarrow T_p M / T_p B$ surjective. The metric gives a smooth right inverse along the compact intersection track, and hence a smooth choice of $\xi(x, t)$ with

$$q_t(f_t(x)) := v_t(f_t(x)) + df_t(\xi(x, t)) \in T_p B. \quad (7.15)$$

The intersection track stays uniformly away from ∂M , because the two boundary tracks are compact and disjoint over J . Extend ξ to $F \times J$, taking it to be zero near $\partial F \times J$; then q_t is tangent to ∂M along ∂A_t .

Extend q to a time-dependent field X_t on M , tangent to B and to ∂M and vanishing on S . At $A_t \cap B$ take a defining coordinate r for B ; since $A_t \pitchfork B$, $r|_{A_t}$ defines the intersection. By (7.15), $q(r) = 0$ there, so Hadamard's lemma gives $q(r) = rh$ along the A -track. Extend the tangential coefficients and h , and use $r\tilde{h}$ as the normal coefficient: it agrees with q and is tangent to B . Near ∂A_t , which is separated from B , the same boundary defining function construction makes the extension tangent to ∂M ; near ∂B take it to be zero. A partition of unity preserves all these linear tangency conditions, and its supports may be taken disjoint from S .

Integrate X_t : M is compact, so the flow exists, preserves B and ∂M and fixes S . If $x(t)$ solves $x' = \xi(x, t)$ then

$$\frac{d}{dt} f_t(x(t)) = v_t + df_t(\xi) = q_t = X_t(f_t(x(t))),$$

so by uniqueness for the ODE the flow carries A_{t_0} to A_t . $\square$

Step 3: the cyclic-cover output classes are independent of $t > 0$

Fix the reference value $t_* > 0$. Applying the paired isotopy lemma on the compact interval with endpoints t_* and any other $t > 0$, lift the resulting ambient isotopy, starting at the identity, uniquely to the infinite cyclic cover $p : \tilde{E} \rightarrow E(K)$. Since the isotopy fixes S pointwise, the lift fixes the chosen lift S_0 , its deck translate S_1 , and the fundamental domain C between them; uniqueness of the lift also makes it commute with the generator of the deck group. It preserves the chosen lift of B setwise, and carries the chosen lift of A_{t_*} to the chosen lift of A_t .

For any wall, the upper side is the complementary component containing S_1 . As S_1 is fixed, the isotopy cannot interchange the two sides. Hence it carries

$$\text{fr}(M_{\downarrow}^{A_{t_*}} \cap M_{\downarrow}^B) \quad \text{and} \quad \text{fr}(M_{\uparrow}^{A_{t_*}} \cap M_{\uparrow}^B)$$

to the corresponding frontiers at t , and carries a small push-off on the indicated side to an admissible push-off; and any two sufficiently small such push-offs are collar isotopic. Projecting back to $E(K)$,

$$[P_{\uparrow}(t)] = [P_{\uparrow}(t_*)], \quad [P_{\downarrow}(t)] = [P_{\downarrow}(t_*)] \quad (\forall t > 0). \quad (7.16)$$

A diffeomorphism preserves which components have boundary, hence preserves the component carrying the longitude.

At t_* choose, once and for all, a finite compression sequence from each coarse surface to an incompressible descendant with boundary, giving two vertices $w_{\uparrow}, w_{\downarrow}$. Transporting the coarse surfaces and the compressing discs along the paired isotopy gives, at every $t > 0$, the same ambient isotopy descendant vertices. *Hence the coarse classes and the chosen output vertices are fixed before z is chosen.*

Step 4: recovering the required order of quantifiers

Let z be distinct from the endpoints and equal or adjacent to both v_+ and v_- , with relatively least-area representative Z . Entry 2 of §13 gives a representative with $Z \cap (A \cup B) = \emptyset$. These compact sets are at positive distance. Since $A_t \rightarrow A$ in C^0 , *after Z is known* one may choose $t_z > 0$ small enough that $Z \cap A_{t_z} = \emptyset$ (Z and B being disjoint already). At t_z take the two frontier push-offs small enough to avoid Z . By (7.16) they represent the two coarse classes fixed at t_* . Feeding the two compression sequences chosen at t_* and transported to t_z into Corollary 3.3 yields the same fixed descendants $w_\uparrow, w_\downarrow$, now disjoint from Z . If z equals an endpoint, (a) already gives the conclusion. This proves (7.6): *t_z changes the representative, not the output vertex.*

The remaining conclusions are unchanged: the genus computation is topological and is transported by the isotopy. As for the *strict area decrease*:

Since $\partial A_t \cap \partial B = \emptyset$, the set $\Gamma_t := A_t \cap B$ is a compact 1-manifold *without boundary*, that is, a union of finitely many closed curves; and $\Gamma_t \neq \emptyset$, for otherwise A_t and B would give disjoint representatives of v_-, v_+ , contradicting $d(v_-, v_+) = 2$. So Corollary 7.2.C applies to every component of Γ_t , and in particular to *at least one*; and since the slicewise gain in Lemma 7.0.A is non-negative everywhere, the total gain from rounding the two cornered frontiers along Γ_t is at least the contribution of the part of that one component lying in G :

$$\varepsilon_* := g_0(\theta_1) \cdot \eta_1 \cdot \rho_0/8 > 0, \quad \text{independent of } t, \text{ and of the push-off scale } \sigma. \quad (7.17)$$

Order: $A, B \Rightarrow (\rho_0, \rho_2, \theta_1, \eta_1) \Rightarrow \varepsilon_* \Rightarrow \delta \Rightarrow (\hat{B}_\uparrow, \hat{B}_\downarrow) \Rightarrow C \Rightarrow \sigma$. First shrink δ so that

$$\text{Area}(A_t) < \text{Area}(A) + \varepsilon_*/2 \quad (0 < t \leq \delta), \quad (7.18)$$

then, as in (7.3b) of §7.1, choose the push-off scale σ with $C\sigma < \varepsilon_*/4$. Applying (i)–(v) of §7.1 to the transverse pair (A_t, B) for $t \in (0, \delta]$, with Γ replaced by Γ_t and ε_0 by ε_* , gives

$$\begin{aligned} \text{Area}(P_\uparrow) + \text{Area}(P_\downarrow) &\leq \text{Area}(A_t) + \text{Area}(B) - \varepsilon_* + C\sigma \\ &< \text{Area}(A) + \text{Area}(B) - \varepsilon_*/4 = A(v_-) + A(v_+) - \varepsilon_*/4, \end{aligned}$$

so $A(w_\uparrow) + A(w_\downarrow) < A(v_-) + A(v_+)$. Finally use (7.16) to read this inequality as one about the two output vertices fixed at t_* (the isotopy classes agree, and $A(\cdot)$ depends only on the isotopy class). This completes the proof of Theorem 7.1.

Remark (Where the earlier draft broke, and why one plausible repair does not work). The earlier version read: “fix a transverse disc D_- and the rounding gain $\varepsilon > 0$ it provides; the constructed family is stationary on D_- .” The existence of D_- was never proved, and it was at the time the sole load-bearing ingredient of this step. Another apparently equivalent route — “establish the strict inequality at the reference value t_* as in the transverse case, then transport it by (7.16)” — *cannot replace it*: at t_* one obtains $\text{Area}(P_\uparrow) + \text{Area}(P_\downarrow) \leq \text{Area}(A_{t_*}) + \text{Area}(B) - \varepsilon_0(t_*)$, while $\text{Area}(A_{t_*}) > \text{Area}(A)$, so the excess $e(t_*)$ must be beaten by $\varepsilon_0(t_*)$; but $\varepsilon_0(t_*)$ depends on the length of $A_{t_*} \cap B$ and on the angle along it, both of which vary with t_* , while t_* can only be taken small once ε_0 is known — a circle in the quantifiers. Equation (7.16) transports isotopy classes; it cannot transport an order of quantifiers. *Corollary 7.2.C supplies exactly the missing layer*: an ε_* pinned down before t is chosen.

Remark (The existence of this family must be proved, not cited). A family that “works for all small t and preserves the longitudinal leaves” is *false* for an arbitrary pair of smooth surfaces: a counterexample is $y = e^{-1/r^2} \sin(1/r)$ over $y = 0$ with common boundary $r = 0$, whose exceptional translation parameters accumulate at zero. *The finite-order property of minimal sheets, from ellipticity, is essential in step 1.*

8 The interface: two combinatorial axioms

Let X be a connected flag complex. For $d(u, w) = 2$, the *two-interval* is

$$I_X(u, w) = N_X(u) \cap N_X(w),$$

the set of common neighbours. A point $x \in I_X(u, w)$ is called an *apex* if it is equal or adjacent to every member of that interval; equivalently, the full subcomplex on the interval is a cone with apex x .

Let W be a *well-ordered* total order, that is, one in which every non-empty subset has a least element. An *exchange structure* is a map $c : X^{(0)} \rightarrow W$ satisfying:

Remark (The relation between the two formulations, which are not interchangeable). For a total order, “every non-empty subset has a least element” implies “no infinite strictly decreasing sequence”, and this is a theorem of ZF; the *converse implication requires dependent choice* (DC) — one must choose, infinitely often, an element from each non-empty set strictly below — and so is *not* a theorem of ZF. This paper uses the well-ordering theorem (equivalent to AC) in Lemma 9.1, and $AC \Rightarrow DC$, so within the framework of this paper the two formulations are equivalent; *when citing this section, cite that dependency with it*. Below, “every non-empty subset has a least element” is the standard: (N2), Theorem 9.3 and Theorem 10.1 all use that form, while what Lemma 5.1 proves directly is “no infinite descending sequence”. For the W actually used here the bridge can in fact be avoided — see the remark after Lemma 5.1, which produces the least element directly in ZF.

- (N1) every two-interval has an apex;
- (N2) for every pair with $d(u, w) = 2$, *some* apex x satisfies $c(x) < \max\{c(u), c(w)\}$.

A *non-empty connected* flag complex carrying such a function is called an *exchange complex*.

Remark (Non-emptiness is part of the definition and cannot be dropped). The transfinite induction of Theorem 10.1 is based at “ $\lambda = 1$, the complex is a point”; the empty complex has no such base (its order type is 0), and the empty complex is not contractible. Both appeals to non-emptiness in this paper are supplied by Proposition 2.3.

Remark (This is where the obstruction is circumvented). The definition contains *no* canonical choice of apex. This purely existential formulation does *not* solve the problem that “the projection cannot be made well defined on the set of isotopy classes”; it *goes around* it: the combinatorial part needs only that an apex exist, never that one can be selected.

Lemma 8.1 (Local descent). *In the setting of Theorem 7.1, rename the endpoints u, w and the common neighbour v . If*

$$c(v) \geq c(u), \quad c(v) \geq c(w), \tag{8.1}$$

then one of the two exchange outputs, x , satisfies

$$c(x) < c(v), \quad g(x) \leq g(v). \tag{8.2}$$

Proof. Write $g_v = g(v)$, and so on. The first coordinate of (8.1) gives $g_u, g_w \leq g_v$. By (7.2), for the two outputs

$$g(x_1) + g(x_2) \leq g_u + g_w \leq 2g_v. \tag{8.3}$$

If some output has genus $< g_v$, take it; the lexicographic order gives (8.2) at once. Otherwise both outputs have genus $\geq g_v$, so every inequality in (8.3) is an equality and $g(x_1) = g(x_2) = g_u = g_w = g_v$.

Then (7.3) applies. Since all the endpoint genera equal g_v , the second coordinate of (8.1) gives $A(u), A(w) \leq A(v)$, whence

$$A(x_1) + A(x_2) < A(u) + A(w) \leq 2A(v),$$

and the output of smaller area has area $< A(v)$ and genus g_v , which is (8.2). $\square$

Proposition 8.2 (An apex usable inside the truncation). *Let u, w be at distance 2 in $\text{IS}(K)$. Both outputs of Theorem 7.1 are apices of $I_{\text{IS}(K)}(u, w)$. If moreover $g(u), g(w) \leq \ell$, then*

$$\min\{g(w_\uparrow), g(w_\downarrow)\} \leq \frac{g(u) + g(w)}{2} \leq \ell, \quad (8.4)$$

so the output of smaller genus remains in $\text{IS}_\ell(K)$ and still dominates the smaller interval.

Proof. By Theorem 7.1(a) both outputs are common neighbours of u and w ; by (b) each is equal or adjacent to every common neighbour, so both are apices. Inequality (8.4) is (7.2) together with the fact that the minimum of two numbers does not exceed their mean. The interval of a full subcomplex is a subset of the larger interval, so the domination is inherited. $\square$

9 Preparation for the combinatorial part (from here on, no three-dimensional topology)

This section and §10 use only the axioms of §8; they quote no conclusion of §§1–8 other than the definition of an exchange complex.

Lemma 9.1 (Ties may be broken). *If X carries an exchange structure, then it carries one whose complexity well-orders the vertex set injectively.*

Proof. The set of attained values is a well-founded total order. By the well-ordering theorem (equivalently, the axiom of choice) well-order each fibre $c^{-1}(a)$, and order the vertices lexicographically: first by the old complexity value, then by the chosen order within the fibre. A lexicographic sum of well-orders indexed by a well-order is again a well-order, and the new complexity is injective. Axiom (N1) does not mention the complexity. If the old (N2) had a witness x with $c(x) < \max\{c(u), c(w)\}$, then its old value block still precedes the block containing the larger endpoint, so the strict inequality survives the refinement, and (N2) does too. $\square$

Lemma 9.2 (A reduction measure). *Let W be a well-founded total order. For a finite path or finite cycle $\gamma = (v_1, \dots, v_n)$ whose vertices take values in W , call the movable vertices all vertices of a cycle, or the interior vertices of a path with fixed endpoints, and put*

$$\mu(\gamma) = \left(n, \max_i c(v_i), \#\{i : c(v_i) = \max_j c(v_j)\} \right), \quad (9.1)$$

ordered lexicographically. Then μ is well-founded on such finite paths; deleting a vertex lowers μ ; and replacing a vertex attaining the maximal complexity by a vertex of strictly smaller complexity also lowers μ .

Proof. The three coordinates lie respectively in the well-ordered non-negative integers, in the well-founded W , and in the positive integers, and a finite lexicographic product of well-founded orders is well-founded. Deletion lowers the first coordinate. In a replacement the length is unchanged: if the maximum was attained only once, the second coordinate drops; otherwise the second coordinate is unchanged and its multiplicity drops. $\square$

Theorem 9.3 (Complexity-convex geodesics). *Let X be an exchange complex. For any two vertices p, q there is a geodesic from p to q each of whose vertices z satisfies*

$$c(z) \leq \max\{c(p), c(q)\}. \quad (9.2)$$

Proof. Put $m = \max\{c(p), c(q)\}$ and start from a path $\gamma = (p = v_0, \dots, v_n = q)$ of minimal length $n = d(p, q)$; fix the endpoints and apply the measure of Lemma 9.2 to the interior vertices.

If every interior vertex satisfies (9.2) we are done. Otherwise the maximal complexity along the path is attained at some interior vertex v_i , so

$$c(v_i) \geq c(v_{i-1}), \quad c(v_i) \geq c(v_{i+1}). \quad (9.3)$$

If v_{i-1} and v_{i+1} are equal or adjacent, delete v_i : the path gets shorter and the measure drops. Otherwise they are at distance 2 and v_i lies in their interval. By (N2) take an apex x with

$$c(x) < \max\{c(v_{i-1}), c(v_{i+1})\} \leq c(v_i). \quad (9.4)$$

The apex is adjacent to the two endpoints of this length-two subpath and dominates v_i . If x equals one of the neighbouring endpoints, delete v_i ; otherwise replace v_i by x , leaving the length unchanged while one occurrence of the maximal complexity is replaced by a smaller one. In either case we obtain a new path from p to q whose measure is *strictly smaller* than that of the old one.

Now conclude by well-foundedness — *not* by “repeat until the process terminates”; well-foundedness supplies the existence of a least element, not the halting of a procedure. Let $\mathcal{P}$ be the set of all finite paths from p to q ; it is non-empty, containing a geodesic of length $d(p, q)$. By Lemma 9.2, μ takes values in a well-founded total order, so the non-empty set $\{\mu(\gamma) : \gamma \in \mathcal{P}\}$ has a least element; take $\gamma_0 \in \mathcal{P}$ with $\mu(\gamma_0)$ equal to that least value. If γ_0 had an interior vertex violating (9.2), the two preceding paragraphs would produce $\gamma_1 \in \mathcal{P}$ with $\mu(\gamma_1) < \mu(\gamma_0)$, contradicting minimality. So every interior vertex of γ_0 satisfies (9.2). Moreover the first coordinate of μ is the length of the path, and lexicographic minimality minimises the length first, so γ_0 has length $d(p, q)$; that is, γ_0 is a geodesic. $\square$

Corollary 9.3.1 (Truncations are connected, and isometrically embedded). *Let X be a connected exchange complex whose complexity is lexicographic, $c(v) = (g(v), A(v))$, with first coordinate g integer-valued. For an integer ℓ , let $X_{\leq \ell} \subset X$ be the full subcomplex spanned by the vertices with $g(v) \leq \ell$. Then $X_{\leq \ell}$ is connected and the inclusion is isometric on 1-skeleta:*

$$d_{X_{\leq \ell}}(u, w) = d_X(u, w) \quad (u, w \text{ vertices of } X_{\leq \ell}). \quad (9.5)$$

Proof. Let u, w be vertices of $X_{\leq \ell}$. As X is connected, $n := d_X(u, w) < \infty$. By Theorem 9.3 take a geodesic $\gamma = (u = z_0, \dots, z_n = w)$ in X with every z_i satisfying (9.2). We may suppose $\max\{c(u), c(w)\} = c(u)$. The first coordinate of the lexicographic order is g , so $c(z_i) \leq_{\text{lex}} c(u)$ implies $g(z_i) \leq g(u)$ — otherwise $g(z_i) > g(u)$ would give $c(z_i) > c(u)$ at once, contradicting (9.2). Hence $g(z_i) \leq g(u) \leq \ell$ and every z_i is a vertex of $X_{\leq \ell}$. Since $X_{\leq \ell}$ is a *full* subcomplex and each $\{z_i, z_{i+1}\}$ is an edge of X with both ends among its vertices, it is an edge of $X_{\leq \ell}$ as well. So all of γ lies in $X_{\leq \ell}$, giving connectedness and $d_{X_{\leq \ell}}(u, w) \leq n$. The reverse inequality holds trivially for any subcomplex, a path in the subcomplex being a path in X . $\square$

Remark (This is the sole source of the connectedness of $\text{IS}_\ell(K)$). Taking $X = \text{IS}(K)$ in §11, Proposition 2.1 (connectedness of $\text{IS}(K)$, cited) together with (N1) and (N2), verified in §11, makes this corollary yield the connectedness of *every* $\text{IS}_\ell(K)$. The direction is *whole complex* $\Rightarrow$ *truncation*; the warning attached to Proposition 2.1 explains why it cannot be run the other way.

Remark (No circularity). The corollary uses only “ X connected + (N1) + (N2) + well-foundedness”. The verification of those four for $\text{IS}(K)$ — Propositions 2.1, 2.2, Lemma 5.1, Theorem 7.1, Proposition 8.2 and §11 — uses *no* truncation connectedness: in §11 the branch “if X is a truncation” is not triggered when $X = \text{IS}(K)$, and (8.4) of Proposition 8.2 is an extra conclusion serving the truncations, not part of (N1) for $\text{IS}(K)$. So the order of deduction

$\text{IS}(K)$ connected (cited) $\Rightarrow \text{IS}(K)$ is an exchange complex $\Rightarrow$ Theorem 9.3 $\Rightarrow$ Corollary 9.3.1 $\Rightarrow \text{IS}_\ell(K)$ connected contains no cycle.

Remark (The honest cost). Truncation connectedness is thereby *no longer an unconditional topological fact* but a corollary *conditional on the geometric part*: it depends, through (N1) and (N2), on Theorem 7.1, hence on Theorem B.3 and Theorem C.3 (Appendices B and C, which are the (E) and (R1) of the earlier draft), on (U), and on the named-theorem conditions tabulated in §12.2. If the geometric part fails, this corollary and the connectedness of $\text{IS}_\ell(K)$ fail with it. This is the price of a change of source: the earlier draft cited “every truncation is connected” wholesale, at the cost of admitting an unrefereed preprint into the chain.

Lemma 9.4 (Heredity). *Let the exchange complexity be injective. For a vertex v let $X_{<v}$ be the full subcomplex on the vertices with $c(u) < c(v)$, and let $L(v)$ be the full subcomplex on those neighbours of v of smaller complexity (the descending link). Then*

1. *if non-empty, $X_{<v}$ is an exchange complex;*
2. *if non-empty, $L(v)$ is an exchange complex; and unless v is the least vertex, $L(v)$ is non-empty.*

Proof. A full subcomplex of a flag complex is flag. Let u, w be non-adjacent at distance 2 in $X_{<v}$, or two non-adjacent vertices of $L(v)$. In the former case a common neighbour inside the sublevel set shows they are at distance 2 in X ; in the latter v itself is a common neighbour. By (N2) the interval $I_X(u, w)$ has an apex x with

$$c(x) < \max\{c(u), c(w)\} < c(v). \quad (9.6)$$

For $X_{<v}$ this puts x directly in the sublevel set. For $L(v)$ the apex dominates the interval member v , while (9.6) excludes $x = v$; so x is adjacent to v and lies in the descending link. In both cases x lies in the smaller interval and dominates it, which proves (N1) there, and the same strict inequality proves (N2).

Connectedness of $L(v)$: any two non-adjacent members have the common neighbour x just constructed, so the link has graph diameter ≤ 2 . Connectedness of $X_{<v}$: Theorem 9.3 joins u, w in it by a geodesic of complexity $\leq \max\{c(u), c(w)\} < c(v)$, and the whole geodesic stays in the sublevel set.

Finally let v_0 be the least vertex and $v \neq v_0$. Apply Theorem 9.3 to v, v_0 ; the vertex next to v on the resulting geodesic is a neighbour $z \neq v$ with $c(z) \leq c(v)$, and injectivity forces $c(z) < c(v)$, so $z \in L(v)$. $\square$

Lemma 9.5 (Two CW facts). *Throughout, $|X|$ denotes the geometric realisation with the CW weak topology: $C \subset |X|$ is closed exactly when $C \cap \bar{e}$ is closed in $\bar{e}$ for every closed cell $\bar{e}$. For an infinite complex this differs from the metric topology given by barycentric coordinates, and part (2) below is false in the metric topology: take countably many 1-simplices $\{v_0, v_n\}$ ($n \geq 1$) glued along the common vertex v_0 , and let x_n be the point of the n th edge at distance $1/n$ from v_0 ; then $\{x_n\}_{n \geq 1} \cup \{v_0\}$ is compact in the metric topology yet meets infinitely many cells, whereas in the weak topology $\{x_n\}$ is closed and discrete, hence not compact. So the words “weak topology” are not optional phrasing.*

1. If $A \hookrightarrow B$ is both a homotopy equivalence and a cofibration, then for any cofibration $A \hookrightarrow Y$ the canonical map $Y \rightarrow Y \cup_A B$ is a homotopy equivalence.
2. A compact subset of a CW complex (with the weak topology) is contained in a finite subcomplex.

Proof. (1) A cofibration that is also a homotopy equivalence is a strong deformation retract inclusion: the homotopy extension property converts a homotopy inverse and its homotopies into a retraction $r : B \rightarrow A$ and a homotopy from id_B to “inclusion after r ” which is stationary on A . Gluing that deformation to the constant deformation of Y gives a strong deformation retraction of $Y \cup_A B$ onto Y . The inclusion of a simplicial subcomplex is a cofibration, since the larger realisation is obtained from the smaller by attaching one cell for each new simplex, and the cellular homotopy extension construction is exactly the required HEP.

(This first part is *conditional on* (C1) — see the remark below and entry H9 of the table in §12.2 — the statement “a cofibration that is also a homotopy equivalence is a strong deformation retract inclusion” being the one homotopy-theoretic input that is not proved within this paper.)

(2) Suppose a compact set C meets infinitely many open cells. Choose one point of C from each of countably many distinct open cells. As the closure of each cell lies in a finite subcomplex, the chosen set meets each closed cell in a finite set and so is closed in the weak topology; removing any one of its points leaves a closed set, so the set is also discrete. Its intersection with C is then an infinite closed discrete subspace of the compact space C , which is impossible. So C meets only finitely many cells, and the union of their finitely many closures is a finite subcomplex containing C . $\square$

10 Closing the combinatorial part

Theorem 10.1 (Exchange complexes are contractible). The geometric realisation of every exchange complex — non-empty, connected and flag, in the sense of §8 — is contractible. No finiteness, local finiteness, dimension or diameter hypothesis is required.

Proof. By Lemma 9.1 we may assume the complexity is an injective well-ordering of the vertex set; let λ be its order type. We induct transfinitely on λ .

$\lambda = 1$: the complex is a point.

Successor case $\lambda = \alpha + 1$. Let v be the greatest vertex and put $X' = X_{<v}$. Every neighbour of v precedes it, so $L(v)$ is the whole link and the closed star is the cone $v * L(v)$ — here flagness is used: the link equals the full subcomplex on the neighbour set. For $\lambda \geq 2$, Lemma 9.4 says X' and $L(v)$ are non-empty exchange complexes of order types α and $\leq \alpha$ respectively, hence contractible by the inductive hypothesis. Every simplex of X either omits v (and lies in X') or contains v (and lies in $v * L(v)$), and

$$X' \cap (v * L(v)) = L(v).$$

Now $|L(v)|$ is contractible (inductive hypothesis) and $|v * L(v)|$ is contractible (Lemma F.1(b), Appendix F), so the inclusion $L(v) \hookrightarrow v * L(v)$ is a homotopy equivalence; and it has the HEP (Lemma F.8, Appendix F). Also $|X|$ is the topological pushout of $|X'|$ and $|v * L(v)|$ along $|L(v)|$ (Lemma F.9(b), Appendix F). *Conditionally on* (C1) — see entry H9 of the hypothesis table in §12.2, and Lemma F.2(c) — the inclusion $L(v) \hookrightarrow v * L(v)$ is a strong deformation retract inclusion, whence Lemma F.2(b) (Appendix F; this is the gluing step of Lemma 9.5(1)) makes $X' \hookrightarrow X$ a homotopy equivalence. As X' is contractible, so is X .

Remark. (C1), namely “HEP + homotopy equivalence $\Rightarrow$ strong deformation retract inclusion”, is *not proved within this project* and is the sole condition of this step; Lemma 9.5(1) is conditional on

it. It is a standard statement of homotopy theory, and it is listed as a residual hypothesis rather than silently used.

Limit case. For each $\alpha < \lambda$ let Y_α be the full subcomplex on the first $\alpha + 1$ vertices. As λ is a limit ordinal the vertex in position $\alpha + 1$ exists, and Y_α is its strict sublevel set. Lemma 9.4 and the inductive hypothesis make Y_α contractible. The Y_α increase and their union is X .

Given a continuous map $f : S^n \rightarrow |X|$ ($n \geq 1$, with $|X|$ carrying the CW weak topology), its image is compact. Fix a base point s_0 of the sphere and put $x_0 = f(s_0)$; Lemma 9.5(2) — again in the weak topology — places the compact set $f(S^n) \cup \{x_0\}$ inside a finite subcomplex, and we then take some Y_α containing all of that subcomplex’s finitely many vertices. Y_α is contractible, and the subspace topology of a subcomplex agrees with its own weak topology (Lemma F.9(a), Appendix F), so f is freely null-homotopic in $|X|$; Lemma F.6 (Appendix F) corrects this to a null-homotopy preserving x_0 . Hence $[f] = 0 \in \pi_n(|X|, x_0)$. (The appeal to Lemma F.6 cannot be dispensed with: $x_0 \in Y_\alpha$ alone does not guarantee that an arbitrarily chosen contraction fixes x_0 .)

Moreover $|X|$ is *path-connected*: X is non-empty and connected — part of the definition of an exchange complex, see §8 — that is, its 1-skeleton is a connected graph, and each edge gives a path in $|X|$ joining its endpoints; any point lies in the realisation of some simplex and may first be joined to a vertex of that simplex, so any two points are joined by a path. Hence $\pi_0(|X|)$ is trivial.

By the above, $|X|$ is path-connected and $\pi_n(|X|, x_0) = 0$ for all $n \geq 1$ and the fixed x_0 . Lemma F.7 (Appendix F; a self-contained proof of Whitehead’s theorem in the special case of a simplicial complex with a point as target) therefore gives that $|X|$ is contractible. $\square$

Remark (The same conclusion by the textbook route). Alternatively: $|X|$ is a CW complex — its cells are the interiors of the simplices, attached along their boundaries, with the weak topology — and a one-point space is a CW complex, while the constant map $|X| \rightarrow \{*\}$ is an isomorphism on π_0 and on every π_n ($n \geq 1$, all base points), that is, a weak homotopy equivalence; Whitehead’s theorem (a weak homotopy equivalence between CW complexes is a homotopy equivalence) then gives $|X| \simeq \{*\}$. This route is mathematically equivalent to the one above and differs only in bookkeeping: it puts Whitehead’s theorem into the textbook-level ledger B of §13, whereas the route via Lemma F.7 keeps the argument inside the paper. *The route taken above is the one via Lemma F.7*, and ledger B accordingly does not list Whitehead’s theorem.

Remark (What “contractible” means here). Throughout this paper, “ $\text{IS}(K) \simeq *$ ” always means that its *geometric realisation, in the CW weak topology*, is contractible, and not some property of the abstract simplicial complex itself. The π_0 paragraph cannot be omitted: the sphere argument above starts at $n \geq 1$, while both closing routes require π_0 as well.

11 The main theorem

Theorem 11.1 (= Theorem 1.1, conditional form). *Conditionally on (U) , Theorem B.3, Theorem C.3, and the named theorems and residual hypotheses listed in the two tables of §12.2: for every knot $K \subset S^3$ and every integer $\ell \geq g(K)$, both $\text{IS}(K)$ and $\text{IS}_\ell(K)$ are contractible.*

Input	Where it is used	What its removal costs
Theorem B.3 (Appendix B; the (E) of the earlier draft)	§5: it makes $A(v)$ <i>attained</i> by a smooth surface, so that all of §7 has an S to exchange, round and compare areas with; and Proposition 2.2 (flagness), which first fixes one minimal representative per vertex.	$A(v)$ is only an infimum; the strict area decrease of §7 is no contradiction; flagness reverts to citing entry 5 of the original §13.
Theorem C.3 (Appendix C; the (R1) of the earlier draft, in the reading of §C.5)	§7.2 step 1: the graph representation (7.7), the Lipschitz character of the coefficients of the elliptic difference equation (7.8), the reflected similarity principle (7.10)–(7.11) — these give the family (7.13) transverse for <i>all</i> small $t > 0$. Also, through Lemma 5.3, the transversality (R2).	(7.7) cannot be written down; transversality retreats to “for <i>almost every</i> t ”, whereas the paired isotopy lemma of step 2 requires transversality everywhere on a <i>closed interval</i> .
(U)	The opening of Theorem 7.1 (one and the same S for (v, v_+) and (v, v_-)); §7.2 step 4 (for Z); Proposition 2.2 (flagness).	The exchange lemma fails in its first sentence.
The named-theorem conditions (table of §12.2)	Theorem B.3 is conditional on (H-sm) and T3/T4/T7/T9; Theorem C.3 on (H-cov) of Theorem A.1, on GP, on $\Delta 2$ and on S402-G3; the PDE items of §5 and §7 on (T8) of Appendix E or on Remark E.52; Lemma 5.1 on Dehn’s lemma (Appendix D).	The corresponding appendix reverts to a conditional proof, and Theorem 11.1 returns to the (E), (R1) hypothesis form of the earlier draft.
(C1) (combinatorial part)	Lemma 9.5(1); the successor case of §10.	The successor step of Theorem 10.1 breaks; see entry H9 of the table in §12.2.

Remark (An entailment that is easily overlooked). The geometric inputs above support (N1) and (N2) through Theorem 7.1; (N1) and (N2) support Corollary 9.3.1 through Theorem 9.3; and Corollary 9.3.1 is the sole source of the connectedness of *each truncation* $\text{IS}_\ell(K)$. So in this manuscript “truncation connectedness” is *not* an unconditional topological fact but a conclusion conditional on Theorems B.3 and C.3, on (U), and on the tables of §12.2. The earlier draft cited it wholesale (entry 1 of the original §13), where it was indeed unconditional; changing the source changed its status with it.

None of this is used in §9–§10. Theorem 10.1 is unconditional (apart from (C1)).

Proof. The complex of the unknot is a point (Proposition 2.1), so assume K non-trivial.

The proof must be carried out in two stages, $\text{IS}(K)$ before the truncations: the connectedness of a truncation is a product of Corollary 9.3.1, and Corollary 9.3.1 requires knowing beforehand that $\text{IS}(K)$ is already an exchange complex. Reversing the order would be circular (see the warning attached to Proposition 2.1).

Stage one ($X = \text{IS}(K)$). Proposition 2.1 gives connectedness, Proposition 2.2 gives flagness, Proposition 2.3 gives non-emptiness, and Lemma 5.1 makes the genus–area complexity well-founded on the set of attained values. (N1) and (N2) are verified below, with $X = \text{IS}(K)$, none of the branches “if X is a truncation” being triggered. So $\text{IS}(K)$ is a connected non-empty flag exchange complex, and Theorem 10.1 makes it contractible.

Stage two ($X = \text{IS}_\ell(K)$, $\ell \geq g(K)$). By stage one $\text{IS}(K)$ is a connected exchange complex; its complexity (5.1) is lexicographic with integer-valued first coordinate g , and $\text{IS}_\ell(K)$ is exactly the *full* subcomplex on $g \leq \ell$. Corollary 9.3.1 therefore gives that $\text{IS}_\ell(K)$ is *connected*, and that the inclusion

is isometric on 1-skeleta. Non-emptiness comes from Proposition 2.3 (for $\ell \geq g(K)$ the truncation contains a minimal-genus vertex); flagness from the fullness part of Proposition 2.2; well-foundedness by restriction from $\text{IS}(K)$. (N1) and (N2) are verified below, with the truncation branches triggered. So $\text{IS}_\ell(K)$ too is a connected non-empty flag exchange complex, and Theorem 10.1 makes it contractible.

Verification of (N1) and (N2), common to both stages. Let $u, w \in X$ be at distance 2. If $X = \text{IS}_\ell(K)$ then they are at distance 2 in $\text{IS}(K)$ as well: they have a common neighbour inside the truncation, and, the truncation being *full*, an edge between them in $\text{IS}(K)$ would already be an edge of the truncation. So in either case Theorem 7.1 may be applied to any common neighbour.

Remark (This paragraph deliberately avoids the isometry of Corollary 9.3.1). The two lines above use only *fullness*, and are an *unconditional* combinatorial argument. Replacing them by an appeal to the isometry would be shorter, but would make an unconditional statement rest on the whole geometric part, the isometry depending on Theorem 7.1 through Corollary 9.3.1. Brevity is not a reason; *being unconditional is*.

(N1). Proposition 8.2 gives two apices in $\text{IS}(K)$; in the truncated case, (8.4) keeps the one of smaller genus inside X , still dominating the smaller interval.

(N2). Take an apex x_* of $I_X(u, w)$ of least complexity ((N1) and well-foundedness guarantee one exists). Suppose

$$c(x_*) \geq \max\{c(u), c(w)\}. \quad (11.1)$$

An apex is a common neighbour, so Lemma 8.1 applies to the triple $(x_*; u, w)$ and yields an exchange output y with

$$c(y) < c(x_*), \quad g(y) \leq g(x_*). \quad (11.2)$$

By Theorem 7.1(a),(b), y is an apex too. If X is a truncation, the second part of (11.2) gives $g(y) \leq g(x_*) \leq \ell$, so y stays in the truncation. In either case y is an apex of $I_X(u, w)$ of complexity strictly below the minimum: a contradiction. Hence $c(x_*) < \max\{c(u), c(w)\}$, which is (N2). $\square$

Remark. Theorem 10.1 is applied *separately* to $\text{IS}(K)$ and to *each* truncation; the increasing union $\text{IS}(K) = \bigcup_\ell \text{IS}_\ell(K)$ is not used.

Remark (The order of dependence, for checking that there is no cycle).

$\underbrace{\text{IS}(K) \text{ connected} + \text{Prop. 2.2, 2.3} + \text{Lemma 5.1} + \text{Theorem 7.1}}_{\text{entry 1 of §13}} \Rightarrow \text{IS}(K) \text{ exchange} \Rightarrow \text{Theorem 9.3}$

$\Rightarrow \text{Corollary 9.3.1} \Rightarrow \text{IS}_\ell(K) \text{ connected} \Rightarrow \text{IS}_\ell(K) \text{ exchange}.$

Each step uses only what has been established to its left, and *no step uses truncation connectedness to prove a property of $\text{IS}(K)$.*

12 Discussion

12.1 The exact mathematical scope

What is used is: a compact orientable irreducible knot exterior $E(K)$ with a single torus boundary, which for non-trivial K is moreover ∂ -irreducible (Theorems D.15 and D.18, Appendix D; the unknot is handled separately by Corollary D.19 and Proposition 2.1); connected orientable Seifert surfaces with exactly one preferred-longitude boundary curve; each vertex surface incompressible and hence ∂ -incompressible (Corollary D.22, Appendix D) — the earlier draft's statement that

“ ∂ -incompressibility is neither in the definition of a vertex nor used” is *no longer true* once the (AW) replacement proof of Lemma 5.1 is adopted (Theorem D.50, Appendix D), since Haken normalisation (Theorem D.23) presupposes that $E(K)$ is ∂ -irreducible and S is ∂ -incompressible; a relative metric with strictly convex boundary together with a *real-analytic* preferred-longitude foliation of the boundary; the external inputs of the two ledgers of §13; the conditions of the two tables of §12.2; and the well-ordering theorem (the axiom of choice) in Lemma 9.1.

There is *no* hypothesis of atoroidality, of a finite vertex set, of local finiteness, of finite dimension, of finite diameter, or of any cardinality bound. Compactness is used in exactly two places: in Lemma 5.1, for the family of surfaces of fixed topological type and bounded area (reduced here, through Theorems D.50 and D.54, to the finiteness of normal surfaces of bounded weight); and in the limit step of Theorem 10.1, for the compactness of the image of a sphere. *Neither is compactness of $IS(K)$.*

The results do not extend to links. No statement about links is proved here. For a knot, a spanning surface is connected with a single boundary curve, its lift is a separating wall in the infinite cyclic cover, and disjointness gives the required edge relation. For a link of several components, the literature allows disconnected spanning surfaces; disjointness and the distance-one relation need not agree; and the complement of a single lift can have more than two relevant sides. (The hypothesis $(H\partial)$ of Appendix C likewise holds automatically only for knots.)

12.2 The exact form of the residual conditions: three tables

In the earlier draft the geometric part rested on three items: **(U)**, verified and retained, and **(E)** and **(R1)**, which were explicit hypotheses. Here (E) and (R1) are proved, as Theorems B.3 and C.3, at the cost of the two tables of conditions below; (U) is unchanged.

12.2.1 First table: the named theorems

Where an entry is marked *numbering unverified*, the statement or the numbering of that theorem has not been checked verbatim against its source by the present authors, and *no closure claim may be based on it*. Level: research-level entries are counted in ledger A of §13; entries marked C16 are textbook-level and are counted in ledger B.

#	Named theorem	Level	Used in	Note
N1	Schultens, Thm 4 / Thm 12 (this is (U))	research	Prop. 2.2, Thm 7.1, §7.2 step 4	verified verbatim; Thm 4 is a variant of FHS Thm 6.2, so not independent of N3
N2	Kakimizu 1992: $IS(K)$ is connected	research	Prop. 2.1	theorem number <i>unverified</i>
N3	FHS 1983, Lemma 1.2	research	§7.1, absence of a product region	Lemma 1.5 has left the chain
N4	T-C, Hildebrandt (Plateau boundary regularity)	research	Thm A.1 step 9 (C3); Thm C.3 via (I_0)	numbering <i>unverified</i> ; the only research-level dependence of Appendix C
N5	T-A, Gulliver–Osserman (no interior branch points)	research	Thm A.1	interior only; absence of boundary branch points is proved in Lemma A.3

#	Named theorem	Level	Used in	Note
N6	Meeks–Yau (barrier, HS 2.9; HS 6.3)	research	Thm A.1 (T3/T4 in the notation of Appendix A)	stated as reported by Hass–Scott; original numbering <i>unverified</i>
N7	Hass–Scott 6.9 / 6.11 (via 3.3/3.5/3.6), HS 6.6, 6.4, 2.2	research	assembly of Thm A.1; Cor. A.16.3	two sentences, P8(B) and P9, are supplied or identified in Appendix A; the Hass–Scott definition of “sufficiently convex” is <i>unverified</i>
N8	Morrey (T1); Courant–Lebesgue (T5)	research / monograph	Thm A.1	numbering <i>unverified</i>
N9	T3, Schoen curvature estimate (interior form)	research	Lemma B.6	—
N10	T4, do Carmo–Peng / Fischer–Colbrie–Schoen / Pogorelov	research	before Thm B.2, to exclude $\partial\tilde{\Sigma} = \emptyset$	added in Appendix B
N11	T7, Osserman, structure of finite total curvature	monograph	after Prop. B.16	recorded as research-level
N12	T9, Schoen, asymptotics of the ends	research	Lemmas B.18, B.19, B.22	the half-sentence “the end normals are asymptotically collinear” is <i>not reproved</i> ; it is the only input of the (G-b) chain that is not self-contained
N13	Dehn’s lemma (Papakyriakopoulos)	research	Thm D.15 (via Thm D.14)	<i>newly registered</i> , at the cost of adopting the (AW) route
N14	(T8), Bers–Nirenberg (Hölder Beltrami $\Rightarrow C^{1,\alpha}$ diffeomorphism)	monograph	Thm E.51 (V2)	<i>or</i> replaced by the reflection-symmetric form of Thm E.34 (Remark E.52); that replacement has not been independently audited, and both routes are recorded
N15	Alexander/Schoenflies; the π_1 -injective forms of the loop theorem and Dehn’s lemma; the Alexander trick; uniqueness of PL regular neighbourhoods; isotopy extension (Palais–Cerf); Stiefel; the Federer area formula; Sard; Thom transversality; tubular neighbourhoods; Mayer–Vietoris; the classification of compact surfaces	C16	§1.1, Prop. 2.1, Appendices A, D	explicitly flagged; not in ledger A

#	Named theorem	Level	Used in	Note
N16	Haken: a normal surface is determined by its normal coordinates up to normal isotopy; there are finitely many of bounded weight	C16	Thm D.54, Prop. D.53, Lemma 5.1	not proved here
N17	Riesz representation; completeness of H^1 ; Poincaré; Lebesgue differentiation; weak sequential compactness; the spectral theorem and Cholesky; Stampacchia truncation; Cauchy–Pompeiu; Calderón–Zygmund; Morrey embedding; Weyl	C16	Appendix E	explicitly flagged
N18	Reeb stability; Ehresmann; the coarea formula; GT Lemma 3.4 (Hopf)	C16	Appendix C; Appendix A (0c)	explicitly flagged
N19	Whitehead’s theorem	C16	only the alternative closing route of §10	the route taken uses Lemma F.7 instead, so this is not in the chain
N20	The axiom of choice (well-ordering; the dimensionwise selection in Lemma F.7)	axiom	Lemma 9.1, Lemma F.7	Lemma F.8 does not depend on it

12.2.2 Second table: the residual hypotheses H1–H14, none concealed

#	What is still conditional	Where it occurs	Nature and status
H1	(U), in the universal form of Schultens Thm 4 / Thm 12 (exact form in §12.2.4)	§2.2, §7.1, §7.2 step 4	research-level, in the chain; statement verified; retained
H2	The named theorems T-C, T-A, Meeks–Yau, HS 6.9/6.11, Morrey, Courant–Lebesgue (N4–N8)	Appendix A	research-level; statements as reported by Hass–Scott; original numbering <i>unverified</i>
H3	T3, T7, T9 (N9, N11, N12) and T4 (N10)	Appendix B	research-level; the “collinear” half-sentence of T9 is not reproved
H4	(H-cov): the minimising sequence $F_{i,n}$ of assembly step 6 covers the limit F^* naturally	Thm A.1 (Lemma A.11)	assumed by Hass–Scott, not proved here, and written explicitly into the statement of Thm A.1; carries with it GP , the general-position selection over countably many open dense conditions

#	What is still conditional	Where it occurs	Nature and status
H5	(H-sm) and the reading of (R1) : each Σ in the family is separately C^2 up to the boundary, supplied by Thm A.1 together with Thm C.3(b). This presupposes the reading fixed in Appendix C §C.5: throughout the main text, “the minimiser of (E)” is read as <i>the construction F' of Thm A.1 over the optimal leaf ℓ_* (Thm C.1)</i> , and $A(v)$ is read as $I(\ell_*)$	Thm B.3(1), §5, §7, the table of Thm 11.1	<i>This reading is adopted in this manuscript</i> , and §C.5 lists the seven call sites affected. Under the opposite decision, Appendices B and C revert to conditional proofs and the main text reverts to the (E), (R1) hypothesis form of the earlier draft
H6	$\Delta 2$: the source proof of conclusion C3 of Thm A.1 (LEMMA H + T-C) has not been rechecked outside the slice; Appendix A step 9 gives an alternative route (HS 3.5 / T-C + Lemma A.4), itself unaudited within the same family	Thm A.1, Thm C.3(C.3-iii)	to be rechecked; Appendix C uses only its $C^{1,\beta}$ part (I_0)
H7	S402-G3 : whether the appeals to (R1) in Lemmas 5.1 and 5.3 quote only “there exists a minimiser satisfying (R1)”; S402-G1 : the interface with the definition of $\mathcal{M}_a$	Remark C.4.3, §5	G1 disappears once Lemma 5.1 is reproved by the (AW) route; G3 awaits a check by a reader holding the whole text; G2 : Thm C.4(A) is stated in $C^{2,\gamma}$ for a general (E)-minimiser, and disappears for F'
H8	(T8) , Bers–Nirenberg (N14)	Thm E.51 (V2)	monograph-level; or replaced, per Remark E.52, by Thm E.34; the replacement has not been independently audited, and both routes are recorded
H9	(C1) : HEP + homotopy equivalence $\Rightarrow$ strong deformation retract inclusion	Lemma 9.5(1), the successor case of §10, Lemma F.2(c)	<i>Not proved within this project</i> : a hole in the main text itself. Appendix F neither proves it nor derives anything from it. Theorem 10.1 is conditional on it
H10	Dehn’s lemma (N13)	Thm D.15	research-level: one further use point and one further independent source, incurred by adopting the (AW) route
H11	Haken finiteness (N16)	Lemma 5.1, Thm D.54	C16 textbook-level, explicitly flagged; the source statement “normal coordinates determine the surface up to normal isotopy” is not proved here
H12	Same-family audit declaration : all six appendices are the output of one model family, audited only within that family, with zero of three cross-engine audits	the opening of each appendix, and this section	a disciplinary declaration; the cross-engine criterion is not met

#	What is still conditional	Where it occurs	Nature and status
H13	(H ∂): ∂F is connected and isotopic in ∂M to a leaf of J — the quantifier that the statement of (E) should carry (§12.2.4); $\Delta 1$: $I(\ell_*) = A(\ell_*)$, that is, equality of the type-(i) and type-(ii) infima, is not proved	Thms C.1, C.2; Remark C.3.1, §C.5	(H ∂) holds automatically for Seifert surfaces and is not a gap; $\Delta 1$ is a definition or a gap according to the reading fixed in H5
H14	Numerical corrections: the constant $\bar{K}_{\max}^+$ of Lemma B.4; the finite- j form of (B.11) carrying a factor $1 + c_M R/m_j$; the budget of Prop. A.19 tightened to $< I + a/4$	Appendices A, B	already incorporated; they do not change any conclusion

Appendices D and E carry further local conditions not numbered in this table; they are listed one by one in Appendix D §D.8(c) — the setting (S) of Theorem D.50; Theorem D.23 being narrower than its source NT-1; the reading under which an ambient isotopy may move ∂M , which is harmless under the convention of §1 that the boundary longitude may move — and in Appendix E §E.7.1, items E-1 to E-9 — Corollary E.20 being local only; the Lipschitz character of $\det a$; (E-div); the deletion of the Fermi-coordinate sentence; the sentence “connected with boundary $\Rightarrow$ connected interior”; the registration of Corollary E.55; and three undecided points in E.2 which do not affect the main text.

12.2.3 Third table: where each hypothesis of the earlier draft is now supplied

Earlier hypothesis	Now supplied by	Conditional on
(E)	Theorem B.3 (Appendix B)	H3, H5, Lemma 4.1 of the main text
(R1)	Theorem C.3 (Appendix C)	H2 (T-C), H4, H6, H7 (G3), the reading H5
(R2), transversality	Lemma 5.3; Thm C.3(a), Cor. C.8.1	as above; for F' , (R1) is not needed as a premise
(U)	unchanged	H1

12.2.4 The exact statements of (U), (E) and (R1)

(U) Universal disjointness of relative area minimisers — verified verbatim against the source. Entry 2 of ledger A in §13 states it in two places, both in universal form:

Theorem 4 (main text). ... let F_1, F_2 be properly embedded **relative least area** incompressible surfaces in E **with disjoint representatives in their proper isotopy classes**. Then either F_1, F_2 are disjoint or they coincide.

Theorem 12 (appendix). Let $f_i : S_i \rightarrow M$... be incompressible surfaces which are pairwise non-isotopic and pairwise disjoint. Let $g_i : S_i \rightarrow M$ **be** relative area minimizers in the isotopy classes of f_i . Then $g_1(S_1), \dots, g_n(S_n)$ are **also pairwise disjoint**.

It is used here in the following form.

(U) Let $S_1, \dots, S_n$ be pairwise non-isotopic, pairwise separable and incompressible. Then for *every* choice of relative area minimising representatives $g_1, \dots, g_n$ of their respective isotopy classes, those representatives (together with their boundaries) are pairwise disjoint.

Points of verification, all matched verbatim: the hypotheses are exactly the three of incompressibility, pairwise non-isotopy and pairwise disjointness, and boundary incompressibility is not required; ∂M is a union of tori, the metric is relative (∂M strictly convex), and J is a smooth foliation of ∂M by closed curves — word for word the setting of §5 above; the statement carries no exception clause; and the words “knot exterior, preferred longitude” occur after Theorem 12 and impose no restriction. *For the pairwise case, Theorem 4 is cited:* its hypotheses are verbatim the adjacency relation used here, “relative least area” being taken in the reading of Definition 3 of that main text, which is the reading (α) of §5 above, and “or they coincide” is excluded because the two vertices are distinct. For $n \geq 3$ Theorem 12 is cited, subject to the $(\alpha)/(\beta)$ warning recorded as entry 2' of ledger A.

This hypothesis cannot be weakened to an existential form: were one to assert only that “some family of minimisers is pairwise disjoint”, the phrase “one and the same choice” in the opening of Theorem 7.1 would not be licensed; nor could entry 5 of the earlier §13 repair it, since the triple $\{v, v_+, v_-\}$ is not a clique.

An entailment: (U) together with Theorem B.3 makes Proposition 2.2 a three-line argument, and the earlier entry 5 (Neumann–Swarup) has been deleted. This is not a net improvement: flagness thereby becomes a conditional conclusion (see the remark under Proposition 2.2).

(E) and (R1) — the two explicit hypotheses of the earlier draft, here supplied by Theorems B.3 and C.3. The statements are given with the $(H\partial)$ quantifier of H13 restored.

(E) (existence, in the reading (α)). Let $F \subset E(K)$ be an incompressible, properly embedded surface satisfying $(H\partial)$: ∂F is connected and isotopic in $\partial E(K)$ to a leaf of J . Then within the normal isotopy class of F , subject to the constraint that the boundary be a leaf of J , the infimum of area is *attained by a smooth surface*. — **Theorem B.3** (Appendix B), conditional on H3, H5.

(R1) (boundary regularity). That minimiser Σ is, within some collar $\{0 \leq r \leq \rho\}$ of $\partial E(K)$, an embedded $C^{1,1}$ surface with boundary, up to $\{r = 0\}$, and $\Sigma \cap \partial E(K) = \partial \Sigma$ (neat up to the boundary). — **Theorem C.3** (Appendix C) gives, for the construction F' of Theorem A.1, smoothness up to the boundary (in particular $C^{1,1}$) and neatness, conditionally on H2, H4, H6, H7; whether “that minimiser” is F' is settled by the reading H5.

Three points must be declared with them.

1. **Transversality is not a hypothesis.** From (R1) and the strict convexity of $\partial E(K)$, Lemma 5.3 proves that Σ meets $\partial E(K)$ transversally along $\partial \Sigma$ (for F' : Theorem C.3(a)). Without strict convexity the conclusion is *false* — a flat disc in a half-space is everywhere tangent.
2. **Regularity thresholds and the PDE inputs (Appendix E).** A whole passage of the earlier draft here — isothermal coordinates, Beltrami coefficients, the $W^{1,p}$ threshold, the Carleman–Bers–Vekua equation with drift, the Hölder/Dini inclusions — is carried self-containedly by Appendix E, and only the conclusions are retained here: equation (7.9) is reduced to $\operatorname{div}(\sigma \nabla u) = 0$ by Theorem E.34 (isothermal coordinates up to the boundary, for a $C^{0,\alpha}$ metric, not passing through the Beltrami equation) together with Corollary E.40, the Fermi-coordinate sentence having been deleted and $a^{12}|_L = 0$ supplied instead by Lemma E.35 (mollified shear); the interior similarity principle and the finite order of zeros are proved in Theorem E.43 and

Corollary E.45; the reflected forms (7.10) and (7.11) come from Theorem E.51, conditionally on (T8) or on the substitute of Remark E.52 (H8); strong unique continuation is Theorem E.48, which removes Aronszajn from the chain. As for coefficient thresholds, $u \in C^{1,1}$ in the main text gives Lipschitz a^{ij} , meeting every hypothesis of Appendix E ($\det a$ Lipschitz, (E-div)); Theorem E.51 permits the relaxation to $a \in C^{0,\alpha_0}$ with $\sqrt{\det a} \in W^{1,p}$ (E-9), *which is not adopted here*. The correction A-3 of the earlier draft (the equation carries a drift term Aw) is implemented in Lemma E.46.

3. **(R1) is used only when A and B are tangent somewhere along L .** If the two are everywhere transverse along L , then $u \in C^1(\overline{D^+})$ and compactness of L give $|du| > 0$ on some collar directly, and (7.13) follows at once. The entire PDE apparatus is there *solely to handle boundary tangency points of finite order*.

12.3 A finite check that carries no weight

The following *takes no part in any argument*; it is recorded only as a log of falsification pressure.

Among the **996** connected flag complexes on at most 7 vertices, **513** satisfy (N1), **474** carry an exchange structure, and **0** fail to be contractible. Three independent implementations agree with one another; the counts 1, 1, 2, 6, 21, 112, 853 agree term by term with OEIS **A001349**; and the reduced homology of all 474, over $\text{GF}(2)$ and over $\mathbb{Q}$, has vanishing Betti numbers.

Falsifying power: of the 996, **500** have non-vanishing reduced homology over $\text{GF}(2)$ and are therefore definitely not contractible — the theorem had 500 chances to die and did not. All **39** complexes with (N1) but without an exchange structure are non-contractible, the smallest being $C_5 \simeq S^1$, so (N2) *carries weight and is not decoration*. An independent check of Lemma 9.1, over the 47293 ordered set partitions for $n = 7$, produced **0** counterexamples.

Limits: Theorem 10.1 holds for arbitrary cardinality, so a census on at most 7 vertices *can falsify but cannot confirm*, and the infinite case is not machine-checkable in principle.

13 The two ledgers of external inputs

Apart from the entries below, every step is either proved in this paper (including Appendices A–F) or quoted verbatim with its source given. To these is added the axiom of choice (Lemma 9.1, Lemma F.7).

The two ledgers are to be read separately and must not be added together: ledger A records research-level inputs *by use point*, not by reference; ledger B is an explicit flagging of textbook-level facts and is not counted into the chain. The cardinalities are to be read off the rows of the tables below and no aggregate figure is asserted here; in particular no claim of a net reduction in external dependence is made.

13.1 Ledger A: research-level inputs (use point $\rightarrow$ source $\rightarrow$ status)

Use point	Source	Status
Prop. 2.1: $\text{IS}(K)$ is connected	Kakimizu, <i>Hiroshima Math. J.</i> 22 (1992) 225–236 (theorem number <i>unverified</i>)	retained. The connectedness of the truncations $\text{IS}_\ell(K)$ is supplied internally by Cor. 9.3.1, conditionally on the geometric part

Use point	Source	Status
§2.2, §7.1, §7.2 step 4: (U)	Schultens, <i>J. Topology</i> 3 (2010), Thm 4 / Thm 12	retained. The use of “compactness of $\mathcal{M}_a$ ” (Prop. 9, Cor. 10) has <i>left the chain</i> : Lemma 5.1 is now supplied by Thms D.50 and D.54, conditionally on NT-1 (Thm D.23)
The $(\alpha)/(\beta)$ warning	same source: Definition 3 (α) in the main text versus $\mathcal{M}([f])$ (β) in the appendix, which are not equivalent	this paper uses (α) throughout, in the main text and in Appendices B and C
§7.1: absence of a product region	FHS, <i>Invent. Math.</i> 71 (1983), Lemma 1.2	retained, and used only for the pair of minimal surfaces S_+, S_- ; Lemma 1.5 has left the chain. Not independent of Schultens (Thm 4 is a variant of FHS Thm 6.2)
Appendix A, Thm A.1: T-C / T-A / Meeks–Yau / HS 6.9–6.11 / Morrey / Courant–Lebesgue	Hildebrandt; Gulliver–Osserman; Meeks–Yau 1982; Hass–Scott, <i>Trans. AMS</i> 310 (1988) 87–114; Morrey	retained; original numbering <i>unverified</i> (§12.2, N4–N8)
Appendix B: T3 / T4 / T7 / T9	Schoen 1983 (two items); do Carmo–Peng / Fischer–Colbrie–Schoen / Pogorelov; Osserman	retained (§12.2, N9–N12)
Appendix D, Thm D.15: Dehn’s lemma	Papakyriakopoulos 1957	<i>newly registered</i> : one use point and one source, incurred by adopting the (AW) route
Appendix E, Thm E.51 (V2): (T8)	Bers–Nirenberg / Ahlfors (monograph-level)	retained, or replaced by Remark E.52; both routes are recorded

The following entries of the earlier draft have *left* the chain, each replaced by a result proved here: Alessandrini–Magnanini (by Thm E.51); Gilbarg–Trudinger ch. 8, for the divergence-form Dirichlet problem, global $C^{1,\alpha}$ and Hopf (by Thms E.2, E.5, E.13, Cor. E.20, Thms E.29 and E.31); Aronszajn (by Thm E.48); Whitehead’s theorem (by Lemma F.7 — the alternative route of §10 would instead move it to ledger B); Anderson Thm 3.1, reached via Schultens Prop. 9 (by Thm C.3, so that Anderson is no longer cited, the $(\alpha)/(\beta)$ discontinuity noted in the earlier draft being bridged by Thms B.3 and C.1 under $(H\partial)$ and H5); the interior similarity principle (by Thm E.43 and Cor. E.45); and isothermal coordinates with bi-Lipschitz control (by Thm E.34 — the (T8) route still needs its interior form, the route of Remark E.52 does not).

13.2 Ledger B: textbook-level facts, explicitly flagged, not counted into the chain

Alexander/Schoenflies (§1.1, Prop. 2.1, and several places in Appendices A and D); the solid-torus recognition (Cor. D.19, proved in Appendix D, borrowing only Alexander); two minimal surfaces agreeing on an open set coincide (Cor. E.55, a same-family use point, adding no new source); the π_1 -injective forms of the loop theorem and Dehn’s lemma, isotopy extension (Palais–Cerf), uniqueness of PL regular neighbourhoods, the Alexander trick, Stiefel, the Federer area formula, Sard, Thom transversality, tubular neighbourhoods, Mayer–Vietoris, the classification of compact surfaces (Appendix D); Riesz representation, completeness of H^1 , Poincaré, Lebesgue differentiation, weak sequential compactness, the spectral theorem, Stampacchia, Cauchy–Pompeiu, Calderón–

Zygmund, Morrey embedding, Weyl (Appendix E); GT Lemma 3.4 (Appendix A, (0c)); Reeb stability, Ehresmann, the coarea formula (Appendix C); Haken finiteness and “normal coordinates determine a surface up to normal isotopy” (§5, Thm D.54).

Remark (A reminder about omitted dependencies). Four items of ledger B — Alexander/Schoenflies, the solid-torus recognition, the coincidence of minimal surfaces, and Dehn’s lemma — were *dependencies present all along and not recorded* in the earlier draft. Their explicit flagging here is purely bad news, not a new dependence.

Not in the chain: the census on at most 7 vertices (§12.3, explicitly non-load-bearing).

Concluding remarks

The purely combinatorial Theorem 10.1 of §§9–10 — “every non-empty connected flag exchange complex is contractible” — is self-contained given Appendix F: it uses only two combinatorial axioms and the axiom of choice, is conditional on **(C1)** alone (not proved within this project, §12.2, H9), can be quoted and formalised on its own, and has survived 500 chances of falsification in the finite census of §12.3. Combined with the geometric instantiation of §§1–8 it yields Theorem 11.1.

The geometric part is conditional. Of the three explicit hypotheses of the earlier draft, (U) remains a research-level input (ledger A of §13), while (E) and (R1) are proved here as Theorems B.3 and C.3, at the cost of the named theorems and residual conditions listed in the two tables of §12.2 — (H-cov), GP, $\Delta 1$, $\Delta 2$, S402-G3, (T8), the registration of Dehn’s lemma, and the reading H5 of (R1). *Theorem 11.1 is conditional on the two tables of §12.2; no unconditional form is proved in this paper.* Most of the external inputs of the earlier §13 have been made self-contained by Appendices A–E, but the resulting cardinalities are to be recounted from the tables rather than asserted here.

All appendix material was produced within a single model family and audited only within that family; it has received no cross-engine audit. Until such an audit is carried out, every “proved” in this manuscript should be read as “proved, pending audit”. On how the manuscript was produced, see Appendix G.

No claim of novelty or priority is made.

Two things must be stated whenever this work is cited: *Lemma 4.1 preserves the boundary setwise and not pointwise — the pointwise version is false; and the conclusions are for knots, not for links.*

A Area minimisers in an isotopy class in a manifold with boundary (Theorem 6.12')

This appendix does one thing: it transports the Hass–Scott Theorem 6.12 — an area-minimising representative of an isotopy class in a closed manifold — to a compact 3-manifold with boundary, giving the full statement and proof of **Theorem A.1 (6.12')**, and listing one by one, in §A.0 and §A.8, every external named theorem the proof depends on and every hypothesis this appendix is *still conditional on*. Downstream (§5 and §7 of the main text, the hypothesis (H-sm) of Theorem B.3, and Appendix C) cites by the fixed numbering used here.

On provenance: the contents of this appendix, like those of Appendices B–F, were produced within one model family and audited only within that family; they have received no cross-engine audit. See Appendix G.

Notation and named theorems

A.0.1 Notation. M is a compact Riemannian 3-manifold with boundary, d_M its intrinsic distance, $B(x, r) = \{y : d_M(x, y) < r\}$, $\bar{B}$ its closure, $S(x, r) = \partial B(x, r)$. For $X = \bar{B}(x, \varepsilon)$ write $Y := \text{Fr}(X)$, the frontier, that is the closure of $\partial X \setminus \partial M$, and $Y' := X \cap \partial M$, so that $\partial X = Y \cup Y'$, with a corner along $\ell := Y \cap \partial M$. Put $\rho := d_M(x, \cdot)$, $t := \varepsilon - \rho$, and $w := \text{dist}(\cdot, \partial M)$ (smooth in a collar, by T-D). “The least-area disc spanning γ in Z ” means one of least area among all maps of D^2 into Z with boundary γ . “ D is properly embedded in M ” is always written out as $D \cap \partial M \subset \partial D$. $F \subset M$ is a given surface, $\mathcal{F}$ is the class of piecewise smooth surfaces isotopic to F rel ∂F (fixing the boundary pointwise), and $I := \inf\{\text{Area}(G) : G \in \mathcal{F}\}$; G denotes a generic member of $\mathcal{F}$ of area near I .

A.0.2 Convexity of ∂M . “Strictly convex” is taken in the geodesic sense of Hass–Scott: each face of ∂M has $\text{II} > 0$ with respect to the inward normal. “Sufficiently convex” is taken in the sense of §6 of that paper: piecewise smooth, each face with $H \geq 0$, and M isometrically embedded in a smooth N in which each face extends to a smoothly embedded surface (*original numbering unverified*; near p. 110). Strictly convex $\Rightarrow$ sufficiently convex.

A.0.3 Named theorems (the permitted list). All of the following are research- or textbook-level inputs; the statements are as reported by Hass–Scott, and the *original numbering is unverified*. This appendix adds no new research-level input to the chain.

- **T1 (Morrey, the Plateau problem; research-level; HS Thm 2.7 and its §6 extension).** If M is homogeneously regular — every point lies in the image of a chart $\varphi : B(0, 1) \rightarrow M$ with $k\|v\|^2 \leq g_{ij}v^i v^j \leq K\|v\|^2$ for constants $k, K > 0$ independent of the point — and $\gamma \subset M$ is a simple closed curve bounding a disc of finite area, then there is a disc D spanning γ whose area equals the infimum over all discs spanning γ , and D is smooth in its interior.
- **T-A (Gulliver–Osserman; research-level; HS Thm 2.8).** A least-area map of a surface into a 3-manifold is, *when restricted to the interior of the surface*, the composite of a branched covering onto another surface with an immersion of the latter; in particular a least-area disc has no branch point at an *interior* point. *T-A concerns interior points only*: whether there is a branch point at a boundary point is not part of its statement, and is supplied here by Lemma A.3.
- **T-B (strong maximum principle for the minimal surface equation / Hopf boundary point lemma, mean-convex barrier form; textbook-level, GT Lemma 3.4 + Hopf).** Let Σ be an immersed minimal surface and S a C^2 surface whose mean curvature with respect to the normal pointing to the side containing Σ satisfies $H_S > 0$. If Σ is tangent to S at p and lies locally on that side of S , a contradiction results. Boundary form: if Σ and S are C^2 up to the boundary, tangent at a boundary point p , with Σ on one side of S and the base domain satisfying an interior ball condition at p , then $\Sigma \equiv S$ locally.
- **T-C (Hildebrandt, boundary regularity of Plateau solutions; research-level; cited in HS Lemma 3.5).** Let N be a Riemannian 3-manifold *without boundary* and $f : D^2 \rightarrow N$ a Plateau solution — conformal, harmonic, area-minimising — spanning a Jordan curve Γ . If an open arc σ of Γ is $C^{m, \alpha}$ ($m \geq 2$, $0 < \alpha < 1$), then f is $C^{m, \alpha}$ up to the boundary on a neighbourhood of $f^{-1}(\sigma)$. *The essential restriction*: N has no boundary and there is no obstacle; to use it inside a manifold with boundary one must first reduce by Lemma A.4.
- **T-D (tubular neighbourhoods + compactness; textbook-level).** A compact smoothly embedded submanifold has positive normal injectivity radius. **T-E (Hopf–Rinow; textbook-**

level). In a complete locally compact length space, closed bounded sets are compact. **T-F (smooth Alexander theorem; textbook-level)**. A smoothly embedded 2-sphere in a 3-ball bounds a 3-ball.

- **T3 (Meeks–Yau, strictly convex boundary; research-level; HS Thm 2.9)**. Let M be compact with ∂M strictly convex and let $\gamma \subset M$ be a null-homotopic simple closed curve. Then γ bounds a least-area disc D ; if $\gamma \subset \partial M$, any such D is properly embedded in M ; two such discs D, D' satisfy $\text{Int } D \cap D' = \emptyset$ or $D = D'$; and least-area discs spanning disjoint $\gamma, \bar{\gamma}$ are disjoint.
- **T4 (Meeks–Yau, sufficiently convex boundary; research-level; HS Thm 6.3)**. M compact (or homogeneously regular), ∂M sufficiently convex, γ a null-homotopic simple closed curve. Then γ bounds a least-area disc D ; if $\gamma \subset \partial M$ then D is *either* properly embedded in M *or* embedded in ∂M ; the intersection statements are as in T3, with the extra branch that both may be embedded in ∂M with one contained in the interior of the other.
- **T5 (Courant–Lebesgue; textbook-level; HS Lemma 2.4)**. For $f : D^2 \rightarrow M$ smooth with $E(f) \leq K$, $x \in D^2$ and $0 < \delta < 1$, there is $r_0 \in [\delta, \sqrt{\delta}]$ with $\text{length } f(C_{r_0}) \leq \sqrt{2K / \log(1/\sqrt{\delta})} \rightarrow 0$ as $\delta \rightarrow 0$.
- **T7 (maximum principle for minimal surfaces, sufficiently convex boundary; HS Lemma 6.6)**. If ∂M is sufficiently convex and two immersed minimal surfaces F_1, F_2 are tangent at an interior point P , then either they coincide locally, or locally they are $x^3 = 0$ and $x^3 = \text{Re } a(x^1 + ix^2)^n$ with $n \geq 2$. In particular a minimal surface tangent to ∂M at an interior point lies inside ∂M .
- **T8 (isoperimetric inequality; HS Lemma 6.4)**. If M is homogeneously regular with ∂M sufficiently convex, there are $a_0, \alpha > 0$ such that a closed curve Γ contained in a ball of radius a_0 spans in M a disc of area $\leq \alpha \text{Length}(\Gamma)^2$. Put $\beta := 1/(4\alpha)$.
- **T9 (coarea inequality; HS Lemma 2.2)**. With $A(r) = \text{Area}(F \cap B(r))$ and $L(r) = \text{length}(F \cap S(r))$ and r below the injectivity radius, $dA/dr \geq L(r)$; when F is properly embedded and $\partial F \cap B(r) = \emptyset$, the restriction $r < \text{dist}(x, \partial M)$ is not needed.
- **HS 6.9 / 6.11 (convergence lemmas; research-level)**. Hypotheses: X a compact Riemannian 3-manifold with ∂X sufficiently convex; $Y \subset \partial X$ a compact strictly convex subsurface; $Y' = \overline{\partial X \setminus Y}$; Γ a compact 1-manifold properly embedded in Y' ; $D_i \subset X$ properly embedded least-area discs with $D_i \cap Y' = \Gamma$ and uniformly bounded area. Conclusion (6.9): a subsequence converges smoothly, with multiplicity, in $X \setminus Y$ to an embedded minimal surface T with $\partial T = \text{Int } \Gamma$, each sheet of which is either properly embedded in $X \setminus Y$ or a minimal subsurface of Y' . (6.11) is the countable-family version, with the same hypotheses and conclusion.
- **T-W (Hartman–Wintner normal form at a boundary branch point; textbook-level, quoted from memory and not checked against the source)**. A conformal harmonic minimal disc, C^1 up to the boundary, satisfies $f = p + \text{Re}(az^{k+1}) + O(|z|^{k+2})$ at a boundary point p , with $a \in \mathbb{C}^3 \setminus 0$ and $k \geq 0$ ($k \geq 1$ meaning a branch point).

A.0.4 Two sentences omitted in the source. HS 6.9 is obtained by “merging” 3.3 with 3.5, and the source does not write out the two sentences the merge requires: **P8(B)** — after deleting the

ε -neighbourhood of Y , the parallel surfaces are mean convex in the tubular part near ∂Y , so the sweep-out maximum principle still applies — and **P9** — a monotonicity lower bound, with fixed boundary, for the sheets near Y' . Item 6 of Proposition A.5 proves P8(B) for $X^\flat$ and identifies P9 as geometrically identical to the original setting. Both are omissions of 6.9 as a published theorem, not new gaps introduced here; §A.8 lists them separately.

Theorem A.1 (Hass–Scott 6.12' with boundary)

Theorem A.1 (6.12'). *Setting.*

- (A-M) M is a compact, homogeneously regular, P^2 -irreducible Riemannian 3-manifold;
- (A- ∂) ∂M is strictly convex (§A.0.2; in particular every face has $H > 0$ and no face is minimal);
- (A-F) $F \subset M$ is compact, connected, two-sided, incompressible and properly embedded, with $\partial F \neq \emptyset$ smoothly embedded in ∂M ;
- (A-cl s) $\mathcal{F}$ is the class of piecewise smooth surfaces isotopic to F in M rel ∂F (fixing the boundary pointwise), and $I := \inf\{\text{Area}(G) : G \in \mathcal{F}\}$.

Conditions (not proved here; they must be quoted along with the conclusion). (**H-cov**) In step 6 of the assembly (§A.7) the minimising sequence $F_{i,n}$ covers the limit F^* naturally: there are covering maps $F_{i,n} \rightarrow F^*$ compatible with the smooth convergence (assumed by Hass–Scott; see Lemma A.11). (**GP**) General position: the centres x_m and radii r_m are taken from the open interval of the radius condition of Lemma A.10, and (i) each ∂B_m is transverse to ∂F , to ∂M and to F_i (Sard, for almost every value); (ii) for $m' < m''$, every sheet of the limit $L_{m'}$ of step m' is transverse to $\partial B_{m''}$ (Proposition A.17); (iii) in the arc case, the rounding parameter r takes one of the almost-every values of Lemma A.8. GP is a selection over countably many open dense conditions; it is not a mathematical premise, but the assembly depends on it, so it is listed.

Conclusion. There is $F' \in \mathcal{F}$ with $\text{Area}(F') = I$ such that

- C1** F' is a minimal surface, smooth and embedded in $\text{Int } M$;
- C2** $F' \cap \partial M = \partial F$;
- C3** F' extends smoothly to ∂F : near each point of ∂F it is $C^{m,\alpha}$ up to the boundary ($m \geq 2$), and $F' \pitchfork \partial M$ along ∂F ;
- C4** F' is two-sided and isotopic to F rel ∂F (that is, $F' \in \mathcal{F}$, not merely of area I);
- C5** (AM) there is $\varepsilon_0 > 0$ such that for all $x \in M$, all $0 < \varepsilon < \varepsilon_0$ and all sufficiently small $\eta > 0$, $F' \cap X_\eta$ is a disjoint union of embedded least-area discs, where $X = \bar{B}(x, \varepsilon)$ and $X_\eta = X \setminus N_\eta(\text{Fr } X)$, with X replaced by the rounded ball $X^\flat$ of §A.2 when x is near ∂F ; here “least-area” is relative to all disc maps into M with the same boundary.

Remark (Premises discharged). The four premises (H-W1a), (H-G5), (R) and (H-glue) of the source argument are discharged within this appendix: (H-W1a) by Lemma A.18 and Proposition A.19 (§A.6); (H-G5) by Theorem A.14 (§A.5); (R) by Proposition A.2 and Lemma A.3 (the rounding route, §A.2); (H-glue) by Proposition A.17 (§A.5). A further premise (H-E) is needed only if the compactness in (A-M) is relaxed to completeness, which is not done here.

Remark (The interface with (E) of the main text, (H-SET)). The competitor class (A-cls) of this theorem fixes the boundary pointwise, whereas the competitor class of (E) in the main text allows the boundary to slide within the foliation (setwise). Hence $\inf_{(E)} \leq I$, and the F' of this theorem need not realise the $A(v)$ of the main text. The bridge is built in §5 of the main text and in Appendix C, not here.

Proof. See §A.7: a ten-step assembly, each step pointing to a numbered item of this appendix. $\square$

The rounded ball $X^\flat$, corner regularity (R), and the absence of boundary branch points (0c)

A.2.1 The rounded ball. Let $x \in M$ and $X = \bar{B}(x, \varepsilon)$, with ε a common admissible value for the radius conditions of Lemmas A.7, A.8 and A.10, generic in the sense of Sard ($Y \pitchfork \partial M$ along ℓ). In the coordinates $t = \varepsilon - \rho$ and $w = \text{dist}(\cdot, \partial M)$ we have $X = \{t \geq 0, w \geq 0\}$; along ℓ , $\nabla t \nparallel \nabla w$, and by continuity this persists on $U_0 = \{t < r_0, w < r_0\}$. Take $0 < r < \min\{r_0, w_0, \frac{1}{2}, \varepsilon/2\}$ and the profile $\phi(\tau) = e^{-1/(r-\tau)}$ for $\tau < r$, $\phi(\tau) = 0$ for $\tau \geq r$: smooth, decreasing, with $\phi' < 0$ and $\phi'' > 0$ on $[0, r)$ and flat to infinite order at r . Put

$$\Psi := \phi(t) + \phi(w) - \phi(0), \quad X^\flat := \{y \in X : \Psi(y) \leq 0\}, \quad \Sigma^\flat := \partial X^\flat,$$

$$Y'_\flat := X^\flat \cap \partial M, \quad Y_\flat := \overline{\Sigma^\flat \setminus \partial M}, \quad \ell_\flat := Y_\flat \cap \partial M, \quad \gamma_0^\flat := \partial F \cap X^\flat.$$

The geometric facts, one line of proof each:

- (Gb1) the modification takes place only inside $U_r = \{t < r, w < r\}$, and $X^\flat \supset \bar{B}(x, \varepsilon - r)$;
- (Gb2) $Y'_\flat = \bar{B}(x, \varepsilon - r) \cap \partial M$, a disc, and $\ell_\flat = S(x, \varepsilon - r) \cap \partial M$, a circle;
- (Gb3) $\Sigma^\flat$ is a smoothly embedded 2-sphere ($\nabla \Psi \neq 0$ in U_r ; circle by circle, $\{\Psi = 0\} \cap U_r$ is an annulus);
- (Gb4) $\Sigma^\flat$ has $\text{II} > 0$ everywhere, and every level set $\{\Psi = \lambda\} \cap U_r$ likewise has $\text{II} > 0$ with mean curvature vector pointing into $\{\Psi < \lambda\}$; indeed $\nabla^2 \Psi = \phi''(t) dt^2 + \phi''(w) dw^2 + |\phi'(t)| \nabla^2 \rho + |\phi'(w)| (-\nabla^2 w)$ is a sum of four positive semidefinite terms, the last using strict convexity of ∂M ;
- (Gb5) along $\ell_\flat$, $T\Sigma^\flat = T\partial M$: $Y_\flat$ is tangent to ∂M and the corner has disappeared;
- (Gb6) $\text{Area}(\Sigma^\flat) \leq C_\flat \varepsilon^2$;
- (Gb7) for almost every r : $\gamma_0^\flat \pitchfork \ell_\flat$ and $G \pitchfork \Sigma^\flat$ (Lemma A.8);
- (Gb8) rounding is used only for balls of types (III) and (IV) (§A.4); if $\gamma_0^\flat = \emptyset$ after rounding, the ball drops to type (II).

Proposition A.2 ((R), corner regularity, rounded form). *Assume (A-M)–(A-F), let $X^\flat$ be as above, let $c_a^\flat = a_0^\flat \cup \gamma_0^\flat$ be the smooth Jordan curve supplied by Lemma A.8 with $\partial \gamma_0^\flat = \{p_+, p_-\} \subset \ell_\flat$, and let D_a be a least-area disc in M spanning $c_a^\flat$; by Lemma A.10, $D_a \subset X^\flat$ is embedded with $D_a \cap \Sigma^\flat = c_a^\flat$ and $D_a \cap \partial M = \gamma_0^\flat$. Then*

1. D_a is $C^{m, \alpha}$ up to the boundary along all of $c_a^\flat$, including $p_\pm$ ($m \geq 2$);

2. D_a is a C^1 embedding of the closed disc with boundary: there are a neighbourhood W of $p_\pm$ and a C^1 embedding Ψ_D of a half-disc into M with $\Psi_D(0) = p_\pm$, $d\Psi_D$ injective everywhere, image $D_a \cap W$, and the diameter mapped onto $c_a^\flat$;
3. $T_p D_a \supset T_p c_a^\flat$ and $D_a \pitchfork \Sigma^\flat$ along all of $c_a^\flat$; in particular at $p_\pm$ one has $D_a \pitchfork \partial M$ and $D_a \pitchfork Y_b$, the two having the same tangent plane there by (G5).

Proof. (1) By Lemma A.4 (reduction of a one-sided obstacle problem), D_a is still a least-area disc spanning $c_a^\flat$ in the boundaryless manifold $\text{Int } N_\delta \supset M$; take its energy-minimising, conformal harmonic parametrisation f . The hypotheses of T-C hold one by one: $N = \text{Int } N_\delta$ has no boundary; f is conformal harmonic and area-minimising; and $\Gamma = c_a^\flat$ is a smooth Jordan curve (Lemma A.8(iv)). So f is $C^{m,\alpha}$ up to the boundary along all of $c_a^\flat$. By-product: the statement “LEMMA H” required by the source assembly, that D_a is C^2 up to the boundary along the interior of γ_0 , follows directly and is no longer an independent premise.

(2) T-A gives f no interior branch point; Lemma A.3 gives f no branch point on ∂B ; f is injective (D_a embedded, Lemma A.10(c)); and $\bar{B}$ is compact, so an injective immersion is an embedding.

(3) By Lemma A.3(ii),(iii). The corner angles $\alpha_\pm$ have been replaced by π , and the statement (R) of the source holds in this form. $\square$

Lemma A.3 ((0c): no branch point on the boundary). *Let $X^\flat$ be a compact 3-ball with $\Sigma^\flat = \partial X^\flat$ smooth and $\Pi > 0$ with respect to the inward normal, let $c \subset \Sigma^\flat$ be a smooth Jordan curve, and let $f : \bar{B} \rightarrow X^\flat$ satisfy $f \in C^1(\bar{B}) \cap C^2(B)$, be harmonic in the interior (vanishing tension field $\tau(f) = 0$), with $f(\partial B) = c$, f injective and $f(\text{Int } B) \subset \text{Int } X^\flat$. Then*

1. for every $z_0 \in \partial B$ and $p = f(z_0)$, one has $d\rho_p(df_{z_0}(-z_0)) > 0$, where ρ is the signed distance to $\Sigma^\flat$, non-negative inside $X^\flat$, and $-z_0$ is the inward normal. In particular $df_{z_0} \neq 0$: there is no branch point on ∂B ;
2. $T_p D \supset T_p c$, where $D := f(\bar{B})$;
3. $D \pitchfork \Sigma^\flat$ along all of c : $T_p D \cap T_p \Sigma^\flat = T_p c$.

Proof. In four steps.

L1 (the distance function). By T-D, ρ is smooth on a tubular neighbourhood $\{|\rho| < \delta\}$ with $|\nabla \rho| = 1$, so $\nabla^2 \rho(\nabla \rho, \cdot) = 0$; and on $\Sigma^\flat$, $\nabla^2 \rho|_{T\Sigma^\flat} = -\Pi$, on differentiating $\langle \nu, w \rangle \equiv 0$ along $\Sigma^\flat$.

L2 (the convex barrier). Compactness together with $\Pi \geq \lambda_0 g$ ($\lambda_0 > 0$) gives $\delta_1 > 0$ such that $\rho(y) < \delta_1$ implies $\nabla^2 \rho_y(v, v) \leq -\frac{\lambda_0}{2} |v^\parallel|^2 \leq 0$, where $v^\parallel$ is the component of v orthogonal to $\nabla \rho$: on the compact set $K = \{(y, v) : \rho(y) \leq \delta, |v| = 1, d\rho(v) = 0\}$ the function $\Phi(y, v) = \nabla^2 \rho(v, v)$ is continuous and $\leq -\lambda_0$ on $\{\rho = 0\}$, so $\{\Phi \geq -\lambda_0/2\}$ is at positive distance δ_1 from $\{\rho = 0\}$. *Strict convexity of ∂M is used here:* for $\Pi \geq 0$ only, δ_1 degenerates to 0.

L3 (superharmonicity). $u := \rho \circ f$ satisfies $\Delta u = \nabla^2 \rho(f_x, f_x) + \nabla^2 \rho(f_y, f_y) + d\rho(\tau(f))$; since $\tau(f) = 0$, L2 gives $\Delta u \leq 0$ on $f^{-1}\{\rho < \delta_1\} \cap B$.

L5 (main step). Choose $\epsilon \in (0, \frac{1}{2})$ with $f(\bar{\Omega}) \subset \{\rho < \delta_1\}$, where $\Omega := B((1 - \epsilon)z_0, \epsilon) \subset \text{Int } B$ satisfies $\bar{\Omega} \cap \partial B = \{z_0\}$ and the interior ball condition at z_0 . Then $u \in C^2(\Omega) \cap C^1(\bar{\Omega})$, $u(z_0) = 0$, and $f(\text{Int } B) \subset \text{Int } X^\flat$ gives $u > 0$ on Ω . Put $v = -u$: then $\Delta v \geq 0$ and $v < 0 = v(z_0)$ on Ω , so the Hopf boundary point lemma (T-B, GT Lemma 3.4 with $L = \Delta$) gives an outward normal derivative $\partial_{z_0} v(z_0) > 0$, that is $d\rho_p(df_{z_0}(z_0)) < 0$, which is (1). If $df_{z_0} = 0$ that derivative would vanish, a contradiction.

(2),(3). Rotate so that $z_0 = 1$. Then $f_y(z_0) = \frac{d}{dt} f(e^{it})|_0 \in T_p c$, and by (1) df_{z_0} is injective, so $f_y(z_0) \neq 0$ and $T_p c = \mathbb{R} f_y(z_0) \subset T_p D$. Also $d\rho_p(f_x(z_0)) < 0$ gives $f_x(z_0) \notin T_p \Sigma^\flat$, so $T_p D \neq T_p \Sigma^\flat$ and $T_p D \cap T_p \Sigma^\flat = T_p c$. $\square$

Remark. This lemma uses neither T-A, nor area minimisation, nor conformality, nor the Hartman–Wintner normal form at a boundary branch point — *T-W is therefore not used in this appendix* and is listed in §A.0 only for reference. The main line uses only GT Lemma 3.4, T-D and the chain rule. The hypothesis $f(\text{Int } B) \subset \text{Int } X^\flat$ follows from D being embedded with $D \cap \Sigma^\flat = c$ (Lemma A.10(c)).

Lemma A.4 (Reduction of a one-sided obstacle to an obstacle-free problem). *Let ∂M be strictly convex, let $N \supset M$ be the smooth extension of §A.0.2, let w be the signed distance to ∂M in N , and put $N_\delta := M \cup \{-\delta < w < 0\}$. Then for small δ the nearest point retraction $\pi : N_\delta \rightarrow M$ — the identity on M , and the projection along normal geodesics onto ∂M on $\{w < 0\}$ — is 1-Lipschitz. Consequently, for any $W^{1,2} \cap C^0$ disc $f : \bar{B} \rightarrow N_\delta$ with $\partial f \subset c \subset M$, the composite $\pi \circ f$ is a disc in M with the same boundary and $\text{Area}(\pi \circ f) \leq \text{Area}(f)$; hence a least-area disc spanning c in M is still a least-area disc spanning c in the boundaryless manifold $\text{Int } N_\delta$.*

Proof. On $\{w = -\tau\}$, $d\pi$ kills ∂_w and equals, in the tangential directions, the normal projection P_τ of the parallel surface $\{w = -\tau\}$ onto ∂M . The metric on the outer parallel surface is $g_\tau = g_0 + 2\tau \text{II}_{\partial M} + O(\tau^2)$, with $\text{II} > 0$ with respect to the normal pointing into M ; from $\text{II} \geq \lambda_0 g_0$ one gets $g_\tau \geq g_0$ for $\tau < \delta$, whence $|dP_\tau v|_{g_0} \leq |v|_{g_\tau}$, and for $v = v_{\text{sl}} + a\partial_w$, $|d\pi v| \leq |v_{\text{sl}}| \leq |v|$. As N_δ is a length space and π does not increase the length of any curve, π is 1-Lipschitz; the consequence follows from the area formula. *Strict convexity is used:* with $\text{II} \geq 0$ only, the $O(\tau^2)$ term could dominate the first-order one. Take D_a to be an energy minimiser; harmonicity is an interior condition, and on $D_a \subset M$ the metrics of N_δ and M agree. $\square$

Proposition A.5 (The seven items rechecked after rounding). *Let $X^\flat$ be as in §A.2.1 with r an almost-every value. Then the following seven items, stated in the source for the geodesic ball X , hold one by one for $X^\flat$; the precise statements are the $\flat$ forms of the corresponding lemmas in §A.4.*

Item 1 (Lemma A.7b). $\Sigma^\flat$ is a smoothly embedded 2-sphere inside X ((Gb3)), so by T-F, $X^\flat$ is a 3-ball and is contractible; $Y_b^\flat$ is a disc by (Gb2), and Y_b , a sphere minus an open disc, is a disc; strict convexity is (Gb4), which is stronger than in the unrounded case, where there is a corner along ℓ ; the area bound is (Gb6); and the three conditions for sufficient convexity hold: global smoothness, $H > 0$, and extendability, $\{\Psi = 0\}$ being a smooth closed surface in N . Covering: for $r < \varepsilon/2$ an $\varepsilon/2$ -net is still covered by the $\{X_j^\flat\}$. *Caveat:* if the Hass–Scott definition of sufficient convexity also demands that Y and Y' meet along ∂Y at an angle strictly less than π — a condition absent from the report used here — then this item becomes “original to be checked”; but Lemma A.10(c) and the proof of HS 6.9 use only strict convexity, so the verdict is unaffected (item 6).

Item 2 (Lemma A.8b). $\gamma_0^\flat = \partial F \cap \bar{B}(x, \varepsilon - r)$, so Lemma A.8 applies verbatim with radius $\varepsilon - r$. Transversality $G \pitchfork \Sigma^\flat$ splits into four places: (a) on $\gamma_0^\flat$, $T\Sigma^\flat = T\partial M$ while $G \pitchfork \partial M$ by properness — *at the tangency edge transversality does not degenerate but rather becomes automatic*; (b) near $p_\pm$, by extending an open condition; (c) on $\{w \geq r\}$, where $\Sigma^\flat \subset S(x, \varepsilon)$, by Sard applied to $\rho|_G$; and (d) inside U_r , by a parametric Sard argument: with $\Phi(y, r) = \phi_r(t) + \phi_r(w) - \phi_r(0)$ one has $\partial_r \Phi = F(\phi(t) + \phi(w)) - F(\phi(t)) - F(\phi(w)) < 0$, where $F(\phi) = \phi(\ln \phi)^2$ is strictly concave, hence strictly subadditive, on $\phi < e^{-1}$; so 0 is a regular value and Sard applies to the r projection. $(\star)^\flat$: the component containing $\gamma_0^\flat$ is a circle $c_a^\flat = \gamma_0^\flat \cup a_0^\flat$ with $a_0^\flat \subset Y_b^\flat$ the unique arc, the other components being circles in $Y_b^\flat$; at $p_\pm$, $c_a^\flat$ is the transverse intersection $G \cap \Sigma^\flat$ of two smooth surfaces, hence smooth, with interior angle π .

Item 3 (Lemma A.9b). The hypothesis list of LEMMA B — $W = X^\flat$ contractible; $a = a_0^\flat \subset G \cap \partial W$ an arc with $\partial a \subset \partial G$; $b = \gamma_0^\flat \subset \partial G \cap W$ an arc with $\partial b = \partial a$ and $a \cap b = \partial a$ — involves neither the smoothness of ∂W nor its edges, and holds item by item. Incidentally, $E_a \cap N = \{y \in G : \Psi(y) \leq 0\}$ near $c_a^\flat$.

Item 4 (Lemma A.10b). Part (d) is unchanged. Part (a) in two steps: first $D \subset X$, using (a1), the $S(x, R_0)$ barrier, and (a2), the disc P on Σ^b as competitor with $\text{Area}(P) \leq C_b \varepsilon^2 < \beta(R/2)^2$, applying T-B at $R_0 = \varepsilon$ to get $t > 0$ on $\text{Int } D$; second $D \subset X^b$: suppose $\lambda_0 = \max_D \Psi > 0$, attained at $p' \in \text{Int } D \cap U_r$; there D is tangent to the level set $\{\Psi = \lambda_0\}$ and lies on its side $\{\Psi \leq \lambda_0\}$, while (Gb4) gives that level set $\Pi > 0$ with mean curvature vector pointing into $\{\Psi < \lambda_0\}$, contradicting T-B — *the barrier is used on a level set with $\lambda_0 > 0$, where the curvature does not degenerate*. Part (b) is unchanged. Part (c): apply T4 with ambient X^b , the hypothesis clause $\gamma \subset \partial X^b$ being satisfied; the branch “embedded in ∂X^b ” is excluded since $H > 0$ everywhere on Σ^b . The radius condition becomes $C_b \varepsilon^2 < \beta(R/2)^2$.

Item 5 (Lemma A.6). The statement does not mention X and is unchanged; when invoked, $\partial D \subset \Sigma^b \subset \bar{B}(x, \varepsilon)$ and $d_M(x_0, \partial D) \geq R/2$ as before.

Item 6 (HS 6.9/6.11 applied to $X^b, Y_b, Y'_b, \gamma_0^b$). At the level of the statement: X^b is compact with Σ^b sufficiently convex; Y_b is a compact disc with $\Pi > 0$; $Y'_b = \overline{\Sigma^b} \setminus Y_b$; γ_0^b is properly embedded in Y'_b ; the D_i are properly embedded in X^b with $D_i \cap Y'_b = \gamma_0^b$ and $\text{area} \leq \text{Area}(E_a)$; and the branch “embedded in Y' ” is excluded by strict convexity of ∂M . At the level of the proof, the fourteen properties used in HS 3.3/3.5/3.6 were checked one by one: the literal hypotheses of those results are “a compact 3-ball with ∂X strictly convex”, which X^b satisfies verbatim, and which is simpler than the cornered setting of the source; no property special to a geodesic ball, and no property of the angle between Y and Y' , is used anywhere. The two sentences needed for the merge into 6.9 and not written in the source: **P8(B)**, that after removing the ε -neighbourhood of Y_b the parallel surfaces near ℓ_b are tubes with $H = 1/\alpha^* + O(1) > 0$ pointing towards the axis, so the sweep-out maximum principle applies; and **P9**, the monotonicity lower bound for sheets near Y'_b , Y'_b being the original Y' with a changed radius and hence geometrically identical. Both are equally necessary for the original setting (§A.0.4).

Item 7 (Proposition A.2(iii) + the arc form of Lemma A.15). $D_a \pitchfork \partial M$ and $D_a \pitchfork Y_b$ at $p_\pm$ come at one stroke from Lemma A.3(iii), with no need for the graph-plus-Hopf case distinction. A uniform shear: the vector field V along c_a^b is required to satisfy $V \pitchfork G$ and $V \in T\Sigma^b$ — one smooth surface now covering both “tangent to ∂M ” and “tangent to Y ” — and the admissible set is non-empty because $G \pitchfork \Sigma^b$. In step 3, at $p_\pm$, $T_p D_a \cap T_p \Sigma^b = T_p c_a^b \not\supset V$. In step 4 take $\tau := -\Psi|_G \geq 0$, so that $\tau^{-1}(0) = c_a^b$ with non-vanishing gradient there: one formula replaces the original three-piece splice. For $y \in \gamma_0^b$ one has $d = \tau = 0$, so ∂M is fixed pointwise and the isotopy is rel ∂F . In the base case $S_a = E_a \cup D_a$ meets along c_a^b at angle π and is locally flat.

Item 0 ((R) itself). (0a) is Lemma A.4; (0b) is the hypothesis list of T-C (Proposition A.2(i)); (0c) is Lemma A.3. (R) is discharged, and rounding has introduced no new gap.

The replacement lemmas

Lemma A.6 (A^+ : monotonicity up to ∂M). *Let M be homogeneously regular with ∂M sufficiently convex, let α, a_0 be the constants of T8 and $\beta = 1/(4\alpha)$. Let D be a least-area disc in M spanning a simple closed curve γ with $\gamma \not\subset \partial M$, let $x \in D$, and let $0 < r < \min\{a_0, d_M(x, \partial D)\}$. Then $\text{Area}(D \cap B(x, r)) \geq \beta r^2$. Neither embeddedness of D nor $r < \text{dist}(x, \partial M)$ is assumed.*

Proof. (0) $D \cap \partial M \subset \partial D$: T-A makes D a minimal immersion; if an interior point p lay on ∂M , then D could only lie on one side of ∂M near p and would be tangent to it, so by the last sentence of T7, $D \subset \partial M$, whence $\gamma \subset \partial M$, a contradiction.

(i) The coarea inequality $A'(r) \geq L(r)$: by (0) and $r < d_M(x, \partial D)$ we have $\partial(D \cap B(x, r)) \subset D \cap S(x, r)$; the rest is T9, which the source states needs no modification, using the Lipschitz coarea

formula and a frame argument (ρ is only Lipschitz on a manifold with boundary, and Sard is not used).

(ii) Isoperimetric comparison: $\partial(D \cap B(x, r))$ lies inside a ball of radius $r < a_0$, and T8 is global, with no “away from ∂M ” restriction, giving $A(r) \leq \alpha L(r)^2$, since D minimal implies $D \cap B(x, r)$ minimal relative to its own boundary.

(iii) ODE comparison: $A' \geq \sqrt{A/\alpha}$ with $A(0^+) = 0$ gives $A(r) \geq \beta r^2$. $\square$

Remark. The hypothesis of proper embeddedness has been dropped — it would be circular with Lemma A.10 — and replaced by $\gamma \not\subset \partial M$, which T7 checks directly. Balls of type (II) (§A.4.1) are where this is needed: there the replacement disc does not touch ∂M but may come arbitrarily close, and HS 6.5 gives no lower bound.

Lemma A.7 (Small balls). *Let M be homogeneously regular with ∂M strictly convex. There are $R > 0$ and $C < \infty$ such that for all $x \in M$ and $0 < \varepsilon < R$: (1) $X = \bar{B}(x, \varepsilon)$ is contractible and $\pi_1(X) = 1$; (2) $Y = \text{Fr } X$ is strictly convex ($H > 0$ with respect to the inward normal of X); (3) $\text{Area}(\partial X) \leq C\varepsilon^2$; (4) ∂X is sufficiently convex, so X may serve as the ambient manifold in T4. Rounded form: $X^\flat$ satisfies (1)–(4), with (2) strengthened to “ $\Sigma^\flat$ globally smooth with $\Pi > 0$ ” and the constant in (3) replaced by $C_\flat$.*

Proof. (1),(2),(3) are the standard comparison-geometry consequences of homogeneous regularity: the principal curvatures of a geodesic sphere, of size $\sim 1/\varepsilon$, dominate the curvature bounds, and the area follows from volume comparison. When the ball meets ∂M , strict convexity makes geodesics enter $\text{Int } M$ at once, X is of half-ball type and contractible, and the frontier remains convex — the same fact is asserted verbatim in the proof of HS 6.4. For (4): piecewise smoothness is clear; Y has $H > 0$ by (2) and $Y' \subset \partial M$ by strict convexity; and for the extension take the $N \supset M$ of the proof of HS 6.4. The rounded form is item 1 of Proposition A.5. *Honest flag:* the comparison-geometry estimates have not been rederived line by line here; they are the same constants used in the proofs of HS 3.1 and 6.4, and are recorded as a citation dependence. $\square$

Lemma A.8 (Uniform arc structure). *Let $\partial F \subset \partial M$ be a compact smoothly embedded 1-manifold. There is $\varepsilon_1 > 0$ such that for all $x \in M$ and almost every $0 < \varepsilon < \varepsilon_1$, provided $X \cap \partial F \neq \emptyset$:*

1. $\gamma_0 = \partial F \cap X$ is a single embedded arc, properly embedded in Y' , with $\gamma_0 \pitchfork \ell$;
2. $(\star)$ for $G \in \mathcal{F}$ properly embedded with $G \pitchfork \partial X$, the set $G \cap Y$ contains exactly one arc a_0 , with $\partial a_0 = \partial \gamma_0$, all other components being circles;
3. the rounded form: (1) and (2) hold for $X^\flat, Y_\flat, Y'_\flat, \gamma_0^\flat$ for almost every r , and $G \pitchfork \Sigma^\flat$;
4. $c_a^\flat := a_0^\flat \cup \gamma_0^\flat$ is a smooth Jordan curve with interior angle π at $p_\pm$, and $T_p c_a^\flat = T_p G \cap T_p \partial M$.

Proof. (1) T-D gives ∂F a positive reach ρ_0 ; take $\varepsilon_1 < \rho_0/4$, so that $X \subset N_{\rho_0}(\partial F)$ and the nearest point projection $\pi_0(X)$ is a connected arc of ∂F of diameter $< \rho_0$, of which $\partial F \cap X$ is a connected subarc — two components would give two points of ∂F at M -distance $< 2\varepsilon$ but at distance $\geq \rho_0$ along ∂F , contradicting the reach. Transversality follows from Sard applied to $d_M(x, \cdot)|_{\partial F}$.

(2) $\partial(G \cap X) = (G \cap Y) \cup (G \cap Y')$ is a 1-manifold without boundary, and $G \cap Y' = \partial G \cap Y' = \gamma_0$ has exactly two endpoints; count endpoints.

(3),(4) are item 2 of Proposition A.5. $\square$

Lemma A.9 (An arc bounds a disc). *Let $W \subset M$ be contractible, so that $\pi_1(W) \rightarrow \pi_1(M)$ is trivial, let $G \in \mathcal{F}$ be incompressible, let $a \subset G \cap \partial W$ be an arc with $\partial a \subset \partial G$, and let $b \subset \partial G \cap W$ be an arc with $\partial b = \partial a$ and $a \cap b = \partial a$. Then $a \cup b$ bounds a disc E_a in G , with $E_a \cap \partial G = b$. For $W = X^\flat$, $a = a_0^\flat$ and $b = \gamma_0^\flat$ the hypotheses are supplied by Lemmas A.7 and A.8.*

Proof. This is LEMMA B of the source, whose hypothesis list has been checked item by item (item 3 of Proposition A.5); the argument is that $a \cup b$ is null-homotopic in M and G is ∂ -incompressible, by incompressibility and two-sidedness in (A-F) together with the annulus theorem. $\square$

Lemma A.10 (C^+): the replacement disc stays in the ball, is embedded, and does not enter ∂M). Let M be homogeneously regular with ∂M strictly convex, let R, C, C_b be as in Lemma A.7 and $\beta = 1/(4\alpha)$. Radius condition: $\varepsilon < \frac{1}{2} \min\{R, a_0\}$, $C_b \varepsilon^2 < \beta(R/2)^2$, $\varepsilon < \varepsilon_1$, and $r < \min\{r_0, w_0, \varepsilon/2, \frac{1}{2}\}$. Let $Z = X$ or X^b with $\Sigma = \partial Z$, let $\gamma \subset \Sigma$ be a piecewise smooth simple closed curve with $\gamma \not\subset \partial M$, and let D be a least-area disc in M spanning γ . Then

- (a) $D \subset Z$;
- (b) D is also of least area in Z ;
- (c) D is embedded with $D \cap \Sigma = \partial D = \gamma$, that is, properly embedded in Z ;
- (d) $D \cap \partial M = \gamma \cap \partial M \subset \partial D$; in particular D satisfies the hypotheses of Lemma A.6.

Moreover the least-area discs of disjoint $\gamma, \tilde{\gamma} \subset \Sigma$ are disjoint.

Proof. (d) is step (0) of Lemma A.6. (a): let $R_0 = \max_D \rho$ and suppose $R_0 > \varepsilon$. (a1) If $R_0 < R$: the extremal point $p \in \text{Int } D$ lies in $\text{Int } M$ by (d), and D is tangent at p to the strictly convex $S(x, R_0)$, lying on its convex side, contradicting T-B. (a2) If $R_0 \geq R$: take $x_0 \in D$ with $\rho(x_0) = 3R/4$, so $d_M(x_0, \partial D) \geq R/2$ and Lemma A.6 gives $\text{Area}(D) \geq \beta(R/2)^2$; on the other hand γ bounds a disc P on the 2-sphere Σ with $\text{Area}(P) \leq C_b \varepsilon^2 < \beta(R/2)^2$, and P is a legitimate competitor: a contradiction. For $Z = X^b$ there is the further second step of item 4 of Proposition A.5, the level-set barrier. (b) The competitor class is smaller. (c) Apply T4 with ambient Z (the hypothesis clause $\gamma \subset \partial Z$ holds): either D is properly embedded in Z or $D \subset \Sigma$, and the latter is excluded because every face of Σ has $H > 0$ and none is minimal. Disjointness is the last sentence of T4. $\square$

Remark (The arc case). Take $\gamma = c_a^b = a_0^b \cup \gamma_0^b$: then $\gamma \subset \Sigma^b$; $\gamma \not\subset \partial M$, since $\text{Int } a_0^b \subset Y_b \setminus \partial M$; γ is smooth; and γ is null-homotopic. So $D_a \subset X^b$ is embedded with $D_a \cap \partial M = \gamma_0^b \subset \partial F$, discs of different γ are disjoint, and $\text{Area}(D_a) \leq \text{Area}(E_a)$, since $E_a \subset G \subset M$ is a legitimate competitor. This is exactly the input of Proposition A.2.

A.4.1 Four types of ball. Let $\{x_j\}$ be an $(\varepsilon/2)$ -net in M and $B_j = B(x_j, \varepsilon)$ with ε satisfying the radius condition of Lemma A.10. Each ball belongs to exactly one type:

- (I) interior balls, $X \cap \partial M = \emptyset$: circles only; the original case of HS §§4–5 together with Lemma A.10;
- (II) balls meeting ∂M but not ∂F : circles only; each $c \subset Y$ is null-homotopic and bounds a disc in G by incompressibility; the replacement disc satisfies $D_c \subset \text{Int } M$ (Lemma A.10(d)); Lemma A.6 is indispensable here; and the branch “embedded in Y' ” of HS 6.9/6.11 is excluded by strict convexity of ∂M ;
- (III), (IV) balls centred on ∂F , and balls meeting ∂F but not centred on it: a single arc γ_0 , handled with the rounded ball X^b and Lemmas A.8, A.9 and A.10 in their arc forms; (III) and (IV) are treated identically.

No new hypothesis is introduced: (IV) uses only Lemma A.8, and (II) only Lemma A.6 and strict convexity of ∂M .

Lemma A.11 (Covering multiplicity one; conditional on (H-cov)). *Let $F_i \in \mathcal{F}$ be compact, connected, properly embedded, with $\partial F_i = \partial F$ pointwise, converging piecewise smoothly to a compact connected surface F^* with $\partial F^* = \partial F$. Assume (H-cov): for large i , F_i lies in an I -bundle tubular neighbourhood $N(F^*)$ of F^* and the bundle projection π restricted to F_i is a covering map between manifolds with boundary. Then the covering multiplicity is $d = 1$, pushing F_i along the fibres gives an isotopy $F_i \simeq F^* \text{ rel } \partial F$, and hence $F^* \in \mathcal{F}$ and F^* is two-sided.*

Proof. $\pi|_{F^*} = \text{id}$; and $\partial F_i = \partial F = \partial F^* \subset F^*$ gives $\pi|_{\partial F_i} = \text{id}_{\partial F}$. A covering carries boundary to boundary, so $\partial F_i = (\pi|_{F_i})^{-1}(\partial F^*)$ and $\pi|_{\partial F_i}$ is a d -fold covering of 1-manifolds; being the identity, $d = 1$. As F^* is connected, $d \equiv 1$, so F_i is isotopic to F^* along the fibres, by the identity on ∂F ; and $d = 1$ also excludes the branch “ $F_i = \partial N(F^*)$ with F^* one-sided”. Neither ∂M nor any transversality enters. $\square$

Lemma A.12 (The unique boundary sheet). *Let ε satisfy the radius condition of Lemma A.10, let $X^\flat$ be a ball of type (III) or (IV), let $G \in \mathcal{F}$ be properly embedded with $G \cap \Sigma^\flat$, and let G' be the result of the replacement of Theorem A.14 on $X^\flat$. Then*

1. *exactly one component of $G' \cap X^\flat$ meets ∂M , namely $D_a = D_{c_a^\flat}$, and $D_a \cap \partial M = \gamma_0^\flat$;*
2. *every other replaced component D_c satisfies $D_c \cap \partial M = \emptyset$.*

Proof. By Lemma A.8(ii),(iii), $G \cap \Sigma^\flat$ contains exactly one arc $a_0^\flat$, the rest being circles in $Y_\flat$, so $c \cap \partial M = \emptyset$ for those; and Lemma A.10(d) gives $D_\gamma \cap \partial M = \gamma \cap \partial M$. The endpoint count is made on the input G , so there is no circularity. $\square$

Proposition A.13 ((AM): the local Plateau property; conclusions C5 and C1). *Let F^* be the limit of the assembly of §A.7. There is $\varepsilon_0 > 0$, depending only on the homogeneous regularity constants, the convexity constants of ∂M and the reach of ∂F , such that for all $x \in M$, all $0 < \varepsilon < \varepsilon_0$ and all sufficiently small $\eta > 0$ one has $F^* \cap X_\eta = \sqcup_k D_k$, where each D_k is an embedded disc and is a least-area disc in M spanning ∂D_k , relative to all maps $\tilde{D} : D^2 \rightarrow M$ with $\tilde{D}|_\partial = \partial D_k$.*

Proof. The chain of provenance: $F^* = \lim F_{i,n}$, and each sheet of $F_{i,n}$ inside X (or $X^\flat$) is a least-area disc (Theorem A.14(b) and Lemma A.10). Two steps.

Step 1 (a limit of least-area maps is a least-area map). Let $f_i : D^2 \rightarrow Z$ be least-area maps of uniformly bounded area converging to f in C^∞ on compact subsets of $\text{Int } D^2$. Then for a compact subdisc $D' \subset \text{Int } D^2$, $f|_{D'}$ has least area among disc maps with the same boundary. (1) Minimality is inherited by restriction: if $h : D' \rightarrow Z$ had the same boundary and smaller area, then splicing $g' = g$ on $D^2 \setminus D'$ with h on D' gives a legitimate $W^{1,2}$ competitor, contradicting minimality of g . (2) Suppose $\varrho := \text{Area}(f|_{D'}) - \text{Area}(h) > 0$. C^1 convergence gives (2a) $|\text{Area}(f_i|_{D'}) - \text{Area}(f|_{D'})| < \varrho/4$, and (2b) the geodesic straight-line homotopy along $f|_{\partial D'}$ gives an annulus A_i with $\text{Area}(A_i) \leq CL\delta_i < \varrho/4$ (textbook-level). Splicing $\Phi_i = h \cup A_i$, with $\partial \Phi_i = f_i|_{\partial D'}$ and $\text{Area}(\Phi_i) < \text{Area}(f_i|_{D'}) - \varrho/2$, contradicts (1). HS 3.3 writes out only the skeleton (2a),(2b); part (1) is supplied here.

Step 2 (inside X_η the limit consists of discs only, with no planar region). This is the parallel-surface sweep-out of HS 3.3: after removing the η -neighbourhood of Y every parallel surface is mean convex, and a planar region touched by the limit contradicts T-B. For $X^\flat$ this is discharged by P8(B), item 6 of Proposition A.5.

Together the two steps give (AM); and C1, that F^* is a smooth embedded minimal surface in $\text{Int } M$, follows from each D_k being an embedded minimal disc (T-A, HS 6.9) together with the locality of (AM). $\square$

Remark (Why this is exactly enough). What T-C requires is precisely minimality within the class of discs, not absolute minimality among all integral currents. The construction replaces the local picture by least-area discs *before* passing to the limit, and the limit merely inherits the property; (AM) records this as a citable conclusion for the hypothesis (H-sm) of Theorem B.3 and for Appendix C.

Maximal disc replacement

A.5.1 Setting and the meaning of “outermost”. X is a 3-ball (Lemma A.7) and $G \in \mathcal{F}$ is smooth, connected, properly embedded with $G \pitchfork Y$. By Lemma A.8, $G \cap Y$ consists of finitely many circles c together with — exactly when $X \cap \partial F \neq \emptyset$ — exactly one arc a_0 . Treat first the interior case $X \cap \partial F = \emptyset$. Each circle c bounds a disc in Y , hence is null-homotopic in M , hence bounds a disc E_c in G by incompressibility, and E_c is *unique*, the other component of $G \setminus c$ containing $\partial G \neq \emptyset$. Call c *outermost* if E_c is contained in no other $E_{c'}$ — the *maximal* reading. *Honest flag*: this reading is taken from the source as the intended sense of “outermost”, on the ground that under the minimal reading the replacement iteration does not terminate in a family of discs; it is an interpretation, not the published definition.

Lemma A.14.1 (Nesting and the tree structure). (a) For $c \neq c'$: either $E_c \subset \text{Int } E_{c'}$, or $E_{c'} \subset \text{Int } E_c$, or $E_c \cap E_{c'} = \emptyset$.

(b) The graph T whose vertices are the closures of the components of $G \setminus (G \cap Y)$ — the blocks, inner ones inside X and outer ones outside — and whose edges are the intersection curves, is a tree, and all of ∂G lies in a single outer block Q_* .

(c) E_c is the union of the subtree, obtained by deleting the edge c , that does not contain Q_* ; the outermost curves are the edges incident to Q_* ; the outermost discs are pairwise disjoint; $G \setminus \bigcup_{\max} \text{Int } E_c = Q_*$; and $Q_* \cap X = \bigcup_{\max} c \subset Y$.

Proof. (a) If $c' \subset \text{Int } E_c$, the subdisc of E_c cut off by c' equals $E_{c'}$ by uniqueness; if $c' \cap E_c = \emptyset$, then $E_{c'}$, being connected, lies in one component of $G \setminus c'$. (b) Each c separates G , so there is no cycle, and G is connected; and $\text{Int } E_c \cap \partial G = \emptyset$ — points of an embedded open disc have $\mathbb{R}^2$ neighbourhoods, and $c \subset Y$ does not meet ∂M — so the circles of ∂G cannot lie on both sides. (c) is elementary tree theory. $\square$

Theorem A.14 (P^+ : maximal disc replacement). In the above setting let $c_1, \dots, c_k$ be the outermost curves and let D_j be a least-area disc in M spanning c_j (existence by T_4 ; by Lemma A.10, $D_j \subset X$ is embedded with $D_j \cap Y = c_j$, and the D_j are pairwise disjoint). Put $G' := Q_* \cup \bigsqcup_j D_j$. Then

(a) G' is a piecewise smooth embedded surface with $\partial G' = \partial F$;

(b) $G' \cap X = \bigsqcup_j D_j$, a family of pairwise disjoint least-area discs properly embedded in X ;

(c) $\text{Area}(G') \leq \text{Area}(G)$;

(d) $G' \in \mathcal{F}$, that is, ambient isotopic to G rel ∂F .

In the arc case (balls of type (III) or (IV), ambient X^b), c_a^b is counted among the outermost curves — it is the edge containing $\partial G \cap X^b$ — with D_a supplied by the remark after Lemma A.10 and $G' = Q_* \cup D_a \cup \bigsqcup_j D_j$; (a)–(d) hold as before, with (d) using the arc form of Lemma A.15. In particular the hypothesis of HS 6.9/6.11, that one has a family of least-area discs properly embedded in X , is discharged directly, and no second replacement is needed.

Proof. (a),(b) By Lemma A.14.1(c), $G \setminus \bigcup \text{Int } E_{c_j} = Q_*$; and $D_j \cap Q_* \subset D_j \cap X \cap Q_* = D_j \cap \bigcup c_i = c_j$. So G' is embedded, its pieces meeting only along the c_j , and $G' \cap X = (Q_* \cap X) \cup \bigcup D_j = \bigcup D_j$; properness in X holds since $D_j \cap \partial X = c_j = \partial D_j$ (in the interior case $D_j \cap \partial M = \emptyset$).

(c) E_{c_j} is a disc map in M with boundary c_j , so $\text{Area}(D_j) \leq \text{Area}(E_{c_j})$; sum over the pairwise disjoint family. *One does not need $E_{c_j} \subset X$: E_{c_j} may leave X and return, minimality being relative to all disc maps in M with the same boundary.*

(d) Lemma A.15 and Corollary A.15.1. □

Lemma A.15 (T: the topological replacement lemma). *Let M be an irreducible 3-manifold with boundary, let $G \subset M$ be a compact connected properly embedded two-sided incompressible surface with $\partial G \neq \emptyset$; let $E_1, \dots, E_k \subset G \setminus \partial G$ be pairwise disjoint discs with $c_j = \partial E_j$; and let $D_1, \dots, D_k \subset \text{Int } M$ be pairwise disjoint embedded discs with $\partial D_j = c_j$, C^1 up to c_j . Put $Q := G \setminus \bigcup \text{Int } E_j$ and assume*

(T-1) $D_j \cap Q = c_j$;

(T-2) *each c_j has a coordinate neighbourhood (s, t, u) in which $G = \{u = 0\}$, $c_j = \{t = u = 0\}$, $E_j \supset \{u = 0, t \geq 0\}$ and $D_j = \{u = d_j(s, t), 0 \leq t < r\}$ with $d_j \in C^1$ and $d_j(s, 0) = 0$; that is, near c_j both G and D_j are graphs over the same base.*

Then $G' := Q \cup \bigcup D_j$ is ambient isotopic to G rel ∂G (and rel ∂M), by an isotopy supported in any prescribed neighbourhood of $\bigcup (E_j \cup D_j)$. Category: piecewise smooth 2-complexes are converted to PL by Whitehead triangulation (textbook-level), and PL isotopy is the topological isotopy required.

Proof. In four steps.

(i) *Aligning the boundary (a shear isotopy).* Inside the box $\{|t| \leq t_0, |u| \leq u_0\}$ of (T-2) — which contains only one sheet of G and the graph of D_j , and no D_i , with $|d_j| < u_0/8$ — take $\kappa t_0 < u_0/8$ and cut-offs $\phi(t)$, $\varrho(u)$, and set $h_\lambda(s, t, u) = (s, t, u + \lambda \varrho(u) \phi(t)(d_j + \kappa t))$ for $t \geq 0$, the identity for $t \leq 0$. Then h_λ is a homeomorphism ($|\varrho' h| < 1$) rel ∂G , and the resulting $\tilde{E}_j = \{u = d_j + \kappa t\}$ meets D_j inside $\{t \leq t_0/2\}$ only along c_j , at distance $\geq \kappa t_0/4$ on $\{t_0/4 \leq t \leq t_0/2\}$.

(ii) *General position.* A small isotopy supported in a small neighbourhood of $\bigcup (E_j \setminus \{t < t_0/4\})$ makes $\text{Int } E_j \pitchfork \bigcup \text{Int } D_i$, giving (T-3): $G \cap D_i = c_i \sqcup \{\text{finitely many pairwise disjoint circles } \alpha \subset \text{Int } D_i \cap \bigcup \text{Int } E_j, \text{ transverse}\}$.

(iii) *Induction on N , the total number of circles in (T-3).* For $N > 0$: take an innermost circle α in some D_i , bounding D_α in D_i with $\text{Int } D_\alpha \cap G = \emptyset$; by incompressibility α bounds $E_\alpha \subset \text{Int } E_j$ in G ; then $S_\alpha = E_\alpha \cup D_\alpha \subset \text{Int } M$ is an embedded sphere, bounding a 3-ball V_α ; and since $G \setminus E_\alpha$ is connected, contains ∂G and misses S_α , we get $G \cap V_\alpha = E_\alpha$, $V_\alpha \cap Q = \emptyset$ and $V_\alpha \cap \partial M = \emptyset$.

(The α model.) By (T-3) take coordinates (θ, x, z) at α with $G = \{z = 0\}$, $D_i = \{x = 0\}$, $E_\alpha \supset \{z = 0, x \leq 0\}$ and $D_\alpha \supset \{x = 0, z \leq 0\}$; then V_α is the quadrant $\{x \leq 0, z \leq 0\}$, the three-quadrant piece containing $\{z = 0, x > 0\} \subset G \setminus E_\alpha$ being excluded, so $D_i \setminus D_\alpha = \{x = 0, z > 0\}$ lies outside V_α .

(Pushing the disc off.) Take a vector field ξ along D_α transverse to D_i and pointing away from V_α (near α , $\xi = \partial_x$), with flow Φ , and the slab $P = \Phi(D_\alpha \times [0, \varepsilon])$: then $P \cap D_i = D_\alpha$, $P \cap D_{i'} = \emptyset$, $P \cap G = A_\alpha$ (the collar of α on the $G \setminus E_\alpha$ side), and $P \cap V_\alpha = D_\alpha$. Put $D_\alpha^{++} = \Phi(D_\alpha \times \{\varepsilon\})$ and $E_\alpha^+ = E_\alpha \cup A_\alpha$; then $D_\alpha^{++} \cap \bigcup D = \emptyset$ and $D_\alpha^{++} \cap G = \partial D_\alpha^{++} = \partial E_\alpha^+$.

(Pushing across.) $V' = V_\alpha \cup P$ is two 3-balls glued along a disc, hence a 3-ball, with $\partial V' = E_\alpha^+ \cup D_\alpha^{++}$, $G \cap V' = E_\alpha^+$ and $V' \cap Q = V' \cap \partial M = \emptyset$. Take a PL regular neighbourhood N of V' relative to G , itself a 3-ball, with $N \cap G = E_\alpha^+ \cup A'$. N may contain $D_i \setminus D_\alpha$ or pieces of other $D_{i'}$, and this is harmless: all that is used is that $E^{+2} = E_\alpha^+ \cup A'$ and $D^{+2} = D_\alpha^{++} \cup A'$ are properly embedded discs in N with the same boundary. The textbook fact of T-F type — two properly

embedded PL discs in a 3-ball with the same boundary are isotopic rel ∂ — gives an isotopy of N rel ∂N carrying E^{+2} to D^{+2} ; extend by the identity. The new surface $G^{(1)} = (G \setminus E_\alpha^+) \cup D_\alpha^{++}$ is rel ∂G ; Q , the c_j and all the D 's are untouched; (T-1) is unchanged; $G^{(1)} \cap D_{i''} = (G \cap D_{i''}) \setminus \text{Int } E_\alpha^+$, so (T-3) persists and N strictly decreases; and the collars of step (i) are untouched, α meeting none of them.

Base case $N = 0$. $S_j = E_j \cup D_j$ is an embedded sphere — along c_j it is a wedge of two graphs by step (i), hence locally flat — bounding a 3-ball V_j with $G \cap V_j = E_j$, the V_j being pairwise disjoint. With regular neighbourhoods N_j and $N_j \cap G = E_j \cup A_j$, the discs $E_j^+ = E_j \cup A_j$ and $D_j^+ = D_j \cup A_j$ have the same boundary, and the T-F type fact gives an isotopy whose result is exactly $(G \setminus \text{Int } E_j) \cup D_j$. Do this for each j . $\square$

Corollary A.15.1 (= Theorem A.14(d)). *Interior case: (T-1) is Theorem A.14(a); for (T-2) take t a defining function for Y , u a defining function for G and s arc length, the coordinates being supplied by $G \pitchfork Y$; $D_j \in C^1$ up to c_j by T-C, since $c_j \subset \text{Int } M$ is smooth, $D_j \cap \partial M = \emptyset$ and D_j is a Plateau solution in the open manifold $\text{Int } M$; and $D_j \pitchfork Y$ along c_j by the boundary form of T-B (Y strictly convex with D_j on its concave side), so D_j is a C^1 graph over (s, t) near c_j . Irreducibility of M is (A-M), two-sidedness of G is (A-F). The support is not inside X : E_j may leave X . Arc form: replace (T-2) by the uniform shear coordinates of item 7 of Proposition A.5 (V tangent to Σ^b , $\tau = -\Psi|_G$); that D_a is a C^1 graph at c_a^b is Proposition A.2(ii),(iii); h_λ is the identity pointwise on γ_0^b , so the isotopy is rel ∂F ; the remaining steps are as in the interior case.*

A.5.2 The ball-by-ball scheme. Balls $B_1, \dots, B_n$ cover M , each radius adjusted within the open interval of the radius condition of Lemma A.10 (GP(i): $F_{i,m-1} \pitchfork \partial B_m$, and ∂B_m avoids the edges created at earlier steps — Sard, for countably many i). Put $F_{i,0} := F_i$ and let $F_{i,m}$ be the result of the replacement of Theorem A.14 applied to $F_{i,m-1}$ on B_m ; the hypotheses hold at each step, $F_{i,m-1} \in \mathcal{F}$ being connected, incompressible, properly embedded and piecewise smooth, and Lemma A.15 applying to piecewise smooth surfaces in the PL category. Write $G = F_{i,m-1}$ and $G' = F_{i,m} = Q_* \cup D_a \cup \bigsqcup D_j$. A single simultaneous replacement is not possible — replacement discs from different balls may intersect and the result is not a surface — and the minimal reading of “outermost” is not possible either: under it the very first step is already a fixed point.

Lemma A.16 (Mono: outside the ball, one only deletes). *$G' \setminus B_m = Q_* \setminus B_m$, and $Q_* \setminus \partial B_m$ is exactly the component of $G \setminus \partial B_m$ containing $\partial G \setminus \gamma_0$. That is: among the components of $G \setminus B_m$, the one containing ∂F is kept and all others are deleted, and nothing is added outside B_m .*

Proof. $D_a, D_j \subset B_m$ by Lemma A.10; and by Lemma A.14.1, Q_* is the unique outer block containing ∂G , every other outer block K lying in the far subtree of some maximal edge c , so that $K \subset \text{Int } E_c$ is deleted. $\square$

Corollary A.16.1 (Mono-1). *$F_{i,m} \setminus B_m \subset F_{i,m-1}$; $\text{Area}(F_{i,m}) \leq \text{Area}(F_{i,m-1})$; and $F_{i,m} \in \mathcal{F}$ by Theorem A.14(c),(d). Hence $I \leq \text{Area}(F_{i,n}) \leq \text{Area}(F_i) \rightarrow I$.*

Corollary A.16.2 (Mono-2). *Let $m' < m$ and $U_{m'} := \text{Int } B_{m'} \setminus \bigcup_{m'' > m'} B_{m''}$. Then $F_{i,n} \cap U_{m'}$ is a union of whole components of $F_{i,m'} \cap U_{m'}$, while $F_{i,m'} \cap B_{m'} = \bigsqcup_k D_k^{(i)}$ is a family of least-area discs properly embedded in $B_{m'}$. (Induct on $m'' = m' + 1, \dots, n$: $U_{m'} \cap B_{m''} = \emptyset$, and Lemma A.16 keeps or deletes whole components.)*

Corollary A.16.3 (Mono-3). *Suppose a subsequence has been extracted, using HS 6.9/6.11 and Lemma A.6, for which $F_{i,m'} \cap \text{Int } B_{m'}$ converges smoothly, with multiplicity, on compact subsets to $L_{m'}$. Then a further subsequence makes $F_{i,n} \cap U_{m'}$ converge on compact subsets, the limit being a*

union of whole sheets of $L_{m'} \cap U_{m'}$. (Sketch: compact exhaustion; each sheet is kept or deleted whole; finitely many patterns; diagonal. Finiteness of the number of sheets rests on the multiplicity bound of Lemma A.6. Honest flag: the convergence definition of HS 6.11 has not been checked verbatim.)

Proposition A.17 ((H-glue): gluing across balls, conditional on GP(ii)). *The argument of step 5 of the assembly (Proposition A.17.2; the cross-ball gluing of HS Thm 4.1, Subcase 2(ii)) remains valid under the weakened hypotheses: (α) after step m , $F_{i,m} \cap \text{Int } B_m$ is a family of properly embedded least-area discs and a subsequence converges in $\text{Int } B_m$; (β) the change outside B_m at step m is the deletion of whole components as in Lemma A.16; (γ) the limit on $U_{m'}$ is a union of whole sheets of the original limit. The general-position condition required is GP(ii): for $m' < m''$, every sheet of $L_{m'}$ — determined by the subsequence of step m' , and unchanged by later subsequences — is transverse to $\partial B_{m''}$; transversality is an open dense condition on $r_{m''}$, handled as the countably many conditions of §A.5.2.*

Proof. Notation: a gluing point $x \in \partial B_{m''}$ with $m'' = \max\{m : x \in B_m\}$ and $m' = \max\{m < m'' : x \in \text{Int } B_m\}$; K a small compact neighbourhood of x with $K \subset \text{Int } B_{m'}$ and $K \cap B_{m''} = \emptyset$ for $m''' \notin \{m', m''\}$. Points lying on the intersection curve of two spheres are a case the source itself does not treat separately (*inherited item I-1*, §A.8).

Tool lemma Mono-2⁺. (i) $F_{i,n} \cap K$ is a union of whole components of $F_{i,m''} \cap K$; (ii) $F_{i,m''} \cap (K \setminus \text{Int } B_{m''})$ is a union of whole components of $F_{i,m'} \cap (K \setminus \text{Int } B_{m''})$; (iii) $F_{i,m''} \cap B_{m''} = \bigsqcup D_c$, where ∂D_c are exactly the boundary curves of the retained outer block Q_* within $F_{i,m''-1} \cap \partial B_{m''}$. (Induct step by step using Lemma A.16.)

Tool lemma Mono-3⁺. Under GP(ii) one may choose K and a subsequence so that $F_{i,n} \cap (K \setminus \text{Int } B_{m''})$ converges smoothly to a union T' of whole sheets of $L_{m'} \cap (K \setminus \text{Int } B_{m''})$. ($L_{m'} \cap K$ has finitely many discs, each cut transversally by $\partial B_{m''}$ into two half-discs; each sheet is connected on a half-disc, so by Mono-2⁺ it is kept or deleted whole; finitely many patterns; extract a subsequence.)

The cross-ball part of HS 4.1 uses only the following nine points, checked one by one under (α), (β), (γ): P1, the outer limit T' is smooth with $T' \pitchfork \partial B_{m''}$ (Mono-3⁺ and GP(ii); the source uses only smoothness of T' itself, never $T' = L_{m'}$); P2, the inner part $F_{i,n} \cap B_{m''}$ is a family of least-area discs (Mono-2⁺(i),(iii); a subfamily is still a family of discs); P3, the boundary arcs converge, $\Gamma_i \rightarrow \Gamma$ in C^1 (restriction of the retained sheets); P4, inside and outside are the same surface: Γ_i is at once a boundary arc of the inner disc D_c and one of the outer sheet — an identity of sets; the deletion at step m'' removes the curve too, and steps $m''' > m''$ delete the single component containing both halves, so “kept together, deleted together”; P5, HS 3.5 uses only P2 and P3; P6, $F_{i,n} \in \mathcal{F}$ with $\text{Area}(F_{i,n}) \rightarrow I$ (Corollary A.16.1), and both the cut-and-paste and the no-bending contradiction of Subcase 2(ii) produce a $G \in \mathcal{F}$ of area $< I$; P7, the cut-and-paste is supported in a ball $B \ni x$ with $B \cap \partial F = \emptyset$ (Proposition A.17.2), and irreducibility gives $\text{rel } \partial F$; the elimination of double curves is Lemma A.18; P8, Proposition A.17.2 applies Lemma A.12 and HS 6.9 to $F_{i,n} \cap X^b$: in the subfamily the component containing γ_0^b is never deleted (Lemma A.16), and the others do not meet ∂M and remain proper; P9, global assembly: F^* is defined region by region, each general-position point being determined by the limit at $m(x)$; Fatou gives $\text{Area}(F^*) \leq I$; and $\partial F \subset F_{i,n}$ gives $F^* \supset \partial F$. An empty subunion inside some ball is harmless, and connectedness comes from (H-cov) with Lemma A.11. Nowhere is $T' = L_{m'}$ used, nor pointwise agreement of the limits from two balls on the overlap. $\square$

Proposition A.17.2 (The pinch point does not lie on ∂F). *The multiplicity pinch point q of Subcase 2(ii) does not lie on ∂F ; hence a small ball $B \ni q$ disjoint from ∂F may be chosen, and the entire cut-and-paste takes place in $\text{Int } M$ and is $\text{rel } \partial F$.*

Proof. Suppose $q \in \partial F$, and take a ball X^b of type (III)/(IV) with $q \in \text{Int } \gamma_0^b$. By HS 6.9 applied to $F_{i,n} \cap X^b$, a subsequence converges smoothly to T with $\partial T = \text{Int } \gamma_0^b$. By Lemma A.12(ii), every component other than D_a misses ∂M , and if the limit of such a sheet touched ∂M at an interior point it would lie in ∂M by T7, contradicting $H > 0$; so both sheets would have to come from D_a . But T has only one sheet with boundary $\text{Int } \gamma_0^b$, and the second sheet would have q as an interior point, contradicting T7 again. $\square$

This also discharges the transfer premise (H0) of Proposition A.19.

Singular branches: eliminating double curves

A.6.1 Where the gap is. The elimination of double curves in the branch “If $S'_{i,2}$ is singular” of HS Thm 4.1 takes as its base case “every double curve on a sphere bounds two discs”; in an isotopy class of genus ≥ 1 that base case does not exist. This section replaces the source of the contradiction by “the surface after replacement is still in $\mathcal{F}$ and has area $< I$ ”.

Lemma A.18 (Replacement lemma). *Let M be an irreducible Riemannian 3-manifold, possibly with boundary, and let $S \subset M$ be compact, connected, properly embedded, two-sided and incompressible with $\emptyset \neq \partial S \subset \partial M$. Let $G \subset S$ be an embedded 2-disc with $\partial G = c_0$ and $G \cap \partial S = \emptyset$; let $Y \subset M$ be an embedded 2-disc with $\partial Y = c_0$ and $\text{Int } Y \cap \partial S = \emptyset$; and assume **(A-N)**: there are $\epsilon > 0$ and $x \in M$ with $Y \cap \partial M \subset \bar{B}(x, \epsilon)$ and $\text{diam}(\partial S) > 4\epsilon$. Then for every $\eta > 0$ there is a compact, connected, properly embedded, piecewise smooth surface $S'' \subset M$, isotopic to S rel ∂S , with*

$$\text{Area}(S'') \leq \text{Area}(S) - \text{Area}(G) + \text{Area}(Y) + \eta.$$

It is not required that $\text{Int } Y \cap G = \emptyset$ (the corresponding lemma of the source does require it, and both of its uses in HS presuppose it without verification — in particular there is no reason for the interiors of H_i and of $L_i \cup H'$ to be disjoint).

Proof, the load-bearing steps. Induct on $n' := \#$ components of $\Gamma' := \text{Int } Y \cap \text{Int } S$, with the disc family $\mathcal{D}' := \{E \subset Y : \partial E \in \Gamma'\} \cup \{\Delta \subset S : \partial \Delta \in \Gamma'\}$. The preparatory facts hold verbatim: Γ' is a finite union of disjoint simple closed curves; a least-area disc has no double curve in its interior; when $D \subset Y$, the curve ∂D bounds a unique disc Δ in S ; the exclusion lemma (S does not fall into the 3-ball used for the exchange, by (A-N)); the exchange lemma (which really removes one double curve, at an area budget of $\eta/4$); and the normalisation (pushing Y off ∂M , at an area cost $\leq 3C_M t \text{Area}(Y)$, see Proposition A.19). Each $c \in \Gamma'$ is disjoint from $c_0 = \partial Y$, so either $c \subset \text{Int } G$ or $c \subset \text{Int}(S \setminus G)$.

Base case $n' = 0$. $\text{Int } Y \cap S = \emptyset$, so $G \cup Y$ is automatically an embedded 2-sphere, bounding a 3-ball by irreducibility; replace G by Y across it, isotopically rel ∂S . (This also covers the embedded branch of HS.)

Inductive step. Take $D \in \mathcal{D}'$ of least area, with $c = \partial D$ and $\text{Int } D \cap \Gamma' = \emptyset$.

- $D \subset S$ (branch B'): then $D \cap \partial M = \emptyset$; let $D' \subset Y$ be the disc bounded by c in Y , so $\text{Area}(D) \leq \text{Area}(D')$; replace Y by $Y^* = (Y \setminus D') \cup D$ and push off. Then n' strictly decreases, $\text{Area}(Y^*) \leq \text{Area}(Y) + \eta/4$, and S and G are untouched. The hypothesis “ $D \subset S \setminus G$ ” is no longer needed.
- $D \subset Y$: with the unique $\Delta \subset S$, the sphere $\Sigma = D \cup \Delta \subset \text{Int } M$ is embedded and bounds a 3-ball V with $\text{Int } V \cap S = \emptyset$. According to the position of c : if $c \subset \text{Int}(S \setminus G)$ and $G \subset \Delta$, branch (T) terminates at once; if $c \subset \text{Int}(S \setminus G)$ and $G \cap \Delta = \emptyset$, branch (A) is the exchange

lemma; and in the *new branch* (G) , $c \subset \text{Int } G$: then c bounds Δ_G in the disc G and, by uniqueness, $\Delta = \Delta_G \subset \text{Int } G$. Applying the exchange lemma to (Y, S) gives an isotopy h_t of S rel ∂S , supported in V and a small neighbourhood of c and disjoint from c_0 ; the surfaces $S^* = h_1(S)$ and $G^* = h_1(G)$ still have boundary c_0 , both S and G lose Δ and gain $\hat{D}$, so $\text{Area}(S^*) - \text{Area}(G^*) = \text{Area}(S) - \text{Area}(G)$ with Y untouched, and n' strictly decreases.

Exhaustiveness: either $D \subset S$ or $D \subset Y$, and in the latter case the position of c and the containment relations between G and Δ exhaust the possibilities. $\square$

The point of branch (G): the double curves created by Y passing through G can be removed at no cost in the area budget, because S and G gain and lose together.

Proposition A.19 (Residual: the area cost of pushing off the boundary, and the tightened budget). *In the invocation $(S, G, Y) = (S_{i,2}, H_i, L_i \cup H')$ at the end of the bending argument (with $x \in T_1 \cap \partial B_2$, $B \ni x$ a small ball, H the disc of $(T_1 \cup T_2) \cap B$ containing x , H' the least-area disc in B with the same boundary as H and independent of i , $H_i \subset S_{i,2} \cap B$ with $H_i \rightarrow H$, and $L_i \subset \partial B$ the annulus between ∂H and ∂H_i), assume the transfer premise **(H0)**: $x \notin \partial F$, discharged by Proposition A.17.2. Then*

- (R-i) *the normalisation hypotheses hold: $\partial H_i \cup H_i \subset \text{Int } M$; Lemma A.18 does not require $\text{Int } Y \cap G = \emptyset$, and $d < \frac{1}{2}d_0$ suffices in the normalisation;*
- (R-ii) *the push-off increment satisfies $\Delta_i(t) \leq 3C_M(i)t \text{Area}(Y)$, which holds even when H' lies wholly against ∂M ; that $C_M(i)$ depends on i is harmless, t being chosen only after i is fixed;*
- (R-iii) *the budget must be tightened: the estimate “ $\text{Area}(S_{i,2}) < I + a/2$ ” must be tightened to “ $< I + a/4$ ”, which a minimising sequence satisfies automatically for large i ; taking $\eta = a/8$ gives $\text{Area}(S'') < I + \frac{a}{4} - \frac{a}{2} + \frac{a}{8} = I - \frac{a}{8} < I$, a contradiction, whereas the untightened $I + a/2 - a/2 + a/8 = I + a/8$ does not suffice;*
- (R-iv) *when ∂M is strictly convex, the least-area disc H' in B cannot lie wholly inside ∂M , by the first variation $\frac{d}{ds} \text{Area}(\Phi_s H') = -\int H_{\partial M} \phi < 0$.*

Proof. (R-i) (H0) gives $B \cap \partial F = \emptyset$, so $H_i \subset S_{i,2} \setminus \partial S_{i,2} \subset \text{Int } M$. (R-ii) In collar coordinates $\Psi_t(p, s) = (p, s + t\chi(s))$ one has $\text{Lip}(\Psi_t) \leq 1 + C_M t$, and the area formula for the 2-rectifiable set Y gives $\text{Area}(\Psi_t Y) \leq (1 + 3C_M t) \text{Area}(Y)$, independently of how Y meets ∂M . (R-iii) The source gives $\text{Area}(L_i) < a/4$, $|\text{Area}(H_i) - \text{Area}(H)| < a/4$ and $\text{Area}(H') < \text{Area}(H) - a$, so $\text{Area}(Y) < \text{Area}(G) - a/2$; substitute into Lemma A.18 with the tightened bound. (R-iv) As stated; it fails for $H \geq 0$ only, but (R-ii) has already made it unnecessary. $\square$

A.6.2 Feeding back into the assembly. First place (Subcase 2(ii)): apply Lemma A.18 to $(F_{i,2}, G_i, G'_i)$ with $\eta = \delta/8$, obtaining $\text{Area}(S'') < I$, a contradiction; the embedded and singular branches merge into a single invocation. Second place (the bending argument): Proposition A.19 shows $T_1 \cup T_2$ does not bend along $T_1 \cap \partial B_2$. The closed-surface version ($\partial F = \emptyset$) is not used here.

Proof of Theorem A.1 (a ten-step assembly)

Proof. Each step records where its conclusion is discharged and where its conditions are used.

Step 1 (minimising sequence). Take smooth properly embedded $\{F_i\} \subset \mathcal{F}$ with $\text{Area}(F_i) \rightarrow I$ and $\text{Area}(F_i) \leq I + 1$.

Step 2 (ball cover). Take ε satisfying the radius condition of Lemma A.10, including $\varepsilon < \varepsilon_1$ from Lemma A.8; M is compact, so finitely many balls $B_1, \dots, B_n$ cover M , classified into four

types by §A.4.1, with types (III) and (IV) replaced by rounded balls X^b (§A.2.1, (Gb8)). Radii and rounding parameters are taken in general position by GP(i),(iii). (For complete non-compact M one would need (H-E) here; no such relaxation is made.)

Step 3 (ball-by-ball maximal disc replacement). Apply Theorem A.14 ball by ball as in §A.5.2: discs for circles come from incompressibility, discs for arcs from Lemma A.9 with hypotheses supplied by Lemma A.8; that the replacement discs lie in the ball, are embedded, are pairwise disjoint and do not fall into ∂M is Lemma A.10 (arc case: the remark after it); the isotopy rel ∂F is Lemma A.15 with Corollary A.15.1 (arc form via Proposition A.2 and item 7 of Proposition A.5). By Lemma A.16, step m deletes, outside B_m , only whole components not containing ∂F . Here (H-G5) is discharged by Theorem A.14 and (R) by Proposition A.2 with Lemma A.3.

Step 4 (extracting a convergent subsequence). For each ball apply HS 6.9/6.11, whose hypotheses are discharged by Theorem A.14(b) (rounded balls: item 6 of Proposition A.5), and extract a subsequence; the “finitely many components” of that proof is supplied by Lemma A.6 (balls of type (II) being the critical case); and for $m' < m$ the convergence of $F_{i,m}$ on $U_{m'}$ is preserved by Corollary A.16.3, the limit being a union of whole sheets of the original limit.

Step 5 (gluing across balls, and Subcase 2(ii)). Proposition A.17, conditional on GP(ii), makes the gluing of HS 4.1 valid verbatim under deletion of whole components; that the cut-and-paste ball misses ∂F , so that the isotopy is rel ∂F , is Proposition A.17.2; the elimination of double curves is Lemma A.18 (first place) and the exclusion of bending is Proposition A.19 (second place, with the budget $< I + a/4$). Here (H-W1a) and (H-glue) are discharged.

Step 6 ($F^* \neq \emptyset$). $\partial F \subset F_{i,n}$ for every i , the component containing ∂F never being deleted (Lemma A.16), so $F^* \supset \partial F \neq \emptyset$.

Step 7 (C2). If a limit sheet touched ∂M at an interior point it would lie in ∂M by T7, contradicting $H > 0$ in (A- ∂) — the same argument as in Proposition A.17.2; and the sheet along ∂F is the limit of D_a , with $D_a \cap \partial M = \gamma_0^b$ (Lemma A.12(i)). Hence $F^* \cap \partial M = \partial F$.

Step 8 (C4). Lemma A.11; (H-cov) is used here, and gives $F^* \in \mathcal{F}$ two-sided.

Step 9 (C1, C3, C5). C5 and C1 are Proposition A.13. For C3: the limit sheet along ∂F is the surface T of HS 6.9, whose boundary regularity comes from HS 3.5 via T-C, applied inside the boundaryless extension $\text{Int } N_\delta$ of Lemma A.4 with the boundary arc γ_0^b held fixed; so T is $C^{m,\alpha}$ up to the boundary along $\text{Int } \gamma_0^b$; and transversality to ∂M follows from Lemma A.3(iii) applied to the limit sheet, itself a least-area disc spanning a smooth boundary arc (Proposition A.13). (The source’s own “LEMMA H + T-C” argument lies outside the slice examined here; see $\Delta 2$ in §A.8.)

Step 10 (area). $\text{Area}(F^*) \leq \liminf \text{Area}(F_{i,n}) = I$ by smooth convergence and Fatou (Corollary A.16.1); and $F^* \in \mathcal{F}$ gives $\geq I$. Take $F' = F^*$. $\square$

What this appendix is still conditional on

A.8.1 The list of conditions.

1. **(H-cov)** (H4): $F_{i,n}$ covers F^* naturally. Assumed by Hass–Scott, not proved here, written explicitly into Theorem A.1; used only in step 8 (Lemma A.11).
2. **GP**: the general-position selections (GP(i), radii by Sard; GP(ii), $L_{m'} \pitchfork \partial B_{m''}$; GP(iii), the rounding parameter r for almost every value). Countably many open dense conditions, on which the assembly depends; listed explicitly.
3. **Named theorems** (H2): T1 Morrey, T-A Gulliver–Osserman (interior only), T-C Hildebrandt, T3/T4 Meeks–Yau, HS 6.9/6.11 (via 3.3/3.5/3.6), T7 HS 6.6, T8 HS 6.4, T9 HS 2.2, T5 Courant–Lebesgue; textbook-level: T-B Hopf/GT 3.4, T-D, T-E, T-F smooth Alexander,

Whitehead triangulation, and the isotopy of discs with common boundary in a 3-ball. *All original numbering unverified*; statements as reported by Hass–Scott. No new research-level input is added here; T-W is listed but not used.

4. **P8(B)/P9**: the two sentences omitted when 6.9 is merged from 3.3/3.5 (§A.0.4); P8(B) is proved here for X^b and P9 identified as geometrically identical to the original setting. If a closure audit requires 6.9 to be self-contained, both must be brought into the text.
5. **The published definition of “sufficiently convex”** lies outside the slice examined and is used as reported. If the definition requires Y and Y' to meet at a strict angle along ∂Y , then Lemma A.7b(4) becomes “original to be checked”, but the proofs use only strict convexity and the verdict is unchanged (items 1 and 6 of Proposition A.5).
6. **Inherited item I-1**: gluing points lying on the intersection curve of two or more spheres are treated neither in HS 4.1 nor in the source, and are not treated here either.
7. **$\Delta 2$ (H6)**: the source’s own argument for conclusion C3 (LEMMA H + T-C) has not been rechecked; what step 9 gives is the alternative route through HS 3.5 / T-C and Lemma A.4, from the same family and unaudited.
8. **The comparison-geometry constants of Lemma A.7, the convergence definition of HS 6.11 in Corollary A.16.3, and the source text of LEMMA B in Lemma A.9**: flagged honestly as citation dependences, not as new gaps.
9. **“Outermost” = the maximal reading** (§A.5.1) is an interpretation of the source, not its published definition.
10. **(H-SET)**: the competitor class of Theorem A.1 fixes the boundary pointwise and does not directly yield (E) of the main text; the bridge is in §5 and Appendix C.
11. **Same-family declaration** (H12): all of this appendix comes from one model family, with no cross-engine audit.

A.8.2 Discharged, no longer conditions. (H-W1a) (Lemma A.18 with Proposition A.19); (H-G5) (Theorem A.14); (R) (Proposition A.2 with Lemma A.3, the rounding route); (H-glue) (Proposition A.17); LEMMA H (a by-product of Proposition A.2(i)); (H0) (Proposition A.17.2). (H-E) does not arise, (A-M) being compact.

A.8.3 Index of call sites in the main text.

Call site	What is used	Note
§5 (attainment of $A(v)$; Lemmas 5.1, 5.3)	Theorem A.1, conclusions C1–C5	bridged by Theorem C.3 of Appendix C and (H-SET); the reading of (R1) is H5
§7 (choice of representatives in Thm 7.1; the graph representation of §7.2) (H-sm) of Theorem B.3	Theorem A.1; (AM), Proposition A.13	cite as “Theorem A.1 (Appendix A)”
Theorem C.3	C3 of Theorem A.1 (C^2 up to the boundary) and Proposition A.13 the construction F' of Theorem A.1, and C2, C3	supplied by this appendix together with Theorem C.3 $\Delta 1$ is closed by the E- β Theorem C.2; $\Delta 2$ is item 7 above
The constants of Appendix B and the (G-b) chain	Lemma A.6 (monotonicity up to ∂M)	the constant $\beta = 1/(4\alpha)$

B Compactness and blow-up analysis for the family of fixed-boundary minimisers

This appendix does one thing: it supplies the compactness and blow-up analysis needed by §5 and §7.1 of the main text, changing no mathematical statement of the main text. Fixed numbering: Theorem B.1 is the compactness statement (G-b); Theorem B.2 is the instability of the half-surface; Theorem B.3 is (G-b) $\Rightarrow$ (E); everything else is numbered from B.4 on. *Notion of stability*: only **Dirichlet stability** — non-negativity of the second variation of area under fixed boundary — is used here; no other notion of stability is introduced.

On provenance: as in Appendix A, all of this material was produced within one model family and audited only within that family; it has received no cross-engine audit. Conditional conclusions carry their conditions explicitly in the statements. See Appendix G.

Setting, the hypothesis (H-sm), and named theorems

B.0.1 Setting. $M = E(K)$ with the metric g of §2.1 of the main text, and $\ell \subset \partial M$ a fixed leaf of J . Three metric facts from §2.1 are used:

- (a) ∂M is a flat torus, totally umbilic, $\Pi = \mu g|_{\partial M}$ with $\mu = 1$; and ℓ is a straight line of ∂M ($D_T T = 0$);
- (b) in a collar of ∂M , the distance r to ∂M satisfies $\Delta_\Sigma r \leq -c_0 < 0$ along any minimal surface Σ , with $c_0 = 1$;
- (c) writing $\bar{K}_{\max}^+ := \max\{0, \sup \bar{K}(\Pi)\}$, the supremum over all tangent 2-planes Π in M , compactness of M gives $\bar{K}_{\max}^+ < \infty$.

For $a > 0$ and $h \geq 0$, let $\mathcal{M}_a^\ell(h)$ be the set of $\Sigma \subset M$ that are connected, orientable, properly embedded, with $\partial\Sigma = \ell$ *pointwise*, of genus h , with $\text{Area}(\Sigma) \leq a$, and *of least area in their own isotopy class rel $\partial\Sigma$ with the boundary fixed pointwise*. This last clause is convention (α) of the main text; this appendix and Appendix C use (α) throughout, never (β) . Since M and Σ are orientable, Σ is two-sided; fix a global unit normal N . As $\partial\Sigma = \ell$ is connected,

$$\chi(\Sigma) = 2 - 2h - 1 = 1 - 2h \quad (\text{the same for every } \Sigma \text{ in the family}). \quad (\text{B.0})$$

B.0.2 The hypothesis (H-sm).

(H-sm) Each Σ in the family $\mathcal{M}_a^\ell(h)$ is *individually* (no uniformity required) C^2 up to $\partial\Sigma = \ell$.

(H-sm) is a qualitative property imposed member by member and contains no uniform estimate; the entire content of (G-b) (Theorem B.1) is precisely the uniformity. (H-sm) is supplied by Theorem A.1 (Appendix A: the construction is smooth up to the boundary) together with Theorem C.3 (Appendix C); the reading of (R1) it presupposes is H5 of the table in §12.2.2. Lemma B.4 and Theorem B.7 use (H-sm), both Gauss–Bonnet and the boundary term of the first variation requiring it.

B.0.3 Stability. “Stable” always means *Dirichlet stable*: for all $u \in C_c^\infty(\Sigma^\circ)$,

$$\int_\Sigma \left(|\nabla u|^2 - (|A|^2 + \overline{\text{Ric}}(N, N))u^2 \right) \geq 0.$$

B.0.4 Named theorems. Research-level items all reappear in the list of §B.4.

- **T3 (Schoen, curvature estimate for stable minimal surfaces; research-level, J. Diff. Geom. 1983, original not checked).** Let (M^3, g) be compact and $\Sigma \subset M$ minimal, two-sided and Dirichlet stable for variations compactly supported in the interior. Then there are $C_1 = C_1(M, g)$ and $r_0 = r_0(M, g)$ with $|A|(x) \leq C_1 / \min\{d_\Sigma(x, \partial\Sigma), r_0\}$ for all $x \in \Sigma^\circ$.
- **T4 (do Carmo–Peng / Fischer–Colbrie–Schoen / Pogorelov; research-level, originals not checked).** A complete, two-sided, Dirichlet stable minimal surface in $\mathbb{R}^3$ is a plane.
- **T6 (Schwarz reflection; textbook-level).** If a minimal surface contains a straight line segment, its image under rotation by π about that line glues to it smoothly along the line, giving a minimal surface.
- **T7 (Osserman’s structure theorem for finite total curvature; monograph-level, Osserman, *A Survey of Minimal Surfaces*; recorded as a research-level dependence).** A complete minimal surface in $\mathbb{R}^3$ with $\int |A|^2 < \infty$ is conformally a compact Riemann surface $\bar{\Sigma}_\gamma$ with finitely many points $p_1, \dots, p_k$ removed, and the Gauss map G extends holomorphically to $\bar{G} : \bar{\Sigma}_\gamma \rightarrow S^2$.
- **T9 (Schoen, asymptotics of the ends; research-level, J. Diff. Geom. 18 (1983), original not checked).** Each end of a properly embedded minimal surface of finite total curvature is asymptotic to a plane or to one end of a catenoid: over the orthogonal complement of the asymptotic normal w it is a graph $\langle x, w \rangle = a \log |y| + b + O(|y|^{-1})$; and the asymptotic normals of the different ends are mutually parallel. This last clause is *not proved here*, and is the one research-level input of the (G-b) chain that this manuscript does not prove for itself.

Textbook-level tools, named only: the Gauss equation; Gauss–Bonnet with boundary; the strong maximum and minimum principles; the Hopf boundary point lemma; the second variation (Jacobi) formula; Federer–Fleming rectifiable compactness; Allard compactness for integral varifolds; interior and boundary Schauder estimates for the minimal surface equation; Arzelà–Ascoli; the monotonicity formula for minimal surfaces (first variation for rectifiable sets); Riemann’s removable singularity theorem; the vanishing of the capacity of a point in dimension two; Jordan–Brouwer (a properly embedded surface is two-sided).

Three lemmas, multiplicity one, and Theorem B.1

Throughout this section $\{\Sigma_j\} \subset \mathcal{M}_a^\ell(h)$ is an arbitrary sequence, A_j and N_j its second fundamental form and unit normal, and $L(\ell)$ the length of ℓ .

Lemma B.4 (Uniform total curvature). *Under (H-sm),*

$$\int_{\Sigma_j} |A_j|^2 dA \leq \Lambda := 2\bar{K}_{\max}^+ a + 4\pi(2h - 1) + 2L(\ell), \quad (\text{B.1})$$

with Λ independent of j . Stability is not used.

Proof. (i) Gauss equation: $K_\Sigma = \bar{K}(T_p\Sigma) + \det A$; minimality $\text{tr } A = 0$ gives $\det A = -\frac{1}{2}|A|^2$, so $|A|^2 = 2(\bar{K}(T_p\Sigma) - K_\Sigma)$.

(ii) Gauss–Bonnet with boundary (Σ_j compact and, by (H-sm), C^2 up to the boundary): $\int_{\Sigma_j} K_{\Sigma_j} dA + \oint_\ell k_g ds = 2\pi\chi(\Sigma_j) = 2\pi(1 - 2h)$, where $k_g = \langle \nabla_T T, \nu_{\text{in}} \rangle$, T the unit tangent of ℓ and ν_{in} the inward conormal of Σ_j .

(iii) The key point: $\partial\Sigma_j = \ell$ is one and the same fixed curve, so after arc-length parametrisation $\nabla_T T$ is the curvature vector of ℓ in M , depending only on ℓ and g — not on j , and not on the

contact angle between Σ_j and ∂M . By B.0.1(a), $\nabla_T T = -\Pi(T, T)n = -\mu n$ with $|\nabla_T T| = 1$, so pointwise $|k_g| \leq 1$ and $|\oint_\ell k_g ds| \leq L(\ell)$.

Combining, $\int |A_j|^2 = 2 \int_{\Sigma_j} \bar{K} - 4\pi\chi + 2 \oint k_g \leq 2\bar{K}_{\max}^+ a + 4\pi(2h-1) + 2L(\ell)$, using $\int_{\Sigma_j} \bar{K} \leq \bar{K}_{\max}^+ \text{Area}(\Sigma_j) \leq \bar{K}_{\max}^+ a$ (the positive part is taken because $\bar{K}$ may change sign). $\square$

Remark. What is used is not fixed genus as such, but fixed χ *together with* a pointwise fixed boundary, the term $\oint k_g$ being controlled by the curvature of ℓ itself.

Lemma B.5 (Neatness and boundary transversality). *Let $\Sigma \subset M$ be minimal, C^2 in the interior, properly embedded, with $\partial\Sigma = \ell$. Then (a) $\Sigma \cap \partial M = \partial\Sigma$; and (b) if moreover (H-sm) holds, Σ is transverse to ∂M along ℓ : $T_p \Sigma \neq T_p \partial M$ for every $p \in \ell$.*

Proof. (a) By B.0.1(b), $r|_\Sigma$ is strictly superharmonic and ≥ 0 in the collar. If an interior point p had $r(p) = 0$, then $r|_\Sigma$ would attain the minimum 0 at an interior point, so by the strong minimum principle $r \equiv 0$ on that component, which would then lie in ∂M ; but ∂M is totally umbilic with $\Pi = g$ and is not minimal, contradicting minimality of Σ . Hence $r > 0$ on Σ° .

(b) From $r|_\Sigma \geq 0$, $r|_\ell = 0$, $\Delta_\Sigma r \leq -c_0 < 0$ and Σ of class C^2 at ℓ , the Hopf boundary point lemma gives $\partial_\nu r < 0$ for the outward conormal ν ; so $\nabla^\Sigma r \neq 0$ on ℓ , and $T_p \Sigma \not\subset T_p \partial M$. $\square$

Part (b) gives transversality pointwise, not a uniform lower bound on the contact angle; the latter comes for free from the uniform curvature bound in Theorem B.1.

Lemma B.6 (Interior stability and the interior curvature estimate). *Let $\Sigma_j \in \mathcal{M}_a^\ell(h)$. Then*

- (a) Σ_j is Dirichlet stable: for all $u \in C_c^\infty(\Sigma_j^\circ)$, $\int_{\Sigma_j} (|\nabla u|^2 - (|A_j|^2 + \bar{\text{Ric}}(N_j, N_j))u^2) \geq 0$;
- (b) there are C_1, r_0 depending only on (M, g) with $|A_j|(x) \leq C_1 / \min\{d_{\Sigma_j}(x, \ell), r_0\}$;
- (c) if $m_j := \max_{\Sigma_j} |A_j|$ is attained at q_j , then $m_j d_{\Sigma_j}(q_j, \ell) \leq C_1$: after rescaling, the curvature blow-up point stays at bounded intrinsic distance from the boundary.

Proof. (a) Extend uN_j to a vector field on M compactly supported in a neighbourhood of Σ_j° and take its flow ψ_t ; ψ_t is an ambient isotopy, the identity near ℓ , so $\psi_t(\Sigma_j)$ lies in the isotopy class of Σ_j rel ℓ with the boundary fixed pointwise. Least area in that class means $t \mapsto \text{Area}(\psi_t \Sigma_j)$ is minimal at $t = 0$, so its second derivative is ≥ 0 , which is the displayed inequality by the second variation formula. (b) From (a) and T3. (c) Take $x = q_j$ in (b); when $m_j \rightarrow \infty$ the minimum is attained by the first term. $\square$

Theorem B.7 (Multiplicity one, at the original scale). *Assume (H-sm); let V_j be the integral varifold induced by Σ_j and $T_j = \llbracket \Sigma_j \rrbracket$ the associated rectifiable 2-current. Then*

- (i) $\|\delta V_j\| = \mathcal{H}^1 \llcorner \ell$, an equality independent of j ;
- (ii) a subsequence has $T_j \rightarrow T$ (rectifiable, with $\partial T = \llbracket \ell \rrbracket$) and $V_j \rightarrow V$ (an integral varifold);
- (iii) $\|\delta V\| \leq \mathcal{H}^1 \llcorner \ell$; and if V is, near $p \in \ell$, a C^1 surface with boundary of multiplicity m and boundary ℓ , then $m = 1$;
- (iv) if moreover $\sup_j \max |A_j| < \infty$, then the support of V is a smooth surface with boundary, of multiplicity one everywhere.

Proof. (i) By minimality and (H-sm), for any C^1 field X , $\delta V_j(X) = \int_{\Sigma_j} \operatorname{div}_{\Sigma_j} X = \oint_{\ell} \langle X, \nu_j \rangle ds$; $|\nu_j| = 1$ gives $\|\delta V_j\| \leq \mathcal{H}^1 \llcorner \ell$, and taking X locally equal to ν_j gives the reverse.

(ii) $\mathbf{M}(T_j) \leq a$ with $\partial T_j = \llbracket \ell \rrbracket$ fixed gives Federer–Fleming; $\text{area} \leq a$ with uniformly bounded first variation measure gives Allard compactness.

(iii) Varifold convergence gives $\delta V_j(X) \rightarrow \delta V(X)$ for each fixed X , hence lower semicontinuity of the total variation and $\|\delta V\| \leq \mathcal{H}^1 \llcorner \ell$. If V is near p a C^1 surface with boundary of multiplicity m and boundary ℓ , then there $\|\delta V\| = m \mathcal{H}^1 \llcorner \ell$, so $m \leq 1$; and $\partial T = \llbracket \ell \rrbracket \neq 0$ gives $m \geq 1$.

(iv) A uniform curvature bound gives, by step (1) of the proof of Theorem B.1, convergence in C^∞ up to the boundary, so the support of V is a smooth surface with boundary and the multiplicity is locally constant on each component; the component containing ℓ has multiplicity one by (iii). If there were a further component $\hat{S}$ missing ℓ — a closed minimal surface of multiplicity m' — then for large j the surface Σ_j would contain, inside a tubular neighbourhood of $\hat{S}$, m' boundaryless normal graphs, which would be closed components of Σ_j , contradicting that Σ_j is connected with boundary. $\square$

Remark. Part (iii) uses only the first variation measure and presupposes nothing about connectedness of the limit; the clause “if V is near p a C^1 surface with boundary” is a boundary regularity statement, supplied by the uniform curvature bound in (iv). Multiplicity one at the blow-up limit has an independent proof (Proposition B.14); the two routes agree.

Lemma B.8 (Boundary monotonicity and a uniform boundary area ratio). *There are $C, \rho_3 > 0$, depending only on (M, g) and ℓ , such that for all j , all $p \in \ell$ and all $0 < \rho < \rho_3$, the quantity $\rho \mapsto e^{C\rho} \|\Sigma_j\|(B_\rho(p)) / \rho^2$ is non-decreasing. Setting*

$$\Theta^* := \sup_j \sup_{p \in \ell} \sup_{0 < \rho < \rho_3} \frac{e^{C\rho} \|\Sigma_j\|(B_\rho(p))}{\pi \rho^2}, \quad (\text{B.2})$$

one has, unconditionally, $\frac{1}{2} \leq \Theta^ \leq e^{C\rho_3} a / (\pi \rho_3^2) < \infty$.*

Proof. Take the radial field $X = \phi(|Z|/\rho)Z$, with Z the geodesic radial field centred at p . Since Σ_j is rectifiable of finite area, the first variation identity for rectifiable sets applies and no boundary regularity is needed. In the boundary term $\rho^{-2} \oint_{\ell \cap B_\rho} \langle Z, \nu_j \rangle ds$, smoothness of ℓ and $\nu_j \perp T$ give $\langle Z, \nu_j \rangle = O(\rho^2)$, so the term is $O(\rho)$ and is absorbed by the factor $e^{C\rho}$; the interior term is handled by the standard monotonicity formula. The upper bound comes from $\text{Area} \leq a$, the lower bound $\frac{1}{2}$ from the density of a half-disc at a boundary point. $\square$

Theorem B.1 ((G-b): compactness). *Assume (H-sm). If there is $C_2 = C_2(M, g, \ell, a, h)$ with $\sup_j \max_{\Sigma_j} |A_j| \leq C_2$ for every sequence $\{\Sigma_j\} \subset \mathcal{M}_a^\ell(h)$, then every sequence in $\mathcal{M}_a^\ell(h)$ has a subsequence converging in C^∞ , up to the boundary and with multiplicity one, to a member of $\mathcal{M}_a^\ell(h)$. The uniform curvature bound itself is supplied by Corollary B.26 (under the conditions listed there), so (G-b) holds under those conditions.*

Proof. (1) $|A_j| \leq C_2$, together with the fixed smooth boundary curve ℓ and the uniform boundary density bound of Lemma B.8, gives at every point — boundary points included — a graph or half-graph representation of uniform size, with uniformly bounded $C^{2,\alpha}$ norm; interior and boundary Schauder estimates for the minimal surface equation bootstrap this to uniform $C^{k,\alpha}$ bounds. Arzelà–Ascoli then gives a subsequence converging in C^∞ to a minimal surface Σ_∞ with $\partial \Sigma_\infty = \ell$.

(2) Theorem B.7(iv) gives multiplicity one.

(3) Multiplicity one together with C^∞ convergence gives, for large j , $\Sigma_j = \text{graph}(u_j)$ in a tubular neighbourhood of the normal bundle of Σ_∞ , with $\|u_j\|_{C^1} \rightarrow 0$ and $u_j|_\ell = 0$; so $\Sigma_j \cong \Sigma_\infty$

diffeomorphically, and $s \mapsto su_j$ is an isotopy rel ℓ fixing the boundary pointwise. Hence Σ_∞ is connected, orientable, of genus h , with $\partial\Sigma_\infty = \ell$.

(4) $\text{Area}(\Sigma_\infty) = \lim \text{Area}(\Sigma_j) \leq a$.

(5) Minimality passes to the limit: if Σ' were isotopic to Σ_∞ rel ℓ pointwise with $\text{Area}(\Sigma') < \text{Area}(\Sigma_\infty)$, then by (3) Σ' would lie in the same class as Σ_j for large j , while $\text{Area}(\Sigma_j) \rightarrow \text{Area}(\Sigma_\infty) > \text{Area}(\Sigma')$, contradicting minimality of Σ_j in its class. So $\Sigma_\infty \in \mathcal{M}_a^\ell(h)$. $\square$

All that remains of (G-b) is therefore a uniform curvature estimate; §B.2 reduces it, by blow-up analysis, to Theorem B.2.

Boundary blow-up analysis

B.2.1 Rescaling. Suppose $m_j = \max_{\Sigma_j} |A_j| \rightarrow \infty$ (otherwise Theorem B.1 applies directly), attained at q_j , with $p_j \in \ell$ a nearest boundary point and $p_j \rightarrow p \in \ell$. In normal coordinates at p_j put $\tilde{\Sigma}_j := m_j(\Sigma_j - p_j)$, $\tilde{\ell}_j := m_j(\ell - p_j)$, $\tilde{g}_j := m_j^2 g$ and $\tilde{q}_j := m_j(q_j - p_j)$; by Lemma B.6(c), $|\tilde{q}_j| \leq C_1$. In the blow-up limit $M \rightarrow \mathbb{H} = \{x_3 \geq 0\}$ and $\ell \rightarrow$ a straight line $L \subset \partial\mathbb{H}$ (Proposition B.13(c)).

Lemma B.10 (Uniform area ratio at every scale along L). *There is a constant c_M depending only on (M, g) such that for all j , all $0 < R < m_j \rho_3/2$ and all $\tilde{p} \in \tilde{\ell}_j$ (with preimage $p \in \ell$), the coordinate ball $B_R(\tilde{p})$ satisfies*

$$\frac{\|\tilde{\Sigma}_j\|_{\tilde{g}_j}(B_R(\tilde{p}))}{\pi R^2} \leq \left(1 + \frac{c_M R}{m_j}\right) e^{-CR/m_j} \Theta^* \leq \left(1 + \frac{c_M R}{m_j}\right) \Theta^*. \quad (\text{B.3})$$

In the C_{loc}^∞ limit (Proposition B.13) one has $\|\tilde{\Sigma}\|(B_R(\tilde{p})) \leq \Theta^* \pi R^2$ for every $R > 0$ and every $\tilde{p} \in L$.

Proof. The area ratio is scale invariant in dimension two: computed in $\tilde{g}_j$, $\|\tilde{\Sigma}_j\|(B_R^{\tilde{g}_j}(\tilde{p})) / (\pi R^2)$ equals $\|\Sigma_j\|(B_{R/m_j}(p)) / (\pi (R/m_j)^2)$ at the original scale, which by (B.2) is $\leq e^{-CR/m_j} \Theta^*$ (as $R/m_j < \rho_3$). Coordinate balls differ from geodesic balls, and $\tilde{g}_j$ from the Euclidean metric, by a factor $1 + O(R/m_j)$ within radius R (in normal coordinates $g = \delta + O(|x|^2)$); write that factor as $1 + c_M R/m_j$ to get (B.3). Letting $j \rightarrow \infty$ with R fixed gives the limiting form. $\square$

Lemma B.12 (Moving the centre off L). *Let μ be a Radon measure on $\mathbb{R}^3$ with $\mu(B_R(\tilde{p})) \leq \Theta \pi R^2$ for some point $\tilde{p}$ and all $R > 0$. Then for any q and all $R > 0$, $\mu(B_R(q)) \leq \mu(B_{R+d}(\tilde{p})) \leq \Theta \pi (R+d)^2$ with $d = |q - \tilde{p}|$; hence $\Theta_\infty(\mu) := \lim_{R \rightarrow \infty} \mu(B_R(q)) / (\pi R^2) \leq \Theta$, independently of the centre q .*

Consequently $\Theta_\infty(\tilde{\Sigma}) \leq \Theta^*$, also when computed at the base point $\tilde{q} = \lim \tilde{q}_j$. Only $\Theta^* < \infty$ is used, which is unconditional (Lemma B.8).

Proposition B.13 (C_{loc}^∞ convergence up to the boundary, and properness). *Under (H-sm) and the setting of B.2.1, a subsequence satisfies $\tilde{\Sigma}_j \rightarrow \tilde{\Sigma} \subset \overline{\mathbb{H}}$ in $C_{\text{loc}}^\infty(\overline{\mathbb{H}})$ — locally, convergence of finitely many graphs or half-graphs — where $\tilde{\Sigma}$ is complete, properly embedded, minimal, with $|\tilde{A}| \leq 1$, $|\tilde{A}|(\tilde{q}) = 1$, $\int_{\tilde{\Sigma}} |\tilde{A}|^2 \leq \Lambda$, Dirichlet stable in the interior, $\partial\tilde{\Sigma} \subset L$, and $\|\tilde{\Sigma}\|(B_R(\cdot)) \leq 4\Theta^* \pi R^2$ for all centres and radii.*

Proof. Three inputs.

(a) *Curvature bounded uniformly including the boundary:* m_j is the maximum over all of Σ_j , boundary included — the maximum exists by (H-sm) — so $|\tilde{A}_j| \leq 1$ on all of $\tilde{\Sigma}_j$ with $|\tilde{A}_j|(\tilde{q}_j) = 1$. This is the pivot of the whole argument: the usual difficulty of boundary blow-up analysis, uncontrolled curvature at the boundary, does not arise here by construction.

(b) *Local area uniformly bounded, including at interior centres:* for $x \in \tilde{\Sigma}_j$ put $\sigma := d(x, \tilde{\ell}_j)$ with nearest point $\tilde{p}$; then $B_\sigma(x) \subset B_{2\sigma}(\tilde{p})$ gives $\|\tilde{\Sigma}_j\|(B_\sigma(x))/(\pi\sigma^2) \leq 4(1 + 2c_M\sigma/m_j)\Theta^*$, and interior monotonicity propagates this to all $R \leq \sigma$; for $R > \sigma$ use $B_R(x) \subset B_{2R}(\tilde{p})$. For large j the factor tends to 1, so at every centre and every fixed scale the bound is $4\Theta^* + o(1)$.

(c) *The boundary curve converges uniformly to a straight line:* ℓ is a C^∞ closed curve with $|\nabla_T T| = 1$, so $\tilde{\ell}_j$ has curvature $1/m_j \rightarrow 0$, and likewise for all higher derivatives; hence $\tilde{\ell}_j \rightarrow L$ in C_{loc}^∞ , the line through the origin with direction $\lim T(p_j)$. As ∂M is totally umbilic with $\text{II} = g$, after rescaling $\text{II} \rightarrow 0$ and $\partial M \rightarrow \partial \mathbb{H}$, so $L \subset \partial \mathbb{H}$.

Now: 1. *Uniform (half-)graph representation.* By (a) and the local graph lemma for minimal surfaces, at an interior point x (with $d(x, \tilde{\ell}_j) \geq c_1$) the set $\tilde{\Sigma}_j \cap B_{c_1}(x)$ is a $C^{1,1}$ graph over $T_x \tilde{\Sigma}_j$; near the boundary, adding the boundary data of (c), $\tilde{\Sigma}_j \cap B_{c_1}(x)$ is a graph over a half-disc in T_x whose straight edge is the graph of $\tilde{\ell}_j$, with uniform $C^{1,1}$ bounds. 2. *Finitely many sheets.* By (b), the number of sheets in $B_{c_1}(x)$ is at most $4\Theta^*c_1^2$ divided by the area lower bound for a single sheet, uniformly in j . 3. *Bootstrap and compactness.* Each sheet solves the minimal surface equation, so interior and boundary Schauder estimates — the boundary data $\tilde{\ell}_j$ being uniformly bounded in $C^{k,\alpha}$ and convergent — give uniform $C^{k,\alpha}$ bounds, and Arzelà–Ascoli with a diagonal argument gives a subsequence converging in $C_{\text{loc}}^\infty(\overline{\mathbb{H}})$.

Transfer of the properties: $|\tilde{A}| \leq 1$ and $|\tilde{A}|(\tilde{q}) = 1$ by C^2 convergence; $\int |\tilde{A}|^2 \leq \Lambda$ by Fatou and the scale invariance of Lemma B.4; $\partial \tilde{\Sigma} \subset L$ by (c); the area bound by (b). Transfer of stability: take $\phi \in C_c^\infty(\tilde{\Sigma}^\circ)$ with support K ; for large j , $\tilde{\Sigma}_j$ is a normal graph over $\tilde{\Sigma}$ near K (take one sheet if the multiplicity exceeds one), and transporting ϕ along the normal gives ϕ_j . The quadratic form is scale invariant in dimension two and the ambient Ricci term scales like $m_j^{-2} \rightarrow 0$; so Lemma B.6(a) gives $Q_j(\phi_j) \geq 0$, C^∞ convergence gives $Q_j(\phi_j) \rightarrow Q(\phi)$, and hence $Q(\phi) \geq 0$. Properness: quadratic area growth, $|\tilde{A}| \leq 1$ and embeddedness bound the number of components of $\tilde{\Sigma} \cap B_R$ uniformly; if $z \notin \tilde{\Sigma}$ were an accumulation point, there would be infinitely many graph pieces of diameter $\geq c_1$ near z and hence infinite area. So $\tilde{\Sigma}$ is a closed subset of $\overline{\mathbb{H}}$, properly embedded. $\square$

Proposition B.14 (Multiplicity one at the blow-up limit). *In the setting of Proposition B.13, let $\tilde{\Sigma}$ denote the component containing the base point $\tilde{q}$. Then, for the fixed subsequence, near each $x \in \tilde{\Sigma}^\circ$ exactly one graph piece of $\tilde{\Sigma}_j$ converges to $\tilde{\Sigma}$: the multiplicity is identically one and $\tilde{\Sigma}$ is embedded.*

Proof. Step 0 (no flat piece contained in $\partial \mathbb{H}$ and passing through L). x_3 is harmonic on a minimal surface and $x_3 \geq 0$ on $\tilde{\Sigma}$. Suppose a sheet S of the limit passes through $p \in L$ and is a full disc graph: then $x_3(p) = 0$ is an interior minimum, so $S \subset \partial \mathbb{H}$. The component S' containing S lies in $\partial \mathbb{H}$, is boundaryless — its boundary points could only come from the half-disc pieces along $\tilde{\ell}_j$, and those converge by step 1 to pieces transverse to $\partial \mathbb{H}$ — and is proper in $\overline{\mathbb{H}}$, so $S' = \partial \mathbb{H} \supset L$. But the base point component is transverse to $\partial \mathbb{H}$ at every point of L (Hopf boundary point lemma, B.2.2(ii)), and the corresponding half-disc of $\tilde{\Sigma}_j$ leaves $\tilde{\ell}_j$ with x_3 rising continuously from 0 to some $\delta > 0$; an almost flat interior piece of height $\varepsilon \ll \delta$ covering a whole disc near $\tilde{p}_j$ would have to meet that half-disc, contradicting embeddedness of $\tilde{\Sigma}_j$.

Step 1 (multiplicity one at the boundary). Let $p \in L \cap \partial \tilde{\Sigma}$ and $\tilde{p}_j \in \tilde{\ell}_j \rightarrow p$. Since $\tilde{\Sigma}_j$ is an embedded manifold with boundary, the arc of $\tilde{\ell}_j$ near $\tilde{p}_j$ is the boundary of exactly one half-disc piece; all other pieces near $\tilde{p}_j$ are interior full-disc pieces, which by step 0 can converge neither to anything through p nor to a full disc of the base point component near p (such a disc would pass through p and is excluded in the same way). So exactly one piece converges to $\tilde{\Sigma}$ near p .

Step 2 (the multiplicity is locally constant on $\tilde{\Sigma}^\circ$). For $x \in \tilde{\Sigma}^\circ$ and a small disc $D \ni x$ with $D \Subset \tilde{\Sigma}^\circ$, for large j the surface $\tilde{\Sigma}_j$ is, inside the ε -tubular neighbourhood of D , exactly $n(D)$ graphs over D ; when $D \cap D' \neq \emptyset$ the graphs on either side restrict to one another on $D \cap D'$ and extend uniquely by the uniform $C^{1,1}$ graph lemma, so $n(D) = n(D')$.

Step 3. $\tilde{\Sigma}$ is connected with boundary, so $\tilde{\Sigma}^\circ$ is connected; $n \equiv 1$ near L therefore gives $n \equiv 1$. $\square$

Remark. This proposition does not pass through varifolds; it uses only that a boundary arc of an embedded manifold with boundary belongs to a single half-disc, and Hopf transversality. The rescaled form of Theorem B.7(iii) ($\|\delta\tilde{V}\| \leq \mathcal{H}^1 \llcorner L$) gives the same conclusion, and the two routes agree. The argument for Theorem B.2 is insensitive to multiplicity — one takes a single sheet when transferring stability — so the present proposition only makes the word “embedded” in Proposition B.13 literally true.

B.2.2 Preliminary structure of the limit. (i) If $\partial\tilde{\Sigma} = \emptyset$: then $\tilde{\Sigma}$ is a complete, two-sided (properly embedded, Jordan–Brouwer), Dirichlet stable minimal surface in $\mathbb{R}^3$, so by T4 it is a plane, whence $|\tilde{A}| \equiv 0$, contradicting $|\tilde{A}|(\tilde{q}) = 1$. (ii) $x_3 \geq 0$ is harmonic on $\tilde{\Sigma}$ and vanishes on $\partial\tilde{\Sigma}$. By the strong maximum principle each component either has $x_3 \equiv 0$ (contained in $\partial\mathbb{H}$, flat, and not containing the base point) or has $x_3 > 0$ in its interior. Taking the base point component, $\tilde{\Sigma} \cap \partial\mathbb{H} = \partial\tilde{\Sigma} \subset L$, and the Hopf boundary point lemma makes $\tilde{\Sigma}$ transverse to $\partial\mathbb{H}$ along $\partial\tilde{\Sigma}$.

Lemma B.15 (The boundary is the whole line). *Let $\tilde{\Sigma}$ be the base point component given by Propositions B.13 and B.14. Then $\partial\tilde{\Sigma} \in \{\emptyset, L\}$; combined with B.2.2(i), $\partial\tilde{\Sigma} = L$.*

Proof. 1. $\tilde{\Sigma}$ is a C^∞ manifold with boundary (the half-disc graph representation of step 1 of Proposition B.13), so $\partial\tilde{\Sigma}$ is a boundaryless 1-manifold. 2. $\tilde{\Sigma}$ is proper in $\overline{\mathbb{H}}$, hence closed, so by B.2.2(ii) $\partial\tilde{\Sigma} = \tilde{\Sigma} \cap \partial\mathbb{H}$ is a closed subset of L . 3. A closed subset of $L \cong \mathbb{R}$ that is a boundaryless 1-manifold: let I be one of its components. If I had an endpoint q , then $q \in I$ would contradict boundarylessness and $q \notin I$ would contradict closedness. So $I = L$. $\square$

The whole content of Lemma B.15 lies in “ C^∞ up to the boundary + proper” from Proposition B.13; if that is weakened, this lemma fails with it, and the two cannot be used separately.

Proposition B.16 (The reflected surface $\hat{\Sigma}$). *Choose coordinates with L the x_1 -axis and $\partial\mathbb{H} = \{x_3 = 0\}$, and let $\rho_L = \text{diag}(1, -1, -1)$ be the rotation by π about L . Put $\hat{\Sigma} := \tilde{\Sigma} \cup \rho_L(\tilde{\Sigma})$. Then*

$\hat{\Sigma}$ is complete, connected, properly embedded, minimal, ρ_L -symmetric, non-planar ($|\tilde{A}|(\tilde{q}) = 1$);

$$\int_{\hat{\Sigma}} |A|^2 \leq 2\Lambda < \infty; \quad \hat{\Sigma} \cap \{x_3 = 0\} = L. \quad (\text{B.4})$$

In particular $\hat{\Sigma} \setminus L = \tilde{\Sigma}^\circ \sqcup \rho_L(\tilde{\Sigma}^\circ)$ has exactly two connected components, interchanged by ρ_L .

Proof. $\partial\tilde{\Sigma} = L$ (Lemma B.15), so by T6 (Schwarz reflection) $\hat{\Sigma}$ is smooth across L , complete, boundaryless and minimal. It is properly embedded because $\tilde{\Sigma} \subset \{x_3 \geq 0\}$, $\rho_L\tilde{\Sigma} \subset \{x_3 \leq 0\}$ and the intersection is L . The total curvature bound $\leq 2\Lambda$ is from Proposition B.13, and $\hat{\Sigma} \cap \{x_3 = 0\} = L$ from $\tilde{\Sigma}^\circ \subset \{x_3 > 0\}$ (B.2.2(ii)) together with $\partial\tilde{\Sigma} = L$. Connectedness of $\tilde{\Sigma}$ gives connectedness of $\tilde{\Sigma}^\circ$, hence exactly two components of $\hat{\Sigma} \setminus L$. $\square$

The last clause of (B.4) is the engine of everything that follows in this section. By T7, $\hat{\Sigma}$ is conformally $\overline{\Sigma}_\gamma \setminus \{p_1, \dots, p_k\}$ with $\overline{G}$ holomorphic and non-constant (non-planarity); by T9, each end is asymptotic to a plane or a catenoid with common asymptotic normal $\pm w$, the i -th end being a graph $\langle x, w \rangle = a_i \log |y| + b_i + O(|y|^{-1})$ over $w^\perp$.

Lemma B.18 (The end containing L). *The two rays of L lie in the same end p_* , and $w \perp e_1$ with $(a_*, b_*) = (0, 0)$: that end is asymptotic to the plane $w^\perp$ through the origin containing L .*

Proof. (1) Suppose the two rays tend to ends p_* and p_{**} . In $\bar{\Sigma}_\gamma$ the set $\bar{L} := L \cup \{p_*, p_{**}\}$ is an embedded arc (if $p_* \neq p_{**}$) or an embedded closed curve (if $p_* = p_{**}$). An embedded arc does not separate a closed surface, and neither does it after removing finitely many points; so $p_* \neq p_{**}$ would make $\hat{\Sigma} \setminus L$ connected, contradicting the “exactly two components” of Proposition B.16. Hence $p_* = p_{**}$.

(2) The end p_* is a graph over $w^\perp \setminus B_R$. A point te_1 of L projects to $t(e_1 - \langle e_1, w \rangle w)$ in $w^\perp$ with w -height $t\langle e_1, w \rangle$; substituting into the graph representation gives $t\langle e_1, w \rangle = a_* \log |y| + b_* + O(|y|^{-1})$ with $|y| \sim t$. A linear function equal to a logarithm plus a constant forces $\langle e_1, w \rangle = 0$, $a_* = 0$ and $b_* = 0$. $\square$

B.2.3 The hypotheses of Theorem B.2. All of the following are supplied by B.2.1–B.2.2 whenever blow-up occurs; they are listed explicitly so that Theorem B.2 may be stated independently of the blow-up setting. Let $\hat{\Sigma} \subset \mathbb{R}^3$, L the x_1 -axis, $\rho_L = \text{diag}(1, -1, -1)$ and $\tilde{\Sigma} := \hat{\Sigma} \cap \{x_3 \geq 0\}$.

- **(G0)** $\hat{\Sigma}$ is complete, connected, properly embedded, minimal and non-planar; N is its unit normal (two-sided, being properly embedded), $G = N$ the Gauss map and $G_1 := \langle N, e_1 \rangle$.
- **(G1)** $\int_{\hat{\Sigma}} |A|^2 < \infty$; so by T7, $\hat{\Sigma} \cong \bar{\Sigma}_\gamma \setminus \{p_1, \dots, p_k\}$ with $\bar{G}$ holomorphic and non-constant.
- **(G2)** $\rho_L(\hat{\Sigma}) = \hat{\Sigma}$, $\hat{\Sigma} \cap \{x_3 = 0\} = L$, and $\hat{\Sigma} \setminus L = \tilde{\Sigma}^\circ \sqcup \rho_L(\tilde{\Sigma}^\circ)$ has exactly two components, interchanged by ρ_L .
- **(G3)** Each end is asymptotic to a plane or a catenoid (first half of T9), *and the asymptotic normals are mutually parallel*, written $\pm w$, so that $\bar{G}(p_i) \in \{w, -w\}$ for all i (second half of T9, *not proved*).
- **(G4)** The two rays of L lie in the same end p_* , and $w \perp e_1$ (from (G2), (G3) via Lemma B.18).
- **(G5)** $k \geq 2$.

Theorem B.2 (Instability of the half-surface). *Assume (G0)–(G5). Then $\tilde{\Sigma}^\circ$ is Dirichlet unstable: there is $\psi \in C_c^\infty(\tilde{\Sigma}^\circ)$ with $\int_{\tilde{\Sigma}} (|\nabla \psi|^2 - |A|^2 \psi^2) < 0$.*

The proof runs through Lemmas B.19–B.25 and is assembled at the end.

Lemma B.19 (Analytic properties of ρ_L , and $\Omega_\pm$). *Assume (G0)–(G4).*

- (a) $\rho_L|_{\hat{\Sigma}}$ is an antiholomorphic involution of $\hat{\Sigma}$, and $N \circ \rho_L = -\rho_L \circ N$.
- (b) ρ_L extends to an antiholomorphic involution σ of $\bar{\Sigma}_\gamma$ whose fixed point set is exactly $\bar{L} := L \cup \{p_*\}$, a smooth embedded closed curve.
- (c) $\bar{\Sigma}_\gamma \setminus \bar{L} = \Omega_+^\circ \sqcup \Omega_-^\circ$ with $\Omega_+^\circ = \tilde{\Sigma}^\circ \cup \{p_i \neq p_* : \text{the end } E_i \subset \{x_3 > 0\}\}$ and $\sigma(\Omega_+) = \Omega_-$; and $\Omega_+ := \overline{\Omega_+^\circ}$ is a compact smooth surface with boundary $\partial\Omega_+ = \bar{L}$ and connected interior.
- (d) $k - 1$ is even, the punctures other than p_* being split evenly between $\Omega_\pm^\circ$; and under (G5), Ω_+° contains at least one puncture p_+ .

Proof. (a) ρ_L is an isometry of $\mathbb{R}^3$ preserving $\hat{\Sigma}$, hence an isometry of $\hat{\Sigma}$, hence conformal or anticonformal. At $x \in L$ one has $e_1 \in T_x \hat{\Sigma}$, and $d\rho_L$ fixes e_1 while reversing the sign of the direction of $T_x \hat{\Sigma}$ orthogonal to e_1 , so it is orientation-reversing; $\hat{\Sigma}$ is connected, so it is orientation-reversing throughout, hence antiholomorphic. An orientation-reversing isometry sends the normal to minus

the pushforward of the normal: $N(\rho_L x) = -\rho_L N(x)$ (on L , $N \in \text{span}(e_2, e_3)$ and $-\rho_L N = N$, consistently).

(b) ρ_L permutes the ends and is antiholomorphic and bounded on a punctured disc neighbourhood of each puncture, so Riemann's removable singularity theorem applied to the conjugate extends it across the punctures, giving the involution σ . In $\mathbb{R}^3$, ρ_L fixes only L , so $\text{Fix}(\sigma) \cap \hat{\Sigma} = L$. If $\sigma(p_i) = p_i$ for some $i \neq *$, then the end E_i , after removing a compact set, is connected and disjoint from L (by (G4) both rays of L lie in E_*), hence lies in one component of $\hat{\Sigma} \setminus L$, while ρ_L interchanges the two components (G2) — a contradiction. So $\text{Fix}(\sigma) = \bar{L}$. Smoothness: at $q \in \text{Fix}(\sigma)$ take a coordinate z with $\sigma(z) = \bar{h}(z)$, h holomorphic, $h(0) = 0$; from $\sigma^2 = \text{id}$, $|h'(0)| = 1$, and after a rotation $h'(0) = 1$; then $\zeta := z + h(z)$ is a local coordinate ($\zeta'(0) = 2$) with $\zeta \circ \sigma = \bar{\zeta}$, so the fixed point set is locally the real axis. That $\bar{L}$ is a closed curve is Lemma B.18(1).

(c) By (G2), $\hat{\Sigma} \setminus L$ has exactly two components; adding back finitely many punctures does not change connectedness, so $\bar{\Sigma}_\gamma \setminus \bar{L}$ has exactly two components; and $\bar{L}$ being a smooth embedded closed curve, the closures on either side are smooth surfaces with boundary, interchanged by σ .

(d) σ acts as an involution on $\{p_i\}$ with the single fixed point p_* , so $k - 1$ is even and the non-fixed punctures are interchanged in pairs, one of each pair in Ω_+° and one in Ω_-° . $\square$

Remark. Part (d) rules out $k = 2$ directly. If $k = 1$, then $\hat{\Sigma}$ has only the planar-type end p_* (Lemma B.18, $a_* = 0$), so the coordinate function $\langle x, w \rangle$ is bounded at that end and bounded on the compact part, hence a bounded harmonic function on $\hat{\Sigma}$; it extends to the compact surface $\bar{\Sigma}_\gamma$ and is therefore constant, making $\hat{\Sigma}$ planar and contradicting (G0). So in the blow-up setting (G5) holds automatically (Corollary B.26).

Lemma B.20 (Conformal change of metric). *Fix on $\bar{\Sigma}_\gamma$ a smooth metric g_0 compatible with the complex structure, with ∇, Δ_0, dA_0 its gradient, Laplacian and area element, and write $\bar{G}^* g_{S^2} = \lambda_0^2 g_0$ with $\lambda_0 \geq 0$ smooth, vanishing exactly at the finitely many branch points of $\bar{G}$. Then*

(a) *for Lipschitz ϕ compactly supported in $\hat{\Sigma}^\circ$,*

$$Q(\phi) := \int_{\hat{\Sigma}} (|\nabla_g \phi|^2 - |A|^2 \phi^2) dA_g = \int_{\Omega_+} (|\nabla \phi|^2 - 2\lambda_0^2 \phi^2) dA_0; \quad (\text{B.5})$$

(b) *globally on $\bar{\Sigma}_\gamma$, $\Delta_0 G_1 = -2\lambda_0^2 G_1$, where $G_1 = x_1 \circ \bar{G} \in C^\infty(\bar{\Sigma}_\gamma)$;*

(c) $G_1|_{\bar{L}} = 0$.

Proof. (a) The Dirichlet integral is conformally invariant in dimension two. For the curvature term: on a minimal surface $|A|^2 = |dN|^2$, and N is anticonformal into S^2 , so if $N^* g_{S^2} = \mu^2 g$ then $|dN|^2 = 2\mu^2$ and $|A|^2 dA_g = 2N^* dA_{S^2} = 2\lambda_0^2 dA_0$, the pullback of the area form being independent of the choice of conformal metric.

(b) $\Delta_{S^2} x_1 = -2x_1$; in dimension two $\Delta_h f dA_h = d^* df$ is independent of the conformal metric, and a holomorphic map commutes with $*$ on 1-forms, so $d^* dG_1 = \bar{G}^*(d^* dx_1) = -2G_1 \bar{G}^* dA_{S^2} = -2\lambda_0^2 G_1 dA_0$, and at the branch points by continuity.

(c) $L \subset \hat{\Sigma}$ gives $e_1 \in T\hat{\Sigma}$ along L , hence $G_1 = 0$ on L ; and $\bar{G}(p_*) = \pm w \perp e_1$ by (G3), (G4). $\square$

Lemma B.22 (G_1 changes sign in Ω_+° and has a regular zero). *Assume (G0)–(G5). Then $\{G_1 > 0\} \cap \Omega_+^\circ$ and $\{G_1 < 0\} \cap \Omega_+^\circ$ are both non-empty, and there is $q \in \Omega_+^\circ$, neither a puncture nor a branch point of $\bar{G}$, with $G_1(q) = 0$ and $\nabla G_1(q) \neq 0$.*

Proof. Sign change. Suppose $G_1 \geq 0$ on Ω_+ . By Lemma B.20(b), $\Delta_0 G_1 \leq 0$, so G_1 is a non-negative superharmonic function on (Ω_+°, g_0) . By Lemma B.19(d), Ω_+° contains a puncture p_+ ,

and $G_1(p_+) = \langle \pm w, e_1 \rangle = 0$ by (G3), (G4): an interior minimum 0. As Ω_+° is connected, the strong minimum principle gives $G_1 \equiv 0$ on Ω_+° , so $\overline{G}(\Omega_+^\circ)$ lies in the great circle $\{x_1 = 0\} \cap S^2$; a non-constant holomorphic map is open and its image cannot lie in a curve, so $\overline{G}$ is constant on Ω_+° , hence constant by the identity theorem, hence $\hat{\Sigma}$ is planar, contradicting (G0).

Regular zero. $Z := \{G_1 = 0\} \cap \Omega_+^\circ$ separates the connected surface Ω_+° , so Z is infinite (removing finitely many points does not disconnect). At $q \in Z$, $\nabla G_1(q) = d(x_1)_{\overline{G}(q)} \circ d\overline{G}_q$; since $\overline{G}(q) \in \{x_1 = 0\}$ it is $\neq \pm e_1$, so $d(x_1)|_{T_{\overline{G}(q)} S^2} \neq 0$, whence $\nabla G_1(q) = 0$ iff q is a branch point. Branch points and punctures together are finitely many, so choose $q \in Z$ avoiding them. $\square$

Lemma B.23 (First-order perturbation). *Assume (G0)–(G5). There is a Lipschitz function $\phi \geq 0$ on Ω_+ with $\phi|_{\partial\Omega_+} = 0$ and $\int_{\Omega_+} (|\nabla\phi|^2 - 2\lambda_0^2\phi^2) dA_0 < 0$. (ϕ is not required to vanish near the punctures.)*

Proof. Step 1. $G_1^+ := \max(G_1, 0)$ is Lipschitz on the compact surface Ω_+ , with $G_1^+|_{\bar{L}} = 0$ (Lemma B.20(c)) and $\nabla G_1^+ = \mathbf{1}_{\{G_1 > 0\}} \nabla G_1$ almost everywhere. Applying the divergence theorem to the Lipschitz field $G_1^+ \nabla G_1$ on the smooth compact surface with boundary (Ω_+, g_0) ,

$$\int_{\Omega_+} \langle \nabla G_1, \nabla G_1^+ \rangle = \int_{\partial\Omega_+} G_1^+ \partial_\nu G_1 - \int_{\Omega_+} G_1^+ \Delta_0 G_1 = 2 \int_{\Omega_+} \lambda_0^2 G_1 G_1^+,$$

that is $N_0 := \int |\nabla G_1^+|^2 = 2M_0$ with $M_0 := \int \lambda_0^2 (G_1^+)^2 > 0$, since $\{G_1 > 0\} \cap \Omega_+^\circ$ is a non-empty open set and λ_0 vanishes only at finitely many points.

Step 2. Take q from Lemma B.22 and a coordinate disc $U \Subset \Omega_+^\circ \setminus \{\text{punctures}\}$ on which $\nabla G_1 \neq 0$. By the implicit function theorem $\Gamma := \{G_1 = 0\} \cap U$ is a smooth embedded arc dividing U into $U^\pm = \{\pm G_1 > 0\} \cap U$. Take $\chi \in C_c^\infty(U)$ with $\chi \geq 0$ and $\chi(q) > 0$, and apply the divergence theorem to $\chi \nabla G_1$ on U^+ :

$$\int_{U^+} \langle \nabla G_1, \nabla \chi \rangle = \int_\Gamma \chi \partial_\nu G_1 - \int_{U^+} \chi \Delta_0 G_1 = -\delta + 2 \int_{U^+} \lambda_0^2 G_1 \chi,$$

with $\delta := \int_\Gamma \chi |\nabla G_1| ds > 0$ (ν being the outward normal of U^+ along Γ , pointing where G_1 decreases, so that $\partial_\nu G_1 = -|\nabla G_1|$).

Step 3. Put $\phi_\varepsilon := G_1^+ + \varepsilon \chi$, Lipschitz, ≥ 0 , vanishing on $\bar{L}$. Expanding, using $\text{supp } \chi \subset U$, $\int \langle \nabla G_1^+, \nabla \chi \rangle = \int_{U^+} \langle \nabla G_1, \nabla \chi \rangle$ and $\int \lambda_0^2 G_1^+ \chi = \int_{U^+} \lambda_0^2 G_1 \chi$:

$$\begin{aligned} \int (|\nabla \phi_\varepsilon|^2 - 2\lambda_0^2 \phi_\varepsilon^2) &= (N_0 - 2M_0) + 2\varepsilon \left[\int_{U^+} \langle \nabla G_1, \nabla \chi \rangle - 2 \int_{U^+} \lambda_0^2 G_1 \chi \right] + \varepsilon^2 \int (|\nabla \chi|^2 - 2\lambda_0^2 \chi^2) \\ &= 0 - 2\varepsilon \delta + O(\varepsilon^2) < 0 \end{aligned} \tag{B.6}$$

for all sufficiently small $\varepsilon > 0$. $\square$

Remark. The mechanism: the Rayleigh quotient of G_1^+ is exactly 2 — it is “half” an eigenfunction on Ω_+ with zero boundary value — and a first-order perturbation at one of its regular zeros strictly lowers the quotient. This is what “ $\lambda_1 < 2$ ” means here.

Lemma B.25 (Back to $C_c^\infty(\tilde{\Sigma}^\circ)$). *Let ϕ be the function of Lemma B.23. Then there is $\psi \in C_c^\infty(\tilde{\Sigma}^\circ)$ with $Q(\psi) < 0$.*

Proof. Write $I(f) := \int_{\Omega_+} (|\nabla f|^2 - 2\lambda_0^2 f^2)$, so $I(\phi) = -c < 0$.

(i) *Cutting off at the punctures.* At each puncture in Ω_+° take a coordinate disc $\{|z| < 1\}$ and the logarithmic cut-off $\eta_r = 0$ for $|z| \leq r^2$, $\eta_r = \log(|z|/r^2)/\log(1/r)$ for $r^2 \leq |z| \leq r$, and $\eta_r = 1$

for $|z| \geq r$, so that $\int |\nabla \eta_r|^2 = 2\pi/\log(1/r) \rightarrow 0$ (the capacity of a point vanishes in dimension two, conformally invariantly). Let η be the product of these cut-offs; then $\phi\eta$ is Lipschitz, vanishes near each puncture and on $\bar{L}$, and $\int |\nabla(\phi\eta)|^2 \leq \int \eta^2 |\nabla \phi|^2 + 2\|\phi\|_\infty \|\nabla \phi\|_{L^2} \|\nabla \eta\|_{L^2} + \|\phi\|_\infty^2 \|\nabla \eta\|_{L^2}^2 \rightarrow \int |\nabla \phi|^2$, while $\int \lambda_0^2 (\phi\eta)^2 \rightarrow \int \lambda_0^2 \phi^2$ by dominated convergence. For small r , $I(\phi\eta) < -c/2$.

(ii) *Moving off $\bar{L}$.* $f := \phi\eta \geq 0$ is continuous and vanishes on $\bar{L}$. The support $\{f \geq \tau\}$ of $f_\tau := (f - \tau)^+$ is closed and disjoint from $\bar{L}$ and from the puncture neighbourhoods, hence a compact subset of $\tilde{\Sigma}^\circ$. Also $\nabla f_\tau = \mathbf{1}_{\{f > \tau\}} \nabla f \rightarrow \nabla f$ almost everywhere (a Lipschitz function has vanishing gradient almost everywhere on $\{f = 0\}$) and $f_\tau \rightarrow f$ uniformly, so $I(f_\tau) \rightarrow I(f)$; choose τ small with $I(f_\tau) < -c/4$.

(iii) *Smoothing.* Mollify in finitely many coordinate charts and patch with a partition of unity to get $\psi \in C_c^\infty(\tilde{\Sigma}^\circ)$ with $\psi \rightarrow f_\tau$ in $W^{1,2}$; I is continuous on $W^{1,2}$ (λ_0 being bounded), so $I(\psi) < 0$, and by (B.5), $Q(\psi) = I(\psi) < 0$. $\square$

Proof of Theorem B.2. (G0)–(G4) give Lemmas B.19 and B.20; adding (G5) gives Lemma B.22, hence Lemma B.23, hence the function ψ of Lemma B.25. $\square$

Corollary B.26 (Boundary curvature blow-up does not occur). *Assume (H-sm) together with T3, T4, T7 and T9. Then for each (a, h) there is $C_2 = C_2(M, g, \ell, a, h)$ with $\max_\Sigma |A| \leq C_2$ for all $\Sigma \in \mathcal{M}_a^\ell(h)$; hence the curvature hypothesis of Theorem B.1 holds and (G-b) holds.*

Proof. Suppose not: there are $\Sigma_j \in \mathcal{M}_a^\ell(h)$ with $m_j \rightarrow \infty$. Then B.2.1–B.2.2, Propositions B.13–B.16 and Lemma B.18 produce $\tilde{\Sigma}$ satisfying (G0) (non-planar, since $|A|(\tilde{q}) = 1$), (G1) ($\leq 2\Lambda$), (G2) ((B.4)), (G3) (T9) and (G4) (Lemma B.18); and $k = 1$ is excluded by the remark after Lemma B.19, giving (G5). Theorem B.2 then makes $\tilde{\Sigma}^\circ$ Dirichlet unstable, contradicting the transfer of stability in Proposition B.13. $\square$

Corollary B.26 uses no density threshold, and neither Schoen’s two-ended theorem, nor flux balancing, nor the Hoffman–Meeks half-space theorem, nor the Fischer–Colbrie–Schoen criterion via positive Jacobi fields.

(G-b) implies (E)

B.3.1 Notation (main text §5). F is the fixed surface of the main text with $\partial F = \ell$; for $n \in \mathbb{Z}$, *class n* means the isotopy class of $\tau_\mu^n(F)$ rel ∂F with the boundary fixed pointwise — this is the meaning of “class” in Theorem A.1, following the convention of the source of 6.12: during the isotopy the boundary curve does not move at all, rather than merely being required to stay in ∂M or in J . Put $I^{(n)} := \inf\{\text{Area}(\Sigma) : \Sigma \in \text{class } n\}$ and $I_{\text{set}} := \inf_n I^{(n)}$. **(X₁)**: for each $a > 0$ the set $\{n \in \mathbb{Z} : I^{(n)} \leq a\}$ is finite. **(E)**: I_{set} is attained by a minimiser of some class (that is, $A(v)$ of §5 is attained).

Theorem B.3 ((G-b) $\Rightarrow$ (X₁) $\Rightarrow$ (E)). *Assume:*

- (1) **(H-sm)** for $\mathcal{M}_a^\ell(h)$ for every $a > 0$, with $h = g(F)$ — supplied by Theorem A.1: each class n contains a minimiser Σ_n of area $I^{(n)}$, smooth up to the boundary; the reading of (R1) this presupposes is H5 of §12.2.2;
- (2) **(G-b)** (Theorem B.1 with Corollary B.26, conditional on T3, T4, T7, T9);
- (3) Lemma 4.1 of the main text: a proper isotopy with pointwise fixed boundary upgrades to an ambient isotopy.

Then (X₁) holds, I_{set} is attained, and (E) follows. No stability hypothesis is required.

Proof. Suppose there were $a > 0$ and pairwise distinct $n_1, n_2, \dots$ with $I^{(n_j)} \leq a$. By (1) each class n_j contains a minimiser Σ_j of area $I^{(n_j)} \leq a$, smooth up to the boundary. Since $\Sigma_j \cong \tau_\mu^{n_j}(F) \cong F$, each Σ_j is connected, orientable, of genus h , with a single boundary component; $\partial\Sigma_j = \ell$ pointwise; $\text{Area}(\Sigma_j) \leq a$; and Σ_j is of least area in class n_j , that is in its own isotopy class rel ℓ with the boundary fixed pointwise (convention (α)). So $\{\Sigma_j\} \subset \mathcal{M}_a^\ell(h)$ and (H-sm) holds.

By (2), pass to a subsequence with $\Sigma_j \rightarrow \Sigma_\infty$ in C^∞ up to the boundary, with multiplicity one and embedded, and $\partial\Sigma_\infty = \ell$. Take a tubular neighbourhood of the normal bundle of Σ_∞ ; for large j , $\Sigma_j = \text{graph}(u_j)$ with $u_j \in C^\infty(\Sigma_\infty)$, $\|u_j\|_{C^1} \rightarrow 0$ and $u_j|_\ell = 0$. For two large indices $j \neq j'$, the linear interpolation $u_s := (1-s)u_j + su_{j'}$ gives a family of normal graphs: all inside the tubular neighbourhood hence embedded, all with $u_s|_\ell = 0$ hence with boundary identically ℓ pointwise, varying smoothly in s . This is a proper isotopy with pointwise fixed boundary carrying Σ_j to $\Sigma_{j'}$. By (3) it upgrades to an ambient isotopy rel ℓ , so Σ_j and $\Sigma_{j'}$ lie in the same class and $n_j = n_{j'}$, contradicting pairwise distinctness. Hence (X₁).

Then $\{n : I^{(n)} \leq I_{\text{set}} + 1\}$ is finite and non-empty, so $I_{\text{set}} = \min_n I^{(n)}$ is attained at some n_0 , and by (1) the minimiser of class n_0 realises it: this is (E). $\square$

Remark (Alignment of the isotopy conventions). Everything the (E) side uses from (G-b) is this: within the subfamily with ℓ fixed, of fixed topological type, with $\partial\Sigma_j \equiv \ell$ pointwise, of area $\leq a$, minimising in the pointwise class, a subsequence converges C^1 up to the boundary, with multiplicity one, embedded, as normal graphs with $u_j|_\ell = 0$. Theorem B.1 gives exactly this (and more, namely C^∞). Every defining clause of $\mathcal{M}_a^\ell(h)$ is supplied by the minimiser of some class n_j ; “minimal in the isotopy class with pointwise fixed boundary” and “minimal in class n ” are the same sentence, provided the “class” of 6.12' is indeed the isotopy class rel ∂F with the boundary fixed pointwise — which is the convention of the source of 6.12 and is inherited by Theorem A.1. An earlier draft stated (G-b) in the broader form “with boundary lying in J ”; the proof uses only the narrow form with ℓ fixed, and Theorem B.1 proves exactly the narrow form, so there is no shortfall.

Remark ((G-a) degenerates to (H-sm)). An earlier draft required “(G-a): the minimiser of each class may be taken to be a smooth *stable* minimal surface”. This appendix uses only Dirichlet stability, and *derives* it from “least area in the class” in Lemma B.6(a) rather than assuming it; what remains of (G-a) is the clause “smooth up to the boundary”, which is (H-sm). So (G-a) degenerates here to the single sentence (H-sm), supplied by Theorem A.1.

What this appendix is still conditional on; index of call sites

B.4.1 Still conditions (matching the table of §12.2.2 of the main text row by row).

#	Condition	Used in	Nature
H3-1	T3 , Schoen’s curvature estimate for stable minimal surfaces	Lemma B.6(b),(c)	research-level; only the interior version is used, not one up to the boundary
H3-2	T7 , Osserman’s finite total curvature structure theorem	after Proposition B.16; (G1)	monograph-level; recorded as a research-level dependence
H3-3	T9 , Schoen’s asymptotics of the ends	(G3), Lemmas B.18, B.19(b), B.22	research-level; the clause “the asymptotic normals of the ends are parallel” is not reproved , and is the one research-level input of the (G-b) chain not proved within this manuscript

#	Condition	Used in	Nature
H3-4	T4 , do Carmo–Peng / Fischer–Colbrie–Schoen / Pogorelov	B.2.2(i), excluding $\partial\tilde{\Sigma} = \emptyset$	research-level; absent from the earlier plan’s list and recorded here
H5	(H-sm) : each Σ in the family is individually C^2 up to the boundary	Lemma B.4, Theorem B.7, Proposition B.13(a), Theorem B.3(1)	supplied by Theorem A.1 (Appendix A) with Theorem C.3 (Appendix C); the reading of (R1) is H5 of §12.2.2, adopted in this manuscript
H14	numerical corrections: the constant $\bar{K}_{\max}^+$ in Lemma B.4; the factor $1 + c_M R/m_j$ in the finite- j form (B.3)	Lemmas B.4, B.10	editorial corrections; the conclusions are unchanged
—	Lemma 4.1 of the main text (proper isotopy upgrades to ambient isotopy)	Theorem B.3(3)	proved in the main text
—	the metric facts (a),(b),(c) of §2.1 of the main text	B.0.1	proved in the main text; $\mu = 1$, $c_0 = 1$

No longer in the chain (relative to earlier drafts): the density threshold $\Theta^* < 3/2$; Schoen’s two-ended theorem; flux balancing; López–Ros; the Hoffman–Meeks half-space theorem; the Jorge–Meeks area growth formula; the Fischer–Colbrie–Schoen criterion via positive Jacobi fields; Allard regularity; and the stability clause of (G-a).

B.4.2 The final list of conditions for Theorem B.3. (i) (H-sm) (from Theorem A.1 with Theorem C.3, under the H5 reading); (ii) T3; (iii) T4; (iv) T7; (v) T9 (the “parallel” clause not reproved); (vi) Lemma 4.1 of the main text. Of these, (ii)–(v) are research-level named theorems, (i) is an internal chain of this manuscript, and (vi) is proved in the main text. No stability hypothesis, no density hypothesis.

B.4.3 Index of call sites in the main text.

Call site	What is used	Supplied by
§5, attainment of $A(v)$ (that is, (E))	$(X_1) \Rightarrow I_{\text{set}}$ attained	Theorem B.3
§7.1, existence of minimisers (choice of representatives)	$\mathcal{M}_a^\ell(h)$ non-empty and sequentially compact; the limit is still a minimiser of its class	Theorem B.1 with Corollary B.26; (H-sm) from Theorem A.1
§12.2.2, rows H3, H5, H14	provenance and nature of the conditions	§B.4.1
Theorem C.3	wherever a uniform curvature bound for the family of minimisers is needed	Corollary B.26

C Optimal leaves and the boundary regularity of the constructed minimiser

This appendix belongs to the same proof family as the main text and Appendices A and B: the notation and the labels of the hypotheses ((E), (R1), (E-fix), (H ∂), Δ_1/Δ_2 , G1–G3) agree with the

main text, and results of Appendices A and B are cited by their fixed numbers (Theorem A.1 = 6.12', Proposition A.2 = (R), Theorems B.1–B.3) without their proofs being repeated. Nothing mathematical is altered here.

On provenance: this appendix was produced within one model family and audited only within that family; it has received no cross-engine audit (criterion ③ counts 0/3). See Appendix G.

C.0 Setting, criteria, and named theorems

C.0.1 Ambient setting. $M := E(K)$ is compact, orientable and irreducible with $\partial M \cong T^2$. The metric g is smooth and, in a collar $U_\varepsilon = \{0 \leq r < \varepsilon\} \cong \partial M \times [0, \varepsilon)$ (r inward, $\partial M = \{r = 0\}$),

$$g = dr^2 + \varphi(r)^2 g_{\partial M}, \quad \varphi(0) = 1, \quad \varphi'(0) < 0. \quad (\text{C.13})$$

With respect to the outward normal $n = -\partial_r$ one has $\Pi(X, X) = -(\varphi'/\varphi)|X|^2 > 0$: ∂M is *strictly convex* (the sign convention of Remark (i) of §5 of the main text). J is a real-analytic foliation of ∂M by closed curves (preferably longitudes), with leaves L_ℓ ; $y : \partial M \rightarrow \mathbb{R}/\mathbb{Z}$ is a real-analytic submersion defining it ($J = \{y = \text{const}\}$, $dy \neq 0$), and $W := \nabla^{\partial M} y / |\nabla^{\partial M} y|^2$ is the transverse field ($dy(W) \equiv 1$, $W \perp$ the leaves).

Let $F \subset M$ be a neatly embedded surface; a *neat isotopy* is a smooth family of neat embeddings. The competitor classes are

$$\mathcal{C} := \{F' \subset M : F' \text{ neatly isotopic to } F, \partial F' \subset \text{some leaf}\}, \quad \mathcal{C}_\ell := \{F' \in \mathcal{C} : \partial F' \subset L_\ell\}. \quad (\text{C.14})$$

Remark (Observation C.0.1: the boundary constraint is rigid). $\partial F'$ is a compact boundaryless 1-manifold embedded in the circle L_ℓ , so $\partial F' = \emptyset$ or $\partial F' = L_\ell$; in particular $\partial F'$ is connected, and the subproblem with ℓ fixed is a Plateau-type problem with the boundary curve *pinned*, not a free boundary problem.

C.0.2 Three variational classes (fixed for the whole appendix). From smallest to largest: (i) fixed boundary curve (rel ∂F pointwise; the class $\mathcal{F}$ of (A-cls) in Theorem A.1); (ii) the *leaf-constrained class* $\mathcal{C}$ — the isotopy may move the boundary, the terminal surface having its boundary on some leaf (the class over which (E) takes its infimum, “the boundary longitude is allowed to move on the boundary torus” of §5 of the main text); (iii) the boundary slides freely on ∂M . Stability is correspondingly ordered: (iii)-stable $\Rightarrow$ (ii)-stable $\Rightarrow$ (i)-stable. Class (iii) is incompatible with the constraint — its first-order condition is $\beta \equiv 0$ pointwise, whereas (ii) yields only the single integral condition (C.24) — and is not used here. The word “stable” in Appendix B means Dirichlet stability of class (i) (B.0.3).

C.0.3 The criterion (E) and the three things taken from it. (E) of the main text: $\inf_{\mathcal{C}} \text{Area}$ is attained by a smooth surface in $\mathcal{C}$. Wherever “by (E)” is invoked, this appendix uses only:

- **(E1)** $\Sigma \subset M$ is a neatly embedded compact surface with boundary (topologically), $\partial \Sigma = L$ is a leaf of J , and $\mathcal{H}^2(\Sigma) < \infty$;
- **(E2)** the interior $\Sigma^\circ := \Sigma \setminus \partial \Sigma$ is a C^∞ embedded surface;
- **(E3)** $\mathcal{H}^2(\Sigma) = \inf\{\mathcal{H}^2(S) : S \text{ neatly isotopic to } \Sigma, \partial S \text{ a leaf of } J\}$ — minimality in class (ii).

No form of boundary regularity is taken from (E): (E) says only that the infimum is attained, and *records nothing about how* it is attained.

C.0.4 The criterion (R1). (R1) of §12.2 of the main text has two halves (the main text reads “the minimiser is $C^{1,1}$ up to $\partial E(K)$ ”):

- **(R1-a) neatness:** $\Sigma \cap \partial E(K) = \partial \Sigma$;
- **(R1-b) boundary regularity:** Σ is $C^{1,1}$ up to the boundary near $\partial \Sigma$.

What is proved here is the stronger C^∞ up to the boundary ($\Rightarrow C^{1,1}$), together with transversality $F' \pitchfork \partial M$ ((R2) of the main text, the conclusion of Lemma 5.3).

C.0.5 The standing hypothesis (H ∂). If ∂F is disconnected then by Observation C.0.1 the boundary of any surface neatly isotopic to F is disconnected too, so $\mathcal{C} = \emptyset$ and (E) is literally false; if ∂F is connected but not isotopic in ∂M to a leaf (a meridian, a trivial curve), again $\mathcal{C} = \emptyset$. Hence throughout this appendix:

(H ∂) ∂F is connected and isotopic in ∂M to a leaf of J .

For the actual objects of the main text (Seifert surfaces, ∂F a preferred longitude; J the preferred-longitude foliation) (H ∂) holds automatically: two essential simple closed curves of the same slope on a torus are isotopic. *This is not a gap but a quantifier that the statement of (E) should carry.* Theorems C.1 and C.2 use only (H ∂), never the incompressibility of F .

C.0.6 Named theorems. No new research-level input enters the chain in this appendix; the single research-level dependence T-C is already on the list of Appendix A.

- **T1 (Reeb local stability; textbook-level).** Let $\mathcal{F}$ be a C^1 foliation of codimension q , L a compact leaf with finite holonomy. Then L has a saturated neighbourhood U and a retraction $p : U \rightarrow L$ such that (U, p) is a fibre bundle with q -disc fibres, every leaf in U being compact and meeting each fibre in finitely many points; *if the holonomy is trivial, U may be taken to be a foliated product $L \times D^q$ with leaves $L \times \{\text{pt}\}$.* Only the trivial-holonomy case is used.
- **T2 (Ehresmann fibration theorem; textbook-level).** A proper smooth surjective submersion between smooth manifolds onto a connected base is a locally trivial smooth fibre bundle.
- **T3 (isotopy extension, Palais–Cerf; Hirsch, *Differential Topology*, Ch. 8 Thm 1.3; textbook-level).** For N compact and $f : N \times [0, 1] \rightarrow X$ an isotopy there is a family of diffeomorphisms H_s with $H_0 = \text{id}_X$ and $H_s \circ f_0 = f_s$.
- **T4 (coarea formula, Federer; textbook-level).** For S rectifiable of dimension m and $u : S \rightarrow \mathbb{R}$ Lipschitz, $\int_S |\nabla^S u| d\mathcal{H}^m = \int_{\mathbb{R}} \mathcal{H}^{m-1}(S \cap u^{-1}(\tau)) d\tau$.
- **T5 (first variation of area with boundary term; textbook-level).** For Σ a compact C^2 immersed surface with boundary, X a C^1 field with flow ϕ_t , $\frac{d}{dt}\big|_0 \text{Area}(\phi_t \Sigma) = -\int_\Sigma \langle X, \vec{H} \rangle + \int_{\partial \Sigma} \langle X, \nu \rangle$, ν the outward unit conormal.
- **T-min (strong minimum principle for superharmonic functions; textbook-level).** A C^2 function with $\Delta u \leq 0$ on a connected Riemannian manifold attaining an interior minimum is constant.
- **T-Hopf (Hopf boundary point lemma; GT Lemma 3.4; textbook-level, original not checked).** Let $L = a^{ij} \partial_{ij} + b^i \partial_i$ be uniformly elliptic in Ω with locally bounded coefficients, $w \in C^2(\Omega)$, $Lw \leq 0$; let $x_0 \in \partial \Omega$ satisfy an interior ball condition, w continuous at x_0 and $w(x_0) < w(x)$ for all $x \in \Omega$. Then the outward normal derivative of w at x_0 , if it exists, is strictly negative.

- **T-Sch (local boundary Schauder regularity for linear equations; GT Lemma 6.18; textbook-level, original not checked).** Let $T \subset \partial\Omega$ be a relatively open $C^{k+2,\alpha}$ boundary portion, $L = a^{ij}\partial_{ij} + b^i\partial_i + c$ strictly elliptic with $a^{ij}, b^i, c, f \in C^{k,\alpha}(\Omega \cup T)$ and $\phi \in C^{k+2,\alpha}(\Omega \cup T)$. If $u \in C^0(\Omega \cup T) \cap C^2(\Omega)$ satisfies $Lu = f$ and $u|_T = \phi$, then $u \in C^{k+2,\alpha}(\Omega \cup T)$.
- **T-LU (Ladyzhenskaya–Ural'tseva boundary gradient Hölder estimate; monograph-level, original not checked).** For uniformly elliptic quasilinear equations in divergence form, a Lipschitz solution taking $C^{1,\alpha}$ boundary values on a $C^{1,\alpha}$ boundary portion is $C^{1,\gamma}$ near that portion, with γ depending only on the ellipticity constants, the gradient bound and the boundary data. Used only in Lemma C.7; Theorem C.3 does not use it.
- **T-Mor (Morrey's analyticity theorem; monograph-level, original not checked).** A C^∞ solution of a second-order analytic elliptic equation is real-analytic up to an analytic boundary portion carrying analytic boundary values. Used only in the real-analytic clause of Lemma C.7.
- **T-C (Hildebrandt boundary regularity for Plateau; research-level; item H2 of the list of Appendix A, original not checked).** A least-area disc spanning a $C^{m,\alpha}$ boundary curve is $C^{m,\alpha}$ up to the boundary at interior points of that curve ($m \geq 1$). It is *not invoked directly* here: it enters only through conclusion C3 of Theorem A.1 (Δ_2 , §C.3.3).

C.1 Theorem C.1: (E) is equivalent to solvability of a fixed-boundary problem on some optimal leaf

For $\ell \in \mathcal{L}$ (the leaf space, see below) put $A(\ell) := \inf\{\text{Area}(F') : F' \in \mathcal{C}_\ell\}$, and let $P(\ell)$ denote the proposition “ $A(\ell)$ is attained by a smooth surface in $\mathcal{C}_\ell$ ”. On an optimal leaf, P is the gap lemma (E-fix), discharged here by Theorem A.1 under the reading of §C.5.

Theorem C.1 (Reduction of (E) to a leaf parameter). *Assume $(H\partial)$. Then the leaf space $\mathcal{L} \cong S^1$ is compact, $A : \mathcal{L} \rightarrow (0, \infty)$ has positive upper and lower bounds and $\log A$ is Lipschitz, $\text{Argmin}(A)$ is non-empty and closed, and*

$$(E) \iff \exists \ell_* \in \text{Argmin}(A) \text{ such that } P(\ell_*) \text{ holds.} \quad (\text{C.15})$$

If $P(\ell)$ holds for every ℓ then (E) holds. Moreover any minimiser Σ of (E) lies in a unique $\mathcal{C}_{\ell_0}$, and $\ell_0 \in \text{Argmin}(A)$.

Proof. Four steps.

Step 1 (the leaf space is a circle). Let $\mathcal{F}$ be a C^∞ foliation of a closed surface with all leaves compact and transversely orientable. Take a leaf L and a transverse arc $\tau \ni x_0$; the holonomy is given by an order-preserving germ h attached to a generator of $\pi_1(L) \cong \mathbb{Z}$. For $x \in \tau$ near x_0 , $\{h^k(x)\}_{k \geq 0} \subset L_x \cap \tau$, a finite set (L_x compact and transverse to τ), so $h^k(x) = x$ for some $k \geq 1$; h increasing then forces $h(x) = x$. Thus every leaf has trivial holonomy, and by T1 has a saturated product neighbourhood $L \times (-\delta, \delta)$. These trivialisations make $\mathcal{L} := \Sigma/\mathcal{F}$ a Hausdorff smooth 1-manifold with π a proper smooth surjective submersion, hence a fibre bundle by T2; $\mathcal{L}$ is compact, connected and boundaryless, so $\mathcal{L} \cong S^1$. The foliation J satisfies the hypotheses (all leaves closed, obtained on T^2 by translating a linear longitudinal foliation, tangent and normal bundles trivial), giving $\pi : \partial M \rightarrow \mathcal{L} \cong \mathbb{R}/\mathbb{Z}$ with $L_\ell = \pi^{-1}(\ell)$.

Step 2 (the leaf-sliding flow: a family of ambient isotopies). Choose a smooth connection for π and let V be the horizontal lift of ∂_ℓ ($\pi_* V = \partial_\ell$). Take a cut-off $\chi \in C^\infty([0, \varepsilon])$ with $\chi \equiv 1$ on $[0, \varepsilon/3]$

and $\equiv 0$ on $[2\varepsilon/3, \varepsilon)$, and set $\tilde{V} := \chi(r)V$ inside the collar (parallel extension along the collar) and $\tilde{V} := 0$ outside. $\tilde{V}$ is smooth, tangent to ∂M , and complete (M compact); its flow $\Phi_t \in \text{Diff}(M)$ preserves ∂M . Three properties: (1) $\Phi_t(L_\ell) = L_{\ell+t}$ (on ∂M , $\frac{d}{dt}\pi \circ \Phi_t = d\pi(V) = 1$); (2) $\{\Phi_s\}_{s \in [0, t]}$ is an ambient isotopy, so for a neat embedding f the family $s \mapsto \Phi_s \circ f$ is a neat isotopy; (3) with $C := \sup_M \|\mathcal{L}_{\tilde{V}}g\|_g < \infty$, from $\frac{d}{dt}\Phi_t^*g = \Phi_t^*(\mathcal{L}_{\tilde{V}}g)$ and Grönwall, $e^{-C|t|}g \leq \Phi_t^*g \leq e^{C|t|}g$, so for every 2-rectifiable S

$$e^{-C|t|} \text{Area}(S) \leq \text{Area}(\Phi_t(S)) \leq e^{C|t|} \text{Area}(S). \quad (\text{C.16})$$

Step 3 (stratification, non-emptiness, two-sided bounds and Lipschitz dependence). By (H ∂) and T3, the curve isotopy in ∂M carrying ∂F to a leaf L_{ℓ_0} extends to a family of diffeomorphisms h_s of ∂M ; cutting off its generating field as in Step 2 gives a time-dependent field on M tangent to ∂M , whose integration is an ambient isotopy H_s with $H_1(F) \in \mathcal{C}_{\ell_0}$; then $\Phi_{\ell-\ell_0}$ carries it into $\mathcal{C}_\ell$. So every $\mathcal{C}_\ell \neq \emptyset$. By Observation C.0.1, ℓ is determined by F' , and $\Phi_t(\mathcal{C}_\ell) = \mathcal{C}_{\ell+t}$ (the reverse inclusion via Φ_{-t}), whence

$$\mathcal{C} = \bigsqcup_{\ell \in \mathcal{L}} \mathcal{C}_\ell, \quad \inf_{\mathcal{C}} \text{Area} = \inf_{\ell \in \mathcal{L}} A(\ell). \quad (\text{C.17})$$

A uniform positive lower bound. Take $\delta \in (0, \varepsilon/2)$ and $T_\rho := \{r = \rho\}$; the induced metric depends continuously on $\rho \in [0, \delta]$, so $\text{sys}(T_\rho) \geq 2s_0 > 0$ uniformly. For $S \in \mathcal{C}_\ell$, by Sard almost every ρ is a regular value of $r|_S$, so $S_\rho := S \cap T_\rho$ is an embedded closed 1-manifold with $[S_\rho] = [L_\ell] \neq 0$ in $H_1(T_\rho; \mathbb{Z}/2)$ ($S \cap \{r \leq \rho\}$ provides the homology, the collar deformation retracting); hence some component is non-contractible, of length $\geq 2s_0$. By T4 (with $|\nabla^S r| \leq 1$), $\text{Area}(S) \geq \int_0^\delta \mathcal{H}^1(S_\rho) d\rho \geq 2s_0\delta =: a_0$.

Two-sided comparison. By (C.16) and the bijection in (C.17), $e^{-C|t|}A(\ell) \leq A(\ell+t) \leq e^{C|t|}A(\ell)$; taking $|t| \leq 1$ gives $a_0 \leq A \leq e^C A(\ell_0) =: a_1 < \infty$ and

$$|\log A(\ell+t) - \log A(\ell)| \leq C|t|, \quad A : \mathcal{L} \rightarrow [a_0, a_1] \text{ Lipschitz}. \quad (\text{C.18})$$

Step 4 (the minimising leaf is attained; both directions). $\mathcal{L}$ is compact and A continuous, so there is ℓ_* with $A(\ell_*) = \min A = \inf_{\mathcal{C}} \text{Area}$, and $\text{Argmin}(A)$ is non-empty and closed.

($\Leftarrow$) Let $\ell \in \text{Argmin}(A)$ and $\Sigma \in \mathcal{C}_\ell$ be smooth with $\text{Area}(\Sigma) = A(\ell)$. Then $\text{Area}(\Sigma) = \min A = \inf_{\mathcal{C}} \text{Area}$ with $\Sigma \in \mathcal{C}$, which is (E).

($\Rightarrow$) Let $\Sigma \in \mathcal{C}$ be smooth with $\text{Area}(\Sigma) = \inf_{\mathcal{C}} \text{Area}$. By Observation C.0.1 there is a unique ℓ_0 with $\Sigma \in \mathcal{C}_{\ell_0}$, and $A(\ell_0) \leq \text{Area}(\Sigma) = \min A \leq A(\ell_0)$, so equality holds throughout: $\ell_0 \in \text{Argmin}(A)$ and $P(\ell_0)$ holds. The last sentence of the statement is a special case of ($\Leftarrow$). $\square$

Remark (C.1.1: compatibility of isotopy classes is not a gap). Φ_t is an ambient isotopy carrying $\mathcal{C}_\ell$ pointwise onto $\mathcal{C}_{\ell+t}$ without changing the neat isotopy class (always $[F]$); there is no possibility of “solutions for different ℓ lying in different classes”.

Remark (C.1.2: the stratification does not lower the difficulty). $P(\ell')$ for g is equivalent to $P(\ell)$ for Φ_t^*g , and the latter is again a smooth metric making ∂M strictly convex; so “prove P for one leaf” and “prove P for all leaves” do not differ in the usual shape of the theorem. What Theorem C.1 buys is the precise localisation of the gap at $P(\ell_*)$, not a discount in difficulty.

Remark (C.1.3: incompressibility is not used beyond (H ∂)). Steps 1–4 use only (H ∂); incompressibility of F enters only in the proof of $P(\ell)$ (through (A-F) of Theorem A.1).

C.2 Theorem C.2: leaf-sliding is eliminated and $\min_t A(t)$ is attained

Theorem C.2 (Elimination of leaf-sliding). *Assume (H ∂) (in the source split into two clauses: ∂F connected; ∂F isotopic in ∂M to a leaf). Then $A : \mathcal{L} \rightarrow [0, \infty)$ is continuous and bounded,*

its minimum is attained at some $t_0 \in \mathcal{L}$, and $A(t_0) = \inf_{\mathcal{C}} \text{Area}$. Consequently: any surface in $\mathcal{C}_{t_0}$ realising $A(t_0)$ simultaneously realises the class-(ii) infimum; and any area-minimising sequence in $\mathcal{C}_{t_0}$ is simultaneously area-minimising in $\mathcal{C}$.

Proof. The argument coincides with Steps 2–4 of Theorem C.1; the source route differs slightly, with the same conclusion, and the differences are recorded: (1) the source takes J to be a parallel foliation ($\partial M \cong \mathbb{R}^2/\Lambda$, leaves $\{y = t\}$) and the sliding field to be $W := \chi(r)\partial_y$, whose flow Φ_s satisfies $\Phi_s(\ell_t) = \ell_{t+s}$ literally; (2) the area distortion is obtained not by Grönwall but from the maximum $\Lambda(s)$ of the 2-dimensional Jacobian over the compact Grassmann bundle $G_2(M)$ (continuous, $\Lambda(0) = 1$) together with the area formula, giving $A(t+s) \leq \Lambda(s)A(t)$ and hence $|A(t+s) - A(t)| \leq (\max_{|\sigma| \leq |s|} \Lambda(\sigma) - 1)A_{\max}$, so that A is uniformly continuous; (3) the rest ($\mathcal{C} = \bigsqcup \mathcal{C}_t$, attainment of the minimum by compactness, transfer of minimising sequences) is verbatim Steps 3–4 of Theorem C.1. $\square$

Remark (C.2.1: audit — no fixed boundary has been smuggled in). The proof of Theorem C.2 uses only (a) compactness of the leaf space, (b) the existence of an ambient isotopy flow permuting the leaves exactly, and (c) a uniform Jacobian distortion bound for area on a compact manifold. Everything afterwards that is carried out with the boundary fixed at ℓ_{t_0} is *licensed* by Theorem C.2, not assumed. Theorems C.1 and C.2 differ only in packaging: C.1 gives the equivalence (C.15) and the quantitative Lipschitz bound (C.18), C.2 gives the transfer of minimising sequences.

C.3 Boundary regularity: auxiliary lemmas and Theorem C.3

C.3.1 Three auxiliary lemmas using only (E1), (E2).

Lemma C.5 (Neatness, (R1-a)). *Under (E1) and (E2) — interior minimality $\vec{H} \equiv 0$ coming from the compactly supported interior variations of (E3) — $\Sigma \cap \partial E(K) = \partial \Sigma$.*

Proof. Inside the collar, (C.13) gives $\text{Hess } r = (\varphi'/\varphi)h_r$ with $h_r = \varphi(r)^2 g_{\partial M}$, so $\text{Hess } r(\partial_r, \cdot) = 0$ and the horizontal directions are negative definite. Taking an orthonormal basis e_1, e_2 of $T_q \Sigma$,

$$\Delta_{\Sigma} r = \sum_i \text{Hess } r(e_i, e_i) + \langle \nabla r, \vec{H}_{\Sigma} \rangle = (\varphi'/\varphi) \sum_i |e_i^{\text{lev}}|^2 \leq 0, \quad (\text{C.19})$$

so $r|_{\Sigma^\circ}$ is superharmonic. If an interior point $q \in \Sigma^\circ$ lay on ∂M , then $r|_{\Sigma} \geq 0$ would attain the minimum 0 at an interior point, so by T-min $r \equiv 0$ on the component of q , which would then lie in the two-dimensional ∂M — contradicting that Σ is a neatly embedded surface in a 3-manifold. Since $\Sigma = \Sigma^\circ \cup \partial \Sigma$ and $\partial \Sigma = L \subset \partial M$, the claim follows. $\square$

This is the argument of Lemma 5.2 of the main text, repeated here for self-containment; it does *not* depend on (R1), only on interior C^2 regularity. The “neat up to the boundary” half of (R1) is therefore not an assumption, and the whole difficulty lies in the $C^{1,1}$ half.

Lemma C.6 (Boundary monotonicity and existence of the density). *Under (E1)–(E3) there are $\rho_3 > 0$ and $C < \infty$, depending only on g , ∂M and the C^2 bounds of L , such that for all $p \in L$ and $0 < \rho < \rho_3$*

$$\frac{d}{d\rho} \left[\frac{e^{C\rho} \|\Sigma\| (B_\rho(p))}{\rho^2} \right] \geq 0; \quad (\text{C.20})$$

in particular $\Theta(p) := \lim_{\rho \downarrow 0} \|\Sigma\| (B_\rho(p)) / (\pi \rho^2)$ exists and is finite, with $\Theta(p) \leq e^{C\rho_3} \mathcal{H}^2(\Sigma) / (\pi \rho_3^2)$ uniformly for $p \in L$.

Proof (skeleton). (a) The first variation vanishes for admissible fields (tangent to ∂M along ∂M and tangent to L along L), a direct consequence of (E3) under sliding variations. (b) The point is that the radial field $Z = \exp_p^{-1}$ can be corrected to an admissible $V = Z + O(|Z|^2)$ with $DV = \text{Id} + O(|Z|)$: subtract $\eta \tilde{h} \tilde{n}$ (with $\tilde{h} := \langle Z, \tilde{n} \rangle \circ \pi_{\partial M} = O(|Z|^2)$) to make it tangent to ∂M , then subtract $\eta_L \tilde{k} \tilde{\partial}_y$ (with $\tilde{k} = O(|Z|^2)$ and $\tilde{\partial}_y$ tangent to ∂M) to make it tangent to L along L . (c) Insert $X := \psi(|Z|/\rho)V$ into (a); expanding $\text{div}_\Sigma X$ is verbatim the interior monotonicity computation, the only new content being the admissibility of X . Letting $\psi \uparrow \chi_{[0,1]}$ gives $\frac{d}{d\rho}(I/\rho^2) \geq -CI/\rho^2 + \frac{2}{\rho^3} \int_{B_\rho} |Z^\perp|^2/|Z|^2$, which is (C.20). $\square$

Remark (C.6.1: what this does and does not give). Lemma C.6 gives weak subsequential convergence of boundary blow-ups to a stable minimal cone in a half-space (with current boundary $\llbracket T_p L \rrbracket$); it does *not* give that the cone is a half-plane, that the density is $\frac{1}{2}$, or any ε -regularity. It therefore remains separated from $C^{1,1}$ by the whole gap, and serves only as an auxiliary result: it does not enter the proof of Theorem C.3.

Lemma C.7 (Lipschitz graph implies C^∞ bootstrap). *Assume (E1)–(E3) and, at $p \in L$, the condition (Lip): there are coordinates at p ($g_{ij}(p) = \delta_{ij}$, $T_p L = \mathbb{R}e_1$) and $\rho > 0$ with $\Sigma \cap B_\rho(p) = \{(x, f(x)) : x \in \Omega\}$, where $f \in C^{0,1}(\bar{\Omega} \cap B'_\rho)$, the portion $\partial\Omega \cap B'_\rho = \Gamma_0$ is the projection of L (a C^∞ arc) and $f|_{\Gamma_0} = \ell \in C^\infty$. Then $f \in C^\infty(\bar{\Omega} \cap B'_{\rho/2})$; and if g and L are real-analytic near p , f is real-analytic up to Γ_0 . In particular Σ is $C^{1,1}$ (indeed $W^{2,p}$ for all $p < \infty$) up to the boundary near p .*

Proof (skeleton, five steps). (1) The area functional is $\mathcal{A}(f) = \int_\Omega F(x, f, Df)$ with $F = \sqrt{\det h_{\alpha\beta}}$, $h_{\alpha\beta} = g_{\alpha\beta} + g_{\alpha 3} q_\beta + g_{3\beta} q_\alpha + g_{33} q_\alpha q_\beta$; on $|q| \leq \Lambda := \text{Lip}(f)$ the matrix F_{qq} is uniformly elliptic, and f is a classical and weak solution of the Euler–Lagrange equation $\partial_\alpha(F_{q_\alpha}) - F_z = 0$. (2) T-LU gives $f \in C^{1,\gamma}$ up to Γ_0 . (3) Freezing coefficients turns the equation into the non-divergence form $a^{\alpha\beta} \partial_{\alpha\beta} f + b = 0$ with $a, b \in C^{0,\gamma}$, and T-Sch gives $C^{2,\gamma}$. (4) The tangential difference quotient w_h satisfies a linear equation with $C^{k-1,\gamma}$ coefficients; uniform Schauder estimates give $\partial_1 f \in C^{k,\gamma}$, the pure normal derivative is solved from the equation using $a^{22} \geq \lambda > 0$, and induction gives C^∞ . (5) T-Mor gives real-analyticity. $\square$

Remark (C.7.1). Step (2) uses only $\|Df\|_{L^\infty} < \infty$, so the real hypothesis of Lemma C.7 is “ Σ is, near p , a graph with bounded gradient”; with only a C^0 graph the minimal surface equation is not uniformly elliptic, T-LU fails, and this step cannot be omitted. Theorem C.3 takes a different input ($C^{1,\beta}$ up to the boundary, supplied by conclusion C3 of Theorem A.1) and depends on neither (Lip) nor T-LU.

C.3.2 The boundary structure of the surface constructed in Theorem A.1: Lemmas C.8–C.10. Below, F' is the minimiser constructed by Theorem A.1 (Appendix A) under its (A-M), (A- ∂), (A-F), (A-cl), (H-cov) and GP, and $\ell := \partial F$ is a leaf of J (a real-analytic closed curve, without corners). Conclusions C1 (smooth minimal interior), C2 ($F' \cap \partial M = \partial F$) and C3 ($C^{m,\alpha}$ up to the boundary, $m \geq 2$) are cited by their Appendix A numbering. The *weakest input* of this subsection is

(I₀) near each $p \in \ell$, F' is an embedded surface with boundary, $C^{1,\beta}$ up to the boundary for some $\beta \in (0, 1)$, with $\partial F' = \ell$, smooth minimal interior ($\vec{H} \equiv 0$; C1) and $F' \cap \partial M = \ell$ (C2).

C3 implies (I₀). Narrowing the input to (I₀) matters in §C.3.3: the dependence of Theorem C.3 on the upstream T-C is thereby reduced to its weakest level.

Lemma C.8 (Transversality). *Under $(A-\partial)$ (∂M strictly convex) and (I_0) , one has $T_p F' \neq T_p \partial M$ for every $p \in \ell$.*

Proof. Suppose $T_p F' = T_p \partial M =: \Pi$. Take coordinates with base Π and inward normal the w -axis, so $\partial M = \{w = \sigma(y)\}$ with $\sigma \in C^\infty$, $\sigma(0) = 0$, $\nabla \sigma(0) = 0$. By (I_0) and the $(C^{1,\beta})$ implicit function theorem, $F' = \{w = \delta(y) : y \in \bar{\Omega}\}$ with $\delta \in C^{1,\beta}(\bar{\Omega}) \cap C^\infty(\Omega)$, $\delta(0) = 0$ and $\nabla \delta(0) = 0$; the boundary $\partial \Omega \ni 0$ is the projection $\tilde{\ell}$ of ℓ , and since $T_p \ell \subset \Pi$ the projection is an immersion on ℓ , so $\tilde{\ell} \in C^\infty$ and Ω satisfies an interior ball condition at 0. From $F' \subset M$, $\delta \geq \sigma$; and C2 gives $\delta = \sigma$ exactly on $\tilde{\ell}$ with $\delta > \sigma$ in Ω . Let $\mathcal{H}[u] = A^{ij}(y, u, \nabla u) \partial_{ij} u + B(y, u, \nabla u)$ be the mean curvature operator of the graph $\{w = u\}$ with respect to the inward normal (A, B smooth, A positive definite; textbook-level). Then $\mathcal{H}[\delta] = 0$ (C1) and $\mathcal{H}[\sigma] > 0$ ($(A-\partial)$). With $v := \delta - \sigma$,

$$0 > \mathcal{H}[\delta] - \mathcal{H}[\sigma] = \left[\int_0^1 A^{ij}(\cdot, \sigma + sv, \nabla \sigma + s \nabla v) ds \right] \partial_{ij} v + \tilde{b}^i \partial_i v =: Lv, \quad (\text{C.21})$$

where $\tilde{b}^i$ involves only second derivatives of σ and $\delta, \nabla \delta$, bounded on $\bar{\Omega}$ by (I_0) , and the principal part is uniformly elliptic ($|\nabla \delta|$ bounded). Thus $v \in C^2(\Omega) \cap C^1(\bar{\Omega})$, $Lv < 0$, $v > 0$ in Ω and $v(0) = 0$; T-Hopf gives $\partial_\nu v(0) < 0$, hence $\nabla \delta(0) \neq \nabla \sigma(0) = 0$, a contradiction. $\square$

The difference from the Hopf argument in the lemmas of Appendix A: here only C^1 up to the boundary is needed, not C^2 — T-Hopf requires only locally bounded coefficients, not C^2 regularity of v at the boundary.

Corollary C.8.1. *F' is transverse to ∂M along ℓ ; combined with C2, F' is neatly embedded ($F' \cap \partial M = \partial F'$ and transverse), which is (R2) of the main text.*

Lemma C.9 (Local graph and the minimal graph equation). *Under (I_0) , for each $p \in \ell$ there are a coordinate neighbourhood $U \ni p$, coordinates (y^1, y^2, z) , and a one-sided region $\Omega \subset \mathbb{R}^2$ bounded by a C^∞ curve $\tilde{\ell} \subset \{z = 0\}$ (with $0 \in \tilde{\ell} = \partial \Omega \cap U$) such that*

$$(i) \ F' \cap U = \{z = u(y) : y \in \bar{\Omega}\} \text{ with } u \in C^{1,\beta}(\bar{\Omega}) \cap C^\infty(\Omega);$$

(ii) u satisfies in Ω the quasilinear uniformly elliptic minimal graph equation

$$Q[u] := A^{ij}(y, u, \nabla u) \partial_{ij} u + B(y, u, \nabla u) = 0, \quad A, B \in C^\infty, \ A \text{ uniformly positive definite on } \{|\nabla u| \leq K\}; \quad (\text{C.22})$$

$$(iii) \ u|_{\tilde{\ell}} = \phi \in C^\infty(\tilde{\ell}).$$

Proof. Take the base plane $\Pi' := T_p F'$ and the z -axis its g -normal. (i) follows from (I_0) and the implicit function theorem; the boundary of the projected image is the projection $\tilde{\ell}$ of ℓ , and the smoothness of $\tilde{\ell}$ is independent of F' : ℓ is real-analytic, $T_p \ell \subset \Pi'$, and the projection restricted to ℓ is an immersion. (ii) follows from C1 and the mean curvature operator of a graph in a Riemannian metric (textbook-level), with $K := \sup_{\bar{\Omega}} |\nabla u| < \infty$ from (I_0) . (iii): ϕ is the z -component of ℓ parametrised along $\tilde{\ell}$. $\square$

Lemma C.9 does *not* need Lemma C.8 (the graph is taken over $T_p F'$, not over $T_p \partial M$); equation (C.22) holds in the interior of Ω and has nothing to do with the obstacle ∂M — F' touches ∂M only along ℓ and leaves transversally.

Lemma C.10 (Schauder bootstrap). *Assume (I_0) . Then for each $p \in \ell$ the function u of Lemma C.9 lies in $C^\infty(\bar{\Omega} \cap U')$ for $U' \Subset U$. In particular F' is C^∞ up to the boundary near ℓ , hence $C^{1,1}$ up to the boundary.*

Proof. Induct on the assertion $P(k)$: $u \in C^{k+1,\beta}(\bar{\Omega} \cap U_k)$. $P(0)$ is (I_0) . Assume $P(k)$ and freeze coefficients: $a^{ij} := A^{ij}(y, u, \nabla u)$, $f := -B(y, u, \nabla u)$; from $A, B \in C^\infty$ and $P(k)$, $a^{ij}, f \in C^{k,\beta}$ (for $k = 0$ using $\nabla u \in C^{0,\beta}$), uniformly elliptic since $|\nabla u| \leq K$. Now $u \in C^2(\Omega) \cap C^0(\bar{\Omega})$ satisfies $a^{ij} \partial_{ij} u = f$ with $u|_{\bar{\ell}} = \phi \in C^\infty$ and $\bar{\ell} \in C^\infty$, so T-Sch (with $\alpha = \beta$, $b^i = c = 0$) gives $u \in C^{k+2,\beta}(\Omega \cup T)$, that is $P(k+1)$. The conclusion of T-Sch holds on all of $\Omega \cup T$, so no shrinking of the neighbourhood is needed at each step. Hence $u \in C^\infty(\bar{\Omega} \cap U')$, and $C^\infty(\bar{\Omega}) \subset C^{1,1}(\bar{\Omega})$. $\square$

Levels: T-Sch, the Hölder continuity of the composed coefficients, and (C.22) are all textbook-level. Lemma C.10 uses neither Hildebrandt nor Meeks–Yau nor (AM); those are used only in discharging (I_0) , that is in the proof of C3.

C.3.3 Theorem C.3: the constructed surface satisfies (R1) and is C^∞ up to the boundary.

Theorem C.3 ((R1) holds for the surface constructed in Theorem A.1). *Conditions (explicit, all needed):*

- **(C.3-i)** the setting (A-M), (A- ∂), (A-F), (A-cl s) of Theorem A.1 together with (H-cov) and GP (items 1–2 of §A.8.1). In the present project $M = E(K)$ is compact, (A- ∂) — strict convexity — is supplied by the collar metric (C.13), and the surface F of (A-F) is the Seifert surface of §5 of the main text with ∂F the leaf ℓ of J ;
- **(C.3-ii)** Proposition A.2 ((R), corner regularity, rounded version), entering through Theorem A.1 and not repeated here;
- **(C.3-iii)** Δ_2 : the source proof of conclusion C3 of Theorem A.1 (LEMMA $H + T-C$) has not been re-verified outside the slice covered by this appendix; only its $C^{1,\beta}$ part, namely (I_0) , is used here. The alternative route given at Step 9 of Appendix A (HS 3.5 / T-C together with Lemma A.4) is same-family and unaudited;
- **(C.3-iv)** (H ∂) (§C.0.5), used only in conclusion (c) and in the alignment of §C.5; (a) and (b) do not use it;
- **(C.3-v)** G3: whether the invocations of (R1) at Lemmas 5.1/5.3 of the main text cite only “there exists a minimiser satisfying (R1)” rather than “the surface Σ given by (E3)” — to be checked by whoever holds the full text (Remark C.4.3 and §C.6). Parts (a) and (b) of Theorem C.3 do not themselves depend on G3; G3 decides whether (a) and (b) can replace the main text’s invocations of (R1).

Conclusions. Let F' be the minimiser constructed by Theorem A.1, with $\ell = \partial F$. Then

- (a) **(R1-a), strengthened:** $F' \cap \partial M = \partial F' = \ell$, and F' is transverse to ∂M along ℓ ((R2) of the main text);
- (b) **(R1-b), strengthened:** F' is C^∞ up to the boundary near ℓ , hence $C^{1,1}$ up to the boundary and $F' \in W^{2,p}$ for all $p < \infty$; the Carleman–Bers–Vekua coefficient condition needed in §7.2 of the main text ($A, B \in L^p$, $p > 2$; Appendix E) holds;
- (c) F' satisfies C1, C4 (two-sided, isotopic to F rel ∂F) and C5 (AM), with $\text{Area}(F') = I = \inf_{\mathcal{F}} \text{Area}$ (class (i), rel ∂F). Whether F' realises the class-(ii) infimum $A(v)$ is Δ_1 ; see §C.5.

Proof. (a) The set equality is C2 verbatim ($\partial F' = \partial F = \ell$ because $F' \in \mathcal{F}$: isotopy rel ∂F preserves the boundary). An independent second proof, using only C1: Lemma C.5 uses only interior

C^∞ regularity, $\vec{H} \equiv 0$ and the collar metric (C.13), all of which C1 supplies. Transversality is Corollary C.8.1 (needing (I_0) and $(A-\partial)$).

(b) C3 gives (I_0) ; Lemma C.9 gives the local graph and (C.22); Lemma C.10 bootstraps to C^∞ . Being C^∞ up to the boundary, the coefficients of the induced metric lie in $C^\infty \subset W^{1,p}$, so the Vekua coefficient condition holds.

(c) These are the conclusions of Theorem A.1 verbatim. $\square$

The dependence chain, item by item. (1) Theorem A.1 as a whole (including (H-cov), GP and Proposition A.2); (2) $(I_0) \Leftarrow C3 (\Delta_2)$ — this appendix narrows the upstream requirement for (R1-b) to (I_0) : even if T-C gave no more than $C^{1,\beta}$, Lemma C.10 would push it to C^∞ ; (3) T-Hopf (textbook-level), used only for the transversality strengthening — *(R1-b) itself does not depend on it*; (4) T-Sch (textbook-level), the core of (R1-b); (5) strict convexity $(A-\partial)$, for Lemma C.8 and for Theorem A.1 itself; (6) Lemma C.5 (unconditional), the second proof of (a). *Not depended on:* (E3); Lemmas C.6 and C.7 and their gap (Lip); the applicability of Hildebrandt in the presence of an obstacle (which is carried by the proof of C3).

Remark (C.3.1: the difference from a minimiser of (E) — (H-SET)). The competitor class $(A\text{-cls})$ of Theorem A.1 is class (i) (rel ∂F pointwise), whereas that of (E) is class (ii), and $\mathcal{F} \subset \mathcal{C}_\ell$, so $\inf_{\mathcal{C}} \leq A(\ell) \leq I$. Thus F' is a minimiser in the sense of (E3) if and only if $I = A(v)$. This appendix does *not* prove that equality: the reverse inequality $I \leq A(v)$ would need the minimiser Σ of (E) to lie in the piecewise smooth rel- ℓ class, which is precisely the boundary regularity of Σ — circular. Theorem C.1 closes only the “choice of leaf” half: taking $\ell = \ell_* \in \text{Argmin}(A)$ gives $A(\ell_*) = \inf_{\mathcal{C}}$, and the residual equality $I(\ell_*) = A(\ell_*)$ is Δ_1 .

Remark (C.3.2: the boundary curve has no corners). The “corners unproved” item of Appendix A (the (R) of Proposition A.2) refers to the corners of the local discs D_a in the construction; the boundary ℓ of F' is a leaf of J (a real-analytic closed curve), so that gap does not fall on F' .

Remark (C.3.3: Δ_3). If the original (R1) of the main text contains a third clause not seen in this appendix (an orientability or two-sidedness requirement, say), two-sidedness is supplied by C4; anything else requires separate checking.

Whether downstream invocations may “take one particular minimiser” (a check in principle; case-by-case verification is left to whoever holds the full text).

Invocation	What is needed	Is taking F' enough?
§5 of the main text, stability (Theorem C.4)	second variation ≥ 0 for some area-minimising surface	class-(i) stability: yes ; class-(ii) stability: conditional on Δ_1 (Remark C.4.2)
(H-sm) of Appendix B	one minimal surface C^2 up to the boundary	yes (Theorem C.3(b) with C1)
§5, Lemma 5.1 (G3)	see (C.3-v)	to be checked: enough if only “there exists a minimiser satisfying (R1)” is cited; not enough if “the Σ given by (E3)” is

C.4 Theorem C.4: the stability assertion of §5 of the main text

C.4.1 Setting. $\Sigma \subset M$ is compact, connected, orientable, neatly embedded, with $\partial\Sigma$ a leaf of J ; M and Σ orientable makes Σ two-sided, and a global unit normal N is fixed. Along $\partial\Sigma$ take the orthonormal frame $\{T, \nu, N\}$ (T tangential, ν the outward unit conormal), and put $A(X, Y) := \langle \nabla_X Y, N \rangle$ and $H = \text{tr } A$. An *admissible variation* is a smooth family $\Phi : \Sigma \times (-\epsilon, \epsilon) \rightarrow M$ with

$\Phi_0 = \iota_\Sigma$ and $\Phi_t(\Sigma) \in \mathcal{C}$ for every t (class (ii): neatly embedded, in the same class, boundary on some leaf — the leaf may change), with variation field $X := \partial_t \Phi_t|_0$. *Minimal* means a critical point of area on $\mathcal{C}$; *stable* means $\frac{d^2}{dt^2}|_0 \text{Area}(\Phi_t \Sigma) \geq 0$ for every admissible variation.

Lemma C.11 (Characterisation of admissible variation fields). *Let Σ be transverse to ∂M along $\partial \Sigma$. Then*

- (a) *a necessary condition for X to be an admissible variation field is that $X|_{\partial \Sigma}$ be tangent to ∂M with $dy(X)|_{\partial \Sigma} \equiv c$ constant; equivalently $X|_{\partial \Sigma} = cW + aT$ with $c \in \mathbb{R}$ and $a \in C^\infty(\partial \Sigma)$;*
- (b) *writing $\alpha := \langle W, N \rangle$ and $\beta := \langle W, \nu \rangle$ (so $W = \alpha N + \beta \nu$), transversality makes α nowhere zero, and the normal component $u := \langle X, N \rangle$ of an admissible X satisfies $u|_{\partial \Sigma} = c\alpha$. Hence the class of admissible normal variations is*

$$\mathcal{A} := \{u \in C^\infty(\bar{\Sigma}) : u|_{\partial \Sigma} = c\alpha \text{ for some constant } c \in \mathbb{R}\}, \quad (\text{C.23})$$

which contains the fixed-boundary class $\{u|_{\partial \Sigma} = 0\}$ as a subspace of codimension one;

- (c) *Σ is minimal (critical in class (ii)) if and only if the equation holds together with one global scalar constraint,*

$$\vec{H}_\Sigma \equiv 0 \text{ on } \text{Int } \Sigma, \quad \oint_{\partial \Sigma} \beta ds = \oint_{\partial \Sigma} \langle W, \nu \rangle ds = 0. \quad (\text{C.24})$$

There is no pointwise natural boundary condition: β need not vanish pointwise, and in general Σ is not orthogonal to ∂M — consistent with the flux balance of Proposition 6.1 of the source of Theorem C.1.

Proof. (a) $\Phi_t(\partial \Sigma) \subset \partial M$ gives $X(p) \in T_p \partial M$; and $\Phi_t(\partial \Sigma)$ being a leaf gives $y(\Phi_t(p)) = \eta(t)$ independently of p , whose derivative is $dy(X) = \eta'(0) =: c$. The vectors of $T_p \partial M$ with $dy = c$ are exactly $cW + aT$. (b) If $\alpha = 0$ then $W \in \text{span}(T, \nu) = T_p \Sigma$, while $\{W, T\}$ spans $T_p \partial M$, so $T_p \partial M = T_p \Sigma$, contradicting transversality; and $u|_{\partial \Sigma} = c\alpha$ is the N -component of (a). (c) By T5 and (a): interior compactly supported variations give $\vec{H} = 0$, and leaf-changing variations ($c \neq 0$) give $c \oint \beta ds = 0$. $\square$

Lemma C.12 (Second variation with boundary term). *Let Σ be minimal, transverse, and $C^{2,\gamma}$ up to the boundary. Then every $u \in \mathcal{A}$ is realised by some admissible variation (an ambient isotopy flow with $\tilde{X}|_{\partial M} = cW$ and $\langle \tilde{X}|_\Sigma, N \rangle = u$), the second derivative depends only on u (the acceleration field is in general not admissible; only differences of such fields are), and, writing it $Q(u, u)$,*

$$Q(u, u) = \int_\Sigma (|\nabla u|^2 - (|A|^2 + \overline{\text{Ric}}(N, N))u^2) dA + c^2 \kappa(\Sigma), \quad (\text{C.25})$$

$$\kappa(\Sigma) = \oint_{\partial \Sigma} \left[\beta W (\log |W|) - \frac{|\alpha|}{|W|} (\text{II}(W, W) + \beta^2 \text{II}(T, T)) - \frac{\lambda \beta (\alpha^2 + |W|^2)}{|W|} \right] ds, \quad (\text{C.26})$$

where $\lambda := \langle \nabla_T T, W/|W| \rangle$ is the geodesic curvature of the leaf $\partial \Sigma$ inside ∂M . The integrand of (C.26) involves only ambient and foliation data together with the values of N and ν along $\partial \Sigma$, and is pointwise well defined and Lipschitz under (R1) ($C^{1,1}$).

Proof (skeleton). Differentiate T5. Since $\vec{H}_0 = 0$, the interior term leaves only $-\int u \dot{H}$, and the Jacobi operator with Green's formula turns it into $\int |\nabla u|^2 - \oint u \partial_\nu u - \int (|A|^2 + \overline{\text{Ric}}(N, N))u^2$. The boundary term $\frac{d}{dt} \oint \langle \tilde{X}, \nu_t \rangle ds_t$ splits into the acceleration $c^2 \langle \nabla_W W, \nu \rangle$, the arc-length term

$c^2\beta\langle\nabla_TW, T\rangle$, and the conormal rotation $c\alpha\partial_\nu u + c^2\alpha\beta A(\nu, \nu)$, in which $c\alpha\partial_\nu u = u\partial_\nu u$ exactly cancels the interior term $-\oint u\partial_\nu u$. This gives (C.25) with the unreduced $\kappa = \oint[\langle\nabla_W W, \nu\rangle + \alpha\beta A(\nu, \nu) + \beta\langle\nabla_TW, T\rangle]$; then minimality ($A(\nu, \nu) = -A(T, T)$), $\nabla_T T = \lambda W/|W| - \Pi(T, T)n$, $\langle n, \nu\rangle = |\alpha|/|W|$ and $\langle n, N\rangle = -\text{sgn}(\alpha)\beta/|W|$ give (C.26) by pure algebra. Three checks made in the source: for $\beta \equiv 0$ the formula degenerates to the classical free-boundary one; for the equatorial disc in the unit ball, direct computation and (C.25)–(C.26) both give -2π ; for a horizontal disc in a cylinder all three give 0. $\square$

Theorem C.4 (“the minimiser is a stable minimal surface” of §5). *(A) Assume (E) holds with minimiser Σ (class (ii)), that Σ is transverse to ∂M (Lemma 5.3 of the main text, needing (R1) and strict convexity) and that Σ is $C^{2,\gamma}$ up to the boundary (G2; Remark C.4.1). Then Σ satisfies (C.24) and $Q(u, u) \geq 0$ for every $u \in \mathcal{A}$: Σ is stable in the leaf-constrained class (ii), and in particular Dirichlet stable in the fixed-boundary class (i) (the convention of B.0.3).*

(B) Let F' be the surface constructed in Theorem C.3 (under (C.3-i)–(C.3-iii) it is C^∞ up to the boundary and transverse, so G2 holds automatically). Then F' is Dirichlet stable in class (i) unconditionally in Δ_1 ; and F' is stable in class (ii) if and only if Δ_1 is supplied (F' realising $\inf_{\mathcal{C}}$; §C.5).

Proof. (A) By Lemma C.11(c), Σ is minimal; by Lemma C.12, any $u \in \mathcal{A}$ is realised by an admissible variation $\Sigma_t = \psi_t(\Sigma) \in \mathcal{C}$. By the minimality in (E), $a(t) := \text{Area}(\Sigma_t) \geq a(0)$ for all small t ; and $a(t) = \int_\Sigma \mathcal{J}(t, p) dA$ with $\mathcal{J}$ smooth and uniformly bounded in t , so $a \in C^\infty$, $a'(0) = 0$ and $a''(0) \geq 0$, while $a''(0) = Q(u, u)$. Class-(i) stability is the case $c = 0$.

(B) F' minimises area in $\mathcal{F}$ (rel ∂F), and fixed-boundary admissible variations ($c = 0$; an ambient isotopy preserving ∂F) stay inside $\mathcal{F}$, so the same argument gives $Q(u, u) \geq 0$ for $u|_{\partial\Sigma} = 0$. Leaf-changing variations ($c \neq 0$) leave $\mathcal{F}$, and comparing areas for them requires $\text{Area}(F') = \inf_{\mathcal{C}}$, which is Δ_1 . $\square$

Remark (C.4.1: G2, the regularity convention). The implication “minimal $\Rightarrow a''(0) \geq 0$ ” of (A) needs only that $a(t)$ be smooth (Σ having finite area, ψ_t smooth) together with interior regularity; the derivation of the explicit formulae (C.25)–(C.26) needs Σ to be C^2 up to the boundary (boundary values of the Jacobi operator, the pointwise value of $A(\nu, \nu)$). (R1) gives only $C^{1,1} = W^{2,\infty}$, where A is meaningless “almost everywhere” on a one-dimensional curve; the reduced form (C.26) avoids pointwise A , but the identification of (C.25) with $a''(0)$ under (R1) alone still requires an approximation or mollification argument, which the source *does not carry out*. So (A) is stated with $C^{2,\gamma}$; for the F' of Theorem C.3 (C^∞) G2 disappears.

Remark (C.4.2: the sign of κ — not yet computed). Under the collar metric ∂M is totally umbilic ($\Pi = \mu g_{\partial M}$ with $\mu = -\varphi'(0) > 0$); if in addition the leaf is a geodesic and $|W|$ is constant (a model case, not verified against the main text in the source), then $\kappa = -\mu \oint |\alpha|(|W|^2 + \beta^2)/|W| ds < 0$: the leaf-changing direction is strictly destabilising, so (ii)-stability is strictly stronger than (i)-stability and Theorem C.4 is not vacuous. *This computation has not been carried out for the actual objects of the main text.*

Remark (C.4.3: G1 and G3, interface gaps to be filled by the main text). **G1**: the main text passes in §5 from “stable minimal surface” to “belongs to $\mathcal{M}_a$ ”, but the membership conditions of $\mathcal{M}_a$ (item 2 of §13, from the appendix of Schultens) are never transcribed into the main text. If the “stable” of the external source is class (i), the corollary of Theorem C.4 covers it; if class (ii), it is exactly (A); if class (iii), the implication is false. The defect is “a definition not transcribed”, not “mathematics not proved”; and if A8 of the main text is reproved along the (AW) route (Appendix D), G1 disappears. **G3**: (R1) is used in Theorem C.4 *only* through transversality, in three places — the $\alpha \neq 0$ of

Lemma C.11(b), the simultaneous straightening of Σ and ∂M in Lemma C.12, and $\langle n, \nu \rangle > 0$ in (C.26). Without strict convexity (a flat disc in a half-space is everywhere tangent) $\alpha \equiv 0$, (C.23) degenerates to $u|_{\partial\Sigma} \equiv 0$ and the argument does fail, consistently with Remark (i) of §5 of the main text. For F' , all three are supplied directly by Theorem C.3(a), without passing through Lemma 5.3 of the main text.

C.5 What “the minimiser of (E)” means in the main text (the reading adopted, H5)

The reading adopted. Throughout the main text, “the minimiser of (E)” and “the surface realising $A(v)$ ” mean *the surface F' obtained by running Theorem A.1 with boundary $\partial F = L_{\ell_*}$ on an optimal leaf $\ell_* \in \text{Argmin}(A)$ given by Theorem C.1; and $A(v)$ is read correspondingly as $I(\ell_*) = \inf\{\text{Area}(G) : G \text{ isotopic to } F \text{ rel } L_{\ell_*}\}$. Under this reading: Theorem C.3 supplies (R1) (C^∞ up to the boundary, transverse), Theorem C.4(B) supplies class-(i) stability, and (H-sm) of Appendix B is supplied by Theorem C.3(b); the clause of (E) that “the infimum is attained” is discharged verbatim by Theorem A.1.*

The cost. $A(v)$ changes from a class-(ii) infimum to a class-(i) infimum ($\inf_C \leq A(\ell_*) \leq I(\ell_*)$); the equality of the two is Δ_1 (Remark C.3.1) and is *not* proved here. The “exchange, corner-rounding and area comparison” of §7 of the main text needs only one smooth surface realising $A(v)$ together with comparison objects in the same class; whether each such invocation is compatible with the class-(i) reading is to be checked place by place by whoever holds the full text (table below). This change alters the statements of the hypotheses (E) and (R1) of the main text.

The alternative, recorded. Had the reading been declined, this appendix would retreat to a *conditional* proof: Theorems C.1 and C.2, Lemmas C.5–C.7 and Theorem C.4(A) would be unaffected (they take (E) as a hypothesis); Theorem C.3 and Theorem C.4(B) would still hold but would be statements about F' only, not replacements for (E) and (R1); and the main text would keep the earlier form in which (E) and (R1) are hypotheses. The seven affected invocation points are:

#	Location	Content	Under the adopted reading	Had it been declined
1	§5	the remark on the definition of $A(v)$ (“a minimum, not an infimum, by (E)”) Lemma 5.1	re-read as $I(\ell_*)$, citing Theorems C.1 and A.1	unchanged
2	§5	(well-foundedness, via “stable minimal surface $\in \mathcal{M}_a$ ”) Lemma 5.3 (transversality, presupposing (R1))	cite Theorem C.4(B); G1 still to be transcribed	cite Theorem C.4(A), conditional on (E), (R1)
3	§5	Theorem 7.1, choice of representative	supplied directly by Theorem C.3(a)	conditional on (R1)
4	§7.1	graph representation (needs $C^{1,1}$ up to the boundary and the Vekua coefficients)	take F' (Theorem A.1 with C.3)	conditional on (E)
5	§7.2	the hypothesis table of Theorem 11.1	Theorem C.3(b)	conditional on (R1)
6	§11		the (E), (R1) rows become “Theorem A.1 with C.3, conditions in §C.6”	unchanged

#	Location	Content	Under the adopted reading	Had it been declined
7	§12.2	the exact form of the hypotheses	(E), (R1) rewritten as discharged, with the list of conditions	unchanged

C.6 What this appendix is still conditional on; index of call sites

C.6.1 The list of conditions.

1. **Proposition A.2 ((R))** together with (H-cov) and GP of Theorem A.1 — entering Theorems C.3 and C.4(B) through Theorem A.1 (§A.8.1).
2. Δ_2 : the proof of conclusion C3 of Theorem A.1 has not been re-verified; only its $C^{1,\beta}$ part (I_0) is used here (condition (C.3-iii)).
3. **G3 and G1**: how Lemma 5.1 of the main text invokes (R1) and $\mathcal{M}_a$, to be checked by whoever holds the full text (Remark C.4.3); if A8 is reproved along the (AW) route, G1 disappears.
4. **(H ∂)**: read out and written into the statements of Theorems C.1 and C.2 (§C.0.5); automatic for the objects of the main text, so not a gap, but a quantifier to be added to the statement of (E) in §12.2.
5. **T-C, Hildebrandt**: the single research-level dependence, entering only through C3; original not checked. All other named theorems are textbook-level or monograph-level (T-LU and T-Mor being used only in Lemma C.7).
6. Δ_1 / **(H-SET)**: $I(\ell_*) = A(\ell_*)$ is unproved; under the reading adopted in §C.5 it is a matter of definition on the $A(v)$ side, and the residual equality is recorded here rather than claimed.
7. **G2**: Theorem C.4(A) is stated with $C^{2,\gamma}$ for a general minimiser of (E); for F' it disappears.
8. **Same-family provenance**: this appendix is entirely Fable 5.1 output, with cross-engine audit 0/3.

C.6.2 Discharged, no longer conditions. (R1-a), neatness (Lemma C.5, using only (E1), (E2)); compactness of the leaf space and the Lipschitz property of A (Theorem C.1, using only (H ∂)); both halves of (R1) for F' (Theorem C.3, conditional on items 1–2 above); class-(i) stability for F' (Theorem C.4(B)).

C.6.3 Index of call sites in the main text.

Call site	What is used	Note
§5, definition of $A(v)$ and Lemma 5.1	Theorem C.1 (existence of ℓ_*), Theorem C.4	the reading is §C.5; G1 still to be transcribed
§5, Lemma 5.3 (transversality)	Theorem C.3(a), Corollary C.8.1	for F' , (R1) is not needed as a hypothesis
§7.1 (choice of representative in Theorem 7.1)	Theorem A.1 with Theorem C.3	conditional on Δ_1 or on the reading H5
§7.2 (graph representation, Vekua coefficients)	Theorem C.3(b)	interface with Appendix E: C^∞ gives Lipschitz coefficients
§11, hypothesis table of Theorem 11.1	the whole of §C.6.1	the (E), (R1) rows follow §C.5
Appendix B, (H-sm)	Theorem C.3(b) with C1	§B.0.2
Appendix A, Remark A.1.2 ((H-SET))	Remark C.3.1, §C.5	the bridge is here; Δ_1 is not closed

D Haken normalisation and the weight–area inequality (AW)

This appendix is same-family with the main text: all notation, criteria and terminological conventions follow the main text (in particular: “ ∂ -irreducible” is used for the 3-manifold $E(K)$, “ ∂ -incompressible” for surfaces, and “ambient isotopy” always means an isotopy preserving the triangulation). Nothing mathematical is altered. The numbering $D.n$ is a single counter.

On provenance: the contents of this appendix were produced by Fable 5.1 and audited along several routes *within that same family*; they have received **no cross-engine audit** (criterion ③ counts 0/3). All conclusions are conditional in the sense of the list in §D.8. See Appendix G.

D.1 Terminology and conventions

Everything is in the smooth (equivalently PL) category. $K \subset S^3$ is a smooth knot, $N = N(K) \cong S^1 \times D^2$ a closed tubular neighbourhood, $E(K) = S^3 \setminus \text{Int } N$ the knot exterior, and $T := \partial E(K)$ the boundary torus. The *unknot* means a knot bounding a smoothly embedded disc in S^3 (this is the only definition used here).

Definition D.1 (∂ -irreducibility of a 3-manifold). Let M be a compact 3-manifold. A *compressing disc* for M is a properly embedded disc $D \subset M$ with $D \cap \partial M = \partial D$ and ∂D essential in ∂M (not bounding a disc in ∂M). M is *∂ -irreducible* if and only if it has no compressing disc (equivalently ∂M is incompressible in M). For $E(K)$ the phrase “ ∂ -irreducible” is used throughout.

Definition D.2 (incompressibility and ∂ -incompressibility of a surface). Let $F \subset M$ be a compact, properly (neatly) embedded surface ($F \cap \partial M = \partial F$).

- (i) A *compressing disc* for F is an embedded disc $D \subset M$ with $D \cap F = \partial D$ and ∂D essential in F ; F is *incompressible* if and only if it has none (“no sphere components” is listed separately as a hypothesis when needed).
- (ii) A *∂ -compressing disc* for F is an embedded disc $D \subset M$ with $\partial D = \alpha \cup \beta$, where $\alpha = D \cap F$ is an *essential arc* of F (α does not cut off a disc of F together with a subarc of ∂F), $\beta = D \cap \partial M$, $\text{int } D \cap F = \emptyset$ and $\partial \alpha = \partial \beta$; F is *∂ -incompressible* if and only if it has none.

The two properties belong to the surface F and to the manifold M respectively, and neither implies the other (a ∂ -parallel annulus in $E(K)$ is incompressible but ∂ -compressible; Remark D.21).

Proposition D.3 (the two definitions of ∂ -irreducibility agree). *Let M be an irreducible compact 3-manifold. Then “ ∂M is incompressible” if and only if “every properly embedded disc in M is ∂ -parallel”.*

Proof. ($\Leftarrow$) The boundary of a compressing disc D is essential in ∂M , while the boundary of a ∂ -parallel disc bounds a disc in ∂M ; so a compressing disc is not ∂ -parallel. ($\Rightarrow$) Let D be a properly embedded disc. Incompressibility of ∂M gives a disc $\Delta \subset \partial M$ with $\partial\Delta = \partial D$; then $D \cup \Delta$ is an embedded sphere, and irreducibility of M makes it bound a ball B , through which D is ∂ -parallel to Δ . $\square$

Irreducibility of $E(K)$ is available from §1 of the main text (its proof uses Alexander’s theorem), so Proposition D.3 applies with $M = E(K)$.

Definition D.4 (triangulation, weight, normal surface). Let $\mathcal{T}$ be a triangulation of a compact 3-manifold M with ∂M a subcomplex, and $\mathcal{T}^{(k)}$ its k -skeleton. A surface $F \subset M$ is in *general position* with respect to $\mathcal{T}$ if and only if it misses $\mathcal{T}^{(0)}$ and is transverse to $\mathcal{T}^{(1)}$ and $\mathcal{T}^{(2)}$. Its *weight* is then

$$w(F) = w_{\mathcal{T}}(F) := \#(F \cap \mathcal{T}^{(1)}) \in \mathbb{N}. \quad (\text{D.5})$$

For a tetrahedron Δ , the connected components of $F \cap \Delta$ are called *pieces*. A *normal triangle* is a disc in Δ cutting off one vertex (its boundary meeting each of three edges once); a *normal quadrilateral* is a disc separating one pair of opposite edges from the other (its boundary meeting each of the remaining four edges once). F is *normal* with respect to $\mathcal{T}$ if and only if it is in general position and every piece is a normal triangle or a normal quadrilateral. A normal surface is determined up to normal isotopy by its *normal coordinates*, the non-negative integer vector $x(F) \in \mathbb{Z}_{\geq 0}^{7t}$ recording, in each of the t tetrahedra, the numbers of pieces of the 4 triangle types and the 3 quadrilateral types. Only the direction “the coordinates determine the surface up to normal isotopy” is used (Theorem D.54), cited as textbook-level.

Remark (Convention D.6: ambient isotopy preserves the triangulation). Here “ambient isotopy” always means an isotopy $\{\Theta_t\}_{t \in [0,1]}$ of M with $\Theta_0 = \text{id}$, acting on surfaces by $F \mapsto \Theta_1(F)$, *with the triangulation $\mathcal{T}$ held fixed*; w is always computed with respect to that fixed $\mathcal{T}$. This is a different notion from the “neat isotopy” (a smooth family of neat embeddings) of §4 of the main text, which is not used in this appendix. Write

$$\mathcal{F}(F) := \{F' : F' \text{ ambient isotopic to } F \text{ and in general position with respect to } \mathcal{T}\}. \quad (\text{D.7})$$

D.2 ∂ -irreducibility of $E(K)$

This section writes out in full the equivalence “ K non-trivial $\iff E(K)$ ∂ -irreducible $\iff E(K) \not\cong S^1 \times D^2$ ” needed in §1 and §12.1 of the main text. Write $i : T \hookrightarrow E(K)$ and $j : T \hookrightarrow N$ for the inclusions.

Lemma D.8 (simple closed curves on a torus). *Let T be a torus and $c \subset T$ a simple closed curve. The following are equivalent: (1) c bounds a disc in T (c is inessential); (2) c separates T ; (3) $[c] = 0 \in H_1(T; \mathbb{Z})$. Consequently c is essential $\iff c$ is non-separating $\iff [c]$ is a primitive element of $H_1(T) \cong \mathbb{Z}^2$; and c is null-homotopic in $T \iff [c] = 0$.*

Proof. (1) $\Rightarrow$ (2): if $c = \partial\Delta$ then Δ and $\overline{T \setminus \Delta}$ separate T . (2) $\Rightarrow$ (3): a separating curve is the boundary of one side, hence null-homologous.

(3) $\Rightarrow$ (1): first, “ c non-separating $\Rightarrow [c]$ primitive (in particular $\neq 0$)”. Cutting T along c gives $T|c$, connected, compact, orientable, with two boundary circles and $\chi(T|c) = \chi(T) = 0$, so its

genus g satisfies $2 - 2g - 2 = 0$, that is $g = 0$: $T|c$ is an annulus. Take $p \in c$; its two preimages $p^\pm$ lie on the two boundary circles of the annulus, and an embedded arc α in the annulus joining p^+ to p^- glues back to a simple closed curve c' with $c \cap c' = \{p\}$ transversally, so $[c] \cdot [c'] = \pm 1$. The intersection form on $H_1(T)$ is unimodular, so if $[c] = nu$ then $n \mid \pm 1$ and $n = \pm 1$: $[c]$ is primitive. Contrapositively, $[c] = 0$ implies c separates, $T = A \cup_c B$ with A, B each having exactly one boundary circle, so $(1 - 2g_A) + (1 - 2g_B) = 0$, giving $g_A + g_B = 1$; one of them has genus 0 with a single boundary circle, that is, is a disc, and c bounds a disc.

Last sentence: null-homotopic gives $[c] = 0$; and $[c] = 0$ gives that c bounds a disc, hence is null-homotopic. $\square$

Lemma D.9 (computation of H_1 and the basis (μ, λ)). *Let $\mu \subset T$ be the meridian $(\{pt\} \times \partial D^2)$ in $N \cong S^1 \times D^2$. Then*

- (1) $H_1(E(K)) \cong \mathbb{Z}$ and $i_* : H_1(T) \rightarrow H_1(E(K))$ is onto;
- (2) $\ker j_* = \langle [\mu] \rangle$;
- (3) $\ker i_*$ is a rank-one direct summand with primitive generator, realised by a simple closed curve $\lambda \subset T$ (the preferred longitude, equivalently characterised by $\text{lk}(\lambda, K) = 0$);
- (4) $\{[\mu], [\lambda]\}$ is a basis of $H_1(T)$ and $[\mu] \cdot [\lambda] = \pm 1$.

Proof. Apply Mayer–Vietoris to $S^3 = E(K) \cup_T N$ (using slightly enlarged open neighbourhoods):

$$0 = H_2(S^3) \rightarrow H_1(T) \xrightarrow{(i_*, -j_*)} H_1(E(K)) \oplus H_1(N) \rightarrow H_1(S^3) = 0. \quad (\text{D.10})$$

So $\Phi := (i_*, -j_*) : \mathbb{Z}^2 \rightarrow H_1(E(K)) \oplus \mathbb{Z}$ is an isomorphism; cancellation for finitely generated abelian groups gives $H_1(E(K)) \cong \mathbb{Z}$, and Φ being an isomorphism makes both i_* and j_* onto, which is (1).

(2) In the model $N = S^1 \times D^2$ one has $H_1(T) = \mathbb{Z}[s] \oplus \mathbb{Z}[\mu]$ with $s = S^1 \times \{pt\}$, and $j_*[s] = 1$, $j_*[\mu] = 0$, so $\ker j_* = \langle [\mu] \rangle$ with $[\mu]$ primitive.

(3) $i_* : \mathbb{Z}^2 \rightarrow \mathbb{Z}$ has free quotient, so the sequence splits and $\ker i_*$ is a rank-one direct summand with primitive generator v . A primitive class is realised by a simple closed curve (take a basis (v, w) ; the corresponding linear homeomorphism sends the standard curve $\ell_{(1,0)}$ to a simple closed curve of class v), giving λ . And $[\lambda] \in \ker i_*$ means λ is null-homologous in $E(K)$, equivalently $\text{lk}(\lambda, K) = 0$.

(4) Take a basis $([\lambda], x)$ with $i_*[\lambda] = 0$ and $i_*x = g$ a generator of $H_1(E(K))$, and write $[\mu] = a[\lambda] + bx$. Surjectivity of Φ gives y with $i_*y = g$ and $j_*y = 0$, so $y = m[\mu]$ and $g = mbg$, whence $mb = 1$, $b = \pm 1$ and $\det([\lambda], [\mu]) = \pm 1$. $\square$

This lemma uses *only* Mayer–Vietoris, not Alexander duality (deliberately, to keep the ledger of external inputs clean).

Theorem D.11 (K trivial $\Rightarrow E(K)$ is not ∂ -irreducible). *If K is the unknot then $\partial E(K)$ is compressible in $E(K)$.*

Proof. Let $K = \partial D$ with $D \subset S^3$ a smoothly embedded disc. By the tubular neighbourhood theorem (taking N adapted to D ; the homeomorphism type of the exterior does not depend on the choice of N) one may arrange that $A := D \cap N$ is the collar annulus of ∂D in D , with $\partial A = K \sqcup \partial D'$ and $D' := D \cap E(K)$. Then D' is a disc, properly embedded, with $D' \cap T = \partial D'$. The annulus $A \subset N$ gives $[\partial D'] = \pm[K]$ in $H_1(N)$, and $[K]$ generates $H_1(N) \cong \mathbb{Z}$, so $j_*[\partial D'] \neq 0$ and hence $[\partial D'] \neq 0$ in $H_1(T)$; by Lemma D.8, $\partial D'$ is essential in T . So D' is a compressing disc. $\square$

(This theorem borrows nothing.)

Lemma D.12 (homological location of the boundary of a compressing disc). *If $D \subset E(K)$ is a compressing disc then $[\partial D] = \pm[\lambda] \in H_1(T)$.*

Proof. ∂D bounds D in $E(K)$, so $i_*[\partial D] = 0$ and $[\partial D] \in \ker i_* = \langle [\lambda] \rangle$ (Lemma D.9(3)); write $[\partial D] = k[\lambda]$. Since ∂D is essential, its class is primitive by Lemma D.8, so $k = \pm 1$. $\square$

The primitivity step is load-bearing: knowing only $k \neq 0$ would give $\lambda^k = 1$ in $\pi_1(E(K))$, and passing to $\lambda = 1$ would need “knot groups are torsion-free” (a sphere-theorem-level input). Lemma D.8 removes that dependence.

Lemma D.13. *If $D \subset E(K)$ is a compressing disc then λ is null-homotopic as a loop in $E(K)$.*

Proof. $\pi_1(T) \cong \mathbb{Z}^2$ is abelian, so free homotopy classes are elements; by Lemma D.12, λ and ∂D (suitably oriented) are freely homotopic within T . And ∂D is null-homotopic in $E(K)$, a property preserved by free homotopy. $\square$

Theorem D.14 (Dehn’s lemma; a research-level input). *Let M be a 3-manifold and $\gamma \subset \partial M$ a simple closed curve which is null-homotopic in M . Then there is a properly embedded disc $D_0 \subset M$ with $D_0 \cap \partial M = \partial D_0 = \gamma$.*

Level and source: research-level (C. D. Papakyriakopoulos, *On Dehn’s lemma and the asphericity of knots*, Ann. of Math. 66 (1957) 1–26); the boundary-curve form above is the standard statement in Hempel, *3-Manifolds*, Ch. 4, and Jaco, *Lectures on Three-Manifold Topology*, Ch. III. *Registration:* adopting it adds **one use site and one independent source** to the ledger of the main text (a source distinct from all present entries of §12); this increment is conditional on adopting the (AW) route (Theorem D.50), and is counted once adopted. *Non-circularity:* the proof of Dehn’s lemma (tower construction plus double covers) uses only covering spaces and surface topology, and no fact about ∂ -irreducibility of knot exteriors.

Theorem D.15 (K non-trivial $\Rightarrow E(K)$ is ∂ -irreducible). *If K is not the unknot then $\partial E(K)$ is incompressible in $E(K)$.*

Proof. Contrapositively, let D be a compressing disc. By Lemma D.13, $\lambda \subset T = \partial E(K)$ is null-homotopic in $E(K)$; by Theorem D.14 (with $M = E(K)$ and $\gamma = \lambda$) there is a properly embedded disc $D_0 \subset E(K)$ with $D_0 \cap T = \partial D_0 = \lambda$.

In $N \cong S^1 \times D^2$ take the trivialisation realising the 0-framing, so that $K = S^1 \times \{0\}$ and $\lambda = S^1 \times \{p\}$ with $p \in \partial D^2$ (the 0-framing being the push-off with $\text{lk}(\cdot, K) = 0$, given by a section of some trivialisation of N). Then $A_\lambda := S^1 \times [0, p] \subset N$ is an embedded annulus with $\partial A_\lambda = K \sqcup \lambda$.

Put $\Delta := D_0 \cup A_\lambda$. Since $D_0 \subset E(K)$, $A_\lambda \subset N$, $E(K) \cap N = T$ and $D_0 \cap T = \lambda = A_\lambda \cap T$, the two pieces meet exactly along λ and match there, so Δ is an embedded disc in S^3 (a disc glued to an annulus along one boundary circle) with $\partial \Delta = K$. Hence K is the unknot. $\square$

This proof uses neither isotopy extension, nor Alexander’s theorem, nor irreducibility of $E(K)$, nor the intermediate step “ $E(K)$ is a solid torus”; the only substantial external input is Theorem D.14.

A second proof (a methodologically independent cross-check, not part of the minimal dependence set). Without Dehn’s lemma, borrow instead two facts of surface topology: (II-a) isotopy classes of essential simple closed curves on a torus correspond bijectively to $\pm$ pairs of primitive elements of $H_1(T)$ (Farb–Margalit, *A Primer on Mapping Class Groups*, Prop. 1.5; behind it lie Baer–Epstein and $\text{MCG}(T^2) \cong \text{SL}_2(\mathbb{Z})$); (II-b) the isotopy extension theorem (Thom–Palais–Cerf; Hirsch, *Differential Topology*, Ch. 8 Thm 1.3). Let D be a compressing disc. By Lemma D.12,

$[\partial D] = \pm[\lambda]$ and both are essential; by (II-a), ∂D and λ are isotopic in T , hence in S^3 . Also $\lambda = S^1 \times \{p\}$ and $K = S^1 \times \{0\}$ are isotopic in N via the radial isotopy $S^1 \times \{tp\}$. Concatenating, ∂D and K are isotopic in S^3 ; by (II-b) there is an ambient isotopy h_t of S^3 with $h_1(K) = \partial D$, and then $h_1^{-1}(D)$ is an embedded disc with boundary K , so K is the unknot. The two borrowings and Theorem D.14 are mutually independent, so this is a methodologically independent cross-check; but taking this route would cost +2 use sites and +2 independent sources, which is not worth it, and only Theorem D.14 is registered.

Proposition D.16 (a compressing disc makes $E(K)$ a solid torus). *If $D \subset E(K)$ is a compressing disc then $E(K) \cong S^1 \times D^2$.*

Proof. Take a regular neighbourhood $W \cong D \times [-1, 1]$ of D in $E(K)$, with $W \cap T = \partial D \times [-1, 1] =: A$ an annulus, and put $E' := E(K) \setminus \text{Int } W$. Then $\partial E' = (T \setminus \text{Int } A) \cup (D \times \{-1\}) \cup (D \times \{+1\})$. As ∂D is essential, it does not separate T (Lemma D.8), so $T \setminus \text{Int } A$ is an annulus, and capping its two ends with discs gives $\partial E' \cong S^2$.

E' is connected (each component has non-empty boundary, else it would be a proper subset of S^3 both open and closed; and $\partial E'$ has just one component). $\text{Int } E'$ is a connected component of $S^3 \setminus \partial E'$, so by **Alexander's theorem (Schoenflies in S^3 ; textbook-level)** its closure satisfies $E' \cong B^3$. Finally $E(K) = E' \cup W$ is a ball with a 1-handle attached (gluing $D^2 \times [-1, 1]$ along two disjoint discs in S^2 , put in standard position by the disc theorem for S^2), and $E(K) \subset S^3$ is orientable, so $E(K) \cong S^1 \times D^2$. $\square$

Proposition D.17 (a solid torus exterior makes K trivial). *If $E(K) \cong S^1 \times D^2$ then K is the unknot.*

Proof. The meridian disc $D = \{pt\} \times D^2$ of the solid torus is properly embedded, and ∂D is a basis vector of $H_1(\partial E(K))$, hence non-zero and, by Lemma D.8, essential; so D is a compressing disc, and the contrapositive of Theorem D.15 gives that K is the unknot. $\square$

Neither Waldhausen's uniqueness of genus-one Heegaard splittings nor Gordon–Luecke is used here; the Mayer–Vietoris computation of Lemma D.9 supplies all the homological information needed.

Theorem D.18 (three equivalences). *For a smooth knot $K \subset S^3$ the following are equivalent: (1) K is the unknot; (2) $E(K)$ is not ∂ -irreducible ($\partial E(K)$ is compressible); (3) $E(K) \cong S^1 \times D^2$. In particular, K is non-trivial $\iff E(K)$ is ∂ -irreducible $\iff E(K) \not\cong S^1 \times D^2$.*

Proof. (1) $\Rightarrow$ (2) is Theorem D.11; (2) $\Rightarrow$ (1) is Theorem D.15; (2) $\Rightarrow$ (3) is Proposition D.16; (3) $\Rightarrow$ (2) is the first sentence of the proof of Proposition D.17. $\square$

Corollary D.19 (the exterior of the unknot is a solid torus). *K trivial $\Rightarrow E(K) \cong S^1 \times D^2$. (Invoked at “the exterior of the unknot is a solid torus” in §1 of the main text; the proof chain is Theorem D.11 with Proposition D.16, borrowing only Alexander's theorem.)*

Dependence list for this section. Proved in full here: Lemmas D.8 and D.9, Theorem D.11, Lemmas D.12 and D.13, Proposition D.17. Research-level input: Theorem D.14 (Dehn's lemma), used only by Theorem D.15. Textbook-level: Alexander's theorem (used only by Proposition D.16 and Corollary D.19, not by Theorem D.15), the tubular neighbourhood theorem, Mayer–Vietoris, the classification of compact surfaces.

D.3 ∂ -incompressibility of vertex surfaces

Theorem D.20 (incompressible $\Rightarrow \partial$ -incompressible). *Let M be an orientable 3-manifold with ∂M a torus, and let $F \subset M$ be compact, properly embedded, two-sided, incompressible, and neither a disc nor a sphere; assume ∂F is connected and an essential simple closed curve in ∂M . Then F is ∂ -incompressible.*

Proof. Let D be a ∂ -compressing disc: $\partial D = \alpha \cup \beta$ with $\alpha = D \cap F$ an essential arc of F , $\beta = D \cap \partial M$, $\text{int } \beta \cap F = \emptyset$, $\partial \alpha = \partial \beta = \{p, q\}$ and $\text{int } D \cap F = \emptyset$.

F is two-sided, so take a product neighbourhood $F \times [-1, 1] \subset M$ with $F = F \times \{0\}$; proper embeddedness gives $(F \times [-1, 1]) \cap \partial M = \partial F \times [-1, 1]$, a collar of ∂F in ∂M , so the two sides of ∂F in ∂M correspond exactly to the two sides of F (labelled $+$ and $-$). Cutting ∂M along ∂F gives A ; ∂F is connected and essential on a torus, hence non-separating by Lemma D.8, so A is one annulus whose two boundary circles lie on the $+$ side and the $-$ side respectively.

(a) *The two ends of β lie on the same boundary circle of A :* as $\text{int } D$ misses F , the disc D approaches F from one side along the connected arc α (the side label is locally constant along α , hence constant), say the $+$ side; near the corners p, q the disc D is a quarter-disc lying in the $+$ region, so β leaves ∂F at p into the $+$ side and arrives at q from the $+$ side: both ends lie on $\partial_+ A$.

(b) An arc properly embedded in an annulus with both ends on the same boundary circle cuts off a disc together with a subarc of that circle (the elementary classification of arcs in an annulus): this gives $E \subset A \subset \partial M$ with $\partial E = \beta \cup \delta$, $\delta \subset \partial F$ and $\text{int } E \cap F = \emptyset$ (since $F \cap \partial M = \partial F$ and $\text{int } E \cap \partial F = \emptyset$).

(c) $c := \alpha \cup \delta$ is an embedded simple closed curve in F , and $D \cup_\beta E$ is a disc in M with boundary c , so c is null-homotopic in M . Since F is incompressible, two-sided and neither a sphere nor a disc, $\pi_1(F) \rightarrow \pi_1(M)$ is injective (*the loop theorem / Dehn's lemma*, cited as textbook-level; see §D.8), so c is null-homotopic in F ; an embedded null-homotopic simple closed curve in a surface bounds a disc, so α together with $\delta \subset \partial F$ cuts off a disc of F , contradicting the essentiality of α . $\square$

An alternative proof in the source avoids π_1 -injectivity: from the disc E of (b), glue D to E and push off along the collar of ∂M to build a compressing disc $D^* = (D \setminus \nu(\beta \times [0, \frac{1}{2}])) \cup \nu(E \times \{\frac{1}{2}\})$ for F , contradicting incompressibility; that route uses no loop theorem, and its conclusion needs neither irreducibility nor ∂ -irreducibility of M . Only its statement is recorded here; the main text invokes the proof above.

Remark (D.21: neither hypothesis can be dropped). (i) Without “ ∂F connected” the statement is false: for K non-trivial, pushing the interior of an annular neighbourhood of a longitude on $\partial E(K)$ into the interior gives a ∂ -parallel annulus A , which is two-sided, incompressible (its core is essential in ∂M and $\pi_1(\partial M) \hookrightarrow \pi_1(E(K))$), neither a disc nor a sphere, with ∂A two parallel essential curves; but a spanning disc in the parallelism region is a ∂ -compressing disc. Here (a) still holds, but the “ $+$ side” corresponds to two *different* circles after cutting, and it is exactly connectedness of ∂F that carries (a) $\Rightarrow$ (b). (ii) Without “ ∂M a torus” the statement is false: in $M = \Sigma_2 \times [0, 1]$, pushing a one-holed-torus subsurface of ∂M into the interior gives F , incompressible (incompressibility of ∂M gives π_1 -injectivity), with ∂F connected and essential, yet ∂ -parallel and hence ∂ -compressible; what fails is “cutting along ∂F gives an annulus”.

Corollary D.22 (vertex surfaces are ∂ -incompressible). *Let K be non-trivial and $M = E(K)$. Then M is ∂ -irreducible; and every compact, connected, orientable, properly embedded, incompressible surface $S \subset M$ whose boundary is a single essential simple closed curve in ∂M (in particular, every vertex surface with boundary a preferred longitude) is ∂ -incompressible.*

Proof. ∂ -irreducibility of M is Theorem D.15. S has boundary, so it is not a sphere; and S is not a disc, for otherwise ∂S would bound a disc in M while being essential in ∂M , so M would not be ∂ -irreducible. Now apply Theorem D.20. $\square$

The hypothesis is stated as “ ∂S essential” rather than “ ∂S is a leaf of J ”: the proof of Theorem D.50 applies this corollary to the translated and perturbed surface S_2 , whose boundary is no longer a leaf but is isotopic in ∂M to ∂S , hence still essential.

D.4 Haken’s normalisation theorem (least-weight form): statement and the first two steps

Theorem D.23 (Haken normalisation, least-weight form). *Let M be a compact, irreducible, ∂ -irreducible 3-manifold and $\mathcal{T}$ a triangulation of M with ∂M a subcomplex. Let $F \subset M$ be properly embedded, two-sided, incompressible, ∂ -incompressible, with no sphere components and no disc components, and in general position with respect to $\mathcal{T}$. Then every representative N of $\mathcal{F}(F)$ (see (D.7)) at which w attains its minimum is a normal surface with respect to $\mathcal{T}$; in particular there is a normal surface N ambient isotopic to F with*

$$w(N) \leq w(F). \quad (\text{D.24})$$

Level: proved here (same-family, awaiting cross-engine audit); the prototype is Haken’s 1961 normalisation theorem (research-level), whose original is not cited — every step is given in §D.4–§D.5. *Difference from the source statement:* the source allowed F to have essential disc components, whereas the skeleton of the third step used here excludes disc components (harmless downstream, since the S of Theorem D.50 is not a disc); this difference is recorded in the issues log.

Relation to the classical statement. The classical Haken normalisation theorem asserts that some ambient isotopy carries F to a normal surface N with $w(N) \leq w(F)$; the least-weight form D.23 is slightly stronger (the minimal representative is itself normal) and immediately implies the classical form. Downstream, only (D.24) is used. The complexity is measured by w alone, with no secondary quantity: each lemma of the third step strictly decreases w , so no lexicographic induction is needed in the assembly; a secondary quantity (the number of components of $F \cap \mathcal{T}^{(2)}$) appears only in the first step.

The proof has three steps: (i) removing closed curves of the intersection of F with 2-simplices (Lemma D.25, w non-increasing); (ii) removing return arcs (Lemma D.26, w drops by exactly 2); (iii) removing non-normal pieces inside tetrahedra (§D.5, w strictly decreasing). The assembly is at the end of §D.5.

Lemma D.25 (closed curves in a 2-simplex; w non-increasing). *Let F be as in Theorem D.23 (only incompressibility of F and irreducibility of M are used), σ a 2-simplex of $\mathcal{T}$, and suppose $F \cap \sigma$ contains a closed curve. Then there is an ambient isotopy $F \mapsto F' \in \mathcal{F}(F)$ with $w(F') \leq w(F)$ and $\#\{\text{components of } F' \cap \mathcal{T}^{(2)}\} < \#\{\text{components of } F \cap \mathcal{T}^{(2)}\}$.*

Proof. Take an innermost closed curve c of $F \cap \sigma$; it bounds a disc $E \subset \text{int } \sigma$ with $\text{int } E \cap F = \emptyset$. Since $c \subset F$ bounds the disc E in M and F is incompressible, c bounds a disc $E' \subset F$. Then $E \cup E'$ is an embedded sphere, and irreducibility of M makes it bound a 3-ball B ; push E' across E through B and slightly off σ — an ambient isotopy supported in a neighbourhood of B (the ball-push model of Lemma D.29 (4)).

The newly created pieces lie in a neighbourhood of $\text{int } \sigma$ and meet no edge, while the removed E' met the edges in ≥ 0 points, so w does not increase; and $F \cap \mathcal{T}^{(2)}$ loses c together with the components inside E' , so the number of components strictly decreases. $\square$

Lemma D.26 (return arcs in a 2-simplex; w drops by exactly 2). *Let F be in general position with respect to $\mathcal{T}$, σ a 2-simplex, and $a \subset F \cap \sigma$ an arc with both endpoints on the same edge e of σ (a return arc). Then there is an ambient isotopy $F \mapsto F' \in \mathcal{F}(F)$, supported in a disc neighbourhood inside σ , with $F' \cap e = (F \cap e) \setminus \{\text{two points}\}$ and the intersections with all other edges unchanged, so that $w(F') = w(F) - 2$. No topological hypothesis is needed.*

Proof. Take an innermost return arc a ; then a together with a subarc b of e cuts off a disc $E \subset \sigma$ with $\text{int } E \cap F = \emptyset$. Push F across e along E : take a small product neighbourhood of E in M and push the band of F near a through it to the other side of e . This is an ambient isotopy supported near E , removing exactly the two intersection points of ∂a with e and creating no new edge intersections (the pushed pieces are, in the faces around e , arcs parallel to b missing the edges). When $e \subset \partial M$, the same push slides ∂F across E within ∂M and is still an ambient isotopy of M (Lemma D.36(ii) gives an explicit model). $\square$

Output of the first two steps. Applying Lemma D.26 repeatedly (each application lowering w by 2, so terminating) and Lemma D.25 repeatedly (w non-increasing, component count decreasing, so terminating) produces $F_1 \in \mathcal{F}(F)$ with $w(F_1) \leq w(F)$ whose intersection with every 2-simplex consists only of arcs running from one edge to a different edge. In particular every component of $F_1 \cap \mathcal{T}^{(2)}$ meets $\mathcal{T}^{(1)}$ — a fact that is load-bearing in Lemma D.29 (5).

D.5 The third step: removing non-normal pieces inside tetrahedra

Standing assumptions (all of §D.5). $M, \mathcal{T}, F$ are as in Theorem D.23 and F has already passed the first two steps: its intersection with each 2-simplex consists only of arcs running between distinct edges (no closed curves in a face, no return arcs). For a tetrahedron Δ take a transverse product collar $\partial\Delta \times [0, 1] \subset \Delta$ (with $\partial\Delta \times \{0\} = \partial\Delta$ and $F \cap (\partial\Delta \times [0, 1]) = (F \cap \partial\Delta) \times [0, 1]$, so that “ F is vertical”); $F \cap \Delta$ is a compact properly embedded surface in Δ , the boundaries of whose pieces are disjoint simple closed curves on $\partial\Delta \cong S^2$. Since $\partial M \subset \mathcal{T}^{(2)}$, one has $\text{int } \Delta \cap \partial M = \emptyset$. The complexity is w alone: each lemma of this section strictly decreases w , so the assembly needs no lexicographic order. The proof splits into two assertions: (A) every piece is a disc (Lemmas D.28–D.29); (B) granting (A), the boundary of each piece meets each edge at most once (Lemmas D.31–D.33, D.37). Lemma D.27 then makes each piece boundary of triangle or quadrilateral type, and Lemma D.35 straightens them.

Lemma D.27 (normal curves meeting each edge at most once). *Let $\gamma \subset \partial\Delta$ be a connected closed curve avoiding the vertices, transverse to the edges, meeting each face only in arcs between distinct edges. If $|\gamma \cap e| \leq 1$ for every edge e , then $|\gamma \cap \mathcal{T}^{(1)}| \in \{3, 4\}$ and γ is of triangle type (encircling a vertex; length 3) or of quadrilateral type (on the four edges other than one pair of opposite edges; length 4). Contrapositively: if the boundary of a piece is neither of triangle nor of quadrilateral type, it meets some edge in at least two points.*

Proof. Put $x_e := |\gamma \cap e| \in \{0, 1\}$ and $L := \sum_e x_e$. For each face f , $\gamma \cap f$ is a disjoint union of arcs, each with its two endpoints on two distinct edges of f , and every point of γ on an edge of f is an endpoint of exactly one such arc; so $s_f := \sum_{e \subset f} x_e = 2\#\{\text{arcs of } \gamma \cap f\}$ is even and ≤ 3 , whence $s_f \in \{0, 2\}$.

Each edge lies in exactly two faces, so $\sum_f s_f = 2L$, that is $L = \#\{f : s_f = 2\} \leq 4$. If $L = 0$, γ lies in one face and is a closed curve in a face, contrary to hypothesis. If $L = 1$, the faces containing that edge have odd s_f , a contradiction. If $L = 2$, both faces with $s_f = 2$ would contain both weighted edges, but two distinct edges lie in at most one common face, a contradiction. If $L = 3$,

exactly one face f_0 has $s_{f_0} = 0$, so the three weight-one edges avoid f_0 , that is, all pass through the vertex v opposite f_0 ; each other face contains two of them and carries one arc cutting off the corner at v , and the three arcs join up into a triangle-type curve.

If $L = 4$, each of the four faces contains exactly two weight-one edges and misses exactly one weight-zero edge, so the two weight-zero edges are not coplanar, that is, form a pair of opposite edges; the weight-one edges form a 4-cycle with one arc per face, joining into a quadrilateral-type curve. Connectedness makes γ exactly the union of these arcs (each face carrying $s_f/2 \leq 1$ arcs). $\square$

A negative result, recorded to prevent it being retried: the statement “a curve of length ≥ 5 has, on some edge, a bigon containing no vertex” is **false**: take an octagon-type curve with $x_{01} = x_{23} = 2$ and the other four edges of weight 1; the segment between the two points on the edge 01 bounds, with the two arcs on either side, two bigons, one containing the vertex 2 and the other the vertex 3, so pushing a bigon would cross a vertex and increase w . Hence this section proceeds instead via ∂ -compressing discs inside Δ (Lemmas D.31–D.33), avoiding vertices entirely.

Lemma D.28 (a non-disc piece gives a compressing disc inside Δ). *Suppose some piece $P \subset \Delta$ is not a disc. Then there is a disc $D' \subset \text{int } \Delta$ with $D' \cap F = \partial D'$, $\text{int } D' \pitchfork F$ along $\partial D'$, and $\partial D'$ essential in the piece P' containing it (not bounding a disc in P'). This lemma does not move F .*

Proof. (1) *Closed pieces are excluded.* If $\partial P = \emptyset$ then P is a subset of F both open and closed, hence a closed component of F ; F is incompressible and two-sided, so the loop theorem gives $\pi_1(P) \rightarrow \pi_1(M)$ injective, and this factors through $\pi_1(\Delta) = 1$, so $\pi_1(P) = 1$ and $P = S^2$, excluded by “no sphere components”. Below, $\partial P \neq \emptyset$.

(2) *An initial disc.* Take a component c of ∂P ; then $A := c \times [0, 1] \subset F \cap \Delta$ is connected and contains c , so $A \subset P$, and $\gamma := c \times \{1\} \subset \text{int } P \cap \text{int } \Delta$. The curve γ is essential in P : indeed $P \setminus \gamma = (A \setminus \gamma) \sqcup R$ with $R := P \setminus A$ connected; if $\gamma = \partial Q$ with $Q \subset P$ a disc, then $\text{int } Q$ is a component of $P \setminus \gamma$, either $\text{int } A$ (an open annulus, not an open disc — a contradiction) or R , in which case $P = A \cup_\gamma Q$ is an annulus with a disc glued along one boundary circle, hence a disc, a contradiction.

That γ bounds: Schoenflies gives $c = \partial G$ with $G \subset \partial \Delta$ a disc, and $D_0 := G \times \{1\} \subset \text{int } \Delta$ is a disc with $\partial D_0 = \gamma$ and $D_0 \cap F = (G \cap F) \times \{1\}$ finitely many circles; and D_0 being horizontal while F is vertical gives $\text{int } D_0 \pitchfork F$.

(3) *Minimising by disc exchange.* Let $\mathcal{D}$ be the set of discs $D \subset \text{int } \Delta$ with (a) $\partial D \subset F$ essential in the piece containing it and (b) $\text{int } D \pitchfork F$; then $D_0 \in \mathcal{D}$. Choose $D \in \mathcal{D}$ minimising the number $n(D)$ of circles of $\text{int } D \cap F$, and claim $n(D) = 0$.

Suppose $n(D) \geq 1$ and take an innermost circle $c' = \partial D_1$ of $\text{int } D \cap F$ in D , with $D_1 \subset \text{int } D$ and $\text{int } D_1 \cap F = \emptyset$, lying in some piece P'' . (i) If c' is essential in P'' then $D_1 \in \mathcal{D}$ with $n(D_1) = 0 < n(D)$, a contradiction. (ii) Otherwise $c' = \partial E''$ with $E'' \subset P'' \cap \text{int } \Delta$ a disc. Every component of $D \cap F$ either misses E'' or falls entirely inside it; and $\partial D \not\subset E''$ (else ∂D would bound a disc inside the disc $E'' \subset P'$, contradicting (a)). Among the circles falling inside E'' , each bounds a subdisc of E'' ; take an innermost one, $c'' = \partial E_1$, so that $E_1 \cap D = c''$; and c'' bounds a subdisc $D_2 \subset \text{int } D$ with $\partial D \cap D_2 = \emptyset$.

The exchange $D^* := (D \setminus \text{int } D_2) \cup E_1$ is a disc with $\partial D^* = \partial D$; taking a product neighbourhood $E_1 \times [-\delta, \delta] \subset \text{int } \Delta$ (with δ small enough that $F \cap (E_1 \times [-\delta, \delta]) = E_1 \times \{0\}$ and $D \cap (E_1 \times [-\delta, \delta]) = c'' \times [-\delta, \delta]$), replacing E_1 by $E_1 \times \{\epsilon\delta\}$ on the side of $D \setminus \text{int } D_2$ and joining along $c'' \times [0, \epsilon\delta]$, gives $D^{**} \in \mathcal{D}$ with $n(D^{**}) \leq n(D) - 1$, a contradiction.

So $D' := D$ works. Transversality along $\partial D'$: D_0 is horizontal against the vertical F , the exchanges do not move a neighbourhood of ∂D , and the disc D_1 of (i) inherits it along c' . $\square$

Lemma D.29 (pushing across D' ; w strictly decreases). *Let D' be as in Lemma D.28. Then there is an ambient isotopy $F \mapsto F' \in \mathcal{F}(F)$, supported in a neighbourhood of a 3-ball $B \subset M$, with*

$$w(F') = w(F) - |E' \cap \mathcal{T}^{(1)}| \leq w(F) - 1, \quad (\text{D.30})$$

where $E' \subset F$ is the disc bounded by $\partial D'$ in F ; and $F' \cap \mathcal{T}^{(2)}$ is $F \cap \mathcal{T}^{(2)}$ with certain whole components deleted, so F' still has no closed curves in faces and no return arcs.

Proof. (1) *Existence and location of E' .* $D' \cap F = \partial D'$ and F is incompressible, so $\partial D' = \partial E'$ with $E' \subset F$ a disc. If $E' \subset \Delta$ then E' , being connected and containing $\partial D' \subset P'$, would satisfy $E' \subset P'$, contradicting essentiality of $\partial D'$ in P' ; so $E' \not\subset \Delta$. Also $\partial E' \subset \text{int } \Delta \subset \text{Int } M$ and $\text{int } E'$ is open in F , so $E' \subset \text{Int } F$ and $E' \cap \partial F = \emptyset$.

(2) *The ball B .* $S := D' \cup E'$ is an embedded sphere in $\text{Int } M$; irreducibility of M makes $S = \partial B$ with $B \subset M$ a 3-ball. And $B \subset \text{Int } M$: points of $\text{int } B$ have $\mathbb{R}^3$ -neighbourhoods, so by invariance of domain they cannot be points of ∂M .

(3) $F \cap B = E'$ (including when F is disconnected). We have $F \cap S = E'$. Let F_0 be the component of F containing E' . (a) For another component F_j : $F_j \cap S = \emptyset$, so either $F_j \subset \text{int } B$ or $F_j \cap B = \emptyset$; in the first case $F_j \subset \text{Int } M$ is closed, and the loop theorem gives $\pi_1(F_j) \hookrightarrow \pi_1(M)$ factoring through $\pi_1(B) = 1$, so $F_j = S^2$, excluded. (b) $G := F_0 \setminus \text{int } E'$ is connected with $G \cap S = \partial E'$, and $G \setminus \partial E'$ is connected and misses S , hence lies entirely in $\text{int } B$ or entirely in $M \setminus B$; the first would make $F_0 \subset B \subset \text{Int } M$ closed and inside a ball, excluded as in (a). So $F \cap B = E'$ and $\text{int } B \cap F = \emptyset$.

(4) *The ball-push isotopy, explicitly. Ball-push model* (uniqueness of PL regular neighbourhoods / smooth collars with corners; textbook-level): there are a neighbourhood $N \subset \text{Int } M$ of B and a homeomorphism $N \rightarrow C := \{(r, \theta, z) : r \leq 1 + 2\eta, -2\eta \leq z \leq 1 + 2\eta\}$ taking $F \cap N$, E' , D' and B to $\{z = 0\}$, $\{z = 0, r \leq 1\}$, $\{z = h(r), r \leq 1\}$ (with $h(1) = 0$ and $h > 0$ for $r < 1$) and $\{r \leq 1, 0 \leq z \leq h\}$ respectively.

Put $\psi(r) = h(r) + \eta$ for $r \leq 1$, decreasing linearly to 0 on $[1, 1 + \eta]$ and $\psi = 0$ for $r \geq 1 + \eta$; and $\phi_t(r, \theta, z) := (r, \theta, \Phi_t(r, z))$ where $\Phi_t(r, \cdot)$ is the piecewise linear homeomorphism fixing the two ends and sending 0 to $t\psi(r)$. Then ϕ_t is the identity on ∂C , extends by the identity to M , and is an ambient isotopy supported in N .

Now $F' := \phi_1(F) = (F \setminus \{z = 0, r < 1 + \eta\}) \cup \{z = \psi(r), r \leq 1 + \eta\}$. Both the inserted part and the removed collar lie in a small neighbourhood of $D' \cup \partial D'$, and $D' \subset \text{int } \Delta$ is compact, so for small η both lie in $\text{int } \Delta$ and miss $\mathcal{T}^{(1)} \cup \mathcal{T}^{(2)}$. Hence $F' \cap \mathcal{T}^{(k)} = (F \cap \mathcal{T}^{(k)}) \setminus E'$ for $k = 1, 2$, with transversality and vertex-avoidance as before, $\partial F' = \partial F$ and $F' \in \mathcal{F}(F)$.

(5) $|E' \cap \mathcal{T}^{(1)}| \geq 1$, and whole components. E' is connected with $\partial E' \subset \text{int } \Delta$ and $E' \not\subset \Delta$, so $E' \cap \partial \Delta \neq \emptyset$. For any component C of $F \cap \mathcal{T}^{(2)}$ one has $C \cap \partial E' = \emptyset$, so $C \cap E'$ is both open and closed in C : either $C \subset E'$ or $C \cap E' = \emptyset$. Hence $E' \cap \mathcal{T}^{(2)}$ is a union of at least one whole component, and by the output of the first two steps every component meets $\mathcal{T}^{(1)}$, so $|E' \cap \mathcal{T}^{(1)}| \geq 1$. And $F' \cap \mathcal{T}^{(2)}$ is the union of the remaining whole components, with no new closed curves and no new return arcs. $\square$

From now on the standing assumption is **(A): inside every tetrahedron every piece is a disc** (obtained from Lemmas D.28–D.29 together with minimality; see the assembly).

Lemma D.31 (an innermost edge segment). *If the boundary of some piece meets some edge in at least two points, then there are a tetrahedron Δ , a disc piece $D \subset \Delta$, an edge $e \subset \Delta$ and points $p, q \in \partial D \cap e$ such that the edge segment $\sigma := [p, q] \subset e$ satisfies $\sigma \cap F = \{p, q\}$.*

Proof. Among all triples $(D, e, \{p, q\})$ with D a disc piece and $p, q \in \partial D \cap e$ adjacent along e within $\partial D \cap e$, choose one minimising $|\text{int } \sigma \cap F|$. If there were $x \in \text{int } \sigma \cap F$, then $x \in \partial D''$ for another disc piece $D'' \neq D$ of the same Δ .

∂D is a simple closed curve on the sphere $\partial \Delta$, and $\text{int } \sigma$ is connected and misses ∂D , so it lies in one disc region U of $\partial \Delta \setminus \partial D$, with σ a chord of U . Now $\partial D''$ is a closed curve in $\text{int } U$ which at x crosses from one side of $U \setminus \sigma$ to the other, so it must cross back: $|\partial D'' \cap \text{int } \sigma| \geq 2$. Taking $p'', q'' \in \partial D'' \cap \text{int } \sigma$ adjacent along e gives $[p'', q''] \subsetneq \sigma$ with strictly smaller $|\text{int}[p'', q''] \cap F|$, contradicting minimality. $\square$

Lemma D.32 (a candidate ∂ -compressing disc inside Δ). *With (D, e, p, q, σ) as in Lemma D.31, there are a properly embedded arc τ from p to q inside D and a disc $E \subset \Delta$ with $\partial E = \tau \cup \sigma$, $E \cap \partial \Delta = \sigma$, $E \cap F = \tau$ and $\text{int } E \pitchfork F$; moreover E is, near σ , the “wall” $\sigma \times [0, \epsilon]$ inside the collar. This lemma does not move F .*

Proof. (1) *An initial disc.* $\partial D \setminus \{p, q\}$ consists of two open arcs, with closures α and α_2 . Then $\alpha \cup \sigma$ is a simple closed curve on $\partial \Delta$; Schoenflies gives two open discs, with α_2° inside one of them, and G denotes the closure of the other: $\partial G = \alpha \cup \sigma$ and $G \cap \partial D = \alpha$.

In the transverse product collar, $\{p\} \times [0, 1]$ and $\alpha \times [0, 1] \subset D$. Put $\tau := \{p\} \times [0, 1] \cup \alpha \times \{1\} \cup \{q\} \times [0, 1] \subset D$, a properly embedded arc from p to q ; and $E_0 := \sigma \times [0, 1] \cup G \times \{1\}$, a disc with $\partial E_0 = \sigma \cup \tau$, $E_0 \cap \partial \Delta = \sigma$, and $E_0 \cap F = \tau$ together with finitely many disjoint circles $c_i \times \{1\}$ in $\text{int } E_0$ (the c_i being the components of $F \cap \partial \Delta$ inside $\text{int } G$), with $\text{int } E_0 \pitchfork F$. There are no arcs with endpoints on $\text{int } \sigma$ (as $\text{int } \sigma \cap F = \emptyset$) and none with endpoints on τ (as τ is already one component).

(2) *Minimising by disc exchange.* Let $\mathcal{E}$ be the set of discs $E \subset \Delta$ with $\partial E = \tau \cup \sigma$, $E \cap \partial \Delta = \sigma$, $\text{int } E \pitchfork F$ and $E \cap F = \tau$ together with finitely many interior circles, and take E minimising the number of circles. If $n(E) \geq 1$, take an innermost circle c' , lying in a piece P'' , which is a disc by (A), so $c' = \partial Q$ with $Q \subset P'' \cap \text{int } \Delta$.

Components of $E \cap F$ either miss Q or fall entirely inside it; and $\tau \ni p \in \partial \Delta$ while $Q \subset \text{int } \Delta$, so everything falling inside Q is a circle. Take an innermost such, $c'' = \partial Q_1$ with $Q_1 \cap E = c''$; then c'' bounds a subdisc E_2 of $\text{int } E$ with $E_2 \cap (\tau \cup \sigma) = \emptyset$. Exchanging and pushing off as in step (3) of Lemma D.28 gives E^{**} ; the modification takes place only inside $Q_1 \times [-\delta, \delta] \subset \text{int } \Delta$, so $E^{**} \cap \partial \Delta = \sigma$ and $\partial E^{**} = \tau \cup \sigma$ with the inserted disc missing σ and τ , and $n(E^{**}) \leq n(E) - 1$, a contradiction. Hence $E \cap F = \tau$, and E is still the initial wall $\sigma \times [0, \epsilon]$ near σ . $\square$

Lemma D.33 (∂ -compressing along E lowers the weight, for $e \not\subset \partial M$). *With $(D, e, p, q, \sigma, \tau, E)$ as in Lemma D.32 and $e \not\subset \partial M$, there is an ambient isotopy $F \mapsto F' \in \mathcal{F}(F)$ supported in a small neighbourhood N of E (meeting only Δ and the tetrahedra around e , with $N \cap \mathcal{T}^{(1)} = \sigma$ and $N \cap \mathcal{T}^{(0)} = \emptyset$) such that*

$$F' \cap e = (F \cap e) \setminus \{p, q\}, \quad F' \cap e_1 = F \cap e_1 \quad (e_1 \neq e), \quad w(F') = w(F) - 2. \quad (\text{D.34})$$

F' has no closed curves in faces; and any return arcs appearing in the faces around e are removed by Lemma D.26, each lowering w by a further 2.

Proof. Let $\Delta = \Delta_1, \dots, \Delta_k$ and $f_1, \dots, f_k$ be the tetrahedra and faces around e (cyclically ordered, $f_1, f_2 \subset \Delta$); since $e \not\subset \partial M$, one has $\text{int } e \subset \text{Int } M$, the link of e is a circle and $k \geq 3$, with $p, q \in \text{int } e$.

(1) *Product coordinates around σ .* There are a closed neighbourhood U of σ in M and a homeomorphism $\Phi : U \rightarrow \sigma \times d$ (d a closed disc with centre 0) with $U \cap \mathcal{T}^{(1)} = \sigma \mapsto \sigma \times \{0\}$, $U \cap f_i \mapsto \sigma \times r_i$ (the r_i being k spokes from 0 to ∂d , cyclically ordered), $U \cap \Delta \mapsto \sigma \times d_\Delta$ (the

sector between r_1 and r_2), $U \cap F \mapsto \{p, q\} \times d$, and $U \cap E \mapsto \sigma \times \rho$ (with $\rho \subset d_\Delta$ an arc from 0 to ∂d and $\rho \cap \partial d_\Delta = \{0\}$).

Construction (textbook-level): take a function h restricting to arc length on e , constant equal to $h(p)$ and $h(q)$ on the p -piece and the q -piece of F , and a submersion on each f_i and on E (the pieces being pairwise transverse with compatible prescribed values; local extension plus a partition of unity); take a vector field Y with $dh(Y) = 1$ tangent to e , to each f_i and to E , whose flow gives the product structure; and use Schoenflies to straighten $r_1, \dots, r_k$ and ρ simultaneously into straight radii, with the radius opposite ρ containing no spoke.

(2) *Extending E across e .* Take the initial part $\hat{\rho}$ of the radius opposite ρ (lying in $d \setminus d_\Delta$ between some adjacent pair of spokes r_j, r_{j+1} with $j \geq 3$, which exists since $k \geq 3$), with endpoint z . Put $C := \Phi^{-1}(\sigma \times \hat{\rho})$; then $E^+ := E \cup C$ is still a disc, with $E^+ \cap \mathcal{T}^{(2)} = E^+ \cap \mathcal{T}^{(1)} = \sigma$ and $E^+ \cap F = \tau \cup \{p, q\} \times \hat{\rho} =: \tau^+$ (an arc from $p^+ = (p, z)$ to $q^+ = (q, z)$), $\partial E^+ = \tau^+ \cup \sigma^+$ with $\sigma^+ := \sigma \times \{z\} \subset \text{int } \Delta_j$, and $F \pitchfork E^+$.

(3) *A product neighbourhood.* Take a product neighbourhood $N := E^+ \times [-2, 2]$ such that (α) $N \cap U = \Phi^{-1}(\sigma \times R)$ with R a thin rectangular neighbourhood in d of the diameter segment $\hat{\rho}^+ := \hat{\rho} \cup \rho$ (convex, containing 0 in its interior, each spoke crossing ∂R transversally exactly once); (β) $N \setminus U \subset \text{int } \Delta$ (since $E \setminus \text{int } U$ is a compact subset of $\text{int } \Delta$, a thin enough neighbourhood suffices); (γ) $F \cap N = \tau^+ \times [-2, 2]$ (exact inside U by the coordinates; outside U , the set $F \setminus \tau^+ \times (-2, 2)$ is a compact set disjoint from E^+ , so a thin enough N misses it). Then $N \cap \mathcal{T}^{(1)} = \sigma$, $N \cap \mathcal{T}^{(0)} = \emptyset$, $N \subset \text{int } \Delta \cup U$, and $F \cap N$ is a properly embedded disc Δ_0 in the 3-ball N .

(4) *Disc exchange inside the ball.* Take $z' \in \text{int } \hat{\rho}$ and put $E' := E \cup \Phi^{-1}(\sigma \times [0, z'])$, with $\tau' \subset \tau^+$ and $\sigma' := \sigma \times \{z'\} \subset \text{int } E^+$ correspondingly, $N' := E' \times [-1, 1] \subset N$ and $R' := [0, z']^+ \times [-1, 1]$. Then $G_0 := F \cap N' = \tau' \times [-1, 1] \subset \partial N'$, and $G_1 := E' \times \{\pm 1\} \cup \sigma' \times [-1, 1] = \overline{\partial N'} \setminus G_0$ is a disc with $\partial G_1 = \partial G_0$. So $\Delta_1 := (\Delta_0 \setminus \text{int } G_0) \cup G_1$ is a properly embedded disc in N with $\partial \Delta_1 = \partial \Delta_0$ and $\text{int } \Delta_1 \subset \text{int } N$.

Citation (textbook-level): two properly embedded PL discs in a 3-ball with the same boundary are ambient isotopic rel boundary (Alexander's theorem plus the Alexander trick). This gives an isotopy Ψ_t of N rel ∂N with $\Psi_1(\Delta_0) = \Delta_1$; extending by $\Psi_t = \text{id}$ on $M \setminus N$ gives an ambient isotopy supported in N , and $F' := \Psi_1(F) = (F \setminus \tau' \times (-1, 1)) \cup G_1$.

(5) *Edge-by-edge weight count.* Write $\lambda := \overline{\partial R' \setminus \partial d}$ (an arc with $0 \notin \lambda$ and $\lambda \cap r_i = \{x_i\}$ a single transverse point). Then $F' \cap U = \Phi^{-1}(\{p, q\} \times (d \setminus \text{int } R') \cup \sigma \times \lambda)$ and $F' \cap (N \setminus U) \subset \text{int } \Delta$. Since $0 \in \text{int } R'$ and $0 \notin \lambda$, one gets $F' \cap \sigma = \emptyset$; and $F' = F$ outside N . This is (D.34). And $F' \in \mathcal{F}(F)$: it avoids the vertices, is transverse to $\mathcal{T}^{(1)}$ (all intersection points are old ones, near which $F' = F$) and transverse to $\mathcal{T}^{(2)}$ (inside U , $\{p\} \times (d \setminus \text{int } R')$ meets $\sigma \times r_i$ in old transverse subarcs, and $\sigma \times \lambda$ meets $\sigma \times r_i$ transversally in $\sigma \times \{x_i\}$).

The new piece meets $\partial \Delta$ in two arcs $\sigma_\pm$ inside $\text{int } f_1$ and $\text{int } f_2$ parallel to σ and missing the edges; the thin piece $\Phi^{-1}(\sigma \times \lambda)$ runs once around e through all k wedges (connectedness of $U \cong \sigma \times d$ being exactly where $e \not\subset \partial M$ is used).

(6) *Arcs in faces.* Fix a face f_i around e and let a_p, a_q be the arcs of $F \cap f_i$ through p and through q . Then $a_p \neq a_q$: otherwise a_p would have both ends on e , a return arc, contrary to the standing assumptions. The components of $F' \cap f_i$ are the arcs of $F \cap f_i$ other than a_p, a_q (unchanged) together with the new arc $b_i := a'_p \cup \Phi^{-1}(\sigma \times \{x_i\}) \cup a'_q$ (with a'_p, a'_q truncated), whose ends $y_p \in e'$ and $y_q \in e''$ lie on edges $e', e'' \neq e$; it is an embedded arc, *not* a closed curve. Faces not around e are untouched. So F' has no closed curves in faces; and when $e' = e''$ the arc b_i is a return arc, removed by Lemma D.26 (an ambient isotopy each time, lowering w by exactly 2 and creating no closed curves), until none remain, giving $F'' \in \mathcal{F}(F)$ with $w(F'') \leq w(F) - 2$ satisfying the standing assumptions. $\square$

Lemma D.35 (straightening pieces with short boundary). *If inside Δ every piece is a disc and every piece boundary is a normal curve of triangle or quadrilateral type, then there is an ambient isotopy supported in $\text{int } \Delta$ turning all pieces simultaneously into standard (linear) normal triangles and quadrilaterals, leaving w and $F \cap \mathcal{T}^{(2)}$ unchanged.*

Proof. The family of boundary curves is given, and a standard realisation exists (arbitrarily many triangles, and at most one quadrilateral type — two quadrilateral curves of different types must intersect — so they can be taken disjoint); for two families of disjoint properly embedded discs with the same boundaries, “families of discs in a ball are unique rel boundary” (Alexander; textbook-level) gives an isotopy of Δ rel $\partial\Delta$, extended by the identity to M . $\square$

(Haken’s definition of a normal surface requires only “pieces are discs with boundary of length 3 or 4”, and downstream only the combinatorial data w and the normal coordinates are used; this lemma is cosmetic.)

Lemma D.36 (boundary faces). *Under the hypotheses of Theorem D.23: (a) every component of ∂F is essential in ∂M ; in particular there are no closed curves of F inside a face of ∂M , so Lemma D.25 on a boundary face is a contradiction rather than a move. (ii) Lemma D.26 holds verbatim on a boundary face $f \subset \partial M$: the push slides ∂F within ∂M and is still an ambient isotopy of M , with w dropping by exactly 2.*

Proof. **Half-ball push model P** (shared with (ii) and with step (4) of Lemma D.37). Let $B \subset M$ be a 3-ball with $\partial B = Q \cup E \cup G$ three discs, $Q \cap E = \tau$, $E \cap G = \sigma$ and $Q \cap G = b$ arcs meeting pairwise in $\{p, q\}$; with $B \cap \partial M = G$, $F \cap B = Q$, and E of the form $\sigma \times \rho$ near σ in the edge neighbourhood model (an interior point of $e \subset \partial M$ has a neighbourhood $s \times H$ with H a half-disc, ∂M corresponding to the diameter, and the faces through e to $s \times r_j$, with $r_0 = f$ and $r_{k+1} = f'$ the two boundary faces), with B locally between ρ and r_0 .

Then for every $\eta > 0$ there is an ambient isotopy Φ_t preserving ∂M , supported in the η -neighbourhood of B , with $F' := \Phi_1(F) = (F \setminus Q^+) \cup E^+$, where Q^+ is Q together with the η -collar of τ , and E^+ is the η -parallel copy of E on the far side of B , with $E^+ \cap \partial M = \sigma^+ \subset \text{int } f'$ parallel to σ , with E^+ meeting each interior face between ρ and r_{k+1} in one arc parallel to σ , and $E^+ \cap \mathcal{T}^{(1)} = \emptyset$. The construction follows step (4) of Lemma D.29, cutting the model to the half-space $\{y \geq 0\}$ (relative uniqueness of regular neighbourhoods; textbook-level); Φ_t preserves y and hence ∂M , and $\partial F'$ arises by sliding the arc b across G inside ∂M to σ^+ .

(a) Essentiality. Suppose a component $c \subset \partial F$ bounds a disc G_1 in ∂M . Take an innermost circle $c_0 = \partial G_0$ of $\partial F \cap G_1$, with $G_0 \cap F = c_0$. Incompressibility of F gives $c_0 = \partial E_0$ with $E_0 \subset F$ a disc; the component F_0 containing c_0 has $\text{int } E_0$ both open and closed in $F_0 \setminus c_0$, which is connected, so $F_0 = E_0$ is a disc component, excluded by “no disc components”. (If the definition of compressing disc requires $\partial D \subset \text{Int } F$, use instead the collar of $G_0 \times \{1\}$ in ∂M and conclude the same way from “annulus $\cup$ disc = disc”.) A closed curve c inside a boundary face is a whole component of ∂F bounding a disc in the face f , contradicting essentiality.

(ii) Let $\beta \subset \partial F \cap f$ be a return arc with $\partial\beta = \{p, q\} \subset e \subset \partial M$, $s = [p, q]$, $\beta \cup s = \partial\Delta_0$ and $\Delta_0 \subset f$ innermost ($\text{int } \Delta_0 \cap F = \emptyset$, hence $\text{int } s \cap F = \emptyset$). Let Δ' be the tetrahedron containing f , take its transverse product collar, and put $Q := \beta \times [0, 1]$, $\tau := \{p\} \times [0, 1] \cup \beta \times \{1\} \cup \{q\} \times [0, 1]$, $E := s \times [0, 1] \cup \Delta_0 \times \{1\}$, $G := \Delta_0$ and $B := \Delta_0 \times [0, 1]$. All the hypotheses of model P hold, with no topological assumption needed. This gives F' with $Q \cap \mathcal{T}^{(1)} = \{p, q\}$ and $E^+ \cap \mathcal{T}^{(1)} = \emptyset$, so $w(F') = w(F) - 2$ and $F' \in \mathcal{F}(F)$. The newly created components in each r_j and in f' are “an arc from p , an arc parallel to s , an arc from q ”, with both ends on edges of the face, hence arcs or return arcs, to be processed further; a closed curve in f' would contradict (a) and does not occur. $\square$

Lemma D.37 (the ball-push version for $e \subset \partial M$). *With $(D, e, p, q, \sigma, \tau, E)$ as in Lemma D.32 and $e \subset \partial M$, there is an ambient isotopy $F \mapsto F' \in \mathcal{F}(F)$ with $w(F') \leq w(F) - 2$ and F' having no closed curves in faces; any return arcs created are removed by Lemma D.26 and Lemma D.36(ii).*

Proof. (1) τ is inessential in F (using ∂ -incompressibility of F). Check that E is a candidate ∂ -compressing disc: $\text{int } \tau \subset \text{int } \Delta \subset \text{Int } M$, so $\tau \cap \partial F = \{p, q\}$; $E \cap F = \tau$; and $E \cap \partial M \subset E \cap \partial \Delta = \sigma \subset \partial M$, so $E \cap \partial M = \sigma$; and $\tau \cap \sigma = \{p, q\}$. ∂ -incompressibility of F therefore gives an arc $b \subset \partial F$ with $\partial b = \{p, q\}$ such that $\tau \cup b = \partial Q$ for a disc $Q \subset F$. Record: (Q1) $Q \cap \partial F = b$; (Q2) $b \cap \sigma = \{p, q\}$ (as $\sigma \cap F = \{p, q\}$); (Q3) $|Q \cap \mathcal{T}^{(1)}| \geq 2$ (as $p, q \in \tau \subset Q$).

(2) $\sigma \cup b$ bounds a disc G in ∂M with $G \cap F = b$ (using ∂ -irreducibility of M and (a)). $S_0 := E \cup Q$ is an embedded disc glued along the arc τ , with $\partial S_0 = \sigma \cup b$ and $S_0 \cap \partial M = \partial S_0$: a properly embedded disc in M (the definition of ∂ -irreducibility does not require general position with respect to F , so no push-off is needed). ∂ -irreducibility of M gives $\sigma \cup b = \partial G$ with $G \subset \partial M$ a disc.

Now $G \cap F = G \cap \partial F$; a component c' of ∂F meeting G but missing ∂G would satisfy $c' \subset \text{int } G$ and bound a disc, contradicting (a); so $c' \cap b \neq \emptyset$ and $c' = c$, the component containing b . Write $c = b \cup b'$; the set $\text{int } b'$ is connected and misses ∂G , and if it lay in $\text{int } G$ then $c \subset G$ would bound a disc in ∂M , contradicting (a). So $G \cap F = b$. Near σ , the disc G lies on one side of e , call it $r_0 = f$.

(3) The sphere $S := E \cup Q \cup G$ bounds a ball B with $F \cap B = Q$ (using irreducibility of M and the loop theorem). S is an embedded 2-sphere, and irreducibility of M gives $S = \partial B$ (if irreducibility is stated only for interior spheres, push G inwards along the collar of ∂M and add back the product block); and $B \cap \partial M = G$, $F \cap S = Q$. For another component F_j : $F_j \cap S = \emptyset$, and if $F_j \subset \text{int } B$ then F_j is closed and the loop theorem gives $F_j = S^2$, excluded. For the component F_0 containing Q : the set $R := F_0 \setminus (Q \setminus \tau)$ is connected with $R \cap S = \tau$ and $R \setminus \tau$ connected and missing S ; were it inside $\text{int } B$ then $F_0 \subset B$ and $\partial F_0 \subset G \cap F = b$ an arc, while ∂F_0 is a union of circles containing b , a contradiction. So $F \cap B = Q$.

(4) The push and the weight count. The data $(B, Q, E, G, \tau, \sigma, b)$ satisfy the hypotheses of model P (B lying locally between r_0 and ρ by the end of step (2)). Take η small enough that the η -collar of τ avoids the finite set $(F \cap \mathcal{T}^{(1)}) \setminus Q$. Model P gives an ambient isotopy preserving ∂M , with $F' = (F \setminus Q^+) \cup E^+$. As Φ_1 is the identity outside the cylinder Q^+ and $E^+ \cap \mathcal{T}^{(1)} = \emptyset$, one has $F' \cap \mathcal{T}^{(1)} = (F \cap \mathcal{T}^{(1)}) \setminus Q$, so by (Q3)

$$w(F') = w(F) - |Q \cap \mathcal{T}^{(1)}| \leq w(F) - 2. \quad (\text{D.38})$$

And $F' \in \mathcal{F}(F)$: it is properly embedded (Φ_1 preserves ∂M), avoids the vertices, is transverse to $\mathcal{T}^{(1)}$ (all intersection points are old) and transverse to $\mathcal{T}^{(2)}$ (E^+ meets each r_j and f' transversally by model P). As for the new components: a component of $F \cap \mathcal{T}^{(2)}$ either lies entirely in Q (deleted), or misses Q , or is an arc $a_p \neq a_q$ issuing from p or from q in a face on the side from ρ to r_{k+1} (equality would be a return arc, contrary to the standing assumptions); the new component $a_p \cup (\text{arc parallel to } \sigma) \cup a_q$ has its two ends on edges $e_1, e_2 \neq e$ and is not a closed curve; when $e_1 = e_2$ it is a return arc, removed by Lemma D.26 in interior faces and by Lemma D.36(ii) in f' , each removal lowering w by a further 2. $\square$

Proof of Theorem D.23 (assembly). Let $F_0 \in \mathcal{F}(F)$ minimise w (which exists, w taking non-negative integer values and $\mathcal{F}(F) \neq \emptyset$). Lemmas D.26 and D.36(ii) strictly lower w , so F_0 has no return arcs; removing the closed curves in faces one by one via Lemma D.25 keeps w non-increasing, hence still minimal (on boundary faces there are none, by Lemma D.36(a)), giving a minimiser F_1 satisfying the standing assumptions of §D.5.

If some piece were not a disc, Lemmas D.28 and D.29 (boundary tetrahedra included: the statements and proofs of D.28–D.32 hold verbatim there, and D.29 has already shown $B \subset \text{Int } M$)

would give $F' \in \mathcal{F}(F)$ with $w(F') < w(F_1)$, a contradiction; so (A) holds. If the boundary of some piece met some edge twice, Lemmas D.31, D.32 and D.33 (or D.37 when $e \subset \partial M$) would give $w(F') \leq w(F_1) - 2$, a contradiction; so (B) holds. By Lemma D.27 every piece boundary is of triangle or quadrilateral type, and by Lemma D.35 they can be straightened without changing w . So $N := F_1$ is normal and ambient isotopic to F , with $w(N) = \min_{\mathcal{F}(F)} w \leq w(F)$, which is (D.24). Any minimiser of w is likewise normal (after straightening). $\square$

Dependence list for §D.5. Where the topological hypotheses are used: incompressibility of F (D.25, D.28(1), D.29(1)(3), D.36(a), D.37(3)); ∂ -incompressibility of F (only D.37(1)); irreducibility of M (D.25, D.29(2), D.37(3)); ∂ -irreducibility of M (only D.37(2)); no sphere components (D.28(1), D.29(3), D.37(3)); no disc components (D.36(a)). Textbook-level citations: the loop theorem (D.28(1), D.29(3), D.37(3), D.53); Schoenflies on a sphere (D.28(2), D.31, D.32(1)); uniqueness of PL regular neighbourhoods / the ball-push model (D.29(4), D.36); Alexander's theorem with the Alexander trick (D.33(4), D.35); the product coordinate construction (D.33(1)). All lemmas are same-family output, awaiting independent audit.

D.6 The weight–area inequality (AW)

Setting (S). $M := E(K)$ is compact and orientable with ∂M a torus and M irreducible (§1 of the main text). g is a smooth Riemannian metric on M (the metric of §5 of the main text will do; convexity of ∂M is nowhere used in this section). J is a C^∞ foliation of ∂M by closed curves (preferred longitudes), with the *sole requirement* that $L_J := \sup\{\text{length}_g(\ell) : \ell \text{ a leaf of } J\} < \infty$ (satisfied by the J constructed in §5 of the main text, the image of a linear longitudinal foliation under a diffeomorphism).

$\mathcal{T}$ is a triangulation of M with ∂M a subcomplex, satisfying **(T1)** every edge is a piecewise C^1 regular embedded arc of finite length, and **(T2)** an edge not contained in ∂M meets ∂M only at its endpoints, transversally. (T1) and (T2) cost no generality: the validity of (AW) and its constants are unchanged under an ambient isotopy of $\mathcal{T}$ (since $\psi_1(N)$ is normal for $\psi_1(\mathcal{T})$ with the same weight), and any smooth or PL triangulation can be moved by an ambient isotopy to satisfy them; (T1) is also the minimal convention making it meaningful to put w and Area in one inequality.

$S \subset M$ is compact, connected, orientable, C^∞ , neatly properly embedded, incompressible, with ∂S a leaf of J (hence connected and essential in ∂M); S is two-sided.

Lemma D.39 (framing subarc by subarc). *M orientable makes TM trivial (Stiefel; textbook-level). Fix a global frame (Y_1, Y_2, Y_3) ; fix a flat metric on ∂M and take two generating fields Z_1, Z_2 of its translation group, extended trivially through the collar $\partial M \times [0, \varepsilon_0)_r$, everywhere tangent to the level sets $\{r = \text{const}\}$. Put $B := \{p \in G \cap \{r \leq \varepsilon_0/2\} : \dot{e}(p) \in \ker dr_p\}$ (with G the union of the interior edges; at a corner it suffices that some one-sided tangent qualifies); B is compact and, by (T2), $B \cap \partial M = \emptyset$, so $\rho_0 := \min_B r > 0$ (take $\varepsilon_0/2$ if $B = \emptyset$).*

Put $\epsilon := \frac{1}{2} \min(\rho_0, \varepsilon_0)$ with cut-offs $\chi_\partial(r) = 1$ for $r \leq \epsilon$ and $= 0$ for $r \geq \frac{3}{2}\epsilon$; $\chi_{\text{int}}(r) = 0$ for $r \leq \frac{1}{4}\epsilon$ and $= 1$ for $r \geq \frac{1}{2}\epsilon$. Define five C^∞ fields

$$X_i := \chi_{\text{int}} Y_i \ (i = 1, 2, 3), \quad X_4 := \chi_\partial Z_1, \quad X_5 := \chi_\partial Z_2, \quad (\text{D.40})$$

all tangent to ∂M along ∂M . For $u \in \mathbb{R}^5$ let Θ_u be the time-one flow of $\sum u_i X_i$: a diffeomorphism preserving ∂M , ambient isotopic to the identity through $t \mapsto \Theta_{tu}$; and on ∂M , Θ_u is exactly the translation by $+(u_4, u_5)$, independently of u_1, u_2, u_3 . Take as parameter domain the open cube $U_\delta := (-\delta, \delta)^5$ (δ small enough that $(-\delta, \delta)^2$ embeds in the flat ∂M). Then there is a subdivision of each interior edge into finitely many C^1 regular subarcs such that each subarc σ carries a pair of indices $I_\sigma = \{i, j\} \subset \{1, \dots, 5\}$ with $\{\dot{e}(s), X_i(e(s)), X_j(e(s))\}$ a basis of $T_{e(s)}M$ for all $s \in \sigma$.

Proof. Let p lie on an interior edge. If $r(p) \geq \frac{1}{2}\epsilon$, then $X_{1,2,3}(p) = Y_{1,2,3}(p)$ is a basis and, expanding $\dot{e}(p) \neq 0$ in it, some coefficient (the k -th, say) is non-zero; take $\{i, j\} = \{1, 2, 3\} \setminus \{k\}$. If $r(p) \leq \epsilon$, then X_4, X_5 span $\ker dr_p$, while $r(p) < \rho_0$ implies $p \notin B$, that is $\dot{e}(p) \notin \ker dr_p$; take $\{4, 5\}$. The two cases cover every p ; being a basis is an open condition, so compactness gives a finite cover and a subdivision into finitely many subarcs, with the corners taken as subdivision points. $\square$

(This construction needs no global 2-plane field along the 1-skeleton — that route is obstructed by $w_1 \in H^1(G; \mathbb{F}_2)$ — but only parallelisability of TM together with a piecewise choice of indices.)

Lemma D.41 (a local diffeomorphism-type Crofton estimate). *Let σ be a subarc of an interior edge as in Lemma D.39 with $I_\sigma = \{i, j\}$, extended at both ends to a C^1 regular embedded arc $\hat{\sigma} : (a, b) \rightarrow M$ on an open interval (at a corner, extend by another C^1 arc; at an endpoint v lying on ∂M , do not cross v , taking (s_0, b') not containing v). Write $u = (u', u'')$ with $u' = (u_i, u_j)$, $U'_\delta = (-\delta, \delta)^2$, $U''_\delta = (-\delta, \delta)^3$ and $\Psi_{u''}(s, u') := \Theta_{(u', u'')}(\hat{\sigma}(s))$. Then there are $\delta > 0$ and $\Lambda_\sigma < \infty$ (depending only on $\mathcal{T}, g, (X_k), \delta$, not on S nor on u'') such that $\Psi_{u''}$ is a local diffeomorphism for every $|u''| < \delta$, and for every compact C^1 surface $S \subset M$*

$$\int_{U'_\delta} \#(S \cap \Theta_{(u', u'')}(\sigma)) du' \leq \Lambda_\sigma \text{Area}(S). \quad (\text{D.42})$$

Proof. (i) At $u = 0$ one has $\partial_s \Psi = \dot{\sigma}$, $\partial_{u_i} \Psi = X_i$ and $\partial_{u_j} \Psi = X_j$, a basis; invertibility is an open condition, so taking a compact interval $[a', b'] \subset (a, b)$ containing $\bar{\sigma}$ (or $\bar{\sigma} \setminus \{v\}$), compactness gives δ with $d\Psi_{u''}$ invertible everywhere on $K := [a', b'] \times \overline{U'_\delta} \times \overline{U''_\delta}$; there are finitely many edges and subarcs, so take the smallest such δ .

(ii) *Uniform constants.* On K the map $d\Psi_{u''}$ is continuous and invertible, so its smallest singular value has a lower bound $\lambda > 0$ independent of u'' . Moreover there is $\eta > 0$ such that for every $|u''| \leq \delta$ the map $\Psi_{u''}$ is injective on every convex subset of diameter $< \eta$: otherwise take $u''_n, x_n \neq y_n$ with $|y_n - x_n| \rightarrow 0$ and $\Psi_{u''_n}(x_n) = \Psi_{u''_n}(y_n)$; first pass to a subsequence with $u''_n \rightarrow u''_*$, $x_n, y_n \rightarrow z$ and $(y_n - x_n)/|y_n - x_n| \rightarrow v$, then fix a chart ϕ at $q := \Psi_{u''_*}(z)$; uniform continuity puts the whole segment $[x_n, y_n]$ into that chart for large n , and applying the fundamental theorem of calculus to $\phi \circ \Psi_{u''_n}$ and dividing by $|y_n - x_n|$, uniform continuity of $d(\phi\Psi)$ on K gives $d(\phi\Psi_{u''_*})(z)v = 0$, contradicting invertibility. Hence each fibre is an η -separated subset of a compact convex set, of cardinality $\leq m$ with m independent of u'' .

(iii) *The area formula twice.* $Z := \Psi_{u''}^{-1}(S) \subset (a', b') \times U'_\delta$ is a boundaryless C^1 surface (this is exactly why the subarcs were extended; and Θ_u preserves $\text{Int } M$ with $\hat{\sigma} \subset \text{Int } M$, so ∂S is never met), and $\Psi_{u''}|_Z : Z \rightarrow S$ is a local diffeomorphism, so by the area formula (Federer; textbook-level) $\mathcal{H}^2(Z) \leq \lambda^{-2} \int_S \#(\Psi_{u''}|_Z)^{-1}(p) d\mathcal{H}^2 \leq \lambda^{-2} m \text{Area}(S)$. The projection $\pi : Z \rightarrow U'_\delta$ is 1-Lipschitz, so the area formula again gives $\int_{U'_\delta} \#\pi^{-1}(u') du' \leq \mathcal{H}^2(Z)$; and $\#\pi^{-1}(u') \geq \#(S \cap \Theta_{(u', u'')}(\sigma))$ (the extension only increases the count). Take $\Lambda_\sigma := \lambda^{-2} m$. $\square$

Lemma D.43 (boundary edges: a translation-family Poincaré formula). (a) *Let $\partial M \cong T^2$ carry the flat structure fixed in Lemma D.39, and let e, γ be two Lipschitz curves of flat lengths L_e, L_γ . Then $\int_{T^2} \#(\gamma \cap (e + v)) dv \leq L_e L_\gamma$.* (b) *Writing $u'' = (u_4, u_5)$ and $U_\delta^{45} = (-\delta, \delta)^2$, there is $D_0 = D_0(\mathcal{T}, g, J) < \infty$ such that, for each fixed (u_1, u_2, u_3) ,*

$$\int_{U_\delta^{45}} \sum_{e \subset \partial M} \#(S \cap \Theta_u(e)) du_4 du_5 \leq D_0. \quad (\text{D.44})$$

Proof. (a) Put $\Phi(s, t) := \gamma(t) - e(s)$ (flat subtraction), so that $\#(\gamma \cap (e + v)) = \#\Phi^{-1}(v)$; Φ is Lipschitz, and the area formula gives $\int_{T^2} \#\Phi^{-1}(v) dv = \iint |\det(\dot{\gamma}, -\dot{e})| dt ds \leq L_e L_\gamma$ (arc-length

parametrisation). No transversality is needed — which is the reason for choosing a family of translations.

(b) By Lemma D.39, $\Theta_u|_{\partial M}$ is the translation by (u_4, u_5) ; neatness gives $S \cap \partial M = \partial S$, and Θ_u preserves ∂M , so for a boundary edge $\#(S \cap \Theta_u(e)) = \#(\partial S \cap (e + (u_4, u_5)))$. And ∂S is a leaf of J , of flat length $\leq \kappa L_J =: L_J^{\text{flat}}$ (κ being the bi-Lipschitz constant between $g|_{\partial M}$ and the flat metric). Applying (a), restricting the integral to U_δ^{45} and summing over the boundary edges gives the claim with $D_0 := L_J^{\text{flat}} \sum_{e \in \partial M} L_e$. $\square$

Proposition D.45 (existence of a good position). *With $|U'_\delta| = |U_\delta^{45}| = (2\delta)^2$, put*

$$C := (2\delta)^{-2} \sum_{\sigma} \Lambda_{\sigma} \quad (\sigma \text{ over all interior edge subarcs}), \quad D_1 := D_0 (2\delta)^{-2}. \quad (\text{D.46})$$

Then there is a set $\mathcal{G} \subset U_\delta$ of positive measure such that for every $u \in \mathcal{G}$: (1) S is transverse to $\Theta_u(e)$ at every point of every closed edge e (endpoints included) and S contains no image under Θ_u of a boundary vertex, so that $\#(S \cap \Theta_u(\mathcal{T}^{(1)})) < \infty$; and (2) $\#(S \cap \Theta_u(\mathcal{T}^{(1)})) \leq C \text{Area}(S) + D_1$.

Proof. (i) *Measurability.* With $\tilde{\Psi}(s, u) := \Theta_u(\hat{\sigma}(s))$, the set $W := \tilde{\Psi}^{-1}(S)$ is closed, hence σ -compact; the sets $\{u : \#W_u \geq n\}$ are countable unions of continuous images of σ -compact sets, hence Borel, so $u \mapsto \#W_u$ is Borel measurable, and likewise for boundary edges. The non-transversality set $\mathcal{N} := \{(s, u) \in \bar{\sigma} \times U_\delta : \tilde{\Psi}(s, u) \in S, d\tilde{\Psi}(\partial_s) \in TS\}$ (restricted to $\bar{\sigma}$, excluding the artificial endpoints a', b') is the preimage of a closed set ($TS \subset TM$ being closed), hence σ -compact, and its u -projection is Borel; so the step “each u'' -slice null + measurable $\Rightarrow$ globally null” is legitimate.

(ii) *The measure part.* For an interior subarc σ , (D.42) with Fubini (Λ_σ being independent of u'' , and $U_\delta = U'_\delta \times U''_\delta$ splitting exactly for each choice of coordinates, with no dimensional constant unaccounted for) gives $\frac{1}{|U_\delta|} \int_{U_\delta} \#(S \cap \Theta_u(\sigma)) du \leq \frac{\Lambda_\sigma}{|U'_\delta|} \text{Area}(S)$; summing over σ covers the interior edges. By (D.44) with Fubini, the average over boundary edges is $\leq D_1$. Adding, the average of the non-negative Borel function $u \mapsto \#(S \cap \Theta_u(\mathcal{T}^{(1)}))$ over U_δ is at most $C \text{Area}(S) + D_1$, so the set where it is at most this average has positive measure.

(iii) *The transversality part.* *Interior points and interior endpoints:* $Z = \Psi_{u''}^{-1}(S)$ is a boundaryless C^1 surface and $\pi : Z \rightarrow U'_\delta$ is a C^1 map from a 2-manifold to a 2-manifold with $T_z Z = (d\Psi_{u''})^{-1}(T_p S)$ and $\ker d\pi = \langle \partial_s \rangle$, so $d\pi|_{T_z Z}$ is onto $\iff \dot{\sigma}_u(s) \notin T_p S \iff$ transversality. Sard’s theorem (C^1 , $2 \rightarrow 2$; textbook-level) gives that regular values have full measure; the interior subarcs satisfy $\bar{\sigma} \subset (a', b')$, so both endpoints are covered. *Edge endpoints v lying on ∂M (boundary vertices included):* the map $(u_4, u_5) \mapsto \Theta_u(v) = v + (u_4, u_5)$ is a global diffeomorphism onto its image, and $\{(u_4, u_5) : \Theta_u(v) \in S\} = (\partial S - v) \cap U_\delta^{45}$ (by neatness), with ∂S a smooth closed curve, of $\mathcal{H}^2$ -measure zero; Fubini gives $\Theta_u(v) \notin S$ for a.e. u , so transversality holds vacuously. *Boundary edges:* apply Sard to the map Φ of (a), so that for a.e. (u_4, u_5) one has $\partial S \pitchfork (e + (u_4, u_5))$ inside ∂M ; and neatness gives $T_p S \cap T_p \partial M = T_p \partial S$, so transversality inside ∂M is equivalent to transversality in M . There are finitely many subarcs, edges and vertices, so a finite intersection of full-measure sets is of full measure, and meets the positive-measure set of (ii) in a set of positive measure. $\square$

Lemma D.47 (restoring general position, at a cost depending only on $\mathcal{T}$). *Let $F \subset M$ be a compact C^∞ neatly embedded surface transverse to every edge of $\mathcal{T}$ (endpoints included). Then there is a C^1 -small ambient isotopy θ (supported in small balls near $\mathcal{T}^{(0)}$) such that $\theta(F)$ is in general position with respect to $\mathcal{T}$ (and $\partial\theta(F)$ with respect to the induced triangulation of ∂M), with*

$$w(\theta(F)) \leq \#(F \cap \mathcal{T}^{(1)}) + \sum_{v \in \mathcal{T}^{(0)}} \deg(v). \quad (\text{D.48})$$

Proof. Suppose $v \in F$ for a vertex v (the other vertices are not moved; by Proposition D.45(1) the case $v \in \partial M$ does not in fact arise here).

(1) Take C^∞ adapted coordinates $(x_1, x_2, x_3) : B_v \rightarrow \mathbb{R}^3$ at v with $F \cap B_v = \{x_3 = 0\}$. Each edge germ γ_k ($k \leq \deg v$, parametrised by arc length) is transverse to F at v , that is $\frac{d}{dt}(x_3 \circ \gamma_k)(0) \neq 0$. Shrink B_v (radius ρ) so that: each edge germ satisfies $|\frac{d}{dt}(x_3 \circ \gamma_k)| \geq c > 0$ and $|\frac{d}{dt}(x_1, x_2) \circ \gamma_k| \leq C_1$ inside B_v ; B_v meets only the edges containing v , each in a single monotone germ (e being embedded, $e([t_0, L_e])$ has positive distance from v , and ρ is taken smaller than that and within the first corner); and the B_v are pairwise disjoint.

(2) *Pushing along x_3 .* Take a cut-off $\chi \in C_c^\infty(B_v)$ with $0 \leq \chi \leq 1$, $\chi \equiv 1$ inside radius $\rho/2$ and $|\nabla \chi| \leq 3/\rho$; let θ_v be the time-one flow of the field $\tau \chi \partial_{x_3}$. The flow moves points only along x_3 , so $\theta_v(F \cap B_v)$ is the graph $\{x_3 = \psi_\tau(x_1, x_2)\}$ with $0 \leq \psi_\tau \leq \tau$. Differentiating the ODE $\dot{z} = \tau \chi(x_1, x_2, z)$ in the parameters (x_1, x_2) , the variational equation $\dot{p} = \tau[\nabla_{(x_1, x_2)} \chi + (\partial_{x_3} \chi)p]$ with $p(0) = 0$ gives

$$|\nabla_{(x_1, x_2)} \psi_\tau| \leq \frac{3\tau}{\rho} e^{3\tau/\rho} \leq 3e^3 \cdot \frac{\tau}{\rho} \quad (\tau \leq \rho), \quad (\text{D.49})$$

where $3e^3$ is an absolute constant. The difference function $\Phi_k(t) := x_3(\gamma_k(t)) - \psi_\tau((x_1, x_2)(\gamma_k(t)))$ satisfies $|\dot{\Phi}_k| \geq c - 3e^3 C_1 \tau / \rho$; first fix ρ (together with c and C_1), then take $\tau < \min\{\rho/2, c\rho/(18e^3 C_1)\}$, so that $|\dot{\Phi}_k| \geq c/2$ and Φ_k is strictly monotone with at most one zero, transverse there: each edge germ meets $\theta_v(F)$ in at most one point inside B_v . And $\chi(0) = 1$ gives $\psi_\tau(0, 0) > 0$, so $\Phi_k(0) < 0$ and $v \notin \theta_v(F)$. The number of intersection points inside B_v therefore goes from 1 to at most $\deg(v)$, and summing over the vertices lying on F gives (D.48). (Transversality does *not* by itself bound the number of intersection points by 1, so a gradient estimate of type (D.49) cannot be dispensed with.)

(3) *Restoring transversality to the 2-skeleton.* “Keeping finitely many transverse edge intersections and creating no new ones” is a C^1 -open condition, “transverse to finitely many 2-simplices” is residual (Thom transversality), so the intersection is non-empty; and a C^1 -small perturbation of a compact embedded surface is realised as an ambient isotopy by the isotopy extension theorem (textbook-level). $\square$

Theorem D.50 (the weight–area inequality (AW); K non-trivial). *In the setting (S), put*

$$C := (2\delta)^{-2} \sum_{\sigma} \Lambda_{\sigma}, \quad D := D_0(2\delta)^{-2} + \sum_{v \in \mathcal{T}^{(0)}} \deg(v), \quad (\text{D.51})$$

both depending only on $\mathcal{T}, g, J$ and on the once-and-for-all choices $(X_k)_{k \leq 5}$ and δ of Lemma D.39, and independent of S and of the genus h of S . Then for every S as in (S) there is a normal surface N with respect to $\mathcal{T}$, ambient isotopic to S , with

$$w(N) \leq C \cdot \text{Area}(S) + D. \quad (\text{D.52})$$

In particular, for fixed h and a , the ambient isotopy class of every S of genus h and area $\leq a$ contains a normal representative of weight $\leq Ca + D$ — and only this weak form is used by Lemma 5.1 of the main text.

Proof. Take $u \in \mathcal{G}$ by Proposition D.45 and put $S_1 := \Theta_u^{-1}(S)$: since Θ_u preserves ∂M and is ambient isotopic to the identity, S_1 is ambient isotopic to S ; and $\#(S_1 \cap \mathcal{T}^{(1)}) = \#(S \cap \Theta_u(\mathcal{T}^{(1)})) \leq C \text{Area}(S) + D_1$, with S_1 transverse to every edge.

By Lemma D.47 take $S_2 := \theta(S_1)$, in general position with respect to $\mathcal{T}$, with $w(S_2) \leq C \text{Area}(S) + D$, still ambient isotopic to S . As S_2 is diffeomorphic and isotopic to S , it is still compact, connected, orientable, two-sided, properly embedded and incompressible, neither a sphere

nor a disc, and ∂S_2 is a single essential curve in ∂M (isotopic to ∂S); by Corollary D.22 it is ∂ -incompressible, and M is irreducible and ∂ -irreducible (K being non-trivial).

Applying Theorem D.23 to S_2 gives an ambient isotopy carrying S_2 to a normal surface N with $w(N) \leq w(S_2)$. Concatenating the three ambient isotopies, N is ambient isotopic to S with $w(N) \leq w(S_2) \leq C \text{Area}(S) + D$. $\square$

The proof in two sentences. (1) The a priori reason for the weight bound is integral geometry, not a local bi-Lipschitz comparison: (AW) allows S to be replaced by any representative of its ambient isotopy class, equivalently allows $\mathcal{T}$ to be moved within an isotopy neighbourhood of itself; averaging over a family of such moves, the area formula gives “average weight $\leq C \cdot \text{area}$ ”, so there *exists* a position in which the weight is linearly controlled by the area — unboundedly many folds occur for some fixed position, but not for the average position. (2) The passage from “a non-normal surface of low weight” to “a normal surface of low weight” is Theorem D.23, normalisation not increasing the weight; incompressibility and the irreducibility and ∂ -irreducibility of $E(K)$ are used only at that step.

The obstruction “a discrete quantity against a two-dimensional integral quantity: the dimensional mismatch cannot be cured by choosing a triangulation” is valid only for “choose one good triangulation”: any fixed $\mathcal{T}$ admits surfaces of small area and large weight. What this proof chooses is not a triangulation but an average position in a two-parameter family, and the dimensional mismatch is bridged by Crofton / the area formula.

Clarification of the quantifiers. (AW) requires $\min\{w(N) : N \text{ normal}, N \simeq S\}$ to be controlled by $\text{Area}(S)$, *not* $w(S)$ to be controlled by $\text{Area}(S)$ — the latter is false for any fixed $\mathcal{T}$ (a surface of small area crossing an edge repeatedly inside a thin neighbourhood).

Dependence list for §D.6. The topological input enters only at the last step: Corollary D.22 (∂ -incompressibility, and ∂ -irreducibility of $E(K)$, conditional through Theorem D.15 on Dehn’s lemma) and Theorem D.23 (normalisation). The analytic inputs are all textbook-level: Stiefel parallelisability (Lemma D.39), the area formula (Lemmas D.41 and D.43), Sard (Proposition D.45), Thom transversality and isotopy extension (Lemma D.47). The dependence graph of C and D is acyclic: ρ_0, ϵ are determined by $(\mathcal{T}, g)$; $\delta, \lambda, m, \Lambda_\sigma$ by $(\mathcal{T}, g, (X_k))$; D_0 additionally by J (through L_J); and $\sum \deg(v)$ only by $\mathcal{T}$. The only things depending on S are the selected $u \in \mathcal{G}$, the isotopy θ of Lemma D.47 and the normalisation isotopy of Theorem D.23, all of which appear only on the conclusion side. *Not used in this section:* the isoperimetric inequality, the monotonicity formula, curvature estimates for stable minimal surfaces, Kneser–Haken finiteness, Gauss–Bonnet, geometrisation, or any entry of the ledger of §12 of the main text (except Dehn’s lemma, entering indirectly through Corollary D.22).

Proposition D.53 (the case of the unknot). *Let K be trivial, so $M = E(K) \cong S^1 \times D^2$ (Corollary D.19), and let S be as in (S). Then S is an essential disc; and, with C, D as in (D.51) and any fixed normal essential disc N_0 , setting $D' := \max\{D, w(N_0)\}$, every such S is ambient isotopic to N_0 and $w(N_0) \leq C \text{Area}(S) + D'$.*

Proof. S incompressible gives $\pi_1(S) \hookrightarrow \pi_1(M) = \mathbb{Z}$ (loop theorem); the fundamental group of a genus- h surface with one boundary component is free of rank $2h$, and injecting into $\mathbb{Z}$ forces $h = 0$, so S is a disc whose boundary is a preferred longitude, that is the meridian of the solid torus, hence essential. Essential discs in a solid torus are pairwise ambient isotopic (textbook-level). Theorem D.23 is *not available* here (a solid torus is not ∂ -irreducible); instead use “in an irreducible manifold with compressible boundary there exists an essential disc normal with respect to $\mathcal{T}$ ” (Haken; textbook-level) to take N_0 . $\square$

D.7 Haken finiteness

Theorem D.54 (finitely many normal surfaces of bounded weight). *Let $\mathcal{T}$ be a triangulation of a compact 3-manifold M with t tetrahedra and $\Delta_{\max}$ an upper bound for the number of tetrahedra containing a given edge. Then for each $W \geq 0$ the normal surfaces with respect to $\mathcal{T}$ of weight $\leq W$ fall into at most $(\lfloor \Delta_{\max} W/3 \rfloor + 1)^{7t}$ normal isotopy classes, hence into finitely many ambient isotopy classes.*

Proof. Each normal disc piece has 3 corners (triangle) or 4 corners (quadrilateral), each corner corresponding to an intersection point of N with an edge; conversely each point $p \in N \cap \mathcal{T}^{(1)}$ contributes at most $\deg(e_p) \leq \Delta_{\max}$ corners (at most one corner in each tetrahedron around the edge of p). So

$$3 \cdot \#\{\text{normal disc pieces}\} \leq \sum_{\text{pieces}} \#\text{corners} = \sum_{p \in N \cap \mathcal{T}^{(1)}} \deg(e_p) \leq \Delta_{\max} w(N). \quad (\text{D.55})$$

Hence when $w(N) \leq W$ every entry of the normal coordinate vector $x(N) \in \mathbb{Z}_{\geq 0}^{7t}$ is at most $\Delta_{\max} W/3$, and there are only finitely many such vectors (the count above). A normal surface is determined up to normal isotopy by its normal coordinates (**textbook-level**, Haken; not proved here), and normal isotopy is a special case of ambient isotopy, so there are also finitely many ambient isotopy classes. $\square$

This is an elementary count and does *not* require the Kneser–Haken finiteness theorem. Two reminders: (1) “normal isotopy” here means an isotopy preserving each simplex and sliding normal pieces along the straight-line families inside the simplices (Haken’s usage); it is a special case of triangulation-preserving ambient isotopy and is unrelated to the “neat isotopy” of §4 of the main text (Convention D.6). (2) The finiteness is in terms of *ambient isotopy classes*; if downstream the unit is the $\text{rel-}\partial$ isotopy class, see the convention condition in §D.8(c).

Corollary D.56 (the finiteness needed by Lemma 5.1). *Let K be non-trivial and (S) hold. For fixed h and a , the incompressible surfaces S of genus h , area $\leq a$ and boundary a leaf of J fall into finitely many ambient isotopy classes.*

Proof. Theorem D.50 gives each class a normal representative of weight $\leq Ca + D$, and Theorem D.54 gives finitely many ambient isotopy classes of such representatives. $\square$

(C and D are independent of h , so “fixed h ” could be dropped; the main text does not use the strengthening.)

D.8 What this appendix is still conditional on; call sites in the main text

(a) Research-level inputs (one only, registered). *Dehn’s lemma* (Theorem D.14, Papakyriakopoulos 1957), used only by Theorem D.15; adopting the (AW) route adds **one use site and one independent source** to the ledger. It does not overlap the other ledger entries of §12 of the main text; non-circularity is explained after Theorem D.14.

(b) Textbook-level citations (to be flagged explicitly; not proved here).

- *The loop theorem / the π_1 -injective form of Dehn’s lemma* (an incompressible two-sided surface that is neither a sphere nor a disc is π_1 -injective; Papakyriakopoulos–Stallings): Theorem D.20(c), Lemmas D.28(1), D.29(3), D.37(3), Proposition D.53.
- *Schoenflies (on S^2)*: Lemmas D.28(2), D.31, D.32(1), D.33(1), D.36(a).

- *Alexander’s theorem (Schoenflies in S^3):* Proposition D.16, Corollary D.19 (not used by Theorem D.15).
- *Alexander’s theorem with the Alexander trick* (properly embedded PL discs in a 3-ball with the same boundary are ambient isotopic rel boundary): Lemmas D.33(4), D.35.
- *Uniqueness of PL regular neighbourhoods / smooth collars with corners (the ball-push model):* Lemma D.29(4), model P of Lemma D.36.
- *The isotopy extension theorem:* Lemma D.47(3).
- *Stiefel (orientable 3-manifolds are parallelisable):* Lemma D.39. *The area formula (Federer):* Lemmas D.41, D.43. *Sard:* Proposition D.45. *Thom transversality:* Lemma D.47(3).
- *Tubular neighbourhoods, Mayer–Vietoris, classification of compact surfaces, classification of arcs in an annulus:* §D.2–§D.3.
- *A normal surface is determined by its normal coordinates up to normal isotopy; an irreducible manifold with compressible boundary contains a normal essential disc (Haken):* Theorem D.54, Proposition D.53.
- *Haken’s normalisation theorem itself:* **not cited** — Theorem D.23 is proved in §D.4–§D.5 (same-family, awaiting audit).

(c) **Mathematical conditions (written into the statements).**

- The setting (S) of Theorem D.50: $L_J < \infty$; $\mathcal{T}$ satisfying (T1) and (T2) (no loss of generality); S a C^∞ neatly embedded incompressible surface with ∂S connected and essential.
- The hypotheses of Theorem D.23 are narrower than those of the source (no disc components); see the issues log.
- **Convention condition:** the ambient isotopies of Theorem D.50 move ∂M , so ∂N is no longer the original leaf. If the vertices of §5 of the main text are taken to be “rel ∂ ” isotopy classes, or classes with the boundary fixed on a given leaf, then passing from the coarse finiteness of Corollary D.56 to the fine finiteness requires one further step (dragging the boundary along ∂M may induce a boundary Dehn twist); if §1 of the main text uses the convention “ambient isotopy, the boundary longitude being allowed to move on the boundary torus”, there is nothing to do. *This appendix does not answer this for the main text; the downstream statement must be confirmed.*

(d) **Audit status.** All contents are same-family Fable 5.1 output; §D.2 was cross-checked by two independent same-family proofs, §D.3 audited along four same-family routes and corrected, and the lemmas of §D.5 and the steps of §D.6 rewritten after multi-route same-family audit; **cross-engine audit 0/3**.

(e) **Index of call sites in the main text.**

Location	Invocation	Supplied by
§1 (“the exterior of the unknot is a solid torus”)	$K \text{ trivial} \Rightarrow E(K) \cong S^1 \times D^2$	Corollary D.19
§1 / §12.1 (list of standing hypotheses)	$K \text{ non-trivial} \Rightarrow E(K) \text{ } \partial\text{-irreducible}$	Theorems D.15 and D.18
§5, Lemma 5.1 (reproved along the (AW) route)	fixed h , area $\leq a \Rightarrow$ a normal representative of weight $\leq Ca + D$ exists; bounded weight $\Rightarrow$ finitely many	Theorems D.50 and D.54, Corollary D.56
§5 (∂ -incompressibility of vertex surfaces, a hypothesis of normalisation)	incompressible + ∂S a single essential curve $\Rightarrow \partial$ -incompressible	Theorem D.20, Corollary D.22
§12.1 / §12.2, hypothesis tables	registration of Dehn’s lemma; the textbook-level list; the convention condition	(a), (b), (c) of this section

E Self-contained items for divergence-form elliptic equations

On provenance: the contents of this appendix were produced by Fable 5.1 (five same-family proof items: the Dirichlet problem, global $C^{1,\alpha}$ regularity, the strong minimum principle with the Hopf lemma, isothermal coordinates up to the boundary, and the similarity principle with unique continuation) and audited within that same family; they have received **no cross-engine audit** (criterion ③ counts 0/3). See Appendix G.

This appendix supplies the divergence-form elliptic inputs of Lemma 5.3 of the main text and of formulas (7.8)–(7.11) and Lemma 7.2.A: each conclusion is given with a complete statement, a proof skeleton and the load-bearing estimates; routine manipulations are compressed to “a direct computation”, but *no hypothesis is dropped*. Facts of textbook level are flagged explicitly and are not entered in the ledger; what remains conditional is stated in the theorems and collected in §E.7. Theorems, lemmas and displays share one counter *E.n*.

E.1 Existence and uniqueness for the Dirichlet problem

Common setting (E-set). $\Omega \subset \mathbb{R}^n$ is a bounded open set and $d := \text{diam } \Omega$. The map $A : \Omega \rightarrow \mathbb{R}^{n \times n}$ is measurable and there are $0 < \lambda \leq \Lambda$ with $A(x)\xi \cdot \xi \geq \lambda|\xi|^2$ and $|A(x)\xi| \leq \Lambda|\xi|$ (a.e. x , all ξ). $H_0^1(\Omega)$ is the closure of $C_c^\infty(\Omega)$ in $H^1(\Omega)$ (this is the definition, not a theorem); $\|\cdot\|$ denotes the $L^2(\Omega)$ norm.

Lemma E.1 (Poincaré, slab version). *Assume (E-set). Then $\|v\| \leq d \|\nabla v\|$ for every $v \in H_0^1(\Omega)$.*

Proof. Ω is bounded, so $\Omega \subset \{a < x_n < a + d\}$. First take $v \in C_c^\infty(\Omega)$, extended by zero; for fixed x' , from $v(x', a) = 0$, the fundamental theorem of calculus and Cauchy–Schwarz, $|v(x', x_n)|^2 \leq d \int_a^{a+d} |\partial_n v(x', t)|^2 dt$; integrating in x_n and then in x' (Fubini, non-negative integrand) gives $\|v\|^2 \leq d^2 \|\partial_n v\|^2 \leq d^2 \|\nabla v\|^2$. A general $v \in H_0^1$ is approximated by C_c^∞ functions, and both sides are H^1 -continuous. $\square$

Theorem E.2 (symmetric, no first-order term: existence and uniqueness). *Assume (E-set) and that $A(x)$ is symmetric a.e. Let $\varphi \in H^1(\Omega)$ and $G \in L^2(\Omega; \mathbb{R}^n)$. Then there is a unique $W \in H^1(\Omega)$ with*

$$W - \varphi \in H_0^1(\Omega), \quad \int_{\Omega} A \nabla W \cdot \nabla \psi \, dx = \int_{\Omega} G \cdot \nabla \psi \, dx \quad \forall \psi \in H_0^1(\Omega), \quad (\text{E.3})$$

and $\|\nabla(W - \varphi)\| \leq \lambda^{-1}(\|G\| + \Lambda \|\nabla \varphi\|)$.

Proof. Facts invoked (**textbook-level, not entered in the ledger**): (C16-R) the Riesz representation theorem — a bounded linear functional ℓ on a real Hilbert space H is represented by a unique $v \in H$ as $\ell(w) = (v, w)_H$, with $\|v\|_H \leq \sup_{\|w\| \leq 1} |\ell(w)|$; (C16-S1) completeness of L^2 implies completeness of H^1 , and H_0^1 is a closed subspace, hence complete.

Step 1 (Hilbert structure). Put $H := H_0^1(\Omega)$ and $((v, w)) := \int_{\Omega} A \nabla v \cdot \nabla w$. It is bilinear, *symmetric* (the only place symmetry of A is used), $((v, v)) \geq \lambda \|\nabla v\|^2$ and $|((v, w))| \leq \Lambda \|\nabla v\| \|\nabla w\|$; and $((v, v)) = 0$ gives $\nabla v = 0$, hence $v = 0$ by Lemma E.1. So $((\cdot, \cdot))$ is an inner product, and by Lemma E.1

$$\lambda \|\nabla v\|^2 \leq \|v\|_H^2 \leq \Lambda \|\nabla v\|^2, \quad \|\nabla v\|^2 \leq \|v\|_{H^1}^2 \leq (1 + d^2) \|\nabla v\|^2, \quad (\text{E.4})$$

that is, $\|\cdot\|_H$ and the H^1 norm are equivalent on H_0^1 , and $(H, ((\cdot, \cdot)))$ is a Hilbert space (S1).

Step 2 (reduction to zero boundary values). Seek $W = \varphi + V$ with $V \in H$. Then (E.3) is equivalent to $((V, \psi)) = \ell(\psi) := \int_{\Omega} (G - A \nabla \varphi) \cdot \nabla \psi$ for all $\psi \in H$, and $|\ell(\psi)| \leq \lambda^{-1/2} (\|G\| + \Lambda \|\nabla \varphi\|) \|\psi\|_H$ is bounded.

Step 3 (existence). (C16-R) gives $V \in H$; put $W := \varphi + V$.

Step 4 (uniqueness, symmetry not used). The difference $Z \in H_0^1$ of two solutions satisfies $\int A \nabla Z \cdot \nabla \psi = 0$; taking $\psi = Z$ gives $\lambda \|\nabla Z\|^2 \leq 0$, and Lemma E.1 gives $Z = 0$.

Step 5 (energy estimate). Take $\psi = V$ in Step 2: $\lambda \|\nabla V\|^2 \leq \ell(V) \leq (\|G\| + \Lambda \|\nabla \varphi\|) \|\nabla V\|$, and divide by $\|\nabla V\|$ (the case 0 being trivial). $\square$

Remark. Theorem E.2 requires no regularity whatever of $\partial\Omega$ and no Hölder condition on A ; the boundary datum is given as $\varphi \in H^1(\Omega)$, so the trace theorem, $H^{1/2}$ extension and Lax–Milgram are never invoked. Parts (b) and (e) of Lemma E.38 in §E.4 apply this theorem with $\Omega = D_1$, $A = a_{\rho}$, $\varphi = x_1$, $G = 0$.

Theorem E.5 (non-symmetric with a small first-order term: uniqueness and an energy estimate in the data alone). *Assume (E-set) (symmetry not assumed), $b \in L^\infty(\Omega; \mathbb{R}^n)$ with $|b| \leq B$ a.e. and $Bd < \lambda$. Let $F \in L^2(\Omega; \mathbb{R}^n)$ and $g \in L^2(\Omega)$. Call $u \in H^1(\Omega)$ a weak solution if*

$$\int_{\Omega} A \nabla u \cdot \nabla \psi = \int_{\Omega} (b \cdot \nabla u) \psi + \int_{\Omega} F \cdot \nabla \psi - \int_{\Omega} g \psi \quad \forall \psi \in H_0^1(\Omega). \quad (\text{E.6})$$

(i) *If u_1, u_2 are weak solutions of (E.6) with $u_1 - u_2 \in H_0^1(\Omega)$ then $u_1 = u_2$.* (ii) *If u is a weak solution of (E.6) with $u - \varphi \in H_0^1(\Omega)$ and $\varphi \in H^1(\Omega)$ then*

$$(\lambda - Bd) \|\nabla(u - \varphi)\| \leq \|F\| + d\|g\| + (\Lambda + Bd) \|\nabla \varphi\|. \quad (\text{E.7})$$

The right-hand side involves the data only, not $\|u\|$. If (E.6) is required only for $\psi \in C_c^\infty(\Omega)$, then, both sides being H^1 -continuous functionals of ψ (as $A \nabla u$, $b \cdot \nabla u$, F and g lie in L^2), it holds for $\psi \in H_0^1$ by the definition of H_0^1 .

Proof. (ii) Put $V := u - \varphi \in H_0^1$, insert it in (E.6) and take $\psi = V$. The left-hand side is $\geq \lambda \|\nabla V\|^2$; the right-hand terms are estimated by Cauchy–Schwarz, $|b| \leq B$, $|A \nabla \varphi| \leq \Lambda |\nabla \varphi|$ and Lemma E.1: $|\int (b \cdot \nabla V) V| \leq Bd \|\nabla V\|^2$, $|\int (b \cdot \nabla \varphi) V| \leq Bd \|\nabla \varphi\| \|\nabla V\|$, $|\int g V| \leq d \|g\| \|\nabla V\|$, $|\int F \cdot \nabla V| \leq \|F\| \|\nabla V\|$ and $|\int A \nabla \varphi \cdot \nabla V| \leq \Lambda \|\nabla \varphi\| \|\nabla V\|$. Combining and dividing by $\|\nabla V\|$ (legitimate since $Bd < \lambda$) gives (E.7). (i) $Z := u_1 - u_2$ satisfies (E.6) with $F = g = 0$ and $Z \in H_0^1$; applying (ii) to Z with $\varphi = 0$ gives $\nabla Z = 0$, and Lemma E.1 gives $Z = 0$. $\square$

Remark. This appendix makes *no claim* of existence for (E.6) in the non-symmetric case or with $b \neq 0$ (Corollary E.19 does not need it). The residual external dependence of §E.1 is (C16-R) and (C16-S1), both textbook-level.

E.2 Global $C^{1,\alpha}$ regularity (divergence form, Hölder coefficients)

Notation and hypotheses. $n \geq 2$ and $\alpha \in (0, 1)$; $B_r^+ := B_r \cap \{x_n > 0\}$ and $\Gamma_r := B_r \cap \{x_n = 0\}$; $[f]_{\alpha;U}$ is the Hölder seminorm, and the $C^{1,\alpha}(\bar{U})$ norm is $\|u\|_\infty + \|\nabla u\|_\infty + [\nabla u]_\alpha$.

- **(E-ell)** $a : \Omega \rightarrow \mathbb{R}^{n \times n}$ is measurable, uniformly elliptic and bounded: $a\xi \cdot \xi \geq \lambda|\xi|^2$ and $|a\xi| \leq \Lambda|\xi|$ (*symmetry not assumed*).
- **(E-Höl)** $[a]_{\alpha;\Omega} < \infty$.
- **(E-b)** $b \in L^\infty(\Omega; \mathbb{R}^n)$ with $|b| \leq B$.
- **(E-data)** $[F]_\alpha < \infty$ and $g \in L^\infty(\Omega)$.

The operator is $Lu := \partial_i(a^{ij}\partial_j u) + b^i\partial_i u$, and $u \in H^1(\Omega)$ is a *weak solution* of $Lu = \operatorname{div} F + g$ if

$$\int_\Omega a^{ij}\partial_j u \partial_i \phi = \int_\Omega (b \cdot \nabla u) \phi + \int_\Omega F \cdot \nabla \phi - \int_\Omega g \phi \quad \forall \phi \in C_c^\infty(\Omega), \quad (\text{E.8})$$

which by H^1 -continuity holds automatically for $\phi \in H_0^1(\Omega')$, $\Omega' \Subset \Omega$. Since $\int F_0 \cdot \nabla \phi = 0$ for a constant vector F_0 , one may always assume $F(x_0) = 0$ at the current centre, so that only $[F]_\alpha$ appears in the estimates. *Boundary weak solutions:* $u \in H^1(B_1^+)$ is said to *take the value zero on Γ_1* if $\zeta u \in H_0^1(B_1^+)$ for every $\zeta \in C_c^\infty(B_1)$; and u is a *weak solution up to Γ_1* if it takes the value zero on Γ_1 and (E.8), integrated over B_1^+ , holds for every $\phi \in C_c^\infty(B_1)$ (test functions are *not* required to vanish near Γ_1). Write $\mathcal{A} := [a]_\alpha$, $\mathcal{B} := B$, $\mathcal{F} := [F]_\alpha$ and $\mathcal{G} := \|g\|_\infty$.

Theorem E.9 (interior $C^{1,\alpha}$). *Let $B_1 \subset \Omega$, let (E-ell), (E-Höl), (E-b) and (E-data) hold, let $u \in H^1(B_1)$ be a weak solution of (E.8) and $F(0) = 0$. Then ∇u has a representative in $C^{0,\alpha}(\overline{B_{1/2}})$ and*

$$\|\nabla u\|_{L^\infty(B_{1/2})} + [\nabla u]_{\alpha;B_{1/2}} \leq C(\|u\|_{L^2(B_1)} + \mathcal{F} + \mathcal{G}), \quad C = C(n, \lambda, \Lambda, \alpha, \mathcal{A}, \mathcal{B}). \quad (\text{E.10})$$

In scaled form, for $B_R(x_0) \subset \Omega$,

$$R\|\nabla u\|_{L^\infty(B_{R/2})} + R^{1+\alpha}[\nabla u]_{\alpha;B_{R/2}} \leq C\left(\left(\frac{1}{|B_R|} \int_{B_R} |u|^2\right)^{1/2} + R^{1+\alpha}[F]_\alpha + R^2\|g\|_\infty\right), \quad (\text{E.11})$$

with $C = C(n, \lambda, \Lambda, \alpha, [a]_\alpha R^\alpha, BR)$: the hypotheses on the coefficients enter only through dimensionless combinations.

Theorem E.12 ($C^{1,\alpha}$ up to a flat boundary). *Let $u \in H^1(B_1^+)$ be a weak solution of (E.8) in B_1^+ up to Γ_1 , let (E-ell), (E-Höl), (E-b) and (E-data) hold on B_1^+ , and let $F(0) = 0$. Then $\nabla u \in C^{0,\alpha}(\overline{B_{1/2}^+})$ and (E.10) holds with $B_{1/2}^+, B_1^+$ in place of $B_{1/2}, B_1$, with a constant of the same type; the scaled form is (E.11).*

Theorem E.13 (global $C^{1,\alpha}$ on a $C^{1,\alpha}$ domain). *Let Ω be a bounded open set with $\partial\Omega \in C^{1,\alpha}$ (near each boundary point a $C^{1,\alpha}$ graph $\{x_n > \gamma(x')\}$ with $\gamma(0) = 0$, $\nabla\gamma(0) = 0$ and $\Xi := \|\nabla\gamma\|_\infty + [\nabla\gamma]_\alpha$), let (E-ell), (E-Höl), (E-b) and (E-data) hold on Ω , let $\phi \in C^{1,\alpha}(\bar{\Omega})$, and let $u \in H^1(\Omega)$ satisfy $Lu = \operatorname{div} F + g$ weakly in Ω with $u - \phi \in H_0^1(\Omega)$. Then $u \in C^{1,\alpha}(\bar{\Omega})$ and*

$$\|u\|_{C^{1,\alpha}(\bar{\Omega})} \leq C(\|u\|_{L^2(\Omega)} + [F]_{\alpha;\Omega} + \|g\|_{L^\infty(\Omega)} + \|\phi\|_{C^{1,\alpha}(\bar{\Omega})}), \quad (\text{E.14})$$

with $C = C(n, \lambda, \Lambda, \alpha, [a]_\alpha, B, \Omega)$, the dependence on Ω being only through the number of local charts of a finite boundary atlas, their radii and Ξ .

Proof skeleton (shared by Theorems E.9, E.12 and E.13; routine manipulations are compressed to “a direct computation”). (1) *Caccioppoli.* For a weak solution u , $B_{2\varrho}(y) \subset B_R$, $k \in \mathbb{R}$ and $2\varrho B \leq 1$, testing with $\eta^2(u - k)$ (η a cut-off with $|\nabla \eta| \leq 2/\varrho$) and absorbing $\frac{3\lambda}{8} \int \eta^2 |\nabla u|^2$ by Cauchy–Schwarz and Young gives

$$\int_{B_{\varrho}(y)} |\nabla u|^2 \leq \frac{C}{\varrho^2} \int_{B_{2\varrho}} |u - k|^2 + C \int_{B_{2\varrho}} |F|^2 + C \varrho^2 \int_{B_{2\varrho}} |g|^2, \quad C = C(n, \lambda, \Lambda). \quad (\text{E.15})$$

Boundary version ($k = 0$, $z \in \Gamma_R$): the same inequality holds on $B_{\varrho}^+(z)$ — η is not required to vanish near Γ , and $\eta^2 u \in H_0^1(B_{2\varrho}^+)$ by the definition of “takes the value zero”.

(2) *Frozen-coefficient comparison (the load-bearing inequality).* Let $U = B_{\varrho}(y)$ or $B_{\varrho}^+(z)$, freeze at $x_{\star}$ (y or z) and put $A_0 := a(x_{\star})$; take $v \in u + H_0^1(U)$ a weak solution of the constant coefficient equation $\operatorname{div}(A_0 \nabla v) = 0$ (existence proved here by the direct method in Hilbert space, using the textbook-level fact (C16-S2): weak sequential compactness of bounded sequences and weak lower semicontinuity of the norm), and put $w := u - v \in H_0^1(U)$. Testing with w , using $F(x_{\star}) = 0$ and Lemma E.1 ($\|w\| \leq 2\varrho \|\nabla w\|$),

$$\lambda \|\nabla w\|_{L^2(U)} \leq ([a]_{\alpha; U} \varrho^{\alpha} + 2B\varrho) \|\nabla u\|_{L^2(U)} + [F]_{\alpha; U} \varrho^{\alpha} |U|^{1/2} + 2\varrho \|g\|_{L^2(U)}. \quad (\text{E.16})$$

Only the term $[a]_{\alpha} \varrho^{\alpha}$ supplies an algebraic rate as $\varrho \rightarrow 0$; if (E-Höl) were weakened to uniform continuity, the iteration would not deliver $C^{0, \alpha}$ (the source item exhibits a family of counterexamples, not reproduced here).

(3) *Decay for constant-coefficient solutions.* After symmetrisation of A_0 (the operator $\operatorname{div}(A_0 \nabla v)$ depends only on the symmetric part of A_0 ; a direct computation), a linear change of variables by a Cholesky or square-root factor (textbook-level (C16-S3)) turns v into a harmonic function; in the interior case one gets $\int_{B_r} |\nabla v|^2 \leq C_*(r/\varrho)^n \int_{B_{\varrho}} |\nabla v|^2$ and $\int_{B_r} |\nabla v - (\nabla v)_r|^2 \leq C_*(r/\varrho)^{n+2} \int_{B_{\varrho}} |\nabla v - (\nabla v)_{\varrho}|^2$. In the boundary case, first use an upper-triangular Cholesky factor, which preserves $\{x_n = 0\}$, then extend the harmonic function v vanishing on Γ by odd reflection ($\tilde{v} \in H^1(B_R)$, $\int_{B_R} |\nabla \tilde{v}|^2 = 2 \int_{B_R^+} |\nabla v|^2$, weakly harmonic — verified directly by splitting test functions into odd and even parts), which reduces to the interior case.

(4) *Iteration.* Put $\Phi_y(r) := \int_{B_r(y)} |\nabla u|^2$, $\Psi_y(r) := \inf_c \int_{B_r(y)} |\nabla u - c|^2$, $\theta(\varrho) := \mathcal{A}\varrho^{\alpha} + 2\mathcal{B}\varrho$ and $D^2 := \mathcal{F}^2 + \mathcal{G}^2$. From the squared form of (E.16) together with (3),

$$\Phi_y(r) \leq C_1 [(r/\varrho)^n + \theta(\varrho)^2] \Phi_y(\varrho) + C_1 D^2 \varrho^{n+2\alpha}, \quad (\text{E.17})$$

$$\Psi_y(r) \leq C_2 (r/\varrho)^{n+2} \Psi_y(\varrho) + C_2 \|\nabla w\|_{L^2(B_{\varrho})}^2, \quad (\text{E.18})$$

with $C_1, C_2 = C(n, \lambda, \Lambda)$. Choose $\varrho_1 := \min\{1/16, (\sqrt{\varepsilon_0}/2\mathcal{A})^{1/\alpha}, \sqrt{\varepsilon_0}/4\mathcal{B}, 1/2\mathcal{B}\}$ so that $\theta(\varrho)^2 \leq \varepsilon_0$, ε_0 being the threshold of the iteration lemma (proved here). Three stages: Morrey decay $\Phi_y(r) \leq Cr^{n-\alpha}$; then (E.18) gives first $\nabla u \in L^{\infty}$ and $\Phi_y(r) \leq Cr^n$; then (E.18) again gives $\Psi_y(r) \leq Cr^{n+2\alpha}$, and the Campanato characterisation (proved here, using the textbook-level Lebesgue differentiation theorem (C16-S1)) gives $\nabla u \in C^{0, \alpha}$. Splitting into two steps is forced by the bookkeeping of exponents: starting directly from $r^{n-\alpha}$ yields only the exponent $\alpha/2$. The boundary case runs word for word in parallel (using the boundary version of (E.15) and the odd extension of (3)).

(5) *Conclusion.* The constants of each stage are linear in $\int_{B_{3/4}} |\nabla u|^2 + D^2$; using (E.15) with $k = 0$ to replace $\|\nabla u\|_{L^2(B_{3/4})}$ by $\|u\|_{L^2(B_1)} + \mathcal{F} + \mathcal{G}$ gives (E.10). And (E.11) is (E.10) applied to $\hat{u}(x) = u(x_0 + Rx)$ (with data $\hat{a} = a(x_0 + R\cdot)$, $\hat{b} = Rb$, $\hat{F} = R(F - F(x_0))(x_0 + R\cdot)$, $\hat{g} = R^2g$), a direct computation.

(6) *Flattening, homogenisation and covering (Theorem E.13).* The flattening map $\Psi(x) = (x', x_n - \gamma(x'))$ has an explicit inverse, is bi-Lipschitz, has $\det D\Psi \equiv 1$ and $D\Psi \in C^{0, \alpha}$, and

satisfies $\|D\Psi^{\pm 1}\| \leq 1 + \Xi$ (no inverse function theorem is needed). The transformation rules $\tilde{a} = (D\Psi a D\Psi^\top) \circ \Psi^{-1}$, $\tilde{b} = (D\Psi b) \circ \Psi^{-1}$, $\tilde{F} = (D\Psi F) \circ \Psi^{-1}$ and $\tilde{g} = g \circ \Psi^{-1}$ preserve (E-ell), (E-Höl), (E-b) and (E-data), the constants depending on Ξ (a direct computation). Setting $\hat{u} := u - \phi$ reduces to zero boundary values ($\phi \in C^{1,\alpha}$ merely replaces F by $F - a\nabla\phi$ and g by $g + b \cdot \nabla\phi$, still satisfying (E-data)). Covering $\bar{\Omega}$ by finitely many interior balls and boundary charts, applying Theorem E.9 or E.12 on each and gluing the Hölder seminorms by the Lebesgue number of the finite cover gives (E.14). $\square$

Corollary E.19 (removing $\|u\|_{L^2}$). *Under the hypotheses of Theorem E.13, if in addition $2B \text{diam } \Omega < \lambda$, then $\|u\|_{L^2(\Omega)}$ may be deleted from the right-hand side of (E.14).*

Proof. Testing with $\hat{u} = u - \phi \in H_0^1$ and absorbing by (E-ell), Lemma E.1 (with $d = \text{diam } \Omega$) and $Bd < \lambda/2$ gives $\|\nabla\hat{u}\| \leq C(\|\nabla\phi\|_{L^2} + \|F\|_{L^2} + \|g\|_{L^2})$ — which is exactly Theorem E.5(ii); then Lemma E.1 gives $\|u\|_{L^2} \leq \|\phi\|_{L^2} + d\|\nabla\hat{u}\|$, to be inserted in (E.14). In general $\|u\|_{L^2}$ cannot be deleted for free (an operator with a general b need not be uniquely solvable). $\square$

Corollary E.20 (the form needed in the main text: $\nabla h \in L^\infty$ up to a flat boundary piece). *Let $n \geq 2$, let D be open and let $\Gamma \subset \partial D$ be a relatively open $C^{1,\alpha}$ boundary piece. Let a satisfy (E-ell) and (E-Höl), let $b, c \in L^\infty(D)$, and let $h \in H^1(D) \cap L^\infty(D)$ satisfy*

$$\partial_i(a^{ij}\partial_j h) + b^i\partial_i h + ch = 0 \quad \text{weakly in } D, \quad h \equiv \text{const} \quad \text{on } \Gamma. \quad (\text{E.21})$$

Then $h \in C^{1,\alpha}(\bar{D}')$ for every $D' \Subset D \cup \Gamma$; in particular

$$\|\nabla h\|_{L^\infty(D')} \leq C(n, \lambda, \Lambda, \alpha, [a]_\alpha, \|b\|_\infty, \|c\|_\infty, D', D) \|h\|_{L^\infty(D)}. \quad (\text{E.22})$$

Limitations of use (explicit conditions): (i) the conclusion is local — D' must stay at positive distance from the endpoints of Γ (in §7.2 of the main text, the two right-angled corners of the half-disc); the corners are not covered by this appendix. The main text uses (7.9) only in a neighbourhood of $p \in L$, and its wording must therefore read “in some neighbourhood of p ” and not “in B_R ”. (ii) $h \in L^\infty$ is necessary (it is what makes $g = -ch$ admissible); the main text supplies it through $\|h - 1\|_\infty = O(R^2)$.

Proof. $\tilde{h} := h - \text{const}$ takes the value zero on Γ and satisfies an equation of type (E.8) with $F := 0$ and $g := -ch \in L^\infty$. At each point of D' apply Theorem E.9 (interior points) or Theorem E.12 together with the flattening of step (6) of the skeleton (points of Γ), then take a finite cover. $\square$

Remark E.23 (the normalisation hypothesis: $\det a$ Lipschitz — a necessary condition before §7.2 of the main text may use §E.4 or §E.5). In two dimensions the transformation rule $\hat{a} = |\det D\Psi|^{-1} D\Psi a D\Psi^\top$ gives $\det \hat{a} = (\det D\Psi)^{2-n} \det a$, so for $n = 2$ the quantity $\det a$ is a coordinate invariant; and if a is symmetric positive definite with $\det a \equiv 1$ and the coordinates are isothermal ($\hat{a} = \sigma I$) and orientation-preserving, then $\sigma \equiv 1$ and $\hat{a} \equiv I$ (pure algebra). But the step “normalise $\det a$ to 1” is *not free*: with $m := (\det a)^{-1/2}$ and $\tilde{a} := ma$,

$$m \partial_i(a^{ij}\partial_j u) = \partial_i(\tilde{a}^{ij}\partial_j u) - (\partial_i m) a^{ij}\partial_j u, \quad (\text{E.24})$$

and the extra first-order coefficient involves ∇m , which is bounded if and only if $\det a$ is Lipschitz; under (E-Höl) alone, with $a \in C^{0,\alpha}$, ∇m is in general not a function and the normalisation is illegitimate. The coefficients of §7.2 of the main text are smooth and those of §5 satisfy $\tilde{a} \in C^{0,1}$, so the condition holds; but any attempt to lower the coefficient class to $C^{0,\alpha}$ breaks first at the normalisation. Moreover, normalising does *not* make the $\bar{w}$ term of the similarity principle (§E.5) disappear: $\sigma \equiv 1$ kills only the term $-\frac{1}{2}\partial_z \log \sigma$, while the drift $\hat{b}$ by itself produces $\partial_{\bar{z}} w = Aw + B\bar{w}$ with $A = -(\hat{b}_1 + i\hat{b}_2)/4$, $B = \bar{A}$ and $|B| = |A| = |\hat{b}|/4$; §E.5 must still handle $Aw + B\bar{w}$.

Residual external dependence of §E.2: (C16-S1) the Lebesgue differentiation theorem; (C16-S2) weak sequential compactness and weak lower semicontinuity of the norm in Hilbert space; (C16-S3) the spectral theorem and Cholesky factorisation; (C16-S4) mollification, dominated convergence, Young's inequality and the Lebesgue number of a finite cover — *all textbook-level, and not one of them a theorem of PDE*. Explicitly *not used*: De Giorgi–Nash–Moser, the weak Harnack inequality, Schauder theory, $W^{2,p}$ estimates, the trace theorem, the maximum principle, and the $C^{1,\alpha}$ inverse function theorem.

E.3 The strong minimum principle and the Hopf boundary point lemma (divergence form)

Hypotheses. $\Omega \subset \mathbb{R}^n$ is open; (E-ell) is as in §E.2 (measurable, uniformly elliptic and bounded, *symmetry not assumed*); (E-b) is as in §E.2 with $B_0 := \|b\|_\infty$. This section does *not* use the Hölder condition (E-Höl), and uses instead

- **(E-div) bounded row divergence.** For each j the distribution $\beta^j := \partial_i a^{ij}$ (that is, $\langle \beta^j, \psi \rangle = -\int a^{ij} \partial_i \psi$) is represented by a function $\beta^j \in L^\infty(\Omega)$; put $B_1 := \|\beta\|_\infty$ and $\tilde{B} := \|\beta + b\|_\infty \leq B_0 + B_1$.

(E-div) is strictly weaker than $a \in C^{0,1}$ (which implies it; a may be discontinuous everywhere and still satisfy (E-div)), and it is *not* implied by $a \in C^{0,\alpha}$. The coefficient $\tilde{a} \in C^{0,1}$ of §5 of the main text satisfies (E-div).

The operator is $\tilde{L}u := \partial_i(a^{ij}\partial_j u) + b^i\partial_i u$, with *no zeroth-order term* (load-bearing: constants are then solutions, so the minimum may be translated to 0, and the comparison principle needs no sign-controlled term cu ; the main text first moves $\tilde{c}w$ to the right-hand side and absorbs it into $-c_0/2$ on a small neighbourhood, which lands in the present case). A function $u \in H_{\text{loc}}^1(\Omega)$ is a *weak supersolution* ($\tilde{L}u \leq 0$) if

$$\int_{\Omega} a^{ij} \partial_j u \partial_i \phi \geq \int_{\Omega} b^i \partial_i u \phi \quad \text{for all } 0 \leq \phi \in C_c^\infty(\Omega); \quad (\text{E.25})$$

weak subsolutions are defined by the reverse inequality. By H^1 -continuity and mollification (which preserves positivity), (E.25) holds for $0 \leq \phi \in H_0^1(\Omega')$ with $\Omega' \Subset \Omega$.

Lemma E.26 (weak comparison principle, thin-domain version; only (E-ell) and (E-b) are needed). *Let $\Omega' \subset \Omega$ be open and bounded, contained in a slab of width d with $dB_0 < \lambda$. Let $u \in H^1(\Omega')$ satisfy $\tilde{L}u \geq 0$ weakly in Ω' and $u^+ \in H_0^1(\Omega')$. Then $u \leq 0$ a.e. in Ω' .*

Proof. Take $\phi = u^+$. By Stampacchia truncation (textbook-level: $Du^+ = \chi_{\{u>0\}} Du$), (E-ell), Cauchy–Schwarz and Lemma E.1, $\lambda \|Du^+\|^2 \leq B_0 \|Du^+\| \|u^+\| \leq B_0 d \|Du^+\|^2$; since $dB_0 < \lambda$ this forces $Du^+ = 0$, and Lemma E.1 gives $u^+ = 0$. $\square$

The thinness restriction is harmless: every comparison below is carried out on balls or annuli that may be shrunk at will, which avoids the weak maximum principle on general domains (that would need Sobolev embedding). Incidentally, if $z \in H^1(\Omega') \cap C^0(\overline{\Omega}')$ and $z \leq 0$ on $\partial\Omega'$, then $z^+ \in H_0^1(\Omega')$ (approximate by the compactly supported $(z^+ - \delta)^+$ and use dominated convergence; a direct computation).

Lemma E.27 (a divergence-form operator acting on a C^2 function is non-divergence form; the only use of (E-div)). *Assume (E-ell) and (E-div), let $U \subset \Omega$ be open and $v \in C^2(U)$. Then the distribution $\partial_i(a^{ij}\partial_j v)$ is represented by an L_{loc}^∞ function:*

$$\partial_i(a^{ij}\partial_j v) = a^{ij}\partial_{ij}v + \beta^j\partial_j v \quad \text{a.e. in } U; \quad \text{hence} \quad \tilde{L}v = a^{ij}\partial_{ij}v + c^j\partial_j v, \quad c := \beta + b, \quad |c| \leq \tilde{B}. \quad (\text{E.28})$$

In particular, if $a^{ij}\partial_{ij}v + c^j\partial_jv \geq 0$ a.e. in U , then $\tilde{L}v \geq 0$ weakly in U .

Proof. For $\phi \in C_c^\infty(U)$ one has pointwise $\partial_jv \partial_i\phi = \partial_i(\partial_jv \phi) - \partial_{ij}v \phi$, and the first term gives $\int \beta^j \partial_jv \phi$ by the definition of (E-div) (transferred to C_c^1 test functions by mollification). Last sentence: non-negativity of an L_{loc}^1 function as a distribution is equivalent to a.e. non-negativity. $\square$

With (E-ell) alone and without (E-div), $\tilde{L}v$ is in general only the distribution $\partial_i(a^{ij}\partial_jv)$ and not a function, so the comparison-function route breaks at its source — *this is exactly why divergence-form operators do not admit the non-divergence comparison functions verbatim, and why (E-div) must appear in the statements.*

Theorem E.29 (strong minimum principle). *Let Ω be open and connected and assume (E-ell), (E-div) and (E-b). Let $w \in H_{\text{loc}}^1(\Omega) \cap C^0(\Omega)$ satisfy $\tilde{L}w \leq 0$ weakly in Ω . If $m := \inf_\Omega w > -\infty$ and is attained at an interior point, then $w \equiv m$. Equivalently: a non-constant weak supersolution of $\tilde{L}$ does not attain its minimum in the interior. Local form: if w attains its minimum over D'' at an interior point of D'' , then w is constant on the connected component containing that point. If moreover $\tilde{L}w \leq -c_0/2 < 0$ weakly, then w cannot be constant, so a strict supersolution never attains its minimum in the interior at all.*

Proof. Assume $m = 0$. The set $Z := \{w = 0\}$ is closed and non-empty, so it suffices to show it is open. Suppose $p \in Z$ is not an interior point of Z .

(a) Choose r with $\overline{B_{4r}(p)} \subset \Omega$ and $4rB_0 < \lambda$; choose $q \in B_r(p)$ with $w(q) > 0$, put $d := \text{dist}(q, Z) \in (0, r)$ and take $x_0 \in Z$ with $|x_0 - q| = d$; then $w > 0$ in $B_d(q)$. The key point: with $\tau := \frac{3}{2}d$ one has $\overline{B_\tau(q)} \subset \Omega$ while x_0 is an interior point of $B_\tau(q)$. (The classical argument would use the Hopf lemma together with “an interior minimum forces a vanishing gradient”, but here w is only $H_{\text{loc}}^1 \cap C^0$ and need not be differentiable; the present proof instead pushes the outer radius beyond x_0 and derives a numerical contradiction, and does not invoke Theorem E.31.)

(b) Put $K := \overline{B_{d/2}(q)}$, $E := B_\tau(q) \setminus K$ and $\mu := \min_K w > 0$; then $w \geq \mu$ on ∂K and $w \geq 0$ on ∂B_τ .

(c) Comparison function $v := \varepsilon(e^{-\alpha\rho^2} - e^{-\alpha\tau^2})$ with $\rho = |x - q|$. By (E.28), a direct computation gives

$$\tilde{L}v = \varepsilon e^{-\alpha\rho^2} [4\alpha^2 a^{ij} \xi_i \xi_j - 2\alpha a^{ii} - 2\alpha c^j \xi_j] \geq 0 \text{ in } E, \text{ provided } \alpha > \frac{2(n\Lambda + 2\hat{B}d)}{\lambda d^2}, \quad (\text{E.30})$$

using $d/2 \leq \rho < 2d$, $a^{ij} \xi_i \xi_j \geq \lambda d^2/4$, $|a^{ii}| \leq n\Lambda$ and $|c^j \xi_j| \leq 2\hat{B}d$; fix such an α and take $\varepsilon e^{-\alpha d^2/4} \leq \mu$, so that $v \leq w$ on ∂E .

(d) Then $z := v - w \in H^1(E) \cap C^0(\overline{E})$ with $z \leq 0$ on ∂E , so $z^+ \in H_0^1(E)$; and $\tilde{L}z \geq 0$ weakly. Since E lies in a slab of width $2\tau < 4r$ and $4rB_0 < \lambda$, Lemma E.26 gives $w \geq v$ in E . At the interior point x_0 this reads $w(x_0) \geq \varepsilon(e^{-\alpha d^2} - e^{-\alpha\tau^2}) > 0$, contradicting $w(x_0) = 0$.

For the last sentence: a constant k has $\tilde{L}k = 0$, contradicting $\tilde{L}k \leq -c_0/2$ tested against $\phi \geq 0$, $\phi \not\equiv 0$. $\square$

Theorem E.31 (Hopf boundary point lemma). *Let Ω be open and assume (E-ell), (E-div) and (E-b). Let $x_0 \in \partial\Omega$ satisfy the interior ball condition: $B_{R_0}(y_0) \subset \Omega$ with $x_0 \in \partial B_{R_0}(y_0)$. Let $w \in H^1(B_{R_0}(y_0)) \cap C^0(B_{R_0}(y_0) \cup \{x_0\})$ satisfy $\tilde{L}w \leq 0$ weakly in $B_{R_0}(y_0)$ and $w(x) > w(x_0)$ for all $x \in B_{R_0}(y_0)$, and put $\nu := (y_0 - x_0)/R_0$. Then for every unit vector μ with $\theta := \mu \cdot \nu > 0$,*

$$\liminf_{t \rightarrow 0^+} \frac{w(x_0 + t\mu) - w(x_0)}{t} \geq 2\alpha\varepsilon R\theta e^{-\alpha R^2} > 0, \quad (\text{E.32})$$

where $R := \min\{R_0/2, \lambda/(4(B_0 + 1))\}$, $\alpha := 4(n\Lambda + \hat{B}R)/(\lambda R^2)$, $\varepsilon := (\min_{\partial B_{R/2}(y)} w - w(x_0))/(e^{-\alpha R^2/4} - e^{-\alpha R^2})$ and $y := x_0 + R\nu$. In particular the inner normal derivative has strictly positive lower limit; and if w is differentiable at x_0 then $\partial_\nu w(x_0) > 0$ and $Dw(x_0) \neq 0$. **The Hopf constant depends only on $n, \lambda, \Lambda, \hat{B}, R$, and w enters linearly, only through a positive lower bound for w on $\partial B_{R/2}(y)$. Neither differentiability of w nor any regularity of $\partial\Omega$ is required (a $C^{1,1}$ domain automatically satisfies the interior ball condition).**

Proof. (a) Assume $w(x_0) = 0$. Shrink the interior ball: with $B := B_R(y)$ one has $\bar{B} \setminus \{x_0\} \subset B_{R_0}(y_0)$ (indeed $|x - y_0| \leq R + (R_0 - R)$, with equality only at $x = x_0$), so $w \in C^0(\bar{B}) \cap H^1(B)$ with $w \geq 0$ on ∂B and $w > 0$ on $\bar{B} \setminus \{x_0\}$. Put $A := B_R(y) \setminus \bar{B}_{R/2}(y)$ and $m := \min_{\partial B_{R/2}(y)} w > 0$.

(b) Use the comparison function of (E.30) (with $\rho = |x - y|$ and R in place of τ): for $\alpha > 2(n\Lambda + \hat{B}R)/(\lambda R^2)$ one has $\tilde{L}v \geq 0$ weakly in A ; choose ε as displayed, so that $v = m \leq w$ on $\partial B_{R/2}$ and $v = 0 \leq w$ on ∂B_R .

(c) $2RB_0 < \lambda$ by the choice of R , so Lemma E.26 gives $w \geq v$ in A .

(d) $|x_0 + t\mu - y|^2 = R^2 - 2Rt\theta + t^2 \in (R^2/4, R^2)$ for $0 < t < \min\{2R\theta, R/2\}$, so $x_0 + t\mu \in A$, and $w(x_0 + t\mu)/t \geq v(x_0 + t\mu)/t \rightarrow \varepsilon e^{-\alpha R^2} \cdot 2\alpha R\theta$. $\square$

Remark. Theorems E.29 and E.31 use the same comparison function and the same comparison principle; they differ only in whether x_0 is an interior point of the comparison region (a numerical contradiction) or a boundary point (a rate). The residual external dependence of §E.3 is Stampacchia truncation, mollification, dominated convergence and Cauchy–Schwarz — all textbook-level. Explicitly *not used*: the weak Harnack inequality and De Giorgi–Nash–Moser.

E.4 Isothermal coordinates up to the boundary ($C^{0,\alpha}$ metric, without the Beltrami equation)

Notation. $D_\rho = \{|z| < \rho\} \subset \mathbb{R}^2$, $D_\rho^+ = D_\rho \cap \{y > 0\}$, $L_\rho = D_\rho \cap \{y = 0\}$; $\sigma(x, y) = (x, -y)$, $R = \text{diag}(1, -1) = D\sigma$ and $J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. A symmetric matrix field a is Λ -uniformly elliptic if $\Lambda^{-1}|\xi|^2 \leq a\xi \cdot \xi \leq \Lambda|\xi|^2$. Under a diffeomorphism $x = \Psi(\zeta)$ with $M = D\Psi$, the divergence-form weak formulation transforms by (a direct computation)

$$\hat{a} = |\det M| M^{-1}(a \circ \Psi)M^{-\top}, \quad \hat{b} = |\det M| M^{-1}(b \circ \Psi); \quad \text{in two dimensions } \det \hat{a} = \det a. \quad (\text{E.33})$$

A diffeomorphism F is an *isothermal coordinate* for a if F_*a is proportional to I ; by the invariance of $\det$ in (E.33), when $\det a \equiv 1$ one necessarily has $F_*a = I$. The normalisation $a \mapsto (\det a)^{-1/2}a$ changes neither the conformal structure nor the $C^{0,\alpha}$ class nor uniform ellipticity, so for the *construction* of isothermal coordinates one may assume $\det a \equiv 1$; this is not an extra hypothesis. Normalising the *equation* as well, however, requires $\det a$ to be Lipschitz (Remark E.23); that is a matter for the downstream Corollary E.40.

Theorem E.34 (isothermal coordinates, up to the boundary). *Let $\alpha \in (0, 1)$, $\Lambda \geq 1$, and let a be a symmetric matrix field on $\overline{D_1^+}$ satisfying (i) Λ -uniform ellipticity; (ii) $a \in C^{0,\alpha}(\overline{D_1^+})$ (up to the boundary, including L_1); (iii) $\det a \equiv 1$. Then there are $\rho_* = \rho_*(\alpha, \Lambda, [a]_\alpha) > 0$ and $F : \overline{D_{\rho_*}^+} \rightarrow \overline{\mathbb{H}}$ with $F(0) = 0$ such that*

- (1) (isothermal) $F_*a = I$; equivalently, $w = v \circ F^{-1}$ satisfies $\text{div}(a\nabla v) = 0 \iff \Delta w = 0$;
- (2) (boundary flattening) $F(L_{\rho_*}) \subset \mathbb{R}$ and $F(D_{\rho_*}^+) \subset \mathbb{H}$;
- (3) (regularity up to the boundary) $F \in C^{1,\alpha}(\overline{D_{\rho_*}^+})$, $F^{-1} \in C^{1,\alpha}$ and $\det DF \geq \frac{1}{2}$;

(4) (quantitative bi-Lipschitz) for every $\varepsilon \in (0, \frac{1}{2})$ one may choose ρ_* so that $(1 - \varepsilon)|z_1 - z_2| \leq |F(z_1) - F(z_2)| \leq (1 + \varepsilon)|z_1 - z_2|$ on $\overline{D_{\rho_*}^+}$.

The interior version (with z_0 an interior point) holds as well; its proof omits the two steps Lemma E.35 and Lemma E.36. If $\partial\Sigma$ is, in some $C^{1,\alpha}$ atlas, a curve $\{y = \gamma(x)\}$, precompose with $\Phi_0(x, y) = (x, y - \gamma(x))$ ($D\Phi_0 \in C^{0,\alpha}$, preserving $C^{0,\alpha}$); formula (7.7) of the main text already gives $\partial E(K) = \{r = 0\}$, so this step is not needed here. **Hypothesis (ii) requires only $C^{0,\alpha}$ with α arbitrarily small**, strictly weaker than the $C^{1,1}$ and $W^{1,p}$ ($p > 2$) of (R1) in the main text; but *this theorem does not claim that (R1) may be downgraded* — the downstream similarity principle still requires $W^{1,p}$. For merely measurable bounded coefficients the conclusion is false (the source item exhibits a counterexample, not reproduced here); whether Dini continuity suffices is *not settled*.

Proof. Three preparatory steps put a in normal form; then one solves a non-degenerate Dirichlet problem and takes a conjugate function. $\square$

Lemma E.35 (mollified shear; a substitute for Fermi coordinates). *Put $c(\xi) := a^{12}(\xi, 0)/a^{22}(\xi, 0)$, so $|c| \leq \Lambda^2$ and $c \in C^{0,\alpha}$ (extended to $\mathbb{R}$ by constants). The naive shear $(\xi + \eta c(\xi), \eta)$ is unusable — its Jacobian involves $\eta c'$ and c' does not exist. Take $\rho_1 \in C_c^\infty((-1, 1))$ with $\rho_1 \geq 0$ and $\int \rho_1 = 1$, put $\tilde{c}(\xi, \eta) := \eta^{-1} \int \rho_1((\xi - s)/\eta) c(s) ds$ for $\eta > 0$ and $\tilde{c}(\xi, 0) := c(\xi)$, and*

$$\Psi(\xi, \eta) := (\xi + \eta \tilde{c}(\xi, \eta), \eta). \quad (\text{E.36})$$

Then there is $C_S = C_S(\alpha, \rho_1)$ such that: (1) $\Psi|_L = \text{id}$ and $\Psi_2 = \eta$ (so $\{\eta = 0\}$ and the upper half-plane are preserved); (2) $\Psi \in C^{1,\alpha}(\{\eta \geq 0\})$, $\|D\Psi - (\begin{smallmatrix} 1 & \tilde{c} \\ 0 & 1 \end{smallmatrix})\|_{L^\infty(\{\eta < \epsilon\})} \leq C_S[c]_\alpha \epsilon^\alpha$, $[D\Psi]_\alpha \leq C_S[c]_\alpha$ and $D\Psi(\xi, 0) = (\begin{smallmatrix} 1 & c \\ 0 & 1 \end{smallmatrix})$; (3) there is $\epsilon_0(\alpha, \Lambda, [c]_\alpha)$ such that for $\epsilon \leq \epsilon_0$ one has $\det D\Psi \geq \frac{1}{2}$, Ψ is injective on $Q_{\delta, \epsilon} = (-\delta, \delta) \times [0, \epsilon)$ and its image contains $(-\delta/2, \delta/2) \times [0, \epsilon)$ (provided $\Lambda^2 \epsilon < \delta/2$), and $\Psi^{\pm 1}$ are Lipschitz; (4) if c is itself Lipschitz (a Lipschitz, the case of (R1)) one may take $\tilde{c} = c$ and obtain $\Psi \in C^{1,1}$.

Proof. Standard estimates for the mollification operator ($|\tilde{c} - c| \leq C[c]_\alpha \eta^\alpha$ and $|\nabla \tilde{c}| \leq C[c]_\alpha \eta^{\alpha-1}$, from $\int \rho'_1 = 0$ and the Hölder difference; a direct computation); (3) follows from (2) and the smallness of $\|D\Psi - (\begin{smallmatrix} 1 & \tilde{c} \\ 0 & 1 \end{smallmatrix})\|$, injectivity and the lower bound on the image coming from the integral mean value identity on a convex domain. $\square$

After the shear, $\hat{a} := |\det D\Psi|(D\Psi)^{-1}(a \circ \Psi)(D\Psi)^{-\top}$ on $Q_{\delta, \epsilon}$ is: of class $C^{0,\alpha}$ (that class being closed under products, inverses and composition with Lipschitz maps); uniformly elliptic (with constants depending only on Λ); of determinant $\det \hat{a} = 1$; and $\hat{a}^{12}(\xi, 0) = a^{12} - c a^{22} = 0$ (a direct computation). Next, a diagonal $M_0 = \text{diag}(\sqrt{\lambda}, 1/\sqrt{\lambda})$ (which commutes with R and preserves $\{\eta = 0\}$ and the upper half-plane) turns $\hat{a}(0) = \text{diag}(\lambda, \lambda^{-1})$ into I . Writing a again for the result, it satisfies on $\overline{D_{r_0}^+}$ the **normal form (E-norm)**: $a \in C^{0,\alpha}$, Λ' -elliptic, $\det a = 1$, $a^{12}|_L = 0$ and $a(0) = I$.

Lemma E.37 (reflection extension). *Let a satisfy (E-norm). Put $\tilde{a}(z) := a(z)$ for $\text{Im } z \geq 0$ and $\tilde{a}(z) := Ra(\sigma z)R$ for $\text{Im } z < 0$. Then $\tilde{a} \in C^{0,\alpha}(\overline{D_{r_0}})$ with $[\tilde{a}]_\alpha \leq 2[a]_\alpha$, symmetric, Λ' -elliptic, $\det \tilde{a} = 1$, $\tilde{a}(0) = I$, and reflection compatible: $\tilde{a}(\sigma z) = R\tilde{a}(z)R$.*

Proof. $M \mapsto RMR$ is a conjugation, preserving symmetry, spectrum and determinant. Continuity: from $a^{12}|_L = 0$ the matrix $a(p)$ is diagonal, so $Ra(p)R = a(p)$ and the two sides agree on L . Hölder continuity: for $z \in \overline{D^+}$ and $w \in D^-$, with p the intersection of the segment with $\mathbb{R}$, $|\tilde{a}(z) - \tilde{a}(w)| \leq [a]_\alpha(|z - p|^\alpha + |p - \sigma w|^\alpha) \leq 2[a]_\alpha|z - w|^\alpha$, using $|p - \sigma w| = |p - w|$. $\square$

The hypothesis $a^{12}|_L = 0$ cannot be dispensed with: $(RaR)^{12} = -a^{12}$, so without it $\tilde{a}$ jumps across L and is not even C^0 — this is precisely where the difficulty at the boundary lies, and it is the reason Lemma E.35 exists.

Lemma E.38 (existence of a non-degenerate solution). *Let $\tilde{a}$ on $\overline{D_{r_0}}$ be symmetric, Λ' -elliptic, $C^{0,\alpha}$, with $\tilde{a}(0) = I$ and reflection compatible. Then there are $\rho_* = \rho_*(\alpha, \Lambda', [\tilde{a}]_\alpha, \varepsilon) \in (0, r_0)$ and $u \in C^{1,\alpha}(\overline{D_{\rho_*}})$ with $\operatorname{div}(\tilde{a}\nabla u) = 0$ in D_{ρ_*} , $\|\nabla u - e_1\|_{L^\infty} \leq \varepsilon$ and $u \circ \sigma = u$; hence $\partial_y u \equiv 0$ on L_{ρ_*} .*

Proof. (a) Rescale: $a_\rho(z) := \tilde{a}(\rho z)$, so that $[a_\rho]_{\alpha; \overline{D_1}} = \rho^\alpha [\tilde{a}]_\alpha \leq [\tilde{a}]_\alpha$ (bounded uniformly in ρ , which is what makes the constants below independent of ρ , so that the argument is not circular) and $\|a_\rho - I\|_\infty \leq [\tilde{a}]_\alpha \rho^\alpha$. (b) Theorem E.2 (with $\Omega = D_1$, $A = a_\rho$, $\varphi = x_1$, $G = 0$) gives a unique $U \in H^1(D_1)$ with $U - x_1 \in H_0^1$ and $\operatorname{div}(a_\rho \nabla U) = 0$. (c) $W := U - x_1$ satisfies $\operatorname{div}(a_\rho \nabla W) = \operatorname{div} G$ with $G := -(a_\rho - I)e_1$ and $\|G\|_{C^{0,\alpha}} \leq 2[\tilde{a}]_\alpha \rho^\alpha$; by Theorem E.13 together with Corollary E.19 (∂D_1 is analytic, the boundary values vanish, $b = 0$, $g = 0$), $\|W\|_{C^{1,\alpha}(\overline{D_1})} \leq C_3(\alpha, \Lambda', [\tilde{a}]_\alpha) \rho^\alpha$; choose ρ_* making the right-hand side $\leq \varepsilon$. (d) Put $u(z) := \rho_*(U(z/\rho_*) - U(0))$. (e) Symmetry: $\sigma(D_1) = D_1$, the boundary datum satisfies $x_1 \circ \sigma = x_1$, and by reflection compatibility and (E.33) (with $M = R$) the σ -pushforward of a_ρ is itself; so $U \circ \sigma$ solves the same problem weakly, and the uniqueness part of Theorem E.2 gives $U \circ \sigma = U$. As u is even and C^1 , $\partial_y u|_L = 0$. $\square$

The conjugate function and isothermality (conclusion of the proof of Theorem E.34).

Put $p := \tilde{a}\nabla u$. Then $\operatorname{div} p = 0$ and D_{ρ_*} is simply connected, so Jp is a gradient field (the Poincaré lemma on a simply connected domain, textbook-level; on a disc, by direct integration): there is $v \in C^{1,\alpha}(\overline{D_{\rho_*}})$ with $v(0) = 0$ and

$$\nabla v = J \tilde{a} \nabla u; \quad F := (u, v). \quad (\text{E.39})$$

(a) $\det DF = \nabla u \cdot \tilde{a}\nabla u \geq (\Lambda')^{-1}(1 - \varepsilon)^2 > 0$. (b) Isothermality: the two-dimensional identity $J^\top A J = (\det A)A^{-1}$ together with $\det \tilde{a} = 1$ (necessary and sufficient) gives $DF \tilde{a} DF^\top = (\det DF)I$, that is $F_* \tilde{a} = I$. (c) $F(L) \subset \mathbb{R}$: on L one has $\partial_x v = -(\tilde{a}^{21} \partial_x u + \tilde{a}^{22} \partial_y u) = 0$ (since $\tilde{a}^{21}|_L = 0$ and $\partial_y u|_L = 0$), and $v(0) = 0$, so $v|_L \equiv 0$. (d) Reflection compatibility: $v^* := -v \circ \sigma$ has the same gradient as v (because $RJR = -J$) and the same initial value, so v is odd in y and $F \circ \sigma = \overline{F}$; on L , $\det DF = \partial_x u \partial_y v$ with $\partial_x u \geq 1 - \varepsilon$, so $\partial_y v > 0$; by injectivity v does not vanish on D^+ , and connectedness gives $v > 0$, that is $F(D^+) \subset \mathbb{H}$. (e) Bi-Lipschitz: $|\nabla v - e_2| \leq \Lambda' \varepsilon + [\tilde{a}]_\alpha \rho_*^\alpha$, so $\|DF - I\|_\infty \leq \varepsilon''$, and the integral mean value identity on a convex domain gives clause (4), in particular injectivity; $\det DF \geq \frac{1}{2}$; and $DF^{-1} = (DF)^{-1} \circ F^{-1}$ is the composition of a $C^{0,\alpha}$ map with a Lipschitz one, so $F^{-1} \in C^{1,\alpha}$. Restricting to $\overline{D_{\rho_*}^+}$ gives clauses (1)–(4). The total map is $F_{\text{tot}} = F \circ M_0 \circ \Psi^{-1}$ (precomposed with Φ_0 if necessary). $\square$

Corollary E.40 (the interface with (7.9) of the main text). *Write (7.9) as $\operatorname{div}(a\nabla v) + b \cdot \nabla v = 0$ in D_1^+ with $v = 0$ on L_1 , where a is symmetric and uniformly elliptic and a is **Lipschitz** (supplied by (R1) of the main text; hence $\det a$ is Lipschitz, and the normalisation of Remark E.23 adds only an L^∞ term to b), and $b \in L^\infty$. Then in the coordinate $\zeta = F_{\text{tot}}(z)$,*

$$\Delta \hat{v} + \hat{b} \cdot \nabla \hat{v} = 0 \quad \text{in } F_{\text{tot}}(D_{\rho_*}^+), \quad \hat{v} = 0 \quad \text{on } F_{\text{tot}}(L_{\rho_*}) \subset \mathbb{R}, \quad \hat{b} \in L^\infty, \quad (\text{E.41})$$

that is, $\sigma \equiv 1$ in the equation $\operatorname{div}(\sigma \nabla u) = 0$ of §12.2 of the main text. After the odd reflection $\hat{v}(x, -y) := -\hat{v}(x, y)$ the leading coefficient is still I and $\hat{b}$ becomes another L^∞ field (a jump across $\mathbb{R}$ being harmless); in the similarity principle the term $B = -\frac{1}{2} \partial_\zeta \log \sigma$ vanishes identically, but the $\bar{w}$ term does not thereby disappear (Remark E.23).

Proof. Theorem E.34 together with (E.33): $\hat{a} = I$ and $\hat{b} = |\det DF_{\text{tot}}^{-1}| DF_{\text{tot}}(b \circ F_{\text{tot}}^{-1}) \in L^\infty$. $\square$

Remark E.42 (the Fermi coordinate sentence must be deleted). The main text originally flattened the boundary by Fermi coordinates and asserted that the coefficients stay Lipschitz; under (R1) ($\Sigma \in C^{1,1}$) Fermi coordinates are only $C^{0,1}$ regular and their Jacobian is not differentiable, so that mechanism is unavailable. This appendix replaces it by the mollified shear of Lemma E.35, which needs only $a \in C^{0,\alpha}$. The corresponding sentence of §7.2 of the main text must be deleted accordingly (item E-4 of the list in §E.7).

Residual external dependence of §E.4: the self-contained items of §E.1 and §E.2 together with the Poincaré lemma on a simply connected domain and mollification estimates (textbook-level). Explicitly *not used*: the measurable Riemann mapping theorem, solvability and Schauder regularity for the Beltrami equation, the Beurling transform, Korn–Lichtenstein, and Morrey’s quasiconformal theory. Incidentally, the bi-Lipschitz constant of F_{tot} is not $1 \pm \varepsilon$ (that belongs to the normalised coordinate) but is bounded by an explicit constant of the form $\text{Lip}(\Psi^{-1})\text{Lip}(M_0^{-1})(1 + \varepsilon)$ depending on Λ — both are bi-Lipschitz, which is all the downstream argument needs.

E.5 The similarity principle and unique continuation

Notation and facts invoked. $z = x + iy$, $\partial_z = \frac{1}{2}(\partial_x - i\partial_y)$, $\partial_{\bar{z}} = \frac{1}{2}(\partial_x + i\partial_y)$ and $\partial_{\bar{z}}\partial_z = \frac{1}{4}\Delta$; $p > 2$ is fixed, $p' = p/(p-1)$ and $\alpha := 1 - 2/p$; the Cauchy transform is $Tf(z) := -\frac{1}{\pi} \iint_{\Omega} f(\zeta)(\zeta - z)^{-1} dA$. The **textbook-level** facts used in this section, stated in full:

- **(T1)** Cauchy–Pompeiu: $\partial_{\bar{z}}(1/\pi z) = \delta_0$; for $f \in L^1_c$, $\partial_{\bar{z}}Tf = f$ in $\mathcal{D}'$.
- **(T3)** The Beurling transform $S = \partial_z T$ is bounded on $L^q(\mathbb{C})$ for $1 < q < \infty$; hence $f \in L^p_c$ implies $Tf \in W^{1,p}_{\text{loc}}$, $\partial_{\bar{z}}Tf = f$ and $\partial_z Tf = Sf$ (Calderón–Zygmund).
- **(T5)** Morrey embedding: $W^{1,p}(\mathbb{R}^2) \hookrightarrow C^{0,1-2/p}$ for $p > 2$.
- **(T6)** Composition with a bi-Lipschitz map preserves $W^{1,q}$ and L^q , and the chain rule holds a.e.
- **(T9)** The C^1 implicit function theorem.
- **(T10)** Standard facts about $W^{1,2}_0$: C^∞_c is dense in $\{f \in W^{1,2} : \text{supp } f \Subset \bar{\Omega}, \text{ tr } f = 0\}$; a function Lipschitz on $\bar{\Omega}$ and vanishing on the boundary lies in $W^{1,2}_0$.
- **(T7)** The measurable Riemann mapping theorem (Ahlfors–Bers): if $\mu \in L^\infty$ has compact support and $\|\mu\|_\infty \leq k < 1$, then $\Psi_{\bar{z}} = \mu\Psi_z$ has a unique principal solution (a $W^{1,2}_{\text{loc}}$ homeomorphism with $\Psi(z) = z + O(1/z)$), which is a K -quasiconformal homeomorphism with $K = \frac{1+k}{1-k}$. Textbook-level, but used only along the (T8) route in Theorem E.51; the alternative route (Remark E.52) does not use it.
- **(T8)** Hölder Beltrami coefficient implies C^{1,α_0} (Bers–Nirenberg / Ahlfors): if $\mu \in C^{0,\alpha_0}_{\text{loc}}$ with $\|\mu\|_\infty \leq k < 1$, then any quasiconformal homeomorphism solution of $\Psi_{\bar{z}} = \mu\Psi_z$ lies in $C^{1,\alpha_0}_{\text{loc}}$ with $J_\Psi > 0$ everywhere, hence is a C^{1,α_0} diffeomorphism and locally bi-Lipschitz. **This one is at monograph level and is not proved in this appendix** (item H8).

Proved here, without external citation, by direct computation: the **Cauchy transform estimates** $\|Tf\|_\infty \leq C_p(\text{diam } \Omega)^{1-2/p}\|f\|_{L^p}$ and $[Tf]_{C^{0,1-2/p}} \leq C_p\|f\|_{L^p}$ (Hölder’s inequality, local integrability of $|\zeta|^{-p'}$, and scaling); **Weyl’s lemma** ($g \in L^1_{\text{loc}}$ with $\partial_{\bar{z}}g = 0$ equals a.e. a holomorphic function; mollification, the area mean value property and Morera); the **weak Leibniz rule** ($\phi \in C^0 \cap L^\infty$, $\partial_{\bar{z}}\phi \in L^p_{\text{loc}}$, $w \in L^q_{\text{loc}}$ with $\frac{1}{p} + \frac{1}{q} \leq 1$ and $\partial_{\bar{z}}w \in L^1_{\text{loc}}$ imply $\partial_{\bar{z}}(\phi w) = (\partial_{\bar{z}}\phi)w + \phi\partial_{\bar{z}}w$); and the **chain rule** ($s \in W^{1,p}_{\text{loc}} \cap L^\infty$ and G entire imply $\partial_{\bar{z}}G(s) = G'(s)\partial_{\bar{z}}s$).

Theorem E.43 ((V1) interior similarity principle, Vekua–Bers type). *Let $D \subset \mathbb{C}$ be a domain, $p > 2$, $A, B \in L^p_{\text{loc}}(D)$ and $w \in L^q_{\text{loc}}(D)$ for some $q \geq p'$, with*

$$\partial_{\bar{z}}w = Aw + B\bar{w} \quad \text{in } \mathcal{D}'(D). \quad (\text{E.44})$$

*Then for every disc $D' \Subset D$ there are $s \in C^{0,1-2/p}(\mathbb{C}) \cap L^\infty \cap W^{1,p}(D')$ and F holomorphic on D' with $w = e^s F$ a.e. in D' , and $\|s\|_\infty \leq C_p(\text{diam } D')^{1-2/p} \| |A| + |B| \|_{L^p(D')}$, $[s]_{C^{0,1-2/p}} \leq C_p \| |A| + |B| \|_{L^p(D')}$. **No gap; everything textbook-level.***

Proof. Put $\alpha(z) := A + B\bar{w}/w$ where $w \neq 0$ and $\alpha(z) := A$ where $w = 0$; it is measurable with $|\alpha| \leq |A| + |B|$, so $\alpha \in L^p(D')$, and $\partial_{\bar{z}}w = \alpha w$ in $\mathcal{D}'(D')$ (both sides agreeing pointwise as L^1_{loc} functions). This is the standard Bers–Vekua device of absorbing the $\bar{w}$ term into A ; Remark E.23 explains why it cannot be avoided. Put $s := T_{D'}\alpha$: by the Cauchy transform estimates and (T1), (T3), $s \in C^{0,1-2/p} \cap L^\infty \cap W^{1,p}(D')$ with $\partial_{\bar{z}}s = \alpha$. With $\phi := e^{-s}$, the chain rule gives $\partial_{\bar{z}}\phi = -e^{-s}\alpha \in L^p$, and the weak Leibniz rule gives $\partial_{\bar{z}}(e^{-s}w) = -e^{-s}\alpha w + e^{-s}\alpha w = 0$; Weyl’s lemma makes $F := e^{-s}w$ holomorphic, and $|e^s| \geq e^{-\|s\|_\infty} > 0$. $\square$

Corollary E.45 (zeros are isolated and of finite integer order; = (7.10) of the main text). *Under the hypotheses of Theorem E.43: (1) w equals a.e. a function in $C^{0,1-2/p}_{\text{loc}}(D)$; (2) on each connected component of D , either $w \equiv 0$, or the zeros of w are isolated and each zero z_0 has a unique finite integer order $m \geq 1$: $c_1|z - z_0|^m \leq |w(z)| \leq c_2|z - z_0|^m$ for $0 < |z - z_0| < \rho_0$; (3) locally $w = e^q G$ with q bounded and G holomorphic with a zero of order m at z_0 .*

Proof. e^s has no zeros, so $\{w = 0\} \cap D' = \{F = 0\}$; the set of points near which w vanishes identically is open and closed (closed: at a limit point F would have a non-isolated zero, forcing $F \equiv 0$); otherwise $F = (z - z_0)^m G_0$ with $c_1 = e^{-\|s\|_\infty} \inf |G_0|$ and $c_2 = e^{\|s\|_\infty} \sup |G_0|$; and (3) holds with $q = s$ and $G = F$. $\square$

Lemma E.46 (reduction to Vekua form, with drift). *Let $\sigma \in W^{1,p}_{\text{loc}}(D)$ with $0 < \sigma_0 \leq \sigma \leq \sigma_1$, let $b \in L^p_{\text{loc}}(D; \mathbb{R}^2)$ with $p > 2$, and let $u \in W^{1,2}_{\text{loc}}(D)$ be a real-valued weak solution of $\text{div}(\sigma \nabla u) + b \cdot \nabla u = 0$. Put $w := \sqrt{\sigma}(u_x - iu_y) = 2\sqrt{\sigma} \partial_z u$ and $\beta := b^1 + ib^2$. Then $w \in L^2_{\text{loc}}$ satisfies (E.44) with*

$$A = -\frac{\beta}{4\sigma}, \quad B = -\frac{1}{2}\partial_z \log \sigma - \frac{\bar{\beta}}{4\sigma}, \quad A, B \in L^p_{\text{loc}}. \quad (\text{E.47})$$

For $b \equiv 0$ one has $A = 0$ and $B = -\frac{1}{2}\partial_z \log \sigma$; the precise content of the correction (A-3) recorded in the main text is that $A = -\beta/(4\sigma) \neq 0$, and Theorem E.43 treats $A \neq 0$ and $A = 0$ alike. When $\sigma \equiv 1$ (Corollary E.40) one gets $B = -\bar{\beta}/4 = \bar{A}$, consistently with Remark E.23.

Proof. Put $v := \sigma(u_x - iu_y)$, so that $2\partial_{\bar{z}}v = (P_x + Q_y) + i(P_y - Q_x)$ with $P = \sigma u_x$ and $Q = \sigma u_y$. The equation gives $P_x + Q_y = -b \cdot \nabla u$, and $P_y - Q_x = \sigma_y u_x - \sigma_x u_y$ in the sense of distributions (by density of C_c^∞ in $W^{1,p}_c$ and continuity of the pairing with $u \in W^{1,2}$). Solving for u_x, u_y and substituting, then applying the weak Leibniz rule to $w = \sigma^{-1/2}v$, gives (E.47); a direct computation (the source item verified it symbolically, with zero residual). $\square$

Theorem E.48 ((V3) interior strong unique continuation, out of the ledger through Theorem E.43, replacing Aronszajn). *Let Ω be a connected two-dimensional C^1 manifold (without boundary, or the interior of a manifold with boundary), and let $u \in W^{1,2}_{\text{loc}}(\Omega)$ be real-valued and, in a covering family of charts, a weak solution of $\text{div}(\sigma \nabla u) + b \cdot \nabla u = 0$ with $\sigma \in W^{1,p}_{\text{loc}}$ bounded above and below by positive constants and $b \in L^p_{\text{loc}}$, $p > 2$. Then (a) (weak UCP) if u is constant, $= \kappa$, on some non-empty open set, then $u \equiv \kappa$ on Ω ; (b) (strong UCP) if $\int_{B_r(z_0)} |u - u(z_0)|^2 = O(r^N)$ for every N*

at some z_0 , then $u \equiv u(z_0)$. **The scope is exactly: two-dimensional, interior, connected, scalar second-order divergence form;** it gives no boundary unique continuation and nothing in dimension ≥ 3 . The coefficient thresholds are those of §12.2 of the main text, so the marginal cost of taking this item out of the ledger is zero.

Proof. Within a chart, Lemma E.46 and Theorem E.43 give $w = 2\sqrt{\sigma}\partial_z u = e^s F$. (a) If u is constant on an open set then F vanishes there, so $F \equiv 0$ and $\nabla u \equiv 0$ on the connected chart. (b) Suppose $w \not\equiv 0$ near z_0 . Corollary E.45 gives $\partial_z u = \frac{1}{2}\Theta z^m$ with $\Theta = e^s G_0/\sqrt{\sigma} \in C^{0,\alpha}$ and $\Theta(0) \neq 0$; since $u \in C^1$, integrating along line segments gives

$$u(z) - u(0) = \operatorname{Re}(c z^{m+1}) + O(|z|^{m+1+\alpha}), \quad c = \Theta(0)/(m+1) \neq 0, \quad (\text{E.49})$$

so on a sector of fixed opening $|u - u(0)| \geq \frac{1}{4}|c||z|^{m+1}$ and hence $\int_{B_r} |u - u(0)|^2 \geq c_4 r^{2m+4}$, contradicting the $O(r^N)$ hypothesis; then apply (a). Globally: $Z := \{u \equiv \kappa \text{ near the point}\}$ is open, closed by (a) within charts, and Ω is connected. $\square$

Check of the call sites in the main text. At (7.10)–(7.11) of §7.2 the function u is defined on a half-disc, and the “open set” is relatively open; it meets the interior in a non-empty open set, Theorem E.48(a) propagates on the connected component of the interior, and $u \in C^0(\overline{D^+})$ extends the conclusion to the closure — *one sentence must be added: a connected C^1 surface with boundary has connected interior*. At Lemma 7.2.A the phrase “used only on interior open sets” is covered word for word by (a) with $\kappa = 0$.

Lemma E.50 (odd reflection). *Let a on $\overline{D_1^+}$ be symmetric and uniformly elliptic (constants λ, Λ) with $a \in C^{0,\alpha_0}(\overline{D_1^+})$, $a^{12} = 0$ on L , $\sqrt{\det a} \in W^{1,p}(D_1^+)$ and $b \in L^p(D_1^+)$, and let $v \in W^{1,2}(D_1^+)$ be a weak solution of $\partial_i(a^{ij}\partial_j v) + b^i\partial_i v = 0$ with $\operatorname{tr} v|_L = 0$. Put $\Phi(x, y) = (x, -y)$ and $\tilde{v} := v$ for $y > 0$, $\tilde{v} := -v \circ \Phi$ for $y < 0$; extend a^{11}, a^{22} evenly, put $\tilde{a}^{12} := \operatorname{sgn}(y) a^{12}(x, |y|)$, extend b^1 evenly and put $\tilde{b}^2 := -\operatorname{sgn}(y) b^2(x, |y|)$. Then $\tilde{v} \in W^{1,2}(D_1)$ is a weak solution of $\partial_i(\tilde{a}^{ij}\partial_j \tilde{v}) + \tilde{b}^i\partial_i \tilde{v} = 0$ in D_1 ; $\tilde{a}$ is uniformly elliptic with the same constants, lies in $C^{0,\alpha_0}(D_1)$, satisfies $\sqrt{\det \tilde{a}} \in W^{1,p}(D_1)$ and $\tilde{b} \in L^p$; and $\tilde{v} \circ R = -\tilde{v}$ with $\tilde{a}(\sigma z) = R\tilde{a}(z)R$ (the reflection compatibility of Lemma E.37).*

Proof. Hölder continuity of $\tilde{a}^{12}$ follows from $a^{12}|_L = 0$ (the segment argument of Lemma E.37); without that hypothesis $\tilde{a}^{12}$ jumps across L and the reflected field is not even a continuous uniformly elliptic coefficient. Next, $\det \tilde{a} = \det a(x, |y|)$ is even, and an even reflection needs no trace condition, so $\sqrt{\det \tilde{a}} \in W^{1,p}$. That $\tilde{v} \in W^{1,2}$ is the standard odd extension ((T10)). Weak solution: for $\phi \in C_c^\infty(D_1)$, changing variables on D^- gives $\mathcal{B}(\tilde{v}, \phi) = 2\mathcal{B}^+(v, \phi_o|_{D^+})$ with ϕ_o the odd part, which is Lipschitz and vanishes near L and near the circular arc, hence lies in $W_0^{1,2}(D^+)$ ((T10)); and $\mathcal{B}^+$ is continuous on $W_0^{1,2}$ ($|\int b \cdot \nabla v \psi| \leq \|b\|_p \|\nabla v\|_2 \|\psi\|_{L^r}$, using $W_0^{1,2} \hookrightarrow L^r$ in two dimensions), so it vanishes. $\square$

Theorem E.51 ((V2) reflected similarity principle and the angular expansion; conditional on (T8), see Remark E.52). *Let a on $\overline{D^+}$ be symmetric and uniformly elliptic (constants λ, Λ), and assume*

- **(E-Höl)** $a \in C^{0,\alpha_0}(\overline{D^+})$ for some $\alpha_0 \in (0, 1]$;
- **(E-det)** $\sqrt{\det a} \in W^{1,p}(D^+)$ for some $p > 2$;
- **(E-a12)** $a^{12} = 0$ on L ;
- **(E-bp)** $b \in L^p(D^+; \mathbb{R}^2)$;

let $v \in W^{1,2}(D^+)$, $v \not\equiv 0$, be a weak solution of $\partial_i(a^{ij}\partial_j v) + b^i\partial_i v = 0$ with $\text{tr } v|_L = 0$, and let $z_0 \in L$. **Assume also that (T8) holds.** Then there are $\rho_0 > 0$, an integer $m_0 \geq 0$ and a C^{1,α_0} diffeomorphism Ψ from a neighbourhood of z_0 onto a neighbourhood of 0 (with $\Psi(L) \subset \mathbb{R}$, preserving the upper half-plane) such that $\hat{v} := v \circ \Psi^{-1}$ satisfies, with $\alpha = 1 - 2/p$ and $c \neq 0$ purely imaginary,

$$\hat{v}(\zeta) = \text{Re}(c\zeta^{m_0+1}) + O(|\zeta|^{m_0+1+\alpha}), \quad \partial_\zeta \hat{v}(\zeta) = \frac{1}{2}c(m_0+1)\zeta^{m_0} + O(|\zeta|^{m_0+\alpha}). \quad (\text{E.53})$$

Consequently (i) (= (7.10)) near z_0 one has $2\partial_\zeta \hat{v} = e^q G$ with q bounded and G holomorphic with a zero of order m_0 at 0; (ii) (= (7.11)) $|\nabla v(z)| \asymp |z - z_0|^{m_0}$ and $|v(z)| = O(|z - z_0|^{m_0+1})$ (in the original coordinate as well); (iii) (angular expansion) in $\overline{D^+} \cap \{|z - z_0| < \rho_0\}$ one has $\{v = 0\} = L \cup \gamma_1 \cup \dots \cup \gamma_{m_0}$, where the γ_k are C^1 simple arcs with endpoint z_0 , mutually disjoint apart from z_0 , with pairwise distinct tangent directions at z_0 (in the Ψ -coordinate, equiangular, $\theta = k\pi/(m_0+1)$); together with the two sides of L exactly $m_0 + 2$ arcs meet at z_0 .

§7.2 of the main text uses only (i) and (ii), **not** (iii). **Regularity claimed for the arcs:** C^1 simple arcs with a well-defined and pairwise distinct tangent direction at the endpoint; it is *not* claimed that the arcs are C^1 at z_0 .

Provenance and minimality of the hypotheses. $a \in W^{1,p}$ with $p > 2$ (the threshold of §12.2 of the main text) implies (E-Höl') and (E-det) by (T5), and so does the $C^{1,1}$ of (R1), so this theorem is a genuine relaxation; (E-det) is not implied by (E-Höl'); and (E-Höl') cannot be weakened to Dini continuity — it is there to serve (T8). Hypothesis (E-a12) is the sole geometric input of the whole reflection route: the main text originally obtained it from Fermi coordinates, a mechanism which does not exist under (R1) (Remark E.42), and it is **now supplied by the mollified shear of Lemma E.35** — a bi-Lipschitz $C^{1,\alpha}$ diffeomorphism, which preserves (E-Höl'), (E-det) and (E-bp) by (E.33) and (T6). For $b \equiv 0$ one could use the stream function together with a Stoilow factorisation and obtain (V3) for merely measurable coefficients; but (7.9) of the main text carries the first-order term $(a^{ij}\partial_j h + hb^i)\partial_i v$, for which no stream function exists, so that route is *unavailable*. One should not attempt to weaken (R1) to “merely measurable coefficients”.

Proof skeleton. Step 1. Reflect to the full disc by Lemma E.50.

Step 2 (a symmetric isothermal chart).

$$\mu := \frac{\tilde{a}^{22} - \tilde{a}^{11} - 2i\tilde{a}^{12}}{\tilde{a}^{11} + \tilde{a}^{22} + 2\sqrt{\det \tilde{a}}}, \quad \|\mu\|_\infty \leq k := \frac{\sqrt{\lambda_2} - \sqrt{\lambda_1}}{\sqrt{\lambda_2} + \sqrt{\lambda_1}} < 1, \quad \mu \in C^{0,\alpha_0} \quad (\text{E.54})$$

(diagonalise and compute directly: $|\mu| = (\sqrt{\lambda_2} - \sqrt{\lambda_1})/(\sqrt{\lambda_2} + \sqrt{\lambda_1})$, and $\mu \circ R = \bar{\mu}$ (since $\tilde{a}^{11}, \tilde{a}^{22}, \det \tilde{a}$ are even and $\tilde{a}^{12}$ is odd). Take a radial cut-off χ and let Ψ be the principal solution of $\Psi_{\bar{z}} = (\chi\mu)\Psi_z$ ((T7)); then $\Psi^*(z) := \overline{\Psi(\bar{z})}$ is also a principal solution, so uniqueness in (T7) gives $\Psi^* = \Psi$, whence $\Psi \circ R = R \circ \Psi$, $\Psi(\mathbb{R}) = \mathbb{R}$ and the two half-planes are preserved. **By (T8)**, Ψ is a C^{1,α_0} diffeomorphism and locally bi-Lipschitz. In the coordinate $\zeta = \Psi(z)$ the equation becomes (by (T6)) $\text{div}(\sigma \nabla \hat{v}) + \hat{b} \cdot \nabla \hat{v} = 0$ with $\sigma = \sqrt{\det \tilde{a}} \circ \Psi^{-1}$ and $\hat{b} = (D\Psi \tilde{b} / \det D\Psi) \circ \Psi^{-1}$ — the key algebraic fact being that $D\Psi \tilde{a} D\Psi^\top / \det D\Psi = \sqrt{\det \tilde{a}} I$ if and only if μ carries the sign of (E.54) (replacing the numerator by $a^{11} - a^{22} + 2ia^{12}$ makes it false; the source item verified this both symbolically and on 4000 numerical samples). Here $\sigma \in W_{\text{loc}}^{1,p}$ is bounded above and below by positive constants ((E-det) and (T6); *no* $\Psi \in W^{2,p}$ is required, since μ depends only on the conformal class $a/\sqrt{\det a}$ and σ is exactly the pullback of $\sqrt{\det a}$ along Ψ^{-1}), and $\hat{b} \in L_{\text{loc}}^p$ ($D\Psi \in L^\infty$ and $\det D\Psi \geq c > 0$ by (T8)); finally $\sigma \circ R = \sigma$ and $\hat{v} \circ R = -\hat{v}$.

Step 3. Lemma E.46, Theorem E.43 and Corollary E.45 give $2\sqrt{\sigma}\partial_\zeta \hat{v} = e^s \zeta^{m_0} G_0$; then $\Theta := e^s G_0 / \sqrt{\sigma} \in C^{0,\alpha}$ (as $\sigma \in W^{1,p} \subset C^{0,\alpha}$ by (T5)) and $\hat{v} \in C^1$, so integrating along line segments gives

(E.53) with $c = \Theta(0)/(m_0 + 1)$; substituting $\hat{v} \circ R = -\hat{v}$ in the expansion and letting $|\zeta| \rightarrow 0$ gives $\operatorname{Re} c = 0$. Part (ii) is obtained by transporting back through the locally bi-Lipschitz Ψ .

Step 4 (part (iii)). Without loss of generality $c = i|c|$, so that $P := \operatorname{Re}(c\zeta^{m_0+1}) = -|c|\rho^{m_0+1}\sin((m_0+1)\theta)$. (A) On a punctured neighbourhood $\nabla\hat{v} \neq 0$. (B) By (T9) the zero set coincides locally with $\mathbb{R}$ near $\mathbb{R} \setminus \{0\}$. (C) The rescalings $\hat{v}_r(\xi) := r^{-(m_0+1)}\hat{v}(r\xi)$ are within C^1 distance $\leq Cr^\alpha$ of P , so on the upper unit semicircle $\hat{v}_r$ has exactly m_0 transverse zeros, with arguments in ε -neighbourhoods of the θ_k . (D) By (A) and (T9) the zero set is a C^1 one-dimensional submanifold, transverse to each circle, so each branch is a graph $\theta = \theta_k(r)$ with $\theta_k(r) \rightarrow \theta_k$. (E) Since Ψ^{-1} is a C^{1,α_0} diffeomorphism with $D\Psi^{-1}(0)$ invertible ((T8)), simple C^1 arcs and distinct tangent directions transport back to the original coordinate; equiangularity is not preserved. $\square$

Remark E.52 (the status of (T8) and an alternative; the check against item H8). In Theorem E.51 hypothesis (T8) supports exactly two things: (1) that Ψ is locally bi-Lipschitz, whence $\hat{b} \in L^p$ and $|\nabla v| \asymp \rho^{m_0}$ can be transported back; and (2) that $D\Psi^{-1}(0)$ is invertible, whence the distinctness of tangent directions can be transported back. It occurs only in its *interior* version (the reflection is performed first, so the isothermal chart lies entirely inside an interior disc). **The alternative, checked:** the map Ψ needed in Step 2 can be supplied directly by §E.4 without any Beltrami equation — the field $\tilde{a}$ of Lemma E.50 satisfies exactly the reflection compatibility $\tilde{a}(\sigma z) = R\tilde{a}(z)R$ of Lemma E.37 and is C^{0,α_0} and uniformly elliptic; put $\tilde{a}_1 := (\det \tilde{a})^{-1/2}\tilde{a}$ (still C^{0,α_0} , reflection compatible, of determinant 1 — *the normalisation acts on the matrix field only and does not differentiate $\det \tilde{a}$*), use a diagonal M_0 to make $\tilde{a}_1(0) = I$, and then run Lemma E.38 and the conjugate construction (E.39) (which work on the full disc and need no boundary condition beyond $\tilde{a}_1^{12}|_L = 0$, supplied by (E-a12)); this produces $F \in C^{1,\alpha_0}$ with $F^{-1} \in C^{1,\alpha_0}$, $\det DF \geq \frac{1}{2}$, bi-Lipschitz, $F \circ \sigma = \overline{F}$, $F(\mathbb{R}) \subset \mathbb{R}$, preserving the upper half-plane, and $F_*\tilde{a}_1 = I$. By linearity in (E.33), $F_*\tilde{a} = (\sqrt{\det \tilde{a}} \circ F^{-1})I$, that is $\sigma = \sqrt{\det \tilde{a}} \circ F^{-1}$ — word for word the same as Step 2(d), with $\sigma \in W_{\text{loc}}^{1,p}$ still supplied by (E-det) and (T6). Both (T7) and (T8) may therefore be replaced by Theorem E.34 (the interior, reflection-symmetric version), and the external dependence of Theorem E.51 drops to the self-contained items of §E.1–§E.4 together with (T1), (T3), (T5), (T6), (T9), (T10), all textbook-level. **Honest statement:** this alternative check was carried out by the writing route; it is not in the source item and *has not been independently audited*. The statement of Theorem E.51 is therefore left “conditional on (T8)”, with the alternative recorded as a remark and entered in §E.7 as two ways of discharging H8, to be adjudicated by the audit route. If neither (T8) nor the alternative is adopted, then (V2) is *not* self-contained — this appendix does not claim that it is; it claims only that (V2) replaces item 4 of §13 of the main text (Alessandrini–Magnanini) by the interior version of item 11, which is already in the ledger, adding no independent source.

E.6 A corollary: two connected minimal surfaces agreeing on an open set agree

Corollary E.55. *Let A, B be two connected embedded minimal surfaces in the same three-manifold (smooth in the interior), agreeing on some non-empty open subset O of A . Then $A = B$ as interior point sets.*

Proof. The **textbook-level** facts used: (1) near each point a minimal surface is a smooth graph over its tangent plane; (2) for two minimal surfaces which are both graphs over $T_q A$, the difference u of the graph functions satisfies a two-dimensional linear divergence-form equation $\operatorname{div}(\sigma \nabla u) + b \cdot \nabla u = 0$ with smooth coefficients (subtract the minimal surface equations for the two graph functions and integrate the linearisation along the segment), hence meets the thresholds of Theorem E.48 ($\sigma \in W_{\text{loc}}^{1,p}$ bounded above and below by positive constants, $b \in L_{\text{loc}}^p$).

Put $Z := \{q \in A : A, B \text{ agree near } q\}$. Then $Z \supset O$ is non-empty, and open by definition. Closed: let $q \in \overline{Z} \cap A$ and take $q_j \in Z$ with $q_j \rightarrow q$. Since B is closed, $q \in B$; and the tangent planes converge in C^1 , so $T_q A = \lim T_{q_j} A = \lim T_{q_j} B = T_q B$. Hence near q both A and B are graphs over $T_q A$, and the difference u satisfies the equation of (2) and vanishes near Z (a non-empty open set); Theorem E.48(a) with $\kappa = 0$ gives $u \equiv 0$ on a connected neighbourhood of q , that is $q \in Z$. Since A is connected, $Z = A$ and $A \subset B$; symmetrically $B \subset A$ (O is an open subset of B as well). $\square$

Remark. This is the explicit proof of the sentence “two connected minimal surfaces agreeing on an open set agree”, passed over in one line after the unique continuation step of §7.2 of the main text; it is a same-family call site of Theorem E.48 and *adds no independent source*. Section 13 of the main text should register it as a call site of the unique-continuation entry, or cite this corollary directly.

E.7 What this appendix is still conditional on; call sites in the main text

E.7.1 The list of residual conditions.

#	Condition	Occurs in	Nature / how it may be discharged
H8	(T8) Hölder Beltrami coefficient implies a C^{1,α_0} diffeomorphism (Bers–Nirenberg / Ahlfors)	Theorem E.51	Monograph level, not proved here; <i>or</i> replaced, as in Remark E.52, by Theorem E.34 (interior, reflection-symmetric version) — that check was carried out by the writing route and has not been independently audited ; the two options stand side by side awaiting the audit route
H12	Same-family declaration: the whole appendix is Fable 5.1 output, with cross-engine audit 0/3	opening paragraph	A statement of discipline; criterion ③ not met
E-1	The boundary conclusion of Corollary E.20 is <i>local</i> : the two right-angled corners of the half-disc are not covered	§E.2	The main text uses (7.9) only near $p \in L$, and its wording must become “in some neighbourhood of p ”; convex corners are expected to be harmless but this is not proved
E-2	The normalisation hypothesis $\det a$ Lipschitz (Remark E.23): normalising $\det a$ to 1 at the level of the equation needs $\nabla(\det a)^{-1/2}$ bounded	§E.2 $\rightarrow$ §E.4, Corollary E.40	Satisfied: the coefficients of §7.2 of the main text are smooth and those of §5 satisfy $\tilde{a} \in C^{0,1}$; any use site lowering the coefficient class to $C^{0,\alpha}$ breaks here first
E-3	(E-div) bounded row divergence $\partial_i a^{ij} \in L^\infty$: a necessary hypothesis for the strong minimum principle and the Hopf lemma (Lemma E.27)	§E.3	Implied by $\tilde{a} \in C^{0,1}$ in §5 of the main text; <i>not</i> implied by $a \in C^{0,\alpha}$
E-4	The Fermi coordinate sentence must be deleted : §7.2 of the main text flattens the boundary by Fermi coordinates to obtain $a^{12} _L = 0$, a mechanism which does not exist under (R1) ($\Sigma \in C^{1,1}$) (Remark E.42)	§E.4, §E.5	(E-a12) is instead supplied by Lemma E.35 (mollified shear, needing only $a \in C^{0,\alpha}$); the corresponding sentence of the main text is deleted, and the “hypotheses of use” column of item 4 of §13 now cites Lemma E.35 or Corollary E.40

#	Condition	Occurs in	Nature / how it may be discharged
E-5	The sentence “a connected C^1 surface with boundary has connected interior”	§E.5, the half-disc call site of Theorem E.48	To be added to the main text (three lines, no new source)
E-6	Corollary E.55 is an unregistered independent step	§E.6	To be registered in §13 of the main text as a same-family call site of the unique-continuation entry; no new source
E-7	The regularity claimed for the arcs in the angular expansion (iii) of Theorem E.51: C^1 simple arcs with distinct tangent directions at the endpoint, not C^1 at the endpoint	§E.5	The main text uses only (i) and (ii); (iii) was written to discharge an audit commitment
E-8	Three questions left open in §E.2: whether Dini coefficients suffice, whether C^1 domains suffice, and whether “measurable bounded with $\operatorname{div} a \in L^\infty$ ” suffices	§E.2, §E.4	All undecided ; none affects the main text, whose coefficients are Lipschitz
E-9	The coefficient threshold “ $a^{ij} \in W^{1,p}$ ” of §12.2 of the main text may be relaxed to “ $a \in C^{0,\alpha_0}$ and $\sqrt{\det a} \in W^{1,p}$ ” (Theorem E.51)	§E.5	A genuine relaxation changing no existing conclusion; whether to adopt it is for the main text to decide, and this appendix alters no mathematics

E.7.2 Index of call sites in the main text.

Location	Fact needed	Supplied by
§5, Lemma 5.3 (strong minimum principle and Hopf boundary point lemma; $\tilde{L}$ without zeroth-order term, $\tilde{a} \in C^{0,1}$)	a non-constant weak supersolution has no interior minimum; a strict supersolution has none at all; at a boundary minimum the inner normal derivative is > 0	Theorems E.29 and E.31 ((E-div) supplied by $C^{0,1}$)
§7.2, (7.8)–(7.9): $Lh = 0$ on a small half-disc with $h _{\partial} = 1$, the assertion that the leading coefficient of (7.9) is Lipschitz and the first-order term bounded	$\nabla h \in L^\infty$ up to the flat boundary (locally)	Corollary E.20 (with limitations E-1, E-2)
§7.2, the normalisation $\det a \rightarrow 1$ in (7.9)	only bounded first-order coefficients change; the solution set does not	Remark E.23 (hypothesis: $\det a$ Lipschitz)
§7.2, the isothermal coordinate for (7.9); the σ of $\operatorname{div}(\sigma \nabla u) = 0$ in §12.2	a $C^{1,\alpha}$ isothermal coordinate up to the boundary, bi-Lipschitz, with $\sigma \equiv 1$	Theorem E.34, Corollary E.40 (Fermi sentence deleted, E-4)
§7.2, (7.10): $2\partial_z v = e^q G$	similarity principle and finite order of zeros	Theorem E.43, Corollary E.45; boundary version Theorem E.51(i)
§7.2, (7.11): $ dv \asymp \rho^m$, $ v = O(\rho^{m+1})$	reflected similarity principle	Theorem E.51(ii) (conditional on H8 or on Remark E.52)
§7.2, the unique continuation branch (“ u vanishes on an open set $\Rightarrow$ the two connected minimal pieces agree”)	interior weak UCP, and connected with boundary $\Rightarrow$ connected interior	Theorem E.48(a), Corollary E.55 (E-5, E-6)
Lemma 7.2.A (“used only on interior open sets”)	interior weak UCP with $\kappa = 0$	Theorem E.48(a)

Location	Fact needed	Supplied by
§12.2, hypothesis table	the list of residual conditions	§E.7.1
§13, items 4 / 7 / 10 / 11 (Alessandrini–Magnanini / Aronszajn / similarity principle / isothermal coordinates)	ledger bookkeeping	item 10 is proved here by Theorem E.43; item 7 leaves the ledger through Theorem E.48 (−1 use site, −1 independent source); item 4 is replaced by Theorem E.51 with the interior version of item 11; item 11 is proved here by Theorem E.34 (the (T8) route still needs its interior version, the route of Remark E.52 does not)

E.7.3 Summary of residual external dependence (all textbook-level; *no research-level source enters the ledger, and this appendix adds none*).

Subsection	Facts invoked	Explicitly not used
§E.1	(C16-R) Riesz representation; (C16-S1) completeness of H^1 and H_0^1	trace theorem, $H^{1/2}$ extension, Lax–Milgram
§E.2	(C16-S1) Lebesgue differentiation; (C16-S2) weak sequential compactness and weak lower semicontinuity; (C16-S3) spectral theorem and Cholesky; (C16-S4) mollification, dominated convergence, Young, Lebesgue number	De Giorgi–Nash–Moser, weak Harnack, Schauder, $W^{2,p}$, maximum principle, $C^{1,\alpha}$ inverse function theorem
§E.3	Stampacchia truncation, mollification, dominated convergence	weak Harnack, the weak maximum principle on general domains (Sobolev embedding)
§E.4	Poincaré lemma on a simply connected domain, mollification estimates, and the self-contained items of §E.1–§E.2	measurable Riemann mapping, solvability and Schauder theory for Beltrami, Beurling transform, Korn–Lichtenstein
§E.5 (V1),(V3)	(T1) Cauchy–Pompeiu; (T3) Calderón–Zygmund; (T5) Morrey; (T6) bi-Lipschitz composition	Aronszajn
§E.5 (V2)	the above together with (T9), (T10); (T7) , (T8) ((T8) at monograph level, item H8) or the alternative of Remark E.52	Alessandrini–Magnanini
§E.6	interior smoothness of minimal surfaces, the local graph representation, and the linear divergence-form equation satisfied by the difference	—

F Self-contained lemmas for the combinatorial part

On provenance: the contents of this appendix were produced by Fable 5.1 and reviewed three times within that same family; they have received **no cross-engine audit** (criterion ③ counts 0/3). See Appendix G.

This appendix transcribes the point-set and homotopy-theoretic lemmas used by §10 of the main text. It introduces no new mathematics and changes no statement of the main text; the four call sites of §10 have already been written to cite the lemmas below. Everything the project has *not*

proved — above all **(C1)** — is carried as an explicit hypothesis in the statements and is collected in §F.11; nothing unproved is presented as proved.

F.0 Conventions

Only the following are used: the definitions of the quotient and product topologies, compactness, paths and homotopies, and the CW weak-topology criterion (the convention of §9 of the main text: $U \subset |X|$ is open if and only if $U \cap |\sigma|$ is open in $|\sigma|$ for every closed simplex, and likewise for closed sets). “Nullhomotopic” without mention of a basepoint means freely homotopic to a constant map. $I = [0, 1]$. The lemmas are numbered F.1–F.9 in the order in which §10 of the main text calls them; forward references inside proofs are flagged where they occur. Lemmas and displayed formulas share the single counter F.n, the displays starting at (F.10).

F.1 The cone is contractible

Lemma F.1 (cone map of a simplicial cone). *Let X be a simplicial complex and v a new vertex not in X ; the simplicial cone $v * X$ has as its simplices those of X together with all $\{v\} \cup \sigma$ for $\sigma \in X$. Then*

- (a) $Q : |X| \times I \rightarrow |v * X|$, $Q(x, t) = (1 - t)x + tv$, is continuous, with $Q(x, 0) = x$ and $Q(x, 1) = v$;
- (b) the same formula $R(y, t) = (1 - t)y + tv$ defines a continuous map $R : |v * X| \times I \rightarrow |v * X|$ with $R_0 = \text{id}$ and $R_1 \equiv v$; hence $|v * X|$ is contractible.

Proof. (a) If $x \in |\sigma|$ has barycentric coordinates $(\lambda_u)_{u \in \sigma}$, the image point has coordinates $\mu_v = t$ and $\mu_u = (1 - t)\lambda_u$ in $|v * \sigma|$; these are non-negative and sum to 1, and the formula agrees on common faces. For each σ the coordinate formula is *polynomial* — affine in λ and affine in t separately, with the product terms $t\lambda_u$, so *not* jointly affine (this correction is recorded) — and therefore defines a continuous map $|\sigma| \times I \rightarrow |v * \sigma|$. By Lemma F.3 (stated below), Q is continuous on $|X| \times I$. The endpoint values are read off the formula.

(b) On each simplex τ of $v * X$ the same formula gives $\mu_v \mapsto (1 - t)\mu_v + t$ and $\mu_u \mapsto (1 - t)\mu_u$ (if $\tau \in X$ then $\mu_v = 0$ and the image lies in $|v * \tau|$; if $v \in \tau$ then $v * \tau = \tau$), continuous on each simplex. Lemma F.3, applied to the complex $v * X$, makes R continuous; the endpoint values are immediate. $\square$

F.2 Gluing (quotient maps and locally compact factors); condition (C1) made explicit

Lemma F.2. (a) *If $q : A \rightarrow B$ is a quotient map and K is locally compact Hausdorff, then $q \times \text{id}_K : A \times K \rightarrow B \times K$ is a quotient map.*

- (b) *Let $A \hookrightarrow B$ be a strong deformation retract inclusion (there are a retraction $r : B \rightarrow A$ and a deformation $H : B \times I \rightarrow B$ rel A , with $H_0 = \text{id}$, $H_1 = i \circ r$ and $H_t|_A = \text{id}$), and let $A \rightarrow Y$ be any continuous map. Then the inclusion $Y \hookrightarrow Y \cup_A B$ into the pushout is a homotopy equivalence.*

- (c) **Condition (C1)** [not proved within this project]: *if an inclusion $i : A \hookrightarrow B$ has the homotopy extension property (HEP) and is a homotopy equivalence, then it is a strong deformation retract inclusion. In the successor case of §10 of the main text (the call to Lemma 9.5(1)) one knows only that $L(v) \hookrightarrow v * L(v)$ has the HEP (Lemma F.8) and is a homotopy equivalence*

(Lemma F.1(b) together with the contractibility of $|L(v)|$); **(C1) is needed to enter (b)**. This appendix does not prove (C1) and derives nothing from it; it is listed in §F.11.

Proof. (a) Two point-set facts first.

First, given $k \in K$ and an open neighbourhood V_0 , there is an open $V \ni k$ with $\bar{V}$ compact and $\bar{V} \subset V_0$. Take a compact neighbourhood C of k and an open R with $k \in R \subset C$. A compact set in a Hausdorff space is closed: if $x \notin C$, separate x from each $c \in C$ by open sets, take a finite subcover by compactness, and intersect finitely many neighbourhoods of x to get one disjoint from C ; hence C is closed. Put $F := C \setminus V_0$, which is compact. For each $y \in F$ take disjoint open $P_y \ni k$ and $Q_y \ni y$; finitely many Q_{y_i} cover F . With $P := \bigcap_i P_{y_i}$ and $Q := \bigcup_i Q_{y_i}$ one has $\bar{P} \cap Q = \emptyset$. Put $V := P \cap R \cap V_0$. From $V \subset C$ and C closed, $\bar{V} \subset C$, a closed subset of a compact set and hence compact; and $\bar{V} \subset \bar{P}$ is disjoint from F , so $\bar{V} \subset V_0$.

Second, if C is compact then the projection $p_A : A \times C \rightarrow A$ is a closed map. Let $D \subset A \times C$ be closed and $a \notin p_A(D)$. For each $c \in C$ take open $U_c \ni a$ and $W_c \ni c$ with $(U_c \times W_c) \cap D = \emptyset$; finitely many W_{c_i} cover C , and $\bigcap_i U_{c_i}$ is an open neighbourhood of a disjoint from $p_A(D)$.

Now let $W \subset B \times K$ have $O := (q \times \text{id}_K)^{-1}(W)$ open. Take $(b, k) \in W$, choose $a \in q^{-1}(b)$, and take open $U_0 \ni a$, $V_0 \ni k$ with $U_0 \times V_0 \subset O$. By the first fact take $V \ni k$ with $C := \bar{V}$ compact and $C \subset V_0$. Put $N := \{x \in A : \{x\} \times C \subset O\}$. Since $(A \times C) \setminus O$ is closed in $A \times C$, the second fact makes $A \setminus N = p_A((A \times C) \setminus O)$ closed, so N is open, and $a \in N$. As O is saturated along the fibres of q in each K -slice ($q(x) = q(x')$ implies $(x, t) \in O \iff (x', t) \in O$), one has $N = q^{-1}(q(N))$, so $q(N)$ is open by the quotient topology, and $(b, k) \in q(N) \times V \subset W$ (using $V \subset C$ and the definition of N). Hence W is open; and $q \times \text{id}_K$ is surjective, so it is a quotient map.

(b) Take the identity on Y and the composite $B \xrightarrow{r} A \rightarrow Y$ on B ; they agree on A and yield $R : Y \sqcup_A B \rightarrow Y$ with $R \circ (Y \hookrightarrow Y \sqcup_A B) = \text{id}_Y$. In the other direction, take the constant homotopy on $Y \times I$ and H on $B \times I$; they agree on $A \times I$ (because $H_t|_A = \text{id}$) and give a map of sets $G : (Y \sqcup_A B) \times I \rightarrow Y \sqcup_A B$. The product of the pushout quotient map $Y \sqcup B \rightarrow Y \sqcup_A B$ with id_I is again a quotient map by (a) ($K = I$ is locally compact Hausdorff), and G pulls back to a continuous map on $(Y \sqcup B) \times I$; hence G is continuous, with $G_0 = \text{id}$ and $G_1 = (Y \hookrightarrow Y \sqcup_A B) \circ R$. $\square$

Remark. The continuity of the gluing step in (b) — “glue H to the constant homotopy on Y ” — is exactly what the last sentence of the proof of Lemma 9.5(1) of the main text needs and does not prove; part (a) supplies it. The (C1) gap is a separate matter, see (c).

F.3 Simplexwise continuity on cylinders

Lemma F.3. *Let X be a simplicial complex and $p : \bigsqcup_{\sigma \in X} |\sigma| \rightarrow |X|$ the surjection given by the natural maps of the closed simplices into the geometric realisation. Then $p \times \text{id}_I : \bigsqcup_{\sigma} (|\sigma| \times I) \rightarrow |X| \times I$ is a quotient map. Consequently a compatible family of maps which is continuous on each $|\sigma| \times I$ induces a continuous map on $|X| \times I$, and conversely.*

Proof. The CW weak-topology criterion says that $U \subset |X|$ is open if and only if $p^{-1}(U)$ is open, so p is a quotient map. As I is locally compact Hausdorff, Lemma F.2(a) makes $p \times \text{id}_I$ a quotient map, and the universal property of quotient maps gives the last sentence. This is precisely the assertion $|X| \times I = \text{colim}_{\sigma} (|\sigma| \times I)$; it is **not** assumed that $|X| \times I$ is itself a CW complex (for X not locally finite it need not be). $\square$

F.4 An extension criterion for spheres

Lemma F.4. *Let $m \geq 1$. A map $f : S^{m-1} \rightarrow Z$ extends to D^m if and only if f is nullhomotopic.*

Proof. Regard D^m as the closed unit ball. If $F : D^m \rightarrow Z$ extends f , then $H(u, t) = F((1 - t)u)$ is a homotopy from f to the constant $F(0)$.

Conversely, let $H : S^{m-1} \times I \rightarrow Z$ satisfy $H(u, 0) = f(u)$ and $H(u, 1) = c$. Define $F(0) := c$ and $F(ru) := H(u, 1 - r)$ for $0 < r \leq 1$. This is continuous away from the origin. At the origin: given an open neighbourhood U of c , each u has an open spherical neighbourhood V_u and some $a_u < 1$ with $H(V_u \times (a_u, 1]) \subset U$. The sphere is compact (it is a closed subset of a Euclidean closed cube, and the finite subcover property of a closed cube follows by coordinatewise bisection: if there were no finite subcover, one could keep choosing half-closed subcubes with no finite subcover, and the nested cubes have edge lengths tending to zero, so the coordinate endpoints converge monotonically to a common point, some open neighbourhood of which contains all sufficiently small subcubes — a contradiction). Hence finitely many V_{u_i} cover the sphere; with $a := \max_i a_{u_i} < 1$ one gets $F(x) \in U$ whenever $\|x\| < 1 - a$. So F is continuous at 0 and $F|_{S^{m-1}} = f$. The formula also covers $m = 1$: the two points of S^0 travel along their nullhomotopies to the same point, and together this extends over the closed interval. $\square$

F.5 From a single basepoint to free nullhomotopy

Lemma F.5. *Let Z be path connected and suppose there is $z_0 \in Z$ with $\pi_n(Z, z_0) = 0$ for all $n \geq 1$. Then every $f : S^n \rightarrow Z$ ($n \geq 0$) is freely nullhomotopic.*

Proof. $n = 0$: take paths from the images of the two points of S^0 to z_0 ; on the two components of $S^0 \times I$ they define a homotopy to a constant map.

$n \geq 1$: write $S^n = D^n / \partial D^n$ and regard f as a map $D^n \rightarrow Z$ taking one value a on the boundary. Take a path $\lambda : I \rightarrow Z$ with $\lambda(0) = z_0$ and $\lambda(1) = a$. With $r = \|x\|$, set $K(x, t) := f(2x)$ for $r \leq \frac{1}{2}$ and $K(x, t) := \lambda(1 - 2t(r - \frac{1}{2}))$ for $r \geq \frac{1}{2}$. Both formulas equal a on $r = \frac{1}{2}$, so K is continuous by gluing over two closed sets (the preimage of a closed set is closed in each, and a union of two closed sets is closed). On the boundary $K(x, t) = \lambda(1 - t)$, independent of x . The product of the quotient map $D^n \rightarrow D^n / \partial D^n$ with id_I is again a quotient map by Lemma F.2(a), so K descends to a continuous homotopy $\bar{K} : S^n \times I \rightarrow Z$.

$\bar{K}_0$ is freely homotopic to f : with $R_s := 1 - s/2$, put $L(x, s) := f(x/R_s)$ for $\|x\| \leq R_s$ and $L(x, s) := a$ for $R_s \leq \|x\| \leq 1$. The two formulas agree on the interface, the same closed-set gluing gives continuity, the value on the boundary is constantly a , and Lemma F.2(a) lets it descend to the sphere; at $s = 0$ it is f and at $s = 1$ it is $\bar{K}_0$.

On the other hand $g := \bar{K}_1$ takes the value z_0 on the collapsed boundary, so it is a based map, and $\pi_n(Z, z_0) = 0$ gives by definition a based homotopy from g to the constant z_0 . Reparametrising L , $\bar{K}$ and this homotopy linearly onto three closed subintervals of I and concatenating gives a free nullhomotopy of f . No identification of the homotopy groups at different basepoints has been made. $\square$

F.6 Basepoint correction

Lemma F.6. *Let $n \geq 1$ and let $f : (S^n, *) \rightarrow (Z, a)$ be a based map. If f is freely nullhomotopic, then it has a based nullhomotopy.*

Proof. Write $S^n = D^n / \partial D^n$, the collapsed boundary being the basepoint. Let $F : S^n \times I \rightarrow Z$ be a free nullhomotopy from f to a constant c and put $\alpha(t) := F(*, t)$, so $\alpha(0) = a$ and $\alpha(1) = c$. Precomposing F with the quotient map, regard it as a map $D^n \times I \rightarrow Z$ equal to $\alpha(t)$ on the boundary. With $r = \|x\|$ define $J(x, t) := F(2x, t)$ for $r \leq \frac{1}{2}$ and $J(x, t) := \alpha(t(2 - 2r))$ for $r \geq \frac{1}{2}$. Both formulas equal $\alpha(t)$ on $r = \frac{1}{2}$, and the value is constantly a on $r = 1$. Closed-set gluing and

Lemma F.2(a) let J descend to a homotopy keeping the spherical basepoint at a throughout. Its initial map J_0 compresses f into the ball of radius $\frac{1}{2}$ and fills the outer shell with a ; the formula L from the proof of Lemma F.5 (whose boundary value is a) gives a based homotopy from f to J_0 .

The terminal map J_1 is constantly c for $r \leq \frac{1}{2}$ and equals $\alpha(2 - 2r)$ on the outer shell. Define $P(x, s) := \alpha(1 - s)$ for $r \leq \frac{1}{2}$ and $P(x, s) := \alpha((1 - s)(2 - 2r))$ for $r \geq \frac{1}{2}$; the formulas agree on the interface and are constantly a on the boundary, with $P_0 = J_1$ and $P_1 \equiv a$; closed-set gluing and Lemma F.2(a) let it descend to a based homotopy. Concatenating L , the descended J and the descended P gives the assertion. $\square$

Remark. Lemmas F.5 and F.6 use the homeomorphism $S^n \cong D^n / \partial D^n$ without proof; the radial homeomorphism $|v * \sigma| \cong D^m$ constructed in the proof of Lemma F.7 (formula (F.10)) supplies it. The internal review classified this as a matter of wording rather than a substantive gap.

F.7 A replacement for Whitehead's theorem

Lemma F.7 (Whitehead's theorem in the special case “simplicial complex, target a point”, proved here). *Let X be a non-empty simplicial complex and give $|X|$ the CW weak topology. If $|X|$ is path connected and there is $x_0 \in |X|$ with $\pi_n(|X|, x_0) = 0$ for all $n \geq 1$, then $|X|$ is contractible.*

Proof. Take a new vertex v and the simplicial cone $v * X$, of which X is a subcomplex. Construct $G : |v * X| \rightarrow |X|$ by dimension $m = 0, 1, 2, \dots$ so that $G|_{|X|} = \text{id}$ and $G(v) = x_0$.

$m = 0$: put $G(v) := x_0$ and send each vertex of X to itself.

Let $m \geq 1$ and suppose G has been defined compatibly on all simplices of dimension at most $m - 1$. The m -simplices not in X are exactly the $v * \sigma$ with σ an $(m - 1)$ -simplex of X , and $\partial(v * \sigma) = \sigma \cup (v * \partial\sigma)$. The maps already defined on the finitely many closed faces are compatible; repeated gluing over finitely many closed sets gives a continuous map $S^{m-1} \rightarrow |X|$ on the boundary. For $m = 1$ apply the case $n = 0$ of Lemma F.5, and for $m \geq 2$ the case $n = m - 1$: the boundary map is freely nullhomotopic, so Lemma F.4 extends it over $|v * \sigma| \cong D^m$. The homeomorphism is this. Write the affine simplex as $\lambda_i \geq 0$, $\sum_{i=0}^m \lambda_i = 1$, put $c_i := 1/(m + 1)$, use the Euclidean norm on $\{\sum u_i = 0\}$, and for a unit vector u set

$$\rho(u) := \frac{1}{(m + 1) \max_i (-u_i)}. \quad (\text{F.10})$$

Since $\sum u_i = 0$ and $u \neq 0$, the denominator is positive; the maximum is continuous, so ρ is continuous. The ray $c + ru$ stays in the simplex exactly for $0 \leq r \leq \rho(u)$. The map $c + ru \mapsto (r/\rho(u))u$ (with $c \mapsto 0$) has inverse $su \mapsto c + s\rho(u)u$. The unit sphere is compact and ρ is continuous and positive, hence bounded above and below by positive constants, so both formulas are continuous at the origin as well; they are mutually inverse and carry the boundary of the simplex to the unit sphere.

For all the σ of this dimension choose an extension each, using the axiom of choice. Distinct new m -simplices meet only along boundaries already fixed, so the choices are compatible; on the m -simplices inside X keep the identity. This completes step m .

Every simplex has finite dimension, so after the induction over the natural numbers G is defined on all closed simplices. The restrictions are continuous and agree on common faces, so the CW weak-topology criterion gives a globally continuous G . No enumeration of simplices has been made, and no finiteness, local finiteness, countability, dimension or cardinality assumption has been placed on X .

Let Q be the cone map of Lemma F.1(a) and put $H := G \circ Q : |X| \times I \rightarrow |X|$. Then H is continuous with $H(x, 0) = G(x) = x$ and $H(x, 1) = G(v) = x_0$, so the identity of $|X|$ is freely homotopic to a constant map, that is, $|X|$ is contractible. $\square$

Remark (scope and external input). The statement of Lemma F.7 covers word for word the use made of Whitehead’s theorem in the limit case of §10 of the main text (“weakly contractible $\Rightarrow$ contractible”). The external facts used by F.1–F.7 are: the definitions of the quotient and product topologies, compactness, paths and homotopies; the CW weak-topology criterion (the convention of §9 of the main text); gluing over finitely many closed sets, closedness of compact sets in a Hausdorff space, and compactness of S^{m-1} (all proved above); and the **axiom of choice** (already registered in the main text). There is no new external source.

F.8 Subcomplex inclusions have the HEP (condition (C2))

Lemma F.8. *For any simplicial subcomplex $A \subset B$ the inclusion of realisations $|A| \hookrightarrow |B|$ has the homotopy extension property. No finiteness is required; the construction is canonical and **does not use the axiom of choice**.*

Proof. Induct over skeleta. Suppose the extension has been constructed on the $(n-1)$ -skeleton times I . For each n -simplex σ not in A the data are already defined on $E_\sigma := (|\sigma| \times \{0\}) \cup (\partial|\sigma| \times I)$. It suffices to have a continuous retraction $r_\sigma : |\sigma| \times I \rightarrow E_\sigma$ (fixing E_σ pointwise): composing the existing data with r_σ extends them over the prism.

Explicit construction of r_σ , using only that a simplex is a compact convex set with non-empty relative interior — *$|\sigma|$ is not identified affinely with a closed Euclidean ball*. Let $\ell_i \geq 0$ ($i = 0, \dots, n$) be the barycentric affine forms of the simplex, c the barycentre, $\beta_i := \ell_i(c) > 0$ and $\delta_i(x) := \ell_i(x) - \beta_i$; put $p := (c, h)$ with $h := 2$. Parametrise the ray from p through (x, t) as $(c, h) + s((x, t) - (c, h))$, $s \geq 0$, and take the *last point at which the ray leaves the prism (the exit point)*, **not** “the first point met in the reverse direction” (the reverse entry point can lie in the interior of the top face). The bottom face gives the constraint $s \leq h/(h-t)$, and where $\delta_i(x) < 0$ the i th lateral face gives $s \leq -\beta_i/\delta_i(x)$. Hence the exit parameter satisfies

$$\frac{1}{s^*(x, t)} = \max \left\{ \frac{h-t}{h}, \max_i \max (0, -\delta_i(x)/\beta_i) \right\} \geq \frac{h-1}{h} > 0. \quad (\text{F.11})$$

The right-hand side is the maximum of finitely many continuous functions and is bounded below by a positive constant, so s^* is continuous and $r_\sigma(x, t) = (c, h) + s^*(x, t)((x, t) - (c, h))$ is continuous. A finite maximum is attained by the bottom term or by some lateral term, so $r_\sigma(x, t)$ lands on the bottom face or on a lateral face. The top face yields only the lower bound $s \geq (h-1)/(h-t) \leq 1$, which is not an exit constraint in the region $s \geq 1$ where the ray continues forward (the height $h + s(t-h)$ is strictly decreasing in s , since $t \leq 1 < h$). If (x, t) already lies on the bottom or a lateral face, the corresponding term of (F.11) equals 1 while no term exceeds 1, so $s^* = 1$ and r_σ fixes E_σ .

On the simplices belonging to A use the given homotopy. The constructions agree on common faces with the previously defined data, hence are compatible. At each step n , Lemma F.3 upgrades continuity on each $|\sigma| \times I$ to continuity on that skeleton times I ; once all steps are done, the definition on each closed simplex was fixed at a finite stage, and Lemma F.3 (the closed-simplex criterion) gives the continuous extension over $|B| \times I$. $\square$

Remark. The internal review also noted that the “ $\Leftarrow$ ” direction of the usual equivalent characterisation of the HEP requires A to be closed in B ; this holds for subcomplexes by Lemma F.9(a), but is stated neither in the main text nor in the source item. This appendix uses the HEP only for subcomplexes, so the point is covered by F.9(a).

F.9 Realisation is a topological pushout; the topology of subcomplexes (condition (C4))

Lemma F.9. *Let $X = X' \cup (v * L)$ with $X' \cap (v * L) = L$, where X' and L are subcomplexes and $v \notin X'$. Then*

- (a) *for any subcomplex $Y \subset X$ the realisation $|Y|$ is a closed subspace of $|X|$, and the subspace topology inherited from $|X|$ coincides with its own CW weak topology;*
- (b) *$v * L$ is a subcomplex of X , with $|X| = |X'| \cup |v * L|$ and $|X'| \cap |v * L| = |L|$, and $|X| \cong |X'| \cup_{|L|} |v * L|$ as a topological pushout, the topology on the right-hand side being the weak topology of $|X|$.*

Proof. (a) For each closed simplex $|\sigma|$ of X the set $|Y| \cap |\sigma|$ is a union of finitely many faces of σ , hence closed; so $|Y|$ is closed by the weak-topology criterion for closed sets. Equality of topologies: one direction is immediate simplexwise. Conversely, if $C \subset |Y|$ has $C \cap |\tau|$ closed for every $\tau \in Y$, then for any $\sigma \in X$, $C \cap |\sigma| = \bigcup \{C \cap |\tau| : \tau \leq \sigma, \tau \in Y\}$; σ has only finitely many faces, and each $C \cap |\tau|$ is closed in the closed face $|\tau|$ and hence in $|\sigma|$, so the right-hand side is a finite union of closed sets and C is closed in $|X|$.

(b) The faces of $\{v\} \cup \sigma$ are $\{v\} \cup \sigma'$ or σ' with $\sigma' \leq \sigma$, so $v * L$ is a subcomplex. The union and intersection identities follow directly from those of the simplex sets. Let continuous maps $f' : |X'| \rightarrow Z$ and $f_v : |v * L| \rightarrow Z$ agree on $|L|$; they define a unique map of sets $f : |X| \rightarrow Z$, and every $\sigma \in X$ lies in X' or in $v * L$, so $f|_{|\sigma|}$ is a restriction of f' or of f_v and is continuous; the weak-topology criterion makes f continuous. This is exactly the universal property of the pushout. $\square$

F.10 Where §10 of the main text uses these lemmas

The following lemmas are *load-bearing* and not scaffolding: the argument of §10 already uses each conclusion, while the main text as it stood proved none of them.

Lemma	Load-bearing position in the main text	Consequence of its absence
F.1 (via F.3 and F.2(a))	successor case: “ $L(v) \hookrightarrow v * L(v)$ is a homotopy equivalence between two contractible complexes” — the <i>only</i> ground for the contractibility of $ v * L(v) $	the hypothesis of that sentence is unproved
F.2(a)	the last sentence of the proof of Lemma 9.5(1), “glue this deformation to the identity deformation of Y ”; and it supports F.3	the gluing map on the pushout cylinder is not shown continuous
F.3	supports F.1; four further direct call sites	the colimit criterion for $ X \times I$ has no source
F.6	limit case: “ Y_α is contractible, hence f is nullhomotopic; so $\pi_n(X , x_0) = 0$ ”	the passage from a free to a based nullhomotopy is missing — knowing only $x_0 \in Y_\alpha$ does not make an arbitrary retraction fix x_0
F.8, F.9	the HEP assertions for $L(v) \hookrightarrow v * L(v)$ and $L(v) \hookrightarrow X'$ in the successor case; “after the pushout along $L(v) \hookrightarrow X'$ ”; and, in the limit case, transporting a nullhomotopy in Y_α to $ X $	the main text gives one summary sentence, or none

Lemma	Load-bearing position in the main text	Consequence of its absence
F.4, F.5	used only by F.7	load-bearing only once Whitehead’s theorem is actually de-linked

Scope statement. The claim made early in the main text that §9–§10 use “only two combinatorial axioms, Whitehead’s theorem and the axiom of choice” does *not* hold as stated: the external inputs of §9–§10 are at least nine — the axiom of choice; Whitehead’s theorem; (C1); continuity of the gluing; basepoint correction; contractibility of the cone; (C2); (C4); the assertion that $|X|$ is a CW complex; together with the ZF facts about well-orderings. Of these, all but (C1) have proofs available within the project: the middle five are Lemmas F.1–F.9 above, while “ $|X|$ is a CW complex” and the well-ordering facts are standard assertions not transcribed here. The assertion that Theorem 10.1 does not depend on §1–§7 does hold for its proof chain.

F.11 What this appendix is still conditional on

#	Condition	Occurs in	Nature / status
1	(C1) HEP + homotopy equivalence $\Rightarrow$ strong deformation retract inclusion	Lemma F.2(c); Lemma 9.5(1) and the successor case of §10 of the main text	Not proved within this project. It is a gap of the main text itself; this appendix neither proves it nor derives anything from it, and lists it explicitly. It is registered in the hypothesis table of §12.2. An earlier internal repair attempt was withdrawn; a genuine repair would need the pushout–product properties of cofibrations
2	Whitehead’s theorem: de-linked through Lemma F.7, or retained as a textbook-level entry	Lemma F.7; the limit case of §10 and §13 of the main text	Adopted: Lemma F.7 covers the required special case word for word and introduces no new external source, so the main text argues through F.7. The alternative — keeping Whitehead’s theorem as a textbook-level entry and treating F.7 as a cross-check — is recorded as equally admissible. No aggregate ledger figure is asserted here; the counts are recomputed after assembly in §13
3	Same-family declaration: F.1–F.9 are all Fable 5.1 output, with cross-engine audit 0/3	the opening paragraph	A statement of discipline; criterion ③ not met

#	Condition	Occurs in	Nature / status
4	The axiom of choice (Lemma F.7 chooses extensions simultaneously in each dimension)	Lemma F.7	Already registered in the main text. After the correction recorded in §F.8, (C2) does <i>not</i> depend on it

Not listed as conditions, but to be kept in view: the assertion that $|X|$ is a CW complex, and the ZF facts about well-orderings, are standard claims of the main text and are not transcribed here; and the body of the successor case (the two-stage strong deformation retraction and its continuity estimates) is likewise not transcribed in this appendix, its own prerequisites being carried with it.

G AI usage

Statement. The mathematical argument of this paper was produced by an artificial-intelligence system, not merely checked or edited by one. The system is an autonomous research fleet of roughly forty agent processes; the language model behind every one of them is Fable 5.1 (model identifier `claude-fable-5-1`). **No cross-engine audit has been performed:** the writing route, the three internal review routes and every proof item cited below belong to that one model family, so the independent-audit criterion under which this project would call a theorem closed counts **0 of 3**, and none of the four closure criteria — complete proof, self-containment, three independent audits, machine-checked formalisation — is met. No formal verification is claimed. The rest of this appendix records what the system did and where its output was wrong.

What the system produced. The two-part architecture is the system’s: the observation that the combinatorial half (Theorem 10.1) can be separated from the geometric half, and that the exchange axiom needs only that an apex *exist* and never that one be *selected*, which is what lets the argument pass the known obstruction to a canonical choice. Both halves, all six technical appendices (A–F), the residual-hypothesis table and the two self-containment ledgers were written by the system. The appendices were produced by parallel routes working from separate proof items and were then assembled; the assembly, and not the individual routes, is where most of the inconsistencies listed below were found.

Reviews performed, and their limits. Each appendix was reviewed by at least one route with a fresh context, and the combinatorial part by three in sequence. These reviews found real errors — the list below is their output — but they are same-family reviews. They are recorded as evidence of what was checked, not as an independent audit, and they are not counted towards the three independent audits.

The system’s output was wrong in identifiable ways. The corrections are part of the record.

- The first route to ∂ -irreducibility argued through a compression, a sphere, a solid torus and a genus-one Heegaard splitting, which silently borrows the uniqueness of that splitting from the literature — a research-level input, and precisely the kind this paper is trying not to take. A second route through Dehn’s lemma reaches the same conclusion on a different dependence; Corollary D.22 is now supplied by Theorem D.15 along that second route, and the first is recorded but not used.
- A source item stated the normalisation theorem in a form allowing essential disc components, while the skeleton of the proof assumed there were none. Theorem D.23 is stated in the narrower form that the proof actually supports; the downstream surface is not a disc, so nothing is lost, but the wider statement was not proved and is not claimed.

- In the compactness appendix, the exclusion of the case $k = 1$ was written with “ x_3 is a bounded harmonic function”. For a single planar-type end only $w \perp e_1$ is known about the asymptotic normal w ; that $w = \pm e_3$ is not proved, so the bounded coordinate function is $\langle x, w \rangle$. The remark after Lemma B.19 is written with $\langle x, w \rangle$; the structure of the argument is unchanged, and the point is flagged for audit.
- A density estimate was written as an equality with the bound $e^{-CR/m_j} \Theta^*$, ignoring the metric distortion between normal coordinates and coordinate balls. Display (B.11) carries the corrected factor $(1 + c_M R/m_j)$ with c_M depending only on the ambient manifold; the limiting form is unaffected.
- An area budget in the double-curve argument was set at $\text{Area} < I + a/2$ against $\eta = a/8$, which yields $I + a/8$ and therefore no contradiction at all — the step did not prove what it claimed. It has been tightened to $I + a/4$, and Proposition A.19 is written in the tightened form.
- The construction of a retraction in the combinatorial part was wrong in three ways at once: it took the first point met in the reverse direction rather than the exit point (the reverse entry point can lie in the interior of the top face), it identified a simplex affinely with a closed Euclidean ball, and it gave no explicit formula. Lemma F.8 is written from the corrected version, with the explicit formula (F.11); the corrected construction is also canonical, so the conclusion that this condition does not need the axiom of choice comes from the correction and not from the original.
- The manuscript contradicted itself about Whitehead’s theorem: one section still invoked it verbatim while the ledger already recorded it as removed from the dependency list. Removal is a statement that the conditions for it are in place, not that it has happened. Lemma F.7 does cover the special case used, without a new external source, and the argument is now routed through F.7; the alternative of retaining Whitehead’s theorem is recorded in §F.11 as equally admissible.
- In the elliptic appendix one source item claimed that, after normalising the leading coefficient, the conjugate-linear term of the similarity principle comes only from the drift and effectively disappears; another item proved that it does not, being of the same size as the linear term. Remark E.23 and Lemma E.46 are written to the second, correct reading. A related claim, that normalising the determinant to 1 is free, holds for the construction of coordinates but not at the level of the equation, where it requires the determinant to be Lipschitz; that hypothesis is now stated where it is used.
- The main text flattened a boundary by Fermi coordinates and asserted that the coefficients stay Lipschitz. Under the regularity actually available that mechanism does not exist. It has been replaced by the mollified shear of Lemma E.35, which needs only Hölder coefficients, and the offending sentence is deleted.
- A task description asserted that a certain gap was closed by a theorem of Appendix C. It is not: that theorem closes half of it, and the remaining identity is unproved. It is recorded as unproved in §C.5 and carried in the residual-hypothesis table, under an explicitly stated reading of the definition rather than a silent repair. A second task description claimed to contain a regularity hypothesis verbatim; it did not, and the hypothesis is reconstructed from a different item and marked as not verbatim.

What is still open, and is recorded as open. Condition (C1) — that an inclusion with the homotopy extension property which is also a homotopy equivalence is a strong deformation retract inclusion — has never been proved within this project; an earlier internal repair was withdrawn. Theorem 10.1 is stated conditionally on it. The universal-disjointness input remains research-level. Several questions raised by the routes themselves are undecided: whether Dini continuity suffices for isothermal coordinates, whether C^1 domains suffice, what happens at the two right-angled corners, and whether the Bers–Nirenberg input can be discharged by the replacement constructed in Remark E.52 — that replacement was carried out by the writing route, is in none of the source items, and has not been independently audited. The residual-hypothesis table and the two ledgers of §13 carry these; no aggregate self-containment figure is asserted, and the two ledgers are not to be added together.

Formalisation. None. No part of this manuscript has been submitted to an interactive theorem prover, which is why no claim of formal verification is made above.